Franz W. Peren

Statistics for Business and Economics

Compendium of Essential Formulas

Springer

Statistics for Business and Economics

Franz W. Peren

Statistics for Business and Economics

Compendium of Essential Formulas

Springer

Franz W. Peren
Bonn-Rhein-Sieg University
Sankt Augustin, Germany

ISBN 978-3-662-64275-7 ISBN 978-3-662-64276-4 (eBook)
https://doi.org/10.1007/978-3-662-64276-4

This Springer imprint is published by the registered company Springer-Verlag GmbH, DE part of Springer Nature.
The registered company address is: Heidelberger Platz 3, 14197 Berlin, Germany

For my mother, Maria.

Preface

The following book is based on the author's expertise in the field of business statistics. After completing his studies in business administration and mathematics, he started his career working for a global bank and the German government. Later he became a professor of business administration, specialising in quantitative methods. He has been a professor at the Bonn-Rhein-Sieg University in Sankt Augustin, Germany since 1995, where he is mainly teaching business mathematics, business statistics, and operations research. He has also previously taught and conducted research at the University of Victoria in Victoria, BC, Canada and at Columbia University in New York City, New York, USA. To the author's best knowledge and beliefs, this formulary presents its statistical contents in a practical manner, as they are needed for meaningful and relevant application in global business, as well as in universities and economic practice.

The author would like to thank his academic colleagues who have contributed to this work and to many other projects with creativity, knowledge and dedication for more than 25 years. In particular, he would like to thank Ms. Eva Siebertz and Mr. Nawid Schahab, who were instrumental in managing and creating this formulary. Special thanks are given to Ms. Camilla Demuth, Ms. Linh Hoang, Ms. Michelle Jarsen, and Ms. Paula Schmidt. Should any mistakes remain, such errors shall be exclusively at the expense of the author. The author is thankful in advance to all users of this formulary for any constructive comments or suggestions.

Bonn, June 2021 — Franz W. Peren

Contents

List of Abbreviations

bln	billion
cm	centimetre(s)
CV	coefficient of variation
e	Euler’s number
ed.	edition, editor
e.g.	exempli gratia
Fig.	figure
G	geometric mean
GDP	gross domestic product
H	harmonic mean
i.e.	id est
lim	limit
max	maximum, maximise
Me	median
min	minimum, minimise
Mo	mode
mph	miles per hour
PCI	Peren-Clement index
QU	quantity units
R	range

rep.	repetition
Tab.	table
Var	variance
Vol.	volume
w/	with
w/out	without

Chapter 1

Statistical Signs and Symbols

General

Signs/Symbols	Meaning
$\mathbb{N}$	set of natural numbers $\{0, 1, 2, \ldots\}$ (formerly $\mathbb{N}_0$)
$\mathbb{Z}$	set of integers
$\mathbb{Q}$	set of rational numbers
$\mathbb{R}$	set of real numbers
$\mathbb{C}$	set of complex numbers
$a \geq b$	a is greater than or equal to b
$a \approx b$	a is approximately equal to b
$\sum_{i=1}^{n} a_i$	$a_1 + a_2 + \ldots + a_n$
$\prod_{i=1}^{n} a_i$	$a_1 \cdot a_2 \cdot \ldots \cdot a_n$
$\frac{\mathrm{d}y}{\mathrm{d}x} = y'(x)$	1st derivative of the function $y = y(x)$ with respect to the variable x
$\frac{\partial y}{\partial x}$	1st partial derivative of the function y with respect to the variable x
$\int$	integral
$\lvert a \rvert$	absolute value of a
$\lim\limits_{x \to a} f(x)$	limit of the function $f(x)$, with x converging towards a

F. W. Peren, *Statistics for Business and Economics*,
https://doi.org/10.1007/978-3-662-64276-4_1

Set Theory

Symbols	**Meaning**
A'	transposed matrix for A
$sgn(x)$	algebraic sign of x
$\{a_1,\ a_2,\ ...,\ a_n\}$	set of the elements $a_1,\ a_2,\ ...,\ a_n$
$\{x \mid B(x)\}$	set of all x, to which $B(x)$ applies
$\varnothing$, also $\{\}$	empty set (includes no elements)
$a \in A$	a is element of A
$a \notin A$	a is not element of A
$A = B$	A equals B
$A \subseteq B$, also $A \subset B$	A is subset of B
$A \subsetneq B$	A is proper subset of B
$A \supseteq B$, also $A \supset B$	A is superset of B
$A \cap B$	intersection of A and B
$A \cup B$	union of A and B
$A \backslash B$	relative complement of A and B
$\bar{A}$	complement of A
$A \times B$	cartesian product of A and B
$\phi(A)$	power set of A

Chapter 2

Descriptive Statistics

2.1 Empirical Distributions

2.1.1 Frequencies

The frequency distribution is a clear and meaningful summary, sorted by frequencies of results in the form of tables, graphs and statistical measurement figures (e.g. mean values, measures of dispersion).

If a statistical characteristic exists in k different characteristic values, $x_1, x_2, ..., x_k$, for which, given a population of N elements or a sample of n observations, the

absolute frequencies, h_i	$h_1, h_2, ..., h_k$	with	$0 \leq h_i \leq N$
		and	$\sum_{i=1}^{k} h_i = N$
		or	$\sum_{i=1}^{k} h_i = n$

are given, this results in the corresponding

relative frequencies, f_i	$f_1, f_2, ..., f_k$	with	$0 \leq f_i \leq 1$
		and	$\sum_{i=1}^{k} f_i = 1$
	$f_i = \frac{h_i}{N}$	or	$f_i = \frac{h_i}{n}$

F. W. Peren, *Statistics for Business and Economics*,
https://doi.org/10.1007/978-3-662-64276-4_2

Example: Height of 100 students

i	Height $[cm]$	h_i	f_i
1	under 160	9	$0.09 = \frac{9}{100}$
2	$[160 - 170[$	28	0.28
3	$[170 - 180[$	35	0.35
4	$[180 - 190[$	24	0.24
5	$190 \leq$	4	0.04
$\sum$	-	100	1.0

2.1.2 Cumulative Frequencies

By continuous summation (cumulation) of the absolute frequencies h_j, one obtains the *absolute* cumulative frequencies H_i.

$$H_i = h_1 + h_2 + ... + h_i \qquad j = 1, ..., i$$

$$= \sum_{j=1}^{i} h_j$$

By continuous summation of the relative frequencies f_j, one obtains the *relative* cumulative frequencies F_i.

$$F_i = f_1 + f_2 + ... + f_i$$

$$= \sum_{j=1}^{i} f_j \qquad j = 1, ..., i$$

$$= \frac{H_i}{N} \quad \Rightarrow \text{ for the population}$$

$$= \frac{H_i}{n} \quad \Rightarrow \text{ for the sample}$$

Cumulative Frequency Function for Ungrouped Data

$$F(x) = \begin{cases} 0 & \text{for} \quad x < x_1 \\ F_i & \text{for} \quad x_i \leq x \leq x_{i+1} \qquad \text{with} \quad i = 1, ..., k-1 \\ 1 & \text{for} \quad x \geq x_k \end{cases}$$

Example: Number of newspapers regularly read by students

i	x_i	h_i	f_i	H_i	F_i
1	0	200	0.160	200	0.160
2	1	510	0.407	710	0.567
3	2	253	0.202	963	0.769
4	3	163	0.130	1,126	0.899
5	4	127	0.101	1,253	1.0
$\sum$	-	1,253	1.0	-	-

Cumulative Frequency Function for Grouped Data

The accumulated (absolute or relative) cumulative frequencies are each assigned to the ends of the class interval.

Example: Height of 100 students

i	Height $[cm]$	h_i	f_i	H_i	F_i
1	under 160	9	0.09	9	0.09
2	$[160 - 170[$	28	0.28	37	0.37
3	$[170 - 180[$	35	0.35	72	0.72
4	$[180 - 190[$	24	0.24	96	0.96
5	$190 \leq$	4	0.04	100	1.0
$\sum$	-	100	1.0	-	-

2.2 Mean Values and Measures of Dispersion

2.2.1 Mean Values

Arithmetic Mean (μ or $\bar{x}$)

Definition for the population N with x_1, x_2, ..., x_N

$$\mu = \frac{1}{N}(x_1 + \ldots + x_N) = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Definition for a sample in the range of n with x_1, x_2, ..., x_n

$$\bar{x} = \frac{1}{n}(x_1 + \ldots + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Frequency Distributions

- absolute frequency distributions

$$\mu = \frac{1}{N}\sum_{i=1}^{k} x_i h_i = \frac{1}{N}(x_1 h_1 + x_2 h_2 + \ldots + x_k h_k)$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{k} x_i h_i = \frac{1}{n}(x_1 h_1 + x_2 h_2 + \ldots + x_k h_k)$$

- relative frequency distributions

$$\mu = \sum_{i=1}^{k} x_i f_i \qquad \text{with} \qquad f_i = \frac{h_i}{N}$$

$$\bar{x} = \sum_{i=1}^{k} x_i f_i \qquad \text{with} \qquad f_i = \frac{h_i}{n}$$

- with a frequency distribution of *grouped data*, the following applies

for the population:

$$\mu = \frac{1}{N}\sum_{i=1}^{k} x'_i h_i \;=\; \sum x'_i f_i$$

Usually the centre of the class interval is chosen.

for the sample:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{k} x'_i h_i \;=\; \sum x'_i f_i$$

Usually the centre of the class interval is chosen.

Example:

i	x_i sleep per night	h_i people	f_i	F_i	H_i
1	8	3	0.176	0.176	3
2	6	1	0.059	0.235	4
3	7	7	0.412	0.647	11
4	10	4	0.235	0.882	15
5	4	2	0.118	1	17
$\sum$	-	17	-	-	-

arithmetic mean

$$\mu = \frac{1}{17}\,(8\cdot 3 \,+\, 6\cdot 1 \,+\, 7\cdot 7 \,+\, 10\cdot 4 \,+\, 4\cdot 2)$$

$= 7.471$ hours

On average, the respondents slept 7.471 hours per night.

Median (Me)

The single values $x_1, x_2, ..., x_N$ are ordered, so that the following applies:

$x_{[1]} \leq x_{[2]} \leq ... \leq x_{[N]}$ with $x_{[j]} =$ the element x at the j^{th} position; $j = 1, ..., N$

Median for **uneven** N: $$Me = x_{\left[\frac{N+1}{2}\right]}$$

<u>Example</u>:

Height of five children in cm: 120, 150, 110, 124, 132

Sorting the values: 110, 120, 124, 132, 150

Calculating median: $\frac{N+1}{2}$

$\Rightarrow \frac{5+1}{2} = 3^{rd}$ observation

$\Rightarrow x_3 = 124$

The median is 124 cm.

Median for **even** N: $$Me = \frac{1}{2}\left(x_{\left[\frac{N}{2}\right]} + x_{\left[\frac{N}{2}+1\right]} \right)$$

<u>Example</u>:

Height of six children in cm: 131, 124, 135, 115, 119, 126

Sorting the values: 115, 119, 124, 126, 131, 135

Calculating median: $\frac{1}{2}\left(\frac{6}{2} + \frac{6}{2} + 1\right)$

$= \frac{1}{2}(x_3 + x_4)$

$= \frac{1}{2}(124 + 126)$

$$= \frac{1}{2} \cdot 250$$

$$= 125$$

The median is 125 cm.

Frequency Distributions

For *ungrouped data*, the median is equal to the characteristic value x_i, for which the cumulative frequency function $F(x)$ exceeds the value 0.5.

For *grouped data*, the median is calculated using the class interval's lower limit x_i^l and the class interval's upper limit x_i^u of the class, in which the cumulative frequency function $F(x)$ exceeds the value 0.5.

$$\begin{aligned}
Me &= x_i^l + \alpha \\
\Rightarrow \quad & \frac{\alpha}{0.5 - F(x_i^l)} = \frac{x_i^u - x_i^l}{F(x_i^u) - F(x_i^l)} \\
\Rightarrow \quad & \alpha = \frac{x_i^u - x_i^l}{F(x_i^u) - F(x_i^l)} \cdot (0.5 - F(x_i^l)) \\
Me &= x_i^l + \frac{x_i^u - x_i^l}{F(x_i^u) - F(x_i^l)} \cdot (0.5 - F(x_i^l))
\end{aligned}$$

Mode (Mo)

The mode is defined as the most frequent characteristic value.

Example: newspapers read regularly

0 newspapers	$\Rightarrow$	19 people
1 newspaper	$\Rightarrow$	45 people
2 newspapers	$\Rightarrow$	24 people
3 newspapers	$\Rightarrow$	8 people

$\Rightarrow$ $Mo = 1$ newspaper, as 45 people form the largest (absolute) frequency

For *grouped data*, the class interval with the highest frequency density is first selected as the modal class. The centre of this class interval is then defined as the mode.

$$\text{density} = \frac{h_i}{\text{class width}} \cdot \text{standard class width}$$

<u>Example</u>: Height of 100 students

i	Height $[\,cm\,]$	h_i	x_i'	Δx	Density
1	$[\,140 - 160\,[$	9	150	20	0.45
2	$[\,160 - 170\,[$	28	165	10	2.8
3	$[\,170 - 180\,[$	35	175	10	3.5
4	$[\,180 - 190\,[$	24	185	10	2.4
5	$[\,190 - 210\,[$	4	200	20	0.2
Σ	-	100	-	-	-

The highest frequency density is located in the third class interval:

$$x_3 = \frac{35}{10} = 3.5 \text{ students per } 10\text{-cm-interval height}$$

The centre of this class is $175\ cm$. $\quad \Rightarrow \quad Mo = 175\ cm$

Geometric Mean (G)

Geometric mean for **single values**:

$$G = \sqrt[N]{x_1 \cdot x_2 \cdot \ldots \cdot x_N}$$

<u>Example</u>:

Percentage p	$7\ \%$	$3\ \%$	$-5\ \%$	$4\ \%$
$x_i = 1 + \frac{p}{100}$	1.07	1.03	0.95	1.04

$G = \sqrt[4]{1.07 \cdot 1.03 \cdot 0.95 \cdot 1.04} = 1.022$

Geometric mean for **frequency distributions**:

$$G = \sqrt[N]{x_1^{h_1} \cdot x_2^{h_2} \cdot \ldots \cdot x_k^{h_k}} \qquad \text{with} \quad i = 1, \ldots, k$$

<u>Example</u>:

Percentage p	$-3\ \%$	$-2\ \%$	$1\ \%$	$3\ \%$
$x_i = 1 + \frac{p}{100}$	0.97	0.98	1.01	1.03
absolute frequencies H_i	3	2	4	1

$n = 3 + 2 + 4 + 1 = 10$

$G = \sqrt[10]{0.97^3 \cdot 0.98^2 \cdot 1.01^4 \cdot 1.03^1} = 0.994 = -0.6\ \%$

Tab. 2.1 shows at which scale level the application of the corresponding mean values is possible.

Mean value	Scale			
	Nominal scale	Ordinal scale	Interval scale	Ratio scale
Mode	×	×	×	×
Median		×	×	×
Arithmetic mean			×	×
Geometric mean				×
Harmonic mean				×

Tab. 2.1: Mean Values in Correspondence to Scale Levels[1]

[1] Cf. Bleymüller, J. & Gehlert, G. (2011), p. 13.

Harmonic Mean (H)

$$H = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_n}}$$

The dimension of the respective characteristic under consideration and the resulting harmonic mean corresponds to a quotient.

<u>Example</u>: A car drives 12 miles: a) 6 miles at 6 mph and
b) 6 miles at 60 mph.

What is the average speed?

$$H = \frac{2}{\frac{1}{6mph} + \frac{1}{60mph}} = \frac{2}{\frac{10}{60} + \frac{1}{60}} = \frac{2}{\frac{11}{60}} = \frac{2 \cdot 60}{11} = 10.91\,mph$$

Remark: The dimension of the characteristic considered here corresponds to the quotient mph.

2.2.2 Measures of Dispersion

Variance σ^2 / Standard Deviation σ

For **single values**:

Definition for the population N

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i^2) - \mu^2$$

$$\sigma = \sqrt{\sigma^2}$$

Example:

US shoe sizes of four people: 9, 8, 10, 11

$$\mu = \frac{9 + 8 + 10 + 11}{4} = 9.5$$

$$\sigma^2 = \frac{(9-9.5)^2 + (8-9.5)^2 + (10-9.5)^2 + (11-9.5)^2}{4} =$$

$$= 1.25$$

$$\sigma = \sqrt{1.25} = 1.12$$

Definition for a sample of n observations with $x_1, x_2, ..., x_n$

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i^2) - \bar{x}^2$$

$$s = \sqrt{s^2}$$

For **frequency distributions**:

- for **absolute** frequency distributions

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{k}(x_i - \mu)^2 h_i = \frac{1}{N}\sum_{i=1}^{k}(x_i^2\, h_i) - \mu^2$$

$$s^2 = \frac{1}{n}\sum_{i=1}^{k}(x_i - \bar{x})^2 h_i = \frac{1}{n}\sum_{i=1}^{k}(x_i^2\, h_i) - \bar{x}^2$$

Example of frequency distribution with absolute frequencies:

i	x_i Number of sold books of a particular book	h_i Number of days	$x_i h_i$	$x_i^2 h_i$
1	0	19	0	0
2	1	34	34	34
3	2	17	34	68
4	3	6	18	54
5	4	12	48	192
Σ	-	88	-	348

$$\begin{aligned}
\mu &= \frac{1}{N}(x_1 \cdot h_1 + \ldots + x_N \cdot h_N) = \frac{1}{N}\sum_{i=1}^{N} x_i h_i = \\
&= \frac{1}{88}(0 \cdot 19 \;+\; 1 \cdot 34 \;+\; 2 \cdot 17 \;+\; 3 \cdot 6 \;+\; 4 \cdot 12) = \\
&= 1.523
\end{aligned}$$

$$\begin{aligned}
\sigma^2 &= \frac{348}{88} \;-\; 1.523^2 = \\
&= 1.635 \text{ books}^2
\end{aligned}$$

$$\sigma = \sqrt{1.635} = 1.279 \text{ books}$$

- for **relative** frequency distributions

$$\sigma^2 = \sum_{i=1}^{k}(x_i - \mu)^2 f_i = \sum_{i=1}^{k}(x_i^2 f_i) - \mu^2$$

$$s^2 = \sum_{i=1}^{k}(x_i - \bar{x})^2 f_i = \sum_{i=1}^{k}(x_i^2 f_i) - \bar{x}^2$$

Example of a frequency distribution for relative frequencies:

i	x_i Number of sold books of a particular books	h_i Number of days	$x_i h_i$	${x_i}^2 h_i$	f_i	$x_i f_i$	$x_i^2 f_i$
1	0	19	0	0	0.216	0	0
2	1	34	34	34	0.386	0.386	0.386
3	2	17	34	68	0.193	0.386	0.772
4	3	6	18	54	0.068	0.204	0.612
5	4	12	48	192	0.136	0.544	2.176
Σ	-	88	-	348	1.000	-	3.946

$$\begin{aligned}
\mu &= \frac{1}{N}(x_1 \cdot h_1 + \ldots + x_N \cdot h_N) = \frac{1}{N}\sum_{i=1}^{N} x_i h_i = \\
&= \frac{1}{88}(0 \cdot 19 \;+\; 1 \cdot 34 \;+\; 2 \cdot 17 \;+\; 3 \cdot 6 \;+\; 4 \cdot 12) = \\
&= 1.523
\end{aligned}$$

$$\begin{aligned}
\sigma^2 &= 3.946 \;-\; 1.523^2 = \\
&= 1.626 \text{ books}^2
\end{aligned}$$

$$\sigma = \sqrt{1.626} = 1.275 \text{ books}$$

With a frequency distribution of *grouped data*, the variance/standard deviation are approximately calculated using the centres of the class intervals x'_i.

- for absolute frequency distributions of grouped data

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{k}(x'_i - \mu)^2 h_i = \frac{1}{N}\sum_{i=1}^{k}(x'^2_i\, h_i) - \mu^2$$

$$s^2 = \frac{1}{n}\sum_{i=1}^{k}(x'_i - \bar{x})^2 h_i = \frac{1}{n}\sum_{i=1}^{k}(x'^2_i\, h_i) - \bar{x}^2$$

<u>Example</u>: Height of 100 students

i	Height $[cm]$	h_i	x'_i	Δx	Density	$x'_i h_i$	$x'^2_i h_i$
1	$[140-160[$	9	150	20	0.45	1,350	202,500
2	$[160-170[$	28	165	10	2.8	4,620	762,300
3	$[170-180[$	35	175	10	3.5	6,125	1,071,875
4	$[180-190[$	24	185	10	2.4	4,440	821,400
5	$[190-210[$	4	200	20	0.2	800	160,000
Σ	-	100	-	-	-	-	3,018,075

with $x'_i =$ class midpoint,
$\Delta x =$ class width,
density of the elements $= \dfrac{h_i}{\Delta x} =$ elements per unit

$$\begin{aligned}
\mu &= \frac{1}{N}(x'_1 \cdot h_1 + \ldots + x'_N \cdot h_N) = \frac{1}{N}\sum_{i=1}^{N} x'_i h_i = \\
&= \frac{1}{100}(150 \cdot 9 \ + \ 165 \cdot 28 \ + \ 175 \cdot 35 \ + \ 185 \cdot 24 \ + \ 200 \cdot 4) = \\
&= 173.35
\end{aligned}$$

$$\begin{aligned}
\sigma^2 &= \frac{3,018,075}{100} \ - \ 173.35^2 = \\
&= 130.5275 \text{ cm}^2
\end{aligned}$$

$$\sigma = \sqrt{130.5275} = 11.425 \text{ cm}$$

- for relative frequency distributions of grouped data

$$\sigma^2 = \sum_{i=1}^{k}(x'_i - \mu)^2 f_i = \sum_{i=1}^{k}(x'^2_i f_i) - \mu^2$$

$$s^2 = \sum_{i=1}^{k}(x'_i - \bar{x})^2 f_i = \sum_{i=1}^{k}(x'^2_i f_i) - \bar{x}^2$$

Example: Height of 100 students

i	Height [cm]	h_i	x'_i	Δx	Density	f_i	$x'_i f_i$	$x_i'^2 f_i$
1	$[140-160[$	9	150	20	0.45	0.09	13.5	2,025
2	$[160-170[$	28	165	10	2.8	0.28	46.2	7,623
3	$[170-180[$	35	175	10	3.5	0.35	61.25	10,718.75
4	$[180-190[$	24	185	10	2.4	0.24	44.4	8,214
5	$[190-210[$	4	200	20	0.2	0.04	8	1,600
Σ	-	100	-	-	-	1	-	30,180.75

with $x'_i =$ class midpoint,
$\Delta x =$ class width,
density of the elements $= \dfrac{h_i}{\Delta x} =$ elements per unit

$$\begin{aligned}
\mu &= \frac{1}{N}(x'_1 \cdot h_1 + \ldots + x'_N \cdot h_N) = \frac{1}{N}\sum_{i=1}^{N} x'_i h_i = \\
&= \frac{1}{100}(150 \cdot 9 \; + \; 165 \cdot 28 \; + \; 175 \cdot 35 \; + \; 185 \cdot 24 \; + \; 200 \cdot 4) = \\
&= 173.35
\end{aligned}$$

$$\begin{aligned}
\sigma^2 &= 30,180.75 \; - \; 173.35^2 = \\
&= 130.5275 \text{ cm}^2
\end{aligned}$$

$$\sigma = \sqrt{130.5275} = 11.425 \text{ cm}$$

If the distribution of the characteristic values is unimodal and the class widths Δx are constant, Sheppard's correction leads to a better approximate value:

$$\sigma^2_{corr.} = \sigma^2 - \frac{(\Delta x)^2}{12}$$

Example: Height of 100 students

i	Height $[cm]$	h_i	x'_i	Δx	Density	$x'_i h_i$	$x'^2_i h_i$	f_i	$x'_i f_i$	$x'^2_i f_i$
1	$[140-150[$	9	145	10	0.9	1,305	189,225	0.09	13.05	1,892.25
2	$[150-160[$	28	155	10	2.8	4,340	672,700	0.28	43.4	6,727
3	$[160-170[$	35	165	10	3.5	5,775	952,875	0.35	57.75	9,528.75
4	$[170-180[$	24	175	10	2.4	4,200	735,000	0.24	42	7,350
5	$[180-190[$	4	185	10	0.4	740	136,900	0.04	7.4	1,369
Σ	-	100	-	-	-	-	2,686,700	1	-	26,867

$$\begin{aligned}
\mu &= \frac{1}{N}\left(x'_1 \cdot h_1 + \ldots + x'_N \cdot h_N\right) = \frac{1}{N}\sum_{i=1}^{N} x'_i h_i = \\
&= \frac{1}{100}\left(145 \cdot 9 \;+\; 155 \cdot 28 \;+\; 165 \cdot 35 \;+\; 175 \cdot 24 \;+\; 185 \cdot 4\right) = \\
&= 163.6
\end{aligned}$$

$$\begin{aligned}
\sigma^2 &= 26,867 \;-\; 163.6^2 = \\
&= 102.04 \text{ cm}^2
\end{aligned}$$

$$\sigma = \sqrt{102.04} = 10.10 \text{ cm}$$

Sheppard's correction:

$$\begin{aligned}
\sigma^2_{korr.} &= \sigma^2 - \frac{(\Delta x)^2}{12} = \\
&= 102.04 \;-\; \frac{(10)^2}{12} = \\
&= 93.706
\end{aligned}$$

The absolute frequencies h_i show that this distribution is unimodal. The maximum of this distribution is within the class $i = 3$.

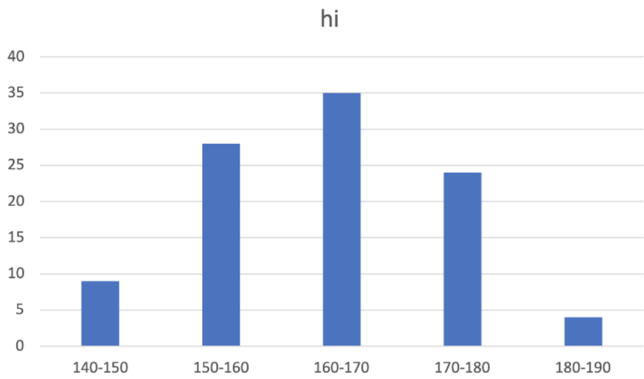


Fig. 2.1: Example of a unimodal frequency distribution with constant class width

Coefficient of Variation (CV)

$$CV = \frac{\sigma}{\mu} \, (\cdot\, 100\,\%) \qquad \text{or} \qquad CV = \frac{s}{\bar{x}} \, (\cdot\, 100\,\%)$$

Example:

Height of six people in cm: 174, 168, 151, 160, 171, 147

$$\mu = \frac{1}{6}\,(174+168+151+160+171+147) =$$
$$= 161.83 \text{ cm}$$

$$\sigma^2 = \frac{(-14.83)^2+(-10.83)^2+(-1.83)^2+6.17^2+9.17^2+12.17^2}{6} =$$
$$= 101.806 \text{ cm}^2$$

$$\sigma = \sqrt{101.806} = 10.09 \text{ cm}$$

$$CV = \frac{\sigma}{\mu \text{ or } \bar{x}} = \frac{10.09}{161.83} = 0.062$$

Range (R)

If the single values x_1, x_2, ..., x_N are arranged according to size, so that:

$$x_{[1]} \leq x_{[2]} \leq ... \leq x_{[N]},$$

the following applies:

$$R = x_{[N]} - x_{[1]} \quad \text{or}$$
$$R = x_{maximum} - x_{minimum}$$

Example:

Person	1	2	3	4	5	6
Age in years	17	24	12	42	60	11

Range: $60 - 11 = 49$ years

- for **grouped data**

  $R =$ upper limit of the largest class interval minus lower limit of the smallest class interval

Example: Height of 100 students

i	Height [cm]	h_i	x'_i	Δx	Density
1	$[140-160[$	9	150	20	0.45
2	$[160-170[$	28	165	10	2.8
3	$[170-180[$	35	175	10	3.5
4	$[180-190[$	24	185	10	2.4
5	$[190-210[$	4	200	20	0.2
$\sum$	-	100	-	-	-

$R = 210 - 140 = 70\ cm$

Tab. 2.2 shows at which scale level the calculation of the corresponding measure of dispersion is possible.

Measures of dispersion	Scale			
	Nominal scale	Ordinal scale	Interval scale	Ratio scale
Range		×	×	×
Mean absolute deviation			×	×
Variance, standard deviation			×	×
Coefficient of variation				×

Tab. 2.2: Measures of Dispersion in Correspondence to Scale Levels[2]

[2] Cf. Bleymüller, J. & Gehlert, G. (2011), p. 17.

2.3 Ratios and Index Figures

2.3.1 Ratios

Ratios are key figures that are formed as quotients (Fig. 2.2).

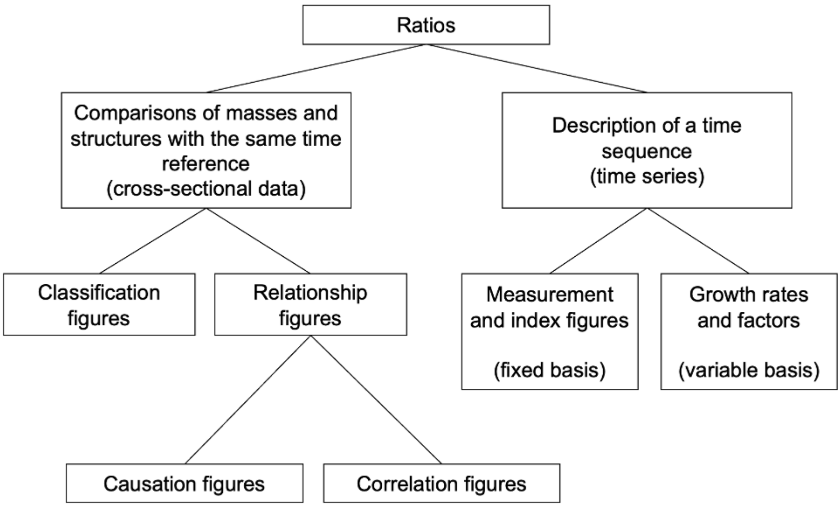


Fig. 2.2: Ratio Figures[3]

Examples:

Classification figures

These relate a partial quantity to a corresponding total quantity

$$\text{e.g.:} \quad \text{consumption rate} = \frac{\text{consumption}}{\text{income}^4} \cdot 100\,\%$$

[3] Cf. Voß, W. (2000), p. 209.

[4] Generally, the disposable income is chosen here.

Relationship figures

These relativise two measures that belong to different sets, i.e. the numerator of the quotient is not a subset of the denominator. In the case of relationship figures, a distinction is made between causation figures and correlation figures.

In the case of *causation figures*, the subset measured in the numerator is "caused" by the mass shown in the denominator:

$$\text{e.g.:}\quad \text{productivity} = \frac{\text{output}}{\text{input}}$$

$$\text{profitability} = \frac{\text{profit}}{\text{capital}}$$

In the case of *correlation figures*, there is no causality between the masses shown in the numerator and denominator:

$$\text{e.g.:}\quad \text{population density} = \frac{\text{population}}{\text{area}}$$

$$\text{density of doctors} = \frac{\text{number of doctors}}{\text{population}}$$

A **measurement figure** m_{0t} describes the ratio of a (usually current) value x_t to the base value x_0, where t is the reporting period or reporting time and 0 is the base period (reference period) or base time (reference time):

$$m_{0t} = \frac{x_t}{x_0} \quad \text{or} \quad \frac{x_t}{x_0} \cdot 100\,\% \qquad \text{with} \qquad \begin{aligned} t &= \text{reporting period or reporting time} \\ 0 &= \text{base period or base time} \end{aligned}$$

Properties of Measurement Figures

(1) If the base and reporting periods or base and reporting time are equal, the following applies: $m_{0t} = 1$.

(2) Measurement figures are dimensionless. The same dimensions of x_t and x_0 cancel each other out.

(3) If base and reporting periods or base and reporting time are swapped reciprocally, the following applies: $m_{t0} = \frac{1}{m_{0t}}$.

(4) Several periods $(0, s$ and $t)$ can be concatenated or uniformly based (see: Operations with Measurement Figures).

(5) If the value W is the product of P and Q for all periods, the following applies analogously to the measurement figures: $m^W_{0t} = m^P_{0t} \cdot m^Q_{0t}$ (factor reversal sampling).

Operations with Measurement Figures

Concatenation: $m_{0t} = m_{0s} \cdot m_{st}$

Rebasing: $m_{st} = \frac{m_{0t}}{m_{0s}}$

Growth rate: $w_t = \frac{x_t - x_{t-1}}{x_{t-1}} \cdot 100\,\%$ or $w_t = \left(\frac{x_t}{x_{t-1}} - 1\right) \cdot 100\,\%$

Examples for Measurement Figures:

1. Index figures (see chapter 2.3.2)

2. Growth rate

The (nominal) gross domestic product (GDP) of an economy in three subsequent years t_i with $i = 1, ..., 3$ is:

$t_1 = \$3,700$ bln, $t_2 = \$3,800$ bln, $t_3 = \$3,900$ bln.

The annual economic growth, measured as the rate of change in nominal GDP, is calculated as follows:

$$w_{t_1-t_2} = \frac{3,800 - 3,700}{3,700} \cdot 100\,\% = 2.70\,\%$$

t_1 corresponds to the base period (reference period)

$$w_{t_2-t_3} = \frac{3,900 - 3,800}{3,800} \cdot 100\,\% = 2.63\,\%$$

t_2 corresponds to the base period (reference period)

2.3.2 Index Figures

Index figures measure aggregated changes.

Symbols for Prices and Quantities

$p_0^{(j)}$ ... price of good j at base period or at base time

$p_t^{(j)}$ ... price of good j at reporting period or at reporting time

$q_0^{(j)}$ ... quantity of good j at base period or at base time

$q_t^{(j)}$ ... quantity of good j at reporting period or at reporting time

Example:

Goods j	Price		Quantity		$p_0 q_0$	$p_t q_t$	$p_0 q_t$	$p_t q_0$
	Base period p_0	Reporting period p_t	Base period q_0	Reporting period q_t				
1	5	7	3	6	15	42	30	21
2	7	9	6	13	42	117	91	54
3	8	4	10	15	80	60	120	40
4	10	12	7	19	70	228	190	96
Σ	-	-	-	-	207	447	431	211

Tab. 2.3: Example with one base period and one reporting period

Sales Index/Value Index

$$U_{0t} = \frac{\sum p_t q_t}{\sum p_0 q_0} \cdot 100\,\%$$

Example:

$$U_{0t} = \frac{447}{207} \cdot 100\,\% =$$
$$= 2.1594 \cdot 100\,\% =$$
$$= 215.94$$

The sales generated within the reporting period more than doubled those generated at the base period, i.e. the sales generated within the reporting period correspond to $215.94\,\%$ of those generated at the base period, i.e. 2.1594 times.

Laspeyres[5] Price Index

$$P^L_{0t} = \frac{\sum_{j=1}^{n} \frac{p_t^{(j)}}{p_0^{(j)}} \cdot p_0^{(j)} q_0^{(j)}}{\sum_{j=1}^{n} p_0^{(j)} q_0^{(j)}} \cdot 100\ \% =$$

$$= \frac{\sum_{j=1}^{n} p_t^{(j)} q_0^{(j)}}{\sum_{j=1}^{n} p_0^{(j)} q_0^{(j)}} \cdot 100\ \% =$$

$$= \frac{\sum p_t q_0}{\sum p_0 q_0} \cdot 100\ \%$$

Example:

$$P^L_{0t} = \frac{211}{207} \cdot 100\ \% =$$
$$= 1.0193 \cdot 100\ \% =$$
$$= 101.93\ \%$$

The prices of this shopping cart within the reporting period increased by 1.93 percent compared to those at the base period, i.e. during the reporting period according to Laspeyres, they amounted to 1.0193 times the average price level within the base period.

Laspeyres Volume Index

$$Q^L_{0t} = \frac{\sum q_t p_0}{\sum q_0 p_0} \cdot 100\,\% = \frac{U_{0t}}{P^P_{0t}}$$

Example:

$$Q^L_{0t} = \frac{431}{207} \cdot 100\ \% = 208.21\ \%$$

[5] Ernst Louis Étienne Laspeyres (1834 - 1913) was a German national economist and statistician.

The quantities of this shopping cart within the reporting period more than doubled those at the base period, i.e. during the reporting period according to Laspeyres, they amounted to 2.0821 times the average quantity level within the base period.

Paasche[6] Price Index

$$P^P_{0t} = \frac{\sum_{j=1}^{n} \frac{p_t^{(j)}}{p_0^{(j)}} \cdot p_0^{(j)} q_t^{(j)}}{\sum_{j=1}^{n} p_0^{(j)} q_t^{(j)}} \cdot 100\ \% =$$

$$= \frac{\sum_{j=1}^{n} p_t^{(j)} q_t^{(j)}}{\sum_{j=1}^{n} p_0^{(j)} q_t^{(j)}} \cdot 100\ \% =$$

$$= \frac{\sum p_t q_t}{\sum p_0 q_t} \cdot 100\ \%$$

Example:

$$P^P_{0t} = \frac{447}{431} \cdot 100\ \% =$$

$$= 1.0371 \cdot 100\ \% =$$

$$= 103.71\ \%$$

The prices of this shopping cart within the reporting period increased by 17.81 percent compared to those at the base period, i.e. during the reporting period according to Paasche, they amounted to 1.1781 times the average price level within the base period.

Paasche Volume Index

$$Q^P_{0t} = \frac{\sum q_t p_t}{\sum q_0 p_t} \cdot 100\,\% = \frac{U_{0t}}{P^L_{0t}}$$

[6] Hermann Paasche (1851 - 1925) was a German statistician.

Example:

$$Q^P_{0t} = \frac{447}{211} \cdot 100\ \% = 211.85\ \%$$

The volumes of this shopping cart within the reporting period more than doubled those at the base period, i.e., during the reporting period according to Paasche, they amounted to 2.1185 times the average volume level within the base period.

Sales Index/Value Index as Index Product

$$U_{0t} = P^L_{0t}\, Q^P_{0t} = P^P_{0t}\, Q^L_{0t}$$

Example:

$$U_{0t} = \frac{447}{431} \cdot \frac{431}{207} \cdot 100\ \% =$$

$$= 215.94\ \%$$

or

$$U_{0t} = \frac{101.93 \cdot 211.85}{100} = \frac{103.71 \cdot 208.21}{100} = 215.94\ \%$$

Fisher[7] Price Index

$$P^F_{0t} = \sqrt{P^L_{0t} \cdot P^P_{0t}} \cdot 100\,\%$$

Example:

$$P^F_{0t} = \sqrt{\frac{211}{207} \cdot \frac{447}{431}} \cdot 100\ \% =$$

$$= \sqrt{1.0193 \cdot 1.0371} \cdot 100\ \% =$$

$$= 102.82\ \%$$

[7] Irving Fisher (1867 - 1947) was an American economist.

Fisher Volume Index

$$Q_{0t}^{F} = \sqrt{Q_{0t}^{L} \cdot Q_{0t}^{P}} \cdot 100\ \%$$

Example:

$$Q_{0t}^{F} = \sqrt{\frac{431}{207} \cdot \frac{447}{211}} \cdot 100\ \% =$$
$$= \sqrt{2.0821 \cdot 2.1185} \cdot 100\ \% =$$
$$= 210.02\ \%$$

Stuvel Price Index[8]

$$P_{0t}^{ST} = \left(\frac{P_{0t}^{L} - Q_{0t}^{L}}{2} + \sqrt{\left(\frac{P_{0t}^{L} - Q_{0t}^{L}}{2} \right)^{2} + U_{0t}} \right) \cdot 100\,\%$$

with $U_{0t} = \frac{\sum p_t\, q_t}{\sum p_0 q_0}$ (sales-/value index)

Example:

$$P_{0t}^{ST} = \left(\frac{1.0193 - 2.0821}{2} + \sqrt{\left(\frac{1.0193 - 2.0821}{2} \right)^{2} + 2.1594} \right) \cdot 100\ \% =$$
$$= 1.0312 \cdot 100\ \% =$$
$$= 103.12\ \%$$

Stuvel Volume Index[9]

$$Q_{0t}^{ST} = \left(\frac{Q_{0t}^{L} - P_{0t}^{L}}{2} + \sqrt{\left(\frac{Q_{0t}^{L} - P_{0t}^{L}}{2} \right)^{2} + U_{0t}} \right) \cdot 100\,\%$$

---

[8] The *Stuvel Price Index* can also be explained as a special case of the *Banerjee* approach. Cf. Banerjee, K.S. (1977): On the Factorial Approach Providing the True Cost of Living Index, Göttingen.

[9] The *Stuvel Volume Index* can also be explained as a special case of the *Banerjee* approach. Cf. Banerjee, K.S. (1977): On the Factorial Approach Providing the True Cost of Living Index, Göttingen.

$$\text{with} \quad U_{0t} = \frac{\sum p_t\, q_t}{\sum p_0 q_0} \quad \text{(sales-/value index)}$$

Example:

$$Q_{0t}^{ST} = \left(\frac{2.0821 - 1.0193}{2} + \sqrt{\left(\frac{2.0821 - 1.0193}{2}\right)^2 + 2.1594} \right) \cdot 100\,\% =$$

$$= 2.094 \cdot 100\,\% =$$

$$= 209.4\,\%$$

Example:

A household consumed in the years 2014 - 2020 products A, B and C in the following volumes:

k	Year	Product A		Product B		Product C	
		volumes	prices per unit	volumes	prices per unit	volumes	prices per unit
0	2014	300	0.25	180	0.70	96	0.80
1	2015	250	0.30	145	0.71	100	1.10
2	2016	340	0.41	290	0.73	142	0.95
3	2017	170	0.28	242	0.72	200	0.96
4	2018	190	0.21	311	0.69	170	0.87
5	2019	245	0.19	196	0.68	164	0.91
6	2020	320	0.31	215	0.74	171	1.02

Tab. 2.4: Example with one base period (year 2014) and five subsequent periods

Lowe[10] Price Index

$$P_{0t}^{LO} = \frac{\sum p_i^{(t)} q_i}{\sum p_i^{(0)} q_i} \cdot 100\,\% \quad \text{with} \quad q_i = \frac{1}{t+1} \sum_{k=0}^{t} q_i^k$$

q_i is the arithmetic mean of the values within the periods 0 to t.

<u>Example</u>:

$$q_1 = \frac{1}{7}\,(300+250+340+170+190+245+320) = 259.29$$
$$q_2 = \frac{1}{7}\,(180+145+290+242+311+196+215) = 225.57$$
$$q_3 = \frac{1}{7}\,(96+100+142+200+170+164+171) = 149.00$$

$$P_{06}^{LO} = \frac{0.31 \cdot 259.29 + 0.74 \cdot 225.57 + 1.02 \cdot 149.00}{0.25 \cdot 259.29 + 0.70 \cdot 225.57 + 0.80 \cdot 149.00} \cdot 100\,\% = $$
$$= \frac{399.28}{341.92} \cdot 100\,\% =$$
$$= 116.78\,\%$$

Lowe Volume Index

$$Q_{0t}^{LO} = \frac{\sum q_i^{(t)} p_i}{\sum q_i^{(0)} p_i} \cdot 100\,\% \quad \text{with} \quad p_i = \frac{1}{t+1} \sum_{k=0}^{t} p_i^k$$

p_i is the arithmetic mean of the values within the periods *0* to t.

<u>Example</u>:

$$p_1 = \frac{1}{7}\,(0.25+0.30+0.41+0.28+0.21+0.19+0.31) = 0.28$$
$$p_2 = \frac{1}{7}\,(0.70+0.71+0.73+0.72+0.69+0.68+0.74) = 0.71$$

[10] Adolph Lowe, born Adolf Löwe, (1893 - 1995) was a German sociologist and national economist.

$$p_3 = \frac{1}{7}\left(0.80+1.10+0.95+0.96+0.87+0.91+1.02\right) = 0.94$$

$$\begin{aligned}
Q_{06}^{LO} &= \frac{320\cdot 0.28+215\cdot 0.71+171\cdot 0.94}{300\cdot 0.28+180\cdot 0.71+96\cdot 0.94}\cdot 100\ \% = \\
&= \frac{402.99}{302.04}\cdot 100\ \% = \\
&= 133.42\ \%
\end{aligned}$$

Laspeyres Price Index

Example:

$$\begin{aligned}
P_{06}^{L} &= \frac{\sum p_6 q_0}{\sum p_0 q_0}\cdot 100\ \% = \\
&= \frac{(0.31\cdot 300)+(0.74\cdot 180)+(1.02\cdot 96)}{(0.25\cdot 300)+(0.70\cdot 180)+(0.80\cdot 96)}\cdot 100\ \% = \\
&= \frac{324.12}{277.80}\cdot 100\ \% = \\
&= 116.67\ \%
\end{aligned}$$

Laspeyres Volume Index

Example:

$$\begin{aligned}
Q_{06}^{L} &= \frac{\sum p_0 q_6}{\sum p_0 q_0}\cdot 100\ \% = \\
&= \frac{(0.25\cdot 320)+(0.70\cdot 215)+(0.80\cdot 171)}{(0.25\cdot 300)+(0.70\cdot 180)+(0.80\cdot 96)}\cdot 100\ \% = \\
&= \frac{367.30}{277.80}\cdot 100\ \% = \\
&= 132.22\ \%
\end{aligned}$$

Alternative calculation using the sales/value index U_{06} and the price index according to Paasche P^P_{06}:

$$
\begin{aligned}
Q^L_{06} &= \frac{U_{06}}{P^P_{06}} = \\
&= \frac{432.72\,/\,277.80}{432.72\,/\,367.30} \cdot 100\ \% = \frac{1.5577}{1.1781} \cdot 100\ \% = \\
&= 132.22\ \%
\end{aligned}
$$

with

$$
\begin{aligned}
U_{06} &= \frac{\sum p_6 q_6}{\sum p_0 q_0} = \\
&= \frac{(0.31\cdot 320)+(0.74\cdot 215)+(1.02\cdot 171)}{(0.25\cdot 300)+(0.70\cdot 180)+(0.80\cdot 96)} = \\
&= \frac{432.72}{277.80} = 1.5577
\end{aligned}
$$

$$
\begin{aligned}
P^P_{06} &= \frac{\sum p_6 q_6}{\sum p_0 q_6} = \\
&= \frac{(0.31\cdot 320)+(0.74\cdot 215)+(1.02\cdot 171)}{(0.25\cdot 320)+(0.70\cdot 215)+(0.80\cdot 171)} = \\
&= \frac{432.72}{367.30} = 1.1781
\end{aligned}
$$

Paasche Price Index

<u>Example</u>:

$$
\begin{aligned}
P^P_{06} &= \frac{\sum p_6 q_6}{\sum p_0 q_6} \cdot 100\ \% = \\
&= \frac{(0.31\cdot 320)+(0.74\cdot 215)+(1.02\cdot 171)}{(0.25\cdot 320)+(0.70\cdot 215)+(0.80\cdot 171)} \cdot 100\ \% = \\
&= \frac{432.72}{367.30} \cdot 100\ \% = \\
&= 117.81\ \%
\end{aligned}
$$

Paasche Volume Index

Example:

$$
\begin{aligned}
Q^P_{06} &= \frac{\sum p_6 q_6}{\sum p_6 q_0} \cdot 100\ \% = \\
&= \frac{(0.31 \cdot 320) + (0.74 \cdot 215) + (1.02 \cdot 171)}{(0.31 \cdot 300) + (0.74 \cdot 180) + (1.02 \cdot 96)} \cdot 100\ \% = \\
&= \frac{432.72}{324.12} \cdot 100\ \% = \\
&= 133.51\ \%
\end{aligned}
$$

Alternative calculation using the sales/value index U_{06} and the price index according to Laspeyres P^L_{06}:

$$
\begin{aligned}
Q^P_{06} &= \frac{U_{06}}{P^L_{06}} = \\
&= \frac{432.72/277.80}{324.12/277.80} \cdot 100\ \% = \frac{1.5577}{1.1667} \cdot 100\ \% = \\
&= 133.51\ \%
\end{aligned}
$$

with

$$
\begin{aligned}
U_{06} &= \frac{\sum p_6 q_6}{\sum p_0 q_0} = \\
&= \frac{(0.31 \cdot 320) + (0.74 \cdot 215) + (1.02 \cdot 171)}{(0.25 \cdot 300) + (0.70 \cdot 180) + (0.80 \cdot 96)} = \\
&= \frac{432.72}{277.80} = 1.5577
\end{aligned}
$$

$$
\begin{aligned}
P^L_{06} &= \frac{\sum p_6 q_0}{\sum p_0 q_0} = \\
&= \frac{(0.31 \cdot 300) + (0.74 \cdot 180) + (1.02 \cdot 96)}{(0.25 \cdot 300) + (0.70 \cdot 180) + (0.80 \cdot 96)} =
\end{aligned}
$$

$$= \frac{324.12}{277.80} = 1.1667$$

Fisher Price Index

Example:

$$P^F_{06} = \sqrt{P^L_{06} \cdot P^P_{06}} \cdot 100\ \% =$$

$$= \sqrt{\frac{324.12}{277.80} \cdot \frac{432.72}{367.30}} \cdot 100\ \% =$$

$$= \sqrt{1.1667 \cdot 1.1781} \cdot 100\ \% =$$

$$= 117.24\ \%$$

Fisher Volume Index

Example:

$$Q^F_{06} = \sqrt{Q^L_{06} \cdot Q^P_{06}} \cdot 100\ \% =$$

$$= \sqrt{\frac{367.30}{277.80} \cdot \frac{432.72}{324.12}} \cdot 100\ \% =$$

$$= \sqrt{1.3222 \cdot 1.3351} \cdot 100\ \% =$$

$$= 132.86\ \%$$

Marshall[11] Edgeworth[12] Price Index

$$P^{ME}_{0t} = \frac{\sum p_{it}(q_{i0}+q_{it})}{\sum p_{i0}(q_{i0}+q_{it})} \cdot 100\,\%$$

[11] Alfred Marshall (1842 - 1924) was a British national economist.

[12] Francis Ysidro Edgeworth (1845 - 1926) was an Irish economist.

Example:

$$
\begin{aligned}
P_{06}^{ME} &= \frac{\sum p_{i6}(q_{i0}+q_{i6})}{\sum p_{i0}(q_{i0}+q_{i6})} \cdot 100\ \% = \\
&= \frac{0.31\ (300+320)+0.74\ (180+215)+1.02\ (96+171)}{0.25\ (300+320)+0.70\ (180+215)+0.80\ (96+171)} \cdot 100\ \% = \\
&= \frac{756.84}{645.10} \cdot 100\ \% = \\
&= 117.32\ \%
\end{aligned}
$$

Walsh Price Index

$$
P_{0t}^{W} = \frac{\sum p_{it}(q_{i0}q_{it})^{0.5}}{\sum p_{i0}(q_{i0}q_{it})^{0.5}} \cdot 100\,\%
$$

Example:

$$
\begin{aligned}
P_{06}^{W} &= \frac{\sum p_{i6}(q_{i0}q_{i6})^{0.5}}{\sum p_{i0}(q_{i0}q_{i6})^{0.5}} \cdot 100\,\% = \\
&= \frac{0.31(300\cdot 320)^{0.5}+0.74(180\cdot 215)^{0.5}+1.02(96\cdot 171)^{0.5}}{0.25(300\cdot 320)^{0.5}+0.70(180\cdot 215)^{0.5}+0.80(96\cdot 171)^{0.5}} \cdot 100\,\% = \\
&= \frac{372.31}{317.67} \cdot 100\,\% = \\
&= 117.20\ \%
\end{aligned}
$$

2.3.3 Peren-Clement Index (PCI)[13]

The PCI[14] is a risk index for the assessment of country risks in *Foreign Direct Investments*.

The PCI is determined by three factors:

- cross-company factors,
- cost and production-oriented factors, and
- sales-oriented factors.

Cross-company factors include:

- political and social stability,
- state influence on business decisions and bureaucratic obstacles,
- general economic policy,
- investment incentives,
- enforceability of contractual agreements, and
- the respect of property rights in the transfer of technology and know-how

Cost and production-oriented factors include:

- legal restrictions on production,
- cost of capital in the host country and possibilities of capital import,
- availability and cost of acquiring land and property,
- availability and cost of labour,
- availability and cost of fixed assets, raw materials and supplies in the host country,
- trade barriers to the import of goods, and
- availability and quality of infrastructure and government services

[13] Reiner Clement (1958 - 2017) was a German economist.

[14] Cf. Pakusch, C.; Peren, F.W. & Shakoor, M.A. (2016): The PCI - A Global Risk Index for the Simultaneous Assessment of Macro and Company Individual Investment Risks. In: Journal of Business Strategies, 33(2), p. 154-173; Clement, R. & Peren, F.W. (2017): Peren-Clement-Index: Bewertung von Direktinvestitionen durch eine simultane Erfassung von Makroebene und Unternehmensebene, Wiesbaden; Pakusch, C.; Peren, F.W. & Shakoor, M.A. (2018): Peren-Clement-Index – eine exemplarische Fallstudie. In: Gadatsch, A. et al. (Ed.): Nachhaltiges Wirtschaften im digitalen Zeitalter: Innovation - Steuerung - Compliance, Wiesbaden, p. 105-117.

Sales-oriented factors include:

- size and dynamics of the market,
- competitive situation,
- reliability, quality of local contractors,
- quality and sales opportunities, and
- trade barriers to exports from the host country

Depending on the type of investment or motives of the respective companies, there are different location factors or differing weightings of the various factors.

PCI | Case Study

Environment	Points	Weight	Sum
Political and social stability	...	4	...
Bureaucratic obstacles	...	2	...
Economic policy	...	3	...
Legal security	...	3	...
Solvency	...	3	...
Sum		**15**	**...**

Localisation	Points	Weight	Sum
Human capital	...	4	...
Transport connections	...	2	...
Management skills	...	2	...
Access to markets	...	2	...
Quality of life	...	3	...
Sum		**13**	**...**

Production	Points	Weight	Sum
Economic and property rights	...	2	...
Manufacturing costs	...	2	...
Capital procurement	...	3	...
Complementary industries	...	2	...
Investment incentives	...	2	...
Sum		**11**	**...**

Sales	**Points**	**Weight**	**Sum**
Size and dynamic of the market	...	3	...
Per capita income	...	2	...
Avoidance of tariff barriers	...	2	...
Reliability of local contractors	...	2	...
Distribution structures	...	2	...
Sum		**11**	**...**
Total score PCI			**...**

Tab. 2.5: Structural Arrangement of PCI

Company A and Company B want to internationalise. Both Countries/ Regions X and Y are eligible for direct investment abroad. Both companies use the index presented here (Tab. 2.5) to measure and make a comparative quantitative assessment of country risks. The PCI shows a total score of 71 points for Country X (Tab. 2.6). This value implies a relatively low investment risk.

Investment Alternative 1: Country/Region X PCI = 71

Environment	Points	Weight	Sum
Political and social stability	1.5	4	6
Bureaucratic obstacles	2	2	4
Economic policy	2	3	6
Legal security	1.5	3	4.5
Solvency	1.5	3	4.5
Sum		**15**	**25**

Localisation	Points	Weight	Sum
Human capital	0.5	4	2
Transport connections	2	2	4
Management skills	0	2	0
Access to markets	1.5	2	3
Quality of life	2	3	6
Sum		**13**	**15**

Production	Points	Weight	Sum
Economic and property rights	2	2	4
Manufacturing costs	2	2	4
Capital procurement	2	3	6
Complementary industries	2	2	4
Investment incentives	2	2	4
Sum		**11**	**22**

Sales	**Points**	**Weight**	**Sum**
Size and dynamic of the market	0	3	0
Per capita income	0.5	2	1
Avoidance of tariff barriers	2	2	4
Reliability of local contractors	0.5	2	1
Distribution structures	1.5	2	3
Sum		**11**	**9**

Total score PCI	**71**

Tab. 2.6: Risk Assessment for Country/Region X

The calculation for the alternative Country/Region Y also results in a total score of 71 points (Tab. 2.7). Company A and Company B now know that both direct investments would be associated with a relatively low risk and thus tend to be assessed positively. However, which region would be the more suitable location for the company cannot be derived from the macroeconomic risk assessment carried out so far.

Investment Alternative 2: Country/Region Y PCI = 71

Environment	Points	Weight	Sum
Political and social stability	2	4	8
Bureaucratic obstacles	1	2	2
Economic policy	1	3	3
Legal security	2	3	6
Solvency	1.5	3	4.5
Sum		**15**	**23.5**

Localisation	Points	Weight	Sum
Human capital	1.5	4	6
Transport connections	2	2	4
Management skills	1.5	2	3
Access to markets	2	2	4
Quality of life	1.5	3	4.5
Sum		**13**	**21.5**

Production	Points	Weight	Sum
Economic and property rights	2	2	4
Manufacturing costs	0.5	2	1
Capital procurement	0	3	0
Complementary industries	0	2	0
Investment incentives	0.5	2	1
Sum		**11**	**6**

Sales	**Points**	**Weight**	**Sum**
Size and dynamic of the market	2	3	6
Per capita income	2	2	4
Avoidance of tariff barriers	1.5	2	3
Reliability of local contractors	1.5	2	3
Distribution structures	2	2	4
Sum		**11**	**20**

Total score PCI	**71**

Tab. 2.7: Risk Assessment for Country/Region Y

The evaluation model is now extended to include another company-specific dimension. The companies pursue different company-specific goals with internationalisation (Tab. 2.8). While for Company A it is most important to access the foreign market and to expand the existing resources through additional resources of the foreign location, Company B focuses on the cost advantages that can be generated with production at the foreign location.

Internationalisation aims of Company A:	**Target weight**
1. Sales in the foreign market	50 %
2. Expand resources through localisation	30 %
3. Secure foreign location/strategic importance (macro environment)	15 %
4. Realise cost advantages (production)	5 %
	100 %
Internationalisation aims of Company B:	**Target weight**
1. Realise cost advantages (production)	70 %
2. Secure foreign location/strategic importance (macro environment)	15 %
3. Expand resources through localisation	10 %
4. Sales in the foreign market	5 %
	100 %

Tab. 2.8: Company-Specific Internationalisation Aims

Accordingly, the companies now assign the individual factors of the PCI in a second dimension with the weights of their individual goals. The factors *macro environment*, *localisation*, *production* and *sales* are to be distributed to 100 % (or 1.0 target weight overall) on a company-specific basis. The result is a company-specific, two-dimensional total score, which now enables a company-specific, target-oriented decision (Fig. 2.3).

In contrast to all currently existing risk indices, the two-dimensional cumulative scores measured in PECLE[15] now clearly show which region is best suited for which investor for a direct investment at the current time. Company A should opt for a direct investment in Country/Region X, as this is where it can best realise its individual company goals at the moment. Country/Region X achieves a company-specific total score of 20.275 PECLE for Company A, while Country/Region Y is only rated with a total of 13.85 PECLE for the individual company. The reason for the now apparent distinction is that Country/Region A offers the better

[15] PECLE as a one-dimensional measurement of a two-dimensional cumulative score.

starting position for the factors *sales* and *localisation*, which are most important to Company A when choosing a location ("sales in the foreign market: 50 % and "expand recources through localisation: 30 %").

Company B primarily wants to internationalise in order to realise cost advantages in production ("realise cost advantages in production: 70 %"). Company B should therefore choose Country/Region Y. Country/Region Y, with a company-specific total score of 21.1 PECLE, is much more suitable for a direct investment for Company B than Country/Region X (total score of 10.875 PECLE).

The two-dimensional graphical representation clearly shows the advantage of such a combinatorial view of the macroeconomic level on the one hand and company-specific objectives on the other, illustrated by the example of Country/Region X versus Country/Region Y (Fig. 2.4).

Company A — Country X

Factor	PCI	Target weights	Sum in PECLE
Macro environment (M)	23.5	0.15	3.525
Localisation (L)	21.5	0.30	6.450
Production (P)	6	0.05	0.300
Sales (S)	20	0.05	10.000
Total Score PCI	**71**	**1.00**	**20.275**

Company B — Country X

Factor	PCI	Target weights	Sum in PECLE
Macro environment (M)	23.5	0.15	3.525
Localisation (L)	21.5	0.10	2.15
Production (P)	6	0.70	4.2
Sales (S)	20	0.05	1
Total Score PCI	**71**	**1.00**	**10.875**

Company A — Country Y

Factor	PCI	Target weights	Sum in PECLE
Macro environment (M)	25	0.15	3.75
Localisation (L)	15	0.30	4.5
Production (P)	22	0.05	1.1
Sales (S)	6	0.50	4.5
Total Score PCI	**71**	**1.00**	**13.85**

Company B — Country Y

Factor	PCI	Target weights	Sum in PECLE
Macro environment (M)	25	0.15	3.75
Localisation (L)	15	0.10	1.5
Production (P)	22	0.70	15.4
Sales (S)	6	0.05	0.45
Total Score PCI	**71**	**1.00**	**21.1**

Fig. 2.3: Individual Assessment of Country Risks

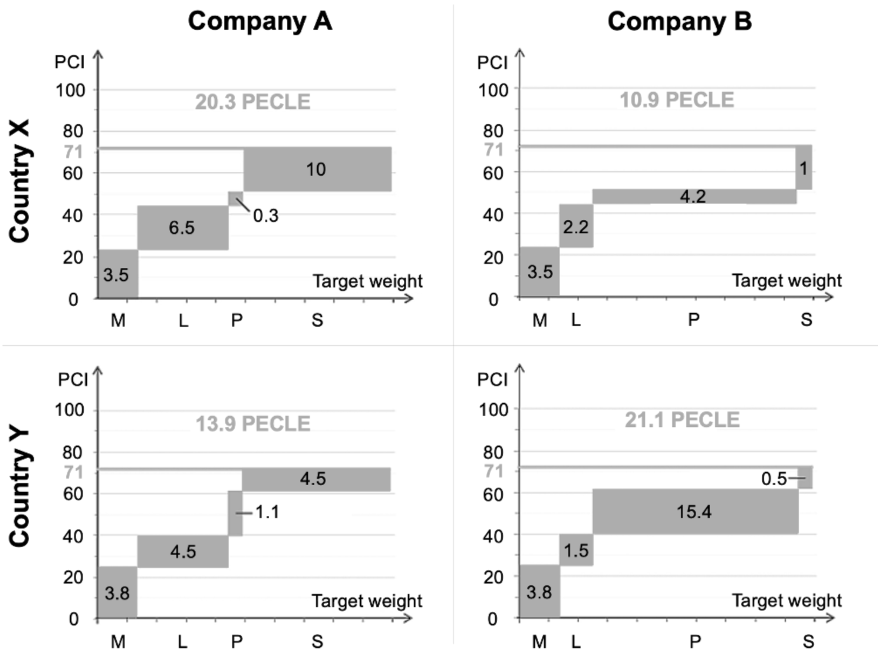


Fig. 2.4: PCI – Simultaneous Assessment of Macroeconomic Macro Level and Individual Company Objectives in Direct Investments

2.4 Correlation Analysis

Simple Linear Coefficient of Determination (r^2)

$$r^2 = \frac{SSE}{SST} = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{SSR}{SST} \qquad 0 \leq r^2 \leq 1$$

with

SST = sum of the squared deviations
= $SSE + SSR$

SSE = sum of the deviation squares explained (by the regression function)

SSR = sum of the deviation squares unexplained (by the regression function)

Simple Linear Correlation Coefficient (r)

$$r = sgn(b_2)\sqrt{r^2} \qquad -1 \leq r \leq 1$$

Pearson's[16] Correlation Coefficient (r)

$$r = \frac{s_{xy}}{s_x \cdot s_y} \qquad -1 \leq r \leq 1$$

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\,(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} =$$

[16] Karl Pearson (1857 - 1936) was a British mathematician.

$$= \frac{\sum\limits_{i=1}^{n} x_i y_i - \frac{\left(\sum\limits_{i=1}^{n} x_i\right)\left(\sum\limits_{i=1}^{n} y_i\right)}{n}}{\sqrt{\sum\limits_{i=1}^{n} x_i^2 - \frac{\left(\sum\limits_{i=1}^{n} x_i\right)^2}{n}}\sqrt{\sum\limits_{i=1}^{n} y_i^2 - \frac{\left(\sum\limits_{i=1}^{n} y_i\right)^2}{n}}}$$

2.5 Regression Analysis

2.5.1 Simple Linear Regression

Description of the relationship (dependence) between two variables x and y using a linear function.

Symbols:

x_i	value of the independent variables x at the i^{th} observation
y_i	value of the dependent variables y at the i^{th} observation
$\hat{y}_i$	value of the dependent variables y estimated by the regression function (regression line) at the point x_i
b_1, b_2	regression coefficients sought, specifying the regression line (b_1: ordinate intercept; b_2: slope of the regression line)
e_i	residual ($e_i = y_i - \hat{y}_i$); deviation of the value estimated by the regression function from the observed (true) value of the dependent variables y

with $i = 1, ..., n$

Least Squares Method (= LS-Method)

The sum of the squared deviation (SAQ) must be minimised:

$$\begin{aligned} SAQ &= \sum_{i=1}^{n} e_i^2 \qquad \rightarrow min \\ &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \qquad \rightarrow min \\ &= \sum_{i=1}^{n} (y_1 - b_1 - b_2 \cdot x_i)^2 \qquad \rightarrow min \end{aligned}$$

$$\Rightarrow \quad \frac{\delta SAQ}{\delta b_1} = \frac{\delta SAQ}{\delta b_2} \overset{!}{=} 0$$ (necessary condition of a local minimum)

Regression coefficients:

$$b_1 = \frac{\sum_{i=1}^{n} x_i^2 \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i y_i}{n \sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i \right)^2}$$

$$b_2 = \frac{n \sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n \sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i \right)^2} \quad \text{or} \quad b_2 = \frac{s_{xy}}{s_{x^2}} \quad \text{or} \quad b_2 = r \cdot \frac{s_y}{s_x}$$

Regression function:

$\hat{y}_i = b_1 + b_2 x_i$ with $i = 1, ..., n$

or $\hat{y} = b_1 + b_2 x$

$\hat{y}_i =$ estimated values of y_i with $i = 1, ..., n$

$\hat{y} =$ estimated regression function

Example:

Following points (observations) are given:

x_i	1	2	3	4	5	6	7	8
y_i	6.36	7.12	8.22	9.55	10.40	11.51	12.58	13.67

with

$$\bar{x} = \frac{1}{8} \sum_{i=1}^{8} x_i = \frac{1}{8} \cdot 36 = 4.5$$

$$\bar{y} = \frac{1}{8} \sum_{i=1}^{8} y_i = \frac{1}{8} \cdot 79.41 = 9.93$$

$$\sum_{i=1}^{8} x_i = 36 \qquad \sum_{i=1}^{8} y_i = 79.41$$

$$\sum_{i=1}^{8} x_i^2 = 204 \qquad \sum_{i=1}^{8} y_i^2 = 835.68$$

$$\sum_{i=1}^{8} x_i y_i = 401.94$$

$$\sum_{i=1}^{8} (x_i - \bar{x})^2 = 12.25 + 6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25 + 12.25 = 42$$

$$b_1 = \frac{204 \cdot 79.41 - 36 \cdot 401.94}{8 \cdot 204 - (36)^2} = 5.1482$$

$$b_2 = \frac{8 \cdot 401.94 - 36 \cdot 79.41}{8 \cdot 204 - (36)^2} = 1.0618$$

$$\hat{y} = 5.1482 + 1.0618x$$

The ordinate value (y-value with $x = 0$) of the estimated regression function $\hat{y}$ is 5.1482 and the slope of $\hat{y}$ is 1.0618.

Properties of the Regression Function

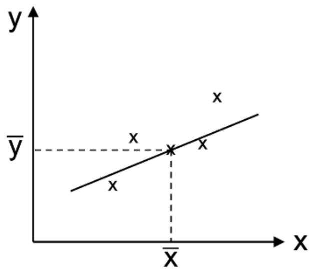
(1) $\sum\limits_{i=1}^{n} e_i = 0$

(2) $\sum\limits_{i=1}^{n} x_i e_i = 0$

(3) $\frac{1}{n} \sum\limits_{i=1}^{n} y_i = \frac{1}{n} \sum\limits_{i=1}^{n} \hat{y}_i$

(4) The regression line passes through the centre of gravity $\bar{P}(\bar{x};\bar{y})$ of the corresponding cluster of points

$\bar{x} = \frac{1}{n} \sum x_i$ or $\bar{y} = \frac{1}{n} \sum y_i$



(5) $s_E =$ standard deviation of the residuals e_i with $i =$ 1, ..., n

$s_E^2 =$ variance of the residuals e_i with $i =$ 1, ..., n

$$s_E^2 = \frac{1}{n-2} \sum_{i=1}^{n} e_i^2 =$$

$$= \frac{1}{n-2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 =$$

$$= \frac{1}{n-2} \left[\sum_{i=1}^{n} y_i^2 - b_1 \sum_{i=1}^{n} y_i - b_2 \sum_{i=1}^{n} x_i y_i \right]$$

2.5.1.1 Confidence Intervals for the Regression Coefficients of a Simple Linear Regression Function

For a simple linear regression function, the confidence intervals for the regression coefficients are shown in Tab. 2.9.

Parameter	Confidence Interval	Standard Error	Applicable Distribution
$\hat{b}_1$	$b_1 - ts_{B_1} \leq \hat{b}_1 \leq b_1 + ts_{B_1}$	$s_{B_1} = s_E \sqrt{\frac{\sum_{i=1}^{n} x_i^2}{n \sum_{i=1}^{n} (x_i - \bar{x})^2}}$	**Student's Distribution** with $v = n - 2$ Requirement: validity of the model assumptions
$\hat{b}_2$	$b_2 - ts_{B_2} \leq \hat{b}_2 \leq b_2 + ts_{B_2}$	$s_{B_2} = \frac{s_E}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}}$	

Tab. 2.9: Confidence Intervals for the Regression Coefficients of a Simple Linear Regression Function[17]

Example:

If $\hat{y} = 5.1482 + 1.0618x$ by considering a confidence level of 95%, the confidence intervals for the regression coefficients b_1 and b_2 are calculated as follows:

When $(1 - \alpha) = 0.95$ and $v = n - 2 = 8 - 2 = 6$, the t-value would be 2.447 (Student's t-distribution, two-sided symmetric confidence interval; see appendix A, statistical tables).

Confidence intervals

(1) for the regression coefficient $\hat{b}_1$:

$$b_1 - ts_{B_1} \leq \hat{b}_1 \leq b_1 + ts_{B_1}$$

$$5.1482 - 2.447 \cdot s_{B_1} \leq \hat{b}_1 \leq 5.1482 + 2.447 \cdot s_{B_1}$$

[17] Cf. Bleymüller, J. & Gehlert, G. (2011), p. 54.

$$s_{B_1} = s_E \sqrt{\frac{\sum\limits_{i=1}^{n} x_i^2}{n \sum\limits_{i=1}^{n} (x_i - \bar{x})^2}}$$

with

$$s_E^2 = \frac{1}{n-2} \left[\sum_{i=1}^{n} y_i^2 - b_1 \sum_{i=1}^{n} y_i - b_2 \sum_{i=1}^{n} x_i y_i \right] =$$

$$= \frac{1}{8-2} \left[835.68 - 5.1482 \cdot 79.41 - 1.0618 \cdot 401.94 \right] =$$

$$= 0.0136$$

$$s_E = \sqrt{0.0136} = 0.1166$$

$$s_{B_1} = 0.1166 \cdot \sqrt{\frac{204}{8 \cdot 42}} = 0.0909$$

$$5.1482 - 2.447 \cdot 0.0909 \leq \hat{b}_1 \leq 5.1482 + 2.447 \cdot 0.0909$$

$$+4.9258 \leq \hat{b}_1 \leq +5.3706$$

With a probability of $95\,\%$ (the selected confidence level), the unknown regression coefficient $\hat{b}_1$, which determines the ordinate value (y-value with $x = 0$) of the estimated (unknown) regression function, assumes a value between $+4.9258$ and $+5.3706$.

(2) for the regression coefficient $\hat{b}_2$:

$$b_2 - t s_{B_2} \leq \hat{b}_2 \leq b_2 + t s_{B_2}$$

$$1.0618 - 2.447 \cdot s_{B_2} \leq \hat{b}_2 \leq 1.0618 + 2.447 \cdot s_{B_2}$$

$$
\begin{aligned}
s_{B_2} &= \frac{s_E}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}} \\
&= \frac{0.1166}{\sqrt{42}} = \\
&= 0.0180
\end{aligned}
$$

$$
\begin{aligned}
1.0618 - 2.447 \cdot 0.018 \leq \hat{b}_2 &\leq 1.0618 + 2.447 \cdot 0.018 \\
+1.0178 \leq \hat{b}_2 &\leq +1.1058
\end{aligned}
$$

With a probability of $95\,\%$ (the selected confidence level), the unknown regression coefficient $\hat{b}_2$, which determines the slope of the estimated (unknown) regression function, assumes a value between $+1.0178$ and $+1.1058$.

2.5.1.2 Student's t-Tests for the Regression Coefficients of a Simple Linear Regression Function

For a simple linear regression function, the Student's t-tests for the regression coefficients are shown in Tab. 2.10.

Parameter	Standard Error			Applicable Distribution
$\hat{b}_1 = 0$	$t = \frac{b_1}{s_{B_1}}$	with	$s_{B_1} = s_E \sqrt{\frac{\sum_{i=1}^{n} x_i^2}{n \sum_{i=1}^{n}(x_i-\bar{x})^2}}$	**Student's Distribution** with $v = n-2$ Requirement: validity of the model assumptions
$\hat{b}_2 = 0$	$t = \frac{b_2}{s_{B_2}}$	with	$s_{B_2} = \frac{s_E}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}}$	

Tab. 2.10: Student's t-Tests for the Regression Coefficients of a Simple Linear Regression Function[18]

[18] Cf. Bleymüller, J. & Gehlert, G. (2011), p. 54.

For testing hypotheses of stochastic parameters like regression coefficients, the practical procedure is as follows:

a. Definition of null hypothesis (H_0) and alternative hypothesis (H_A) as well as significance level (α)

b. Determination of the test statistic

c. Determination of the test distribution

d. Identification of the critical range

e. Calculation of the value of the test statistic

f. Decision and interpretation

(1) Test of the regression coefficient $\hat{b}_1$

a. H_0: $\hat{b}_1 = 0$
   H_A: $\hat{b}_1 \neq 0$
   $\alpha = 0.05 \quad (1-\alpha) = 0.95$ (in the example above)

   H_0 means that the ordinate value (y-value with $x = 0$) of the estimated regression function would be zero. For example, in a cost function, the fixed costs are assumed to be zero.

   H_A can be $\hat{b}_1 \neq 0$, $\hat{b}_1 > 0$ or $\hat{b}_1 < 0$. For example, in a cost function, H_A would be defined as $\hat{b}_1 > 0$.

b. Test statistic

$$t = \frac{b_1}{s_{B_1}}$$ (table 2.10)

with $$s_{B_1} = s_E \sqrt{\frac{\sum_{i=1}^{n} x_i^2}{n \sum_{i=1}^{n} (x_i - \bar{x})^2}}$$

e.g. $s_{B_1} = 0.0909$ (in the example above)
with $b_1 = 5.1482$

c. Determination of the test distribution

Student's t-distribution, two-sided symmetric confidence interval with $(1-\alpha)$ and $\nu = n-2$ (see appendix A, statistical tables)

e.g. with $(1-\alpha) = 0.95$ and $\nu = 8-2 = 6$
(in the example above)

d. Identification of the critical range

For $(1-\alpha) = 0.95$ and $\nu = 6$, the critical t-value, t_c, is 2.447 (Student's t-distribution, two-sided symmetric confidence interval; see appendix A, statistical tables).

If $t = \dfrac{b_1}{s_{B_1}} > 2.447$, the null hypothesis H_0 has to be rejected.

If $t = \dfrac{b_1}{s_{B_1}} \leq 2.447$, the null hypothesis H_0 cannot be rejected.

e. Calculation of the value of the test statistic

$$t = \frac{5.1482}{0.0909} = 56.636 \quad \text{(in the example above)}$$

f. Decision and interpretation

$t > t_c \quad (56.636 > 2.447)$

H_0 has to be rejected.

The observed value for b_1 $(b_1 = 5.1482)$ is statistically valid with a significance level of 0.05 $(= 5\ \%)$.

In case of a cost function, there would be a $95\ \%$ probability that the fixed costs are positive at a level of approximately 5.1482 monetary units (e.g. $\$5.1482$ bln).

(2) Test of the regression coefficient $\hat{b}_2$

a. H_0: $\hat{b}_2 = 0$

H_A: $\hat{b}_2 \neq 0$

$\alpha = 0.05 \quad (1-\alpha) = 0.95$ (in the example above)

H_0 implies that the slope of the real function would be zero, which means there is no correlation between the tested variables (y and x_2 in the example above).

H_A can be $\hat{b}_2 \neq 0$, $\hat{b}_2 > 0$ (positive correlation between x and y) or $\hat{b}_2 < 0$ (negative correlation between x_2 and y).

b. Test statistic

$$t = \frac{b_2}{s_{B_2}}$$ (table 2.10)

with $$s_{B_2} = \frac{s_E}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}}$$

e.g. $s_{B_2} = 0.0180$ (in the example above)

with $b_2 = 1.0618$

c. Determination of the test distribution

Student's t-distribution, two-sided symmetric confidence interval with $(1-\alpha)$ and $\nu = n-2$ (see appendix A, statistical tables)

e.g. with $(1-\alpha) = 0.95$ and $\nu = 8-2 = 6$

(in the example above)

d. Identification of the critical range

For $(1-\alpha) = 0.95$ and $\nu = 6$, the critical t-value, t_c, is 2.447 (Student's t-distribution, two-sided symmetric confidence interval; see appendix A, statistical tables).

If $t = \dfrac{b_2}{s_{B_2}} > 2.447$, the null hypothesis H_0 has to be rejected.

If $t = \dfrac{b_2}{s_{B_2}} \leq 2.447$, the null hypothesis H_0 cannot be rejected.

e. Calculation of the value of the test statistic

$t = \dfrac{1.0618}{0.0180} = 58.989$ (in the example above)

f. Decision and interpretation

$t > t_c \quad (58.989 > 2.447)$

H_0 has to be rejected.

The observed value for b_2 $(b_2 = 1.0618)$ is statistically valid with a significance level of 0.05 $(= 5\ \%)$.

There is a significant correlation between the tested variables (y and x_2 in the example above).

2.5.2 Multiple Linear Regression

Description of the relationship (dependence) between a dependent variable y and multiple independent variables x_i, with $i = 1, ..., n$ using a linear function.

Symbols:

x_{ji} fixed (non-random) value of the independent random variable X_j ($j = 2, ..., k$) at the i^{th} observation $(i = 1, ..., n)$ with $x_{1i} = 1$ for all i

y_i stochastic (random) value of the dependent variable y at the i^{th} observation

$\hat{y}_i$ the estimated value provided by the sampling regression line for y_i

b_j the regression coefficient sought for the independent variable x_j $(j = 2, ..., k)$ for the regression function of the population

e_i residual $(e_i = y_i - \hat{y}_i)$; value of the deviation between the observed value y_i and the value $\hat{y}_i$ estimated by the regression function at the point x_i

$[\hat{y}]$ column vector of dimension $n \times 1$ containing the estimated values of the dependent variables y

$[b]$ column vector of dimension $k \times 1$ containing the sought sample regression coefficients

$[X]$ matrix of the dimension $n \times k$ (n: observations, k: characteristic values considered in the estimation), which contains the observed values of the independent variables

Remark: $x_{1i} = 1$ for all i with $i = 1, ..., n$

$$[\hat{y}] = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix}_{n\times 1} \qquad [b] = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_k \end{bmatrix}_{k\times 1} \qquad [X] = \begin{bmatrix} 1 & x_{21} & x_{31} & \dots & x_{k1} \\ 1 & x_{22} & x_{32} & \dots & x_{k2} \\ 1 & x_{23} & x_{33} & \dots & x_{k3} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{2n} & x_{3n} & \dots & x_{kn} \end{bmatrix}_{n\times k}$$

Least Squares Method

Minimise the sum of squared deviation (SAQ):

$$\begin{aligned}
SAQ &= \sum_{i=1}^{n} e_i^2 \qquad \rightarrow min \\
&= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \qquad \rightarrow min \\
&= \sum_{i=1}^{n} (y_i - b_1 - b_2 x_{2i} - b_3 x_{3i} - \ldots - b_k x_{ki})^2 \qquad \rightarrow min \\
&\Rightarrow \quad \frac{\delta SAQ}{\delta b_1} = \frac{\delta SAQ}{\delta b_2} = \frac{\delta SAQ}{\delta b_3} = \ldots = \frac{\delta SAQ}{\delta b_k} \overset{!}{=} 0
\end{aligned}$$

Regression coefficients:

$$[b]_{k\times 1} = ([X]' \cdot [X])^{-1}_{k\times k} \cdot [X]'_{k\times n} \cdot [y]_{n\times 1} \qquad \text{with} \qquad b = (b_1, \ldots, b_k)'$$

Regression function:

- Using the normal equation

  $$\hat{y}_i = b_1 + b_2 x_{2i} + b_3 x_{3i} + \ldots + b_k x_{ki} \qquad \text{with} \qquad i = 1, \ldots, n$$

- Using the matrix notation

  $$[\hat{y}]_{n\times 1} = [X]_{n\times k} \cdot [b]_{k\times 1}$$

Partial Linear Coefficient of Determination

$$SST = \sum_{i=1}^{n} (y_i - \bar{y}_i)^2 = \sum_{i=1}^{n} y_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)^2$$

$$SSE = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

Multiple Linear Coefficient of Determination

$$r^2_{Y\cdot 2,3,\ldots,k} = \frac{SSE}{SST} = \frac{\sum\limits_{i=1}^{n} (\hat{y}_i - \bar{y})^2}{\sum\limits_{i=1}^{n} (y_i - \bar{y})^2} = 1 - \frac{SSR}{SST} = 1 - \frac{\sum\limits_{i=1}^{n} e_i^2}{\sum\limits_{i=1}^{n} (y_i - \bar{y})^2}$$

$$0 \leq r^2_{Y\cdot 2,3,\ldots,k} \leq 1$$

with

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{1}{n}(\sum y_i)^2$$

$$SSE = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

$$SSR = \sum_{i=1}^{n} (y_i - \hat{y})^2 = \sum_{i=1}^{n} e_i^2$$

$$= \sum_{i=1}^{n} y_i^2 - b_1 \sum_{i=1}^{n} y_i - b_2 \sum_{i=1}^{n} x_{2i} y_i - \ldots - b_k \sum_{i=1}^{n} x_{ki} y_i$$

Multiple Linear Correlation Coefficient

$$r_{Y\cdot 2,3,\ldots,k} = \sqrt{r^2_{Y\cdot 2,3,\ldots,k}} \qquad 0 \leq r_{Y\cdot 2,3,\ldots,k} \leq 1$$

2.5.2.1 Confidence Intervals for the Regression Coefficients of a Multiple Linear Regression Function

For a multiple linear regression function, the confidence intervals for the regression coefficients are shown in Tab. 2.11.

Parameter	Confidence Interval	Standard Error	Applicable Distribution
$\hat{b}_j$ $j = 1, ..., k$	$b_j - ts_{B_j} \leq \hat{b}_j \leq b_j + ts_{B_j}$	$s_{B_j} = \sqrt{\widehat{CV}_{jj}}$ $\widehat{CV} = s_E^2\,([X]' \cdot [X])^{-1}$ s_{Bj}^2 with $j = 1, ... k$ correspond to the elements at the main diagonal of the estimated covariance matrix $\widehat{CV}$.	**Student's Distribution** with $v = n - k$ Requirement: validity of the model assumptions

Tab. 2.11: Confidence Intervals for the Regression Coefficients of a Multiple Linear Regression Function[19]

A covariance matrix (also known as auto-covariance matrix or variance-covariance matrix) is a square matrix which shows the covariances between each pair of elements of a given random vector (of a multiple random variable). Any covariance matrix is symmetric and positive semi-definite. Its main diagonal contains the covariances (the variances of each element with itself).

The estimated cavariance matrix can be calculated as follows:

$$\widehat{CV} = s_E^2\left([X]'_{n\times n} \cdot [X]_{n\times n}\right)^{-1}$$

with $s_E^2 = \dfrac{1}{n-k}\sum e_i^2$

$n =$ number of observations (elements)
$k =$ number of variables

$$s_E^2 = \frac{1}{n-k}\left[\sum_{i=1}^{n} y_i^2 - b_1\sum_{i=1}^{n} y_i - b_2\sum_{i=1}^{n} x_{2i}y_i - \ldots - b_k\sum_{i=1}^{n} x_{ki}y_i\right]$$

[19] Cf. Bleymüller, J. & Gehlert, G. (2011), p. 61.

2.5.2.2 Student's t-Tests for the Regression Coefficients of a Multiple Linear Regression Function

For a multiple linear regression function, the Student's t-tests for the regression coefficients are shown in Tab. 2.12.

Hypothesis	Test Statistic	Applicable Distribution
$\hat{b}_j = 0$ $j = 1, \ldots, k$	$t = \frac{b_j}{s_{B_j}}$ with $s_{B_j} = \sqrt{\widehat{CV}_{jj}}$ $\widehat{CV} = s_E^2\,([X]' \cdot [X])^{-1}$ s_{Bj}^2 with $j = 1, \ldots k$ correspond to the elements at the main diagonal of the estimated covariance matrix $\widehat{CV}$.	**Student's Distribution** with $\nu = n - k$ Requirement: validity of the model assumptions

Tab. 2.12: Student's t-Tests for the Regression Coefficients of a Multiple Linear Regression Function[20]

The Student's t-Tests for the Regression Coefficients of a Multiple Regression Function take place analogously as the Student's t-Tests for the Regression Coefficients of a Linear Regression Function (chapter 2.5.1.2).

2.5.3 Double Linear Regression

Regression function:

- Regression function for the population (normal equation)

$$\hat{y} = b_1 + b_2 x_{2i} + b_3 x_{3i} \qquad i = 1, \ldots, n$$

- Regression function for the population (matrix notation)

$$[\hat{y}]_{n\times 1} = [X]_{n\times 3} \cdot [b]_{3\times 1}$$

[20] Cf. Bleymüller, J. & Gehlert, G. (2011), p. 62.

Regression coefficients:

- Regression coefficients by using equations

$$
\begin{aligned}
b_1 n \quad & + b_2 \sum x_{2i} && + b_3 \sum x_{3i} && = \sum y_i \\
b_1 \sum x_{2i} & + b_2 \sum x_{2i}^2 && + b_3 \sum x_{2i} x_{3i} && = \sum x_{2i} y_i \\
b_1 \sum x_{3i} & + b_2 \sum x_{3i} x_{2i} && + b_3 \sum x_{3i}^2 && = \sum x_{3i} y_i
\end{aligned}
$$

- Regression coefficients by using a matrix notation

$$
[b]_{3\times 1} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \qquad [b]_{3\times 1} = ([X]' \, [X])^{-1}_{3\times 3} \cdot [X]'_{3\times n} \cdot [y]_{n\times 1}
$$

Example:

Following revenues (observations) are given:

i	y_i revenue in $\$1,000$	x_{1i}	x_{2i} price of an offered good USD/unit	x_{3i} sales area in square metres
1	$2,512$	1	112	$1,980$
2	$1,810$	1	108	$1,400$
3	$1,635$	1	104	$1,420$
4	$1,487$	1	100	$1,160$
5	$2,270$	1	110	$1,750$
6	$1,805$	1	108	$1,400$
7	$1,984$	1	109	$1,560$
8	$2,043$	1	110	$1,590$
9	$1,943$	1	108	$1,500$
10	$2,170$	1	111	$1,700$
11	$1,820$	1	108	$1,430$
12	$1,440$	1	102	$1,110$

Tab. 2.13: Example of a double linear regression

$$[X]' \cdot [X] = \begin{pmatrix} 12 & 1290 & 18000 \\ 1290 & 138822 & 1943640 \\ 18000 & 1943640 & 27643600 \end{pmatrix}$$

$$([X]' \cdot [X])^{-1} = \begin{pmatrix} 249.685 & -2.8170 & 0.03549 \\ -2.8170 & 0.03224 & -0.00043 \\ 0.03549 & -0.00043 & 0.0000074 \end{pmatrix}$$

$$[X]' \cdot [Y] = \begin{pmatrix} 22919 \\ 2475344 \\ 35201790 \end{pmatrix}$$

$$
\begin{aligned}
&b_1 n &&+b_2 \sum x_{2i} &&+b_3 \sum x_{3i} &&= \sum y_i \\
&b_1 \sum x_{2i} &&+b_2 \sum x_{2i}^2 &&+b_3 \sum x_{2i} x_{3i} &&= \sum x_{2i} y_i \\
&b_1 \sum x_{3i} &&+b_2 \sum x_{3i} x_{2i} &&+b_3 \sum x_{3i}^2 &&= \sum x_{3i} y_i
\end{aligned}
$$

$$
\begin{aligned}
12\ b_1 + \quad 1290\ b_2 + \quad 18000\ b_3 &= 22919 \\
1290\ b_1 + \quad 138822\ b_2 + \quad 1943640\ b_3 &= 2475344 \\
18000\ b_1 + \quad 1943640\ b_2 + \quad 27643600\ b_3 &= 35201790
\end{aligned}
$$

This is a linear system of equations with three equations and three variables. A linear system of equations can be solved with the substitution method, the equalization method or the addition method.[21] A definite determination of n variables is only possible if n independent equations exist (unambiguously determinable equation system).

$$
\begin{aligned}
b_1 &= -1415.294 \\
b_2 &= 16.099 \\
b_3 &= 1.0631
\end{aligned}
$$

$\hat{y} = -1415.294 + 16.099\, x_2 + 1.0631\, x_3$
with $i = 1, \ldots, 12$

2.5.3.1 Confidence Intervals for the Regression Coefficients of a Double Linear Regression Function

$$s_E^2 = \frac{1}{n-k} \sum e_i^2$$

$n =$ number of observations (elements)
$k =$ number of variables

[21] Cf. Peren, F.W. (2021), pp. 57-60.

$$
\begin{aligned}
s_E^2 &= \frac{1}{n-3}\left[\sum y_i^2 - b_1 \sum y_i - b_2 \sum x_{2i} y_i - b_3 \sum x_{3i} y_i\right] \\
&= \frac{1}{12-3} \cdot (44861817 + 1415.294 \cdot 22919 - 16.099 \cdot 2475344 - \\
&\qquad - 1.0631 \cdot 35201990) \\
&= \frac{1}{9} \cdot 25354.181 \\
&= 2817.13 \\
s_E &= \sqrt{2817.13} = 53.08
\end{aligned}
$$

At a multiple linear regression function, it is recommendable to identify the variances respectively the standard deviations of the regression coefficients by using the (estimated) covariance matrix:

(a) $\widehat{CV} = s_E^2 \left([X]'_{n\times n} \cdot [X]_{n\times n}\right)^{-1}$

$$
= 2817.13 \cdot \begin{pmatrix} 249.685 & -2.8170 & 0.03549 \\ -2.8170 & 0.03224 & -0.00043 \\ 0.03549 & -0.00043 & 0.0000074 \end{pmatrix}
$$

$$
= \begin{pmatrix} 703395.10 & -7935.86 & 99.98 \\ -7935.86 & 90.82 & -1.2114 \\ 99.98 & -1.2114 & 0.02085 \end{pmatrix}
$$

(b) s_{Bj}^2 with $j = 1, \dots k$ correspond to the elements at the main diagonal of the estimated covariance matrix $\widehat{CV}$.

$s_{B1}^2 = 703395.10 \qquad s_{B1} = \sqrt{703395.10} = 838.69$

$s_{B2}^2 = 90.82 \qquad s_{B2} = 9.530$

$s_{B3}^2 = 0.02085 \qquad s_{B3} = 0.1444$

Confidence intervals when $(1-\alpha)=0.95$

(1) for the regression coefficient $\hat{b}_1$:

$$b_1 - ts_{B_1} \leq \hat{b}_1 \leq b_1 + ts_{B_1}$$

$$-1415.294 - 2.228 \cdot 838.69 \leq \hat{b}_1 \leq -1415.294 + 2.228 \cdot 838.69$$

$$-3283.90 \leq \hat{b}_1 \leq +453.31$$

(2) for the regression coefficient $\hat{b}_2$:

$$b_2 - ts_{B_2} \leq \hat{b}_2 \leq b_2 + ts_{B_2}$$

$$16.099 - 2.228 \cdot 9.530 \leq \hat{b}_2 \leq 16.099 + 2.228 \cdot 9.530$$

$$-5.134 \leq \hat{b}_2 \leq +37.332$$

(3) for the regression coefficient $\hat{b}_3$:

$$b_3 - ts_{B_3} \leq \hat{b}_3 \leq b_3 + ts_{B_3}$$

$$1.0631 - 2.228 \cdot 0.1444 \leq \hat{b}_3 \leq 1.0631 + 2.228 \cdot 0.1444$$

$$+0.7414 \leq \hat{b}_3 \leq +1.3848$$

2.5.3.2 Student's t-Tests for the Regression Coefficients of a Simple Linear Regression Function

For testing hypotheses of stochastic parameters like regression coefficients, the practical procedure is as follows:

a. Definition of null hypothesis (H_0) and alternative hypothesis (H_A) as well as significance level (α)

b. Determination of the test statistic

c. Determination of the test distribution

d. Identification of the critical range

e. Calculation of the value of the test statistic

f. Decision and interpretation

(1) Test of the regression coefficient $\hat{b}_1$

a. H_0: $\hat{b}_1 = 0$
H_A: $\hat{b}_1 \neq 0$
$\alpha = 0.05 \quad (1-\alpha) = 0.95$ (in the example above)

H_0 means that the ordinate value of the estimated regression function would be zero.

H_A can be $\hat{b}_1 \neq 0$, $\hat{b}_1 > 0$ or $\hat{b}_1 < 0$.

b. Test statistic

$$t = \frac{b_1}{s_{B_1}} \quad \text{(table 2.12)}$$

with $s_{B_1} = 838.69$ (in the example above)

c. Determination of the test distribution

Student's t-distribution, two-sided symmetric confidence interval with $(1-\alpha)$ and $\nu = n-2$ (see appendix A, statistical tables)

e.g. with $(1-\alpha) = 0.95$ and $\nu = 12-2 = 10$
(in the example above)

d. Identification of the critical range

For $(1-\alpha) = 0.95$ and $\nu = 10$, the critical t-value, t_c, is 2.228 (Student's t-distribution, two-sided symmetric confidence interval; see appendix A, statistical tables).

If $t = \dfrac{b_1}{s_{B_1}} > 2.228$, the null hypothesis H_0 has to be rejected.

If $t = \dfrac{b_1}{s_{B_1}} \leq 2.228$, the null hypothesis H_0 cannot be rejected.

e. Calculation of the value of the test statistic

$$t = \frac{-1415.294}{838.69} = -1{,}6875 \quad \text{(in the example above)}$$

f. Decision and interpretation

$t < t_c \quad (-1{,}6875 < 2.228)$

H_0 cannot be rejected.

The observed value for b_1 $(b_1 = -1415.294)$ is statistically invalid with a significance level of 0.05 $(= 5\,\%)$.

(2) Test of the regression coefficient $\hat{b}_2$

a. H_0: $\hat{b}_2 = 0$
H_A: $\hat{b}_2 \neq 0$

$\alpha = 0.05 \quad (1 - \alpha) = 0.95$ (in the example above)

H_0 implies that there would be no correlation between the tested variables (y and x_2 in the example above).

H_A can be $\hat{b}_2 \neq 0$, $\hat{b}_2 > 0$ (positive correlation between x_2 and y) or $\hat{b}_2 < 0$ (negative correlation between x_2 and y).

b. Test statistic

$t = \dfrac{b_2}{s_{B_2}}$ (table 2.12)

with $s_{B_2} = 9.530$ (in the example above)

c. Determination of the test distribution

Student's t-distribution, two-sided symmetric confidence interval with $(1-\alpha)$ and $\nu = n-2$ (see appendix A, statistical tables)

e.g. with $(1-\alpha) = 0.95$ and $\nu = 12-2 = 10$
(in the example above)

d. Identification of the critical range

For $(1-\alpha) = 0.95$ and $\nu = 6$, the critical t-value, t_c, is 2.228 (Student's t-distribution, two-sided symmetric confidence interval; see appendix A, statistical tables).

If $t = \dfrac{b_2}{s_{B_2}} > 2.228$, the null hypothesis H_0 has to be rejected.

If $t = \dfrac{b_2}{s_{B_2}} \leq 2.228$, the null hypothesis H_0 cannot be rejected.

e. Calculation of the value of the test statistic

$$t = \frac{16.099}{9.530} = 1.6893 \quad \text{(in the example above)}$$

f. Decision and interpretation

$t < t_c \quad (1.6893 < 2.228)$

H_0 cannot be rejected.

The observed value for b_2 $(b_2 = 16.099)$ is statistically invalid with a significance level of 0.05 $(= 5\ \%)$.

There is no significant correlation between the tested variables y (revenue) and x_2 (price) in the example above.

(3) Test of the regression coefficient $\hat{b}_3$

a. H_0: $\hat{b}_3 = 0$

H_A: $\hat{b}_3 > 0$ (a positive correlation between y (revenue) and x_3 (sales area) can be expected)

$\alpha = 0.05 \quad (1-\alpha) = 0.95$ (in the example above)

b. Test statistic

$$t = \frac{b_3}{s_{B_3}}$$ (table 2.12)

with $s_{B_3} = 0.1444$ (in the example above)

c. Determination of the test distribution

Student's t-distribution function with $(1-\alpha) = 0.95$ and $\nu = 12-2 = 10$ (in the example above)

d. Identification of the critical range

For $(1-\alpha) = 0.95$ and $\nu = 10$, the critical t-value, t_c, is 1.812 (Student's t-distribution, two-sided symmetric confidence interval; see appendix A, statistical tables).

If $t = \dfrac{b_3}{s_{B_3}} > 1.812$, the null hypothesis H_0 has to be rejected.

If $t = \dfrac{b_3}{s_{B_3}} \leq 1.812$, the null hypothesis H_0 cannot be rejected.

e. Calculation of the value of the test statistic

$$t = \frac{1.0631}{0.1444} = 7.3622$$ (in the example above)

f. Decision and interpretation

$t > t_c \quad (7.3622 > 1.812)$

H_0 has to be rejected.

The observed value for b_3 $(b_3 = 1.0631)$ is statistically valid with a significance level of 0.05 $(= 5\ \%)$.

There is a significant correlation between the tested variables y (revenue) and x_3 (sales area) in the example above.

Chapter 3
Inferential Statistics

3.1 Probability Calculation

3.1.1 Fundamental Terms/Definitions

Random Experiment

A process that can be repeated as often as desired and whose result depends on chance.

Elementary Event e_i

An elementary event e_i (also called a sample point) is an event which contains only the single outcome e_i in the sample space S.

Sample Space S

The sample space S (of an experiment or random trial) is the set of all elementary events or possible outcomes or results (of that experiment or that random trial):

$$S = \{e_1, e_2, \ldots, e_i, \ldots, e_n\}$$

Event A

Any subset of the sample space S.

F. W. Peren, *Statistics for Business and Economics*,
https://doi.org/10.1007/978-3-662-64276-4_3

Laplace's[1] Definition of Probability

If all elementary events are equally possible, then

$$P(A) = \frac{\text{number of favourable cases}}{\text{number of all cases}}$$

<u>Examples:</u> For the event A that a coin lands on heads in a toss, $P(A) = \frac{1}{2}$

For the event B that a die lands on a two in a roll, $P(B) = \frac{1}{6}$

Von Mises'[2] Definition of Probability

$$P(A) = \lim_{n\to\infty} f_n(A) = \lim_{n\to\infty} \frac{h_n(A)}{n}$$

<u>Example:</u> For the event A that a die lands on a two in infinite rolls, $P(A) = \frac{1}{6}$

The more often a die is rolled, the probability that this die lands on the number two approaches $\frac{1}{6}$.

[1] Pierre-Simon Laplace (1749 - 1827) was a French mathematician, physicist and astronomer.

[2] Richard Edler von Mises (1883 - 1953) was an Austrian-American mathematician.

Kolmogorov's[3] Definition of Probability

Axiom 1:	$0 \leq P(A) \leq 1$	for $A \subset S$	(non-negativity)
Axiom 2:	$P(S) = 1$		(standardisation)
Axiom 3:	$P(A \cup B) = P(A) + P(B)$	for $A \cap B = \varnothing$	(additivity)

Axiom (3) results in the relation

$$P(A_1 \cup A_2 \cup \ldots \cup A_n) = P(A_1) + P(A_2) + \ldots + P(A_n)$$

for $\quad A_i \cap A_j = \varnothing \quad (i \neq j)$

Rules of Calculation for Probabilities

(1) Complementary Probability (Counter Probability)

For $\overline{A}$, the complementary event of A, the following applies:

$P(\overline{A}) = 1 - P(A)$

Example:

The probability of rolling a six in one roll of a die is $\frac{1}{6}$. What is the probability of rolling a six when rolling a die four times?

$\overline{A}$ = rolling at least a six
A = not rolling any sixes

[3] Andrej Nikolaevič Kolmogorov (1903 - 1987) was a Soviet mathematician.

The probability of not rolling any sixes in four rolls is:

$$\frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} = \frac{625}{1,296} = 0.482 = 48.2\,\%$$

The counter probability is therefore:

$$1 - \frac{625}{1,296} = 0.518 = 51.8\,\%$$

(2) Probability of an Impossible Event

$P(\varnothing) = 0$

Example:

The probability of getting an eight in a single roll of a die is zero.
$\varnothing$: "getting eight points in a single roll of a die"

Indeed, a normal die only has the numbers one to six.
$\varnothing$: "number of dots is equal to eight"

$P(\varnothing) = 0$

(3) Addition Theorem for Two Arbitrary Events

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Example:

A = {rolling a number < 4}
B = {rolling an odd number}

$P(A) = \frac{3}{6}$ and $P(B) = \frac{3}{6}$

Probability of $A \cap B$:

$$\begin{aligned} P(A \cap B) &= P(\{\text{rolling an odd number less than } 4\}) \\ &= P(\{\text{rolling } 1 \text{ or } 3\}) \\ &= \frac{2}{6} = \frac{1}{3} \end{aligned}$$

Accordingly:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) = \\ &= \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \\ &= \frac{2}{3} = 66.67\ \% \end{aligned}$$

(4) Condition of Probability for a Sub-Event

$$P(A) \leq P(B) \qquad \text{for} \quad A \subset B$$

<u>Example</u>:

$$\begin{aligned} A &= \{\text{rolling an odd number} < 4\} \\ B &= \{\text{rolling a number} < 4\} \end{aligned}$$

$$P(A) = \frac{2}{6} \quad \text{and} \quad P(B) = \frac{3}{6}$$

$$P(A) < P(B) \qquad \frac{2}{6} < \frac{3}{6}$$

3.1.2 Theorems of Probability Theory

Multiplication Theorem

For two stochastically independent events A and B, the following applies:

$$P(A \cap B) = P(A) \cdot P(B)$$

Example:

What is the probability of getting "heads" twice if a coin is tossed twice?

For each toss, the probability of getting "heads" is $\frac{1}{2}$, so the following applies:

$$\begin{aligned} P(A \cap B) &= P(A) \cdot P(B) = \\ &= \frac{1}{2} \cdot \frac{1}{2} = \\ &= \frac{1}{4} = 25\,\% \end{aligned}$$

For two stochastically dependent events A and B, the following applies:

$$P(A \cap B) = P(A) \cdot P(B/A) = P(B) \cdot P(A/B)$$

Example:

The probability of being a motorcyclist in a traffic accident last year was $31\,\%$. Of these, $46\,\%$ were not wearing a helmet. What is the probability of being a motorcyclist in a traffic accident who is not wearing a helmet?

$$\begin{aligned} P(A) &= P(\text{accident as a motorcyclist}) = 0.31 \\ P(B/A) &= P(\{\text{without helmet}\} \setminus \{\text{accident as a motorcyclist}\}) = 0.46 \end{aligned}$$

$$\begin{aligned} P(A \cap B) &= P(A) \cdot P(B/A) = \\ &= 0.31 \cdot 0.46 = \\ &= 0.1426 = 14.26\,\% \end{aligned}$$

Conditional Probability

For $P(A) > 0$, the conditional probability of event B under the condition A is defined as

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

Example:

Out of 20 students, 4 own a car. 12 of these 20 students are male. 3 of the 12 male students own a car and the remaining 9 do not. What is the probability that a randomly selected student in this group who owns a car is male?

A: The selected student has a car.
B: The selected student is male.

Fourfold table:

	B	$\overline{B}$	
A	3	1	4
$\overline{A}$	9	7	16
	12	8	20

Calculating probabilities:

	B	$\overline{B}$	
A	$\frac{3}{20} = 0.15$	$\frac{1}{20} = 0.05$	$\frac{4}{20} = 0.2$
$\overline{A}$	$\frac{9}{20} = 0.45$	$\frac{7}{20} = 0.35$	$\frac{16}{20} = 0.8$
	$\frac{12}{20} = 0.6$	$\frac{8}{20} = 0.4$	$\frac{20}{20} = 1$

Conditional probability:

$$
\begin{aligned}
P(B/A) &= \frac{P(A \cap B)}{P(A)} = \\
&= \frac{0.15}{0.2} = \\
&= 0.75 = 75\,\%
\end{aligned}
$$

The probability of a randomly selected student in this group who owns a car being male is $75\,\%$.

$$
\begin{aligned}
\text{with} \quad P(A \cap B) &= P(A) \cdot P(B/A) = P(B) \cdot P(A/B) = \\
&= 0.2 \cdot (0.15/0.2) = 0.6 \cdot (0.15/0.6) = \\
&= 0.15 = 15\,\%
\end{aligned}
$$

The proportion of students who A) owns a car and B) is male is $15\,\%$.

Stochastic Independence

Two events A and B are stochastically independent if the following applies:

$$P(A/B) = P(A/\overline{B}) \quad \vee \quad P(B/A) = P(B/\overline{A})$$

or $\quad P(A \cap B) = P(A) \cdot P(B)$

<u>Example</u>:

A die is rolled once. Event A is "even number of dots" and event B is "number of dots less than 5".

A: 2, 4, 6
B: 1, 2, 3, 4

$$P(A) = \frac{3}{6} = \frac{1}{2} \qquad P(\overline{A}) = \frac{3}{6} = \frac{1}{2}$$

$$P(B) = \frac{4}{6} = \frac{2}{3} \qquad P(\overline{B}) = \frac{2}{6} = \frac{1}{3}$$

$$P(A \cap B) = \text{events } 2 \text{ and } 4 = \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6} = \frac{1}{3}$$

$$P(\overline{A} \cap B) = \text{events } 1 \text{ and } 3 = \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6} = \frac{1}{3}$$

Furthermore, the following applies in the case of stochastic independence:

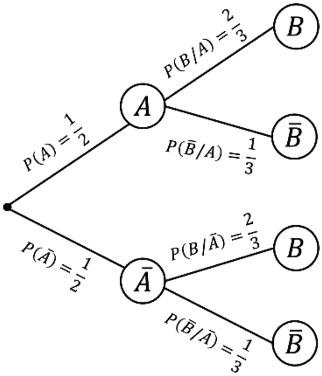
$$P(B/A) = P(B/\overline{A})$$

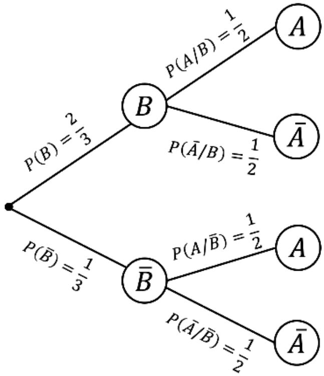
with

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{3} \Big/ \frac{1}{2} = \frac{2}{3}$$

$$P(B/\overline{A}) = \frac{P(\overline{A} \cap B)}{P(A)} = \frac{1}{3} \Big/ \frac{1}{2} = \frac{2}{3}$$

$$P(B/A) = P(B/\overline{A}) \qquad\qquad P(A/B) = P(A/\overline{B})$$
$$\frac{2}{3} = \frac{2}{3} \qquad\qquad \frac{1}{2} = \frac{1}{2}$$





Law of Total Probability

If $A_1 \cup A_2 \cup ... \cup A_n = S$ and $A_i \cap A_j = \varnothing$ for $i \neq j$, the following is true for $E \subset S$:

$$P(E) = \sum_{i=1}^{n} P(A_i) \cdot P(E/A_i) \qquad \text{with} \qquad i = 1, ..., n$$

Example:

A company manufactures $2,000$ units of a product daily. Out of these, the following machines produce:

M_1 500 units with a scrap rate of $5\,\%$,
M_2 800 units with a scrap rate of $4\,\%$,
M_3 700 units with a scrap rate of $2\,\%$.

One unit is randomly selected from a daily production. What is the probability that the selected unit is defective?

A_i : The selected unit was produced by M_i.
E : The selected unit is defective.

$$P(A_1) = \frac{500}{2,000} = 0.25 \quad \text{and} \quad P(E/A_1) = \frac{5}{100} = 0.05$$

$$P(A_2) = \frac{800}{2,000} = 0.4 \quad \text{and} \quad P(E/A_2) = \frac{4}{100} = 0.04$$

$$P(A_3) = \frac{700}{2,000} = 0.35 \quad \text{and} \quad P(E/A_3) = \frac{2}{100} = 0.02$$

Searched probability:

$$\begin{aligned} P(E) &= \sum_{i=1}^{3} P(A_i) \cdot P(E/A_i) = \\ &= 0.25 \cdot 0.05 + 0.4 \cdot 0.04 + 0.35 \cdot 0.02 = \\ &= 0.0355 \end{aligned}$$

The probability of a unit being defective is approximately $3.55\ \%$.

Bayes'[4] Theorem

If $A_1 \cup A_2 \cup ... \cup A_n = S$ and $A_i \cap A_j = \varnothing$ for $i \neq j$, the following is true for $E \subset S$:

$$P(A_j/E) = \frac{P(A_j) \cdot P(E/A_j)}{\sum_{i=1}^{n} P(A_i) \cdot P(E/A_i)} \qquad \text{with} \qquad i, j = 1, ..., n$$

[4] Thomas Bayes (1701 - 1761) was an English mathematician and statistician.

Example:

The example of the "Law of Total Probability" is considered again. One unit is selected from a daily production again. In this case, it is a defective unit.

What is the probability that this unit comes from M_1, M_2 or M_3?

$$
\begin{aligned}
P(A_1/E) &= \frac{P(A_1)\cdot P(E/A_1)}{\sum_{i=1}^{3} P(A_i)\cdot P(E/A_i)} = \\
&= \frac{0.25\cdot 0.05}{0.25\cdot 0.05\ +\ 0.4\cdot 0.04\ +\ 0.35\cdot 0.02} = \\
&= \frac{0.0125}{0.0355} = 0.35
\end{aligned}
$$

$$P(A_2/E) = \frac{0.4\cdot 0.04}{0.0355} = 0.45$$

$$P(A_3/E) = \frac{0.35\cdot 0.02}{0.0355} = 0.20$$

The probability of the randomly selected unit being from machine 1 is $35\ \%$, of it being from machine 2 is $45\ \%$, and of it being from machine 3 is $20\ \%$.

3.2 Probability Distributions

3.2.1 Concept of Random Variables

To express or process the result of a random experiment quantitatively, it must be transformed into a real number, if possible.

The random variable X comprises a certain number of n elementary events e_j with $j = 1, 2, ..., n$ in the sample space S.

Domain: sample space S
Codomain: set of real numbers

A distinction must be made:

1. *Discrete random variables:* Each possible event can be assigned a specific probability of occurrence (e.g. rolling a die).
2. *Continuous random variables:* There is an infinite number of possible manifestations of a characteristic. The possibility of determining the probability of occurrence is almost zero.

3.2.2 Probability, Distribution and Density Function

3.2.2.1 Discrete Random Variables

Probability Function

$$f(x_i) = P(X = x_i) \qquad \text{with} \qquad i = 1, 2, ...$$

Characteristics of a probability function:

(1) $f(x_i) \geq 0$ with $i = 1, 2, ...$

(2) $\sum_i f(x_i) = 1$

Distribution Function

$$F(x) = P(X \leq x)$$

Characteristics of a distribution function:

(1) $F(x)$ is monotonically increasing

(2) $F(x)$ is continuous

(3) $\lim_{x \to -\infty} F(x) = 0$

(4) $\lim_{x \to +\infty} F(x) = 1$

3.2.2.2 Continuous Random Variables

Instead of probabilities, so-called densities are given. This is done in the form of density functions.

Probability Density

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Characteristics of each probability density:

(1) $f(x) \geq 0$

(2) $\int_{-\infty}^{+\infty} f(x)dx = 1$

Distribution Function

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(q)dq$$

$$\Rightarrow F'(x) = f(x)$$

Properties of each distribution function of continuous random variables:

(1) $0 \leq F(x) \leq 1$

(2) $F(x)$ is monotonically increasing,

for $x_1 < x_2$ applies $F(x_1) \leq F(x_2)$

(3) $\lim\limits_{x \to -\infty} F(x) = 0$

(4) $\lim\limits_{x \to +\infty} F(x) = 1$

(5) $F(x)$ is continuous throughout the entire domain.

3.2.3 Parameters for Probability Distributions

Expected Value and Variance of Random Variables

Discrete Random Variables

$$E(X) = \sum_i x_i\, f(x_i)$$

$$\begin{aligned} Var(X) &= E\left[[X - E(X)]^2\right] \\ &= \sum_i [x_i - E(X)]^2\, f(x_i) \\ &= \sum_i x_i^2 \cdot f(x_i) - [E(X)]^2 \end{aligned}$$

Continuous Random Variables

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

$$\begin{aligned} Var(X) &= E\left[[X - E(X)]^2\right] \\ &= \int_{-\infty}^{+\infty} [x - E(X)]^2 f(x) dx \\ &= \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx - [E(X)]^2 \end{aligned}$$

3.3 Theoretical Distributions

3.3.1 Discrete Distributions

Binomial Distribution

Probability Function

$$f_B(x/n;\,\theta) = \begin{cases} \binom{n}{x}\,\theta^x(1-\theta)^{n-x} & \text{for } x = 0, 1, ..., n \\ 0 & \text{for } x \neq 0, 1, ..., n \end{cases}$$

Distribution Function

$$F_B(x/n;\,\theta) = \sum_{v=0}^{x} \binom{n}{v}\theta^v\,(1-\theta)^{n-v}$$

Expected Value

$$E(X) = n \cdot \theta$$

Variance

$$Var(X) = n \cdot \theta(1-\theta)$$

Recursive Formula

$$f_B(x+1/n;\,\theta) = f_B(x/n;\,\theta) \cdot \frac{n-x}{x+1} \cdot \frac{\theta}{1-\theta}$$

Hypergeometric Distribution

Probability Function

$$f_H(x/N;n;M) = \begin{cases} \dfrac{\dbinom{M}{x}\dbinom{N-M}{n-x}}{\dbinom{N}{n}} & \text{for } x = 0, 1, ..., n \\ 0 & \text{for } x \neq 0, 1, ..., n \end{cases}$$

Distribution Function

$$F_H(x/N;n;M) = \sum_{v=0}^{x} \frac{\dbinom{M}{v}\dbinom{N-M}{n-v}}{\dbinom{N}{n}}$$

Expected Value

$$E(X) = n \cdot \frac{M}{N}$$

Variance

$$Var(X) = n \cdot \frac{M}{N} \cdot \frac{N-M}{N} \cdot \frac{N-n}{N-1}$$

Recursive Formula

$$f_H(x+1/N;n;M) = f_H(x/N;n;M) \cdot \frac{(M-x)(n-x)}{(x+1)(N-M-n+x+1)}$$

Poisson[5] Distribution

Probability Function

$$f_p(x/\mu) = \begin{cases} \dfrac{\mu^x e^{-\mu}}{x!} & \text{for } x = 0, 1, ... \\ 0 & \text{for } x \neq 0, 1, ... \end{cases}$$

$e = 2.71828...$ (Euler's Number)

Distribution Function

$$F_p(x/\mu) = \sum_{v=0}^{x} \frac{\mu^v e^{-\mu}}{v!}$$

Expected Value and Variance

$$E(X) = Var(X) = \mu$$

Recursive Formula

$$f_p(x+1/\mu) = f_p(x/\mu)\frac{\mu}{x+1}$$

[5] Siméon Denis Poisson (1781 - 1840) was a French physicist and mathematician.

3.3.2 Continuous Distributions

Normal Distribution

Probability Density

$$f_n(x/\mu;\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Distribution Function

$$F_n(x/\mu;\sigma^2) = \int\limits_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{1}{2}\left(\frac{q-\mu}{\sigma}\right)^2}\,dq$$

Expected Value

$$E(X) = \mu$$

Variance

$$Var(X) = \sigma^2$$

Standard Normal Distribution

If the random variable X is normally distributed with $E(X) = \mu$ and $Var(X) = \sigma^2$, then X becomes the standardised random variable Z with

$$Z = \frac{X-\mu}{\sigma}$$

and the expected value $E(Z) = 0$ and the variance $Var(Z) = 1$

Probability Density

$$f_N(z) = \frac{1}{\sqrt{2\pi}}\,e^{-\frac{1}{2}z^2}$$

Distribution Function

$$F_N(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}q^2}\, dq$$

Expected Value

$$E(Z) = 0$$

Variance

$$Var(Z) = 1$$

Chi-Square Distribution

Probability Density

$$f_{Ch}(\chi^2/v) = \begin{cases} \dfrac{1}{2^{v/2}\,\Gamma\left(\dfrac{v}{2}\right)}\, e^{-\frac{\chi^2}{2}}\, \left(\chi^2\right)^{\left(\frac{v}{2}-1\right)} & \text{for } \chi^2 \geq 0 \\ 0 & \text{for } \chi^2 < 0 \end{cases}$$

Distribution Function

$$F_{Ch}(\chi^2/v) = \frac{1}{2^{v/2}\,\Gamma\left(\dfrac{v}{2}\right)} \int_{0}^{\chi^2} e^{-\frac{q}{2}}\, q^{\left(\frac{v}{2}-1\right)}\, dq$$

Expected Value

$$E(\chi^2) = v$$

Variance

$$Var(\chi^2) = 2v$$

Student's Distribution

Probability Density

$$f_S(t/v) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\,\Gamma\left(\frac{v}{2}\right)} \cdot \frac{1}{\left(1+\frac{t^2}{v}\right)^{(v+1)/2}} \qquad -\infty < t < +\infty$$

Distribution Function

$$F_S(t/v) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\,\Gamma\left(\frac{v}{2}\right)} \cdot \int_{-\infty}^{t} \frac{dq}{\left(1+\frac{q^2}{v}\right)^{(v+1)/2}}$$

Expected Value

$$E(T) = 0 \qquad \text{for } v1$$

Variance

$$Var(T) = \frac{v}{v-2} \qquad \text{for } v > 2$$

F-Distribution

Probability Density

$$f_F(f/v_1;v_2) = \begin{cases} \dfrac{\Gamma\left(\dfrac{v_1+v_2}{2}\right)}{\Gamma\left(\dfrac{v_1}{2}\right)\Gamma\left(\dfrac{v_2}{2}\right)} \quad \dfrac{\left(\dfrac{v_1}{v_2}\right)^{\frac{v_1}{2}} f^{\frac{v_1}{2}-1}}{\left(1+\dfrac{v_1}{v_2}f\right)^{\frac{v_1+v_2}{2}}} & \text{for } f>0 \\ 0 & \text{for } f \leq 0 \end{cases}$$

Distribution Function

$$F_F(f/v_1;v_2) = \begin{cases} \dfrac{\Gamma\left(\dfrac{v_1+v_2}{2}\right)}{\Gamma\left(\dfrac{v_1}{2}\right)\Gamma\left(\dfrac{v_2}{2}\right)} \left(\dfrac{v_1}{v_2}\right)^{\frac{v_1}{2}} \int\limits_0^f \dfrac{q^{\frac{v_1}{2}-1}}{\left(1+\dfrac{v_1}{v_2}q\right)^{\frac{v_1+v_2}{2}}}\, dq & \text{for } f>0 \\ 0 & \text{for } f \leq 0 \end{cases}$$

Expected Value

$$E(F) = \frac{v_2}{v_2-2} \qquad \text{for} \quad v_2 > 2$$

Variance

$$Var(F) = \frac{2\,{v_2}^2\,(v_1+v_2-2)}{v_1(v_2-2)^2\,(v_2-4)} \qquad \text{for} \quad v_2 > 4$$

3.4 Statistical Estimation Methods (Confidence Intervals)

(1) Mean Value of the Population μ

Confidence interval: $\bar{x} - t\hat{\sigma}_{\bar{x}} \leq \mu \leq \bar{x} + t\hat{\sigma}_{\bar{x}}$

with $x =$ mean of the sample

sampling without replacement (w/o rep.)	sampling with replacement (w/ rep.)
σ is unknown	σ is known
$\hat{\sigma}_{\bar{x}} = \frac{s}{\sqrt{n}}\sqrt{\frac{N-n}{N}}$	$\sigma_x = \frac{\sigma}{\sqrt{n}}$

with $s =$ standard deviation of the sample

and $t =$ Student's distribution with $v = n-1$

If $\frac{n}{N} < 0.05$, $\sqrt{\frac{N-n}{N}}$ can be omitted.

(2) Variance of the Population σ^2

Confidence interval: $$\frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2};\,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2};\,n-1}}$$

with $s^2 =$ variance of the sample

and $n =$ sample size

and $\chi^2 =$ chi-square distribution with $v = n-1$

and sampling without replacement $=$ sampling with replacement, i.e. there is no case distinction

(3) Share Values in the Population θ

Confidence interval: $p - z\hat{\sigma}_p \leq \theta \leq p + z\hat{\sigma}_p$

with $p =$ share value in the sample

and $z =$ standard normal distribution

sampling without replacement	sampling with replacement
$\hat{\sigma}_p = \sqrt{\frac{p(1-p)}{n-1}}\sqrt{\frac{N-n}{N}}$	$\hat{\sigma}_p = \sqrt{\frac{p(1-p)}{n-1}}$

If $\frac{n}{N} < 0.05$, $\sqrt{\frac{N-n}{N}}$ can be omitted.

(4) Difference of the Mean Values of Two Populations μ_1 and μ_2

Confidence interval: $(\bar{x}_1 - \bar{x}_2) - t\hat{\sigma}_D \leq \mu_1 - \mu_2 \leq (\bar{x}_1 - \bar{x}_2) + t\hat{\sigma}_D$

with $\quad \bar{x}_1 - \bar{x}_2 =$ difference of the means of the two samples

and $\quad t =$ Student's distribution with $v = \dfrac{\left[\dfrac{s_1{}^2}{n_1} + \dfrac{s_2{}^2}{n_2}\right]^2}{\dfrac{\left[\dfrac{s_1{}^2}{n_1}\right]^2}{n_1 - 1} + \dfrac{\left[\dfrac{s_2{}^2}{n_2}\right]^2}{n_2 - 1}}$

and $\quad \hat{\sigma}_D = \hat{\sigma}_{\mu_1 - \mu_2} = \sqrt{\dfrac{s_1{}^2}{n_1} + \dfrac{s_2{}^2}{n_2}}$

with $\quad s_1{}^2 =$ variance of the first sample

and $\quad s_2{}^2 =$ variance of the second sample

and $\quad$ sampling without replacement $\widehat{=}$ sampling with replacement, i.e. sampling without replacement is equivalent to sampling with replacement

(5) Difference of the Share Values of Two Populations θ_1 and θ_2

Confidence interval: $\quad (p_1 - p_2) - z\hat{\sigma}_D \leq \theta_1 - \theta_2 \leq (p_1 - p_2) + z\hat{\sigma}_D$

with $\quad \bar{x}_1 - \bar{x}_2 =$ difference of the means of the two samples

and $\quad z =$ standard normal distribution

and $\quad \hat{\sigma}_D = \hat{\sigma}_{\mu_1 - \mu_2} = \sqrt{\dfrac{p_1(1 - p_1)}{n_1} + \dfrac{p_2(1 - p_2)}{n_2}}$

with $\quad p_1 =$ share value of the first sample

and $\quad p_2 =$ share value of the second sample

3.5 Determination of the Required Sample Size

Estimation of the arithmetic mean μ:

$$n = \frac{z^2 \cdot \sigma^2}{(\Delta\mu)^2}$$ sampling w/ rep. or $\frac{n}{N} < 0.05$

$$n = \frac{z^2 \cdot N \cdot \sigma^2}{(\Delta\mu)^2(N-1) + z^2 \cdot \sigma^2}$$ sampling w/o rep.

$\Delta\mu$ is defined as the absolute error of the arithmetic mean.

Estimation of the share value θ:

$$n = \frac{z^2 \cdot \theta(1-\theta)}{(\Delta\theta)^2}$$ sampling w/ rep. or $\frac{n}{N} < 0.05$

$$n = \frac{z^2 \cdot N \cdot \theta(1-\theta)}{(\Delta\mu)^2(N-1) + z^2 \cdot \theta(1-\theta)}$$ sampling w/o rep.

$\Delta\theta$ is defined as the absolute error of the share value.

Suitable estimates are used for σ^2 and θ. In practice, these can be determined through expert interviews (Delphi study).

3.6 Statistical Testing Methods

Random sampling is used to test certain assumptions (hypotheses) about unknown populations.

3.6.1 Parameter Tests

Testing hypotheses of unknown parameters of one or two populations.

<u>Practical Procedure</u>

a. Definition of null hypothesis (H_0) and alternative hypothesis (H_A) as well as significance level (α)

b. Determination of the test statistic

c. Determination of the test distribution

d. Identification of the critical range

e. Calculation of the value of the test statistic

f. Decision and interpretation

(1) a. Null hypothesis $\mu = \mu_0$ (σ unknown)

Testing a specific mean value μ_0

b. Test statistic

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

c. Test distribution

$t =$ Student's distribution with $\nu = n - 1$

(2) a. Null hypothesis $\theta = \theta_0$

Testing a specific share value θ_0

b. Test statistic

$$z = \frac{p - \theta_0}{\sqrt{\frac{\theta_0(1-\theta_0)}{n}}}$$

c. Test distribution

$z =$ standard normal distribution

(3) a. Null hypothesis $\sigma^2 = \sigma_0^2$

Testing a specific variance σ_0^2

b. Test statistic

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

c. Test distribution

$\chi^2 =$ chi-square distribution with $\nu = n - 1$

(4) a. Null hypothesis $\mu_1 = \mu_2$ (σ_1, σ_2 unknown and $\sigma_1 \neq \sigma_2$)

Testing the equality of the means of two populations

b. Test statistic

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

c. Test distribution

$z =$ standard normal distribution

(5) a. Null hypothesis $\theta_1 = \theta_2$

Testing the equality of equal share values of two populations

b. Test statistic

$$z = \frac{p_1 - p_2}{\sqrt{p(1-p)}\sqrt{\frac{n_1 + n_2}{n_1 n_2}}}$$

with $p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

c. Test distribution

$z =$ standard normal distribution

(6) a. Null hypothesis $\sigma_1^2 = \sigma_2^2$

Testing the equality of the variance of two populations

b. Test statistic

$$F = \frac{s_1^2}{s_2^2}$$

c. Test distribution

$F = F$-distribution with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$

3.6.2 Distribution Tests (Chi-Square Tests)

Testing hypotheses of unknown distributions of a population.

(1) Chi-square goodness of fit test

a. Null hypothesis

Testing of a sample taken from a population

b. Test statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(h_i^0 - h_i^e)^2}{h_i^e}$$

c. Test distribution

$\chi^2 =$ chi-square distribution with $\nu = k - m - 1$

k: number of classes

m: number of estimated parameters

Condition: $h_i^e \geq 5 \qquad i = 1, ..., k$

(2) Chi-square independence test

a. Null hypothesis

Testing two characteristics of two samples from two different populations

b. Test statistic

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{s} \frac{(h_{ij}^0 - h_{ij}^e)^2}{h_{ij}^e}$$

c. Test distribution

$\chi^2 =$ chi-square distribution with $\nu = (r-1)(s-1)$

Condition: $h_{ij}^e \geq 5 \qquad i = 1, ..., r$ and $j = 1, ..., s$

(3) Chi-square homogeneity test

a. Null hypothesis

Testing of two characteristics of two samples that come from the same population

b. Test statistic

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{s} \frac{\left(h_{ij}^0 - \frac{h_{i.}^0 h_{.j}^0}{n}\right)^2}{\frac{h_{i.}^0 h_{.j}^0}{n}}$$

c. Test distribution

$\chi^2 =$ chi-square distribution with $v = (r-1)(s-1)$

Condition: $h_{ij}^e \geq 5$ $\quad i = 1, ..., r$ and $j = 1, ..., s$

The following meanings apply:

h_i^o observed absolute frequency of a characteristic in its i^{th} occurrence ($i = 1, ..., k$)

h_i^e expected absolute frequency of a characteristic in its i^{th} occurrence ($i = 1, ..., k$)

h_{ij}^{o} observed absolute frequency of the combination of two characteristics.The first characteristic in its i^{th} occurrence ($i = 1, ..., r$) and the second characteristic in its j^{th} occurrence ($j = 1, ..., s$).

$$h_{i.}^{o} = \sum_{j=1}^{s} h_{ij}^{o} \qquad i = 1, ..., r$$

$$h_{.j}^{o} = \sum_{i=1}^{r} h_{ij}^{o} \qquad j = 1, ..., s$$

$$h_{..}^{o} = \sum_{i=1}^{r} h_{i.}^{o} = \sum_{j=1}^{s} h_{.j}^{o} = n \qquad i = 1, ..., r \text{ and } j = 1, ..., s$$

h_{ij}^{e} expected absolute frequency of the combination of two characteristics. The first characteristic in its i^{th} occurrence ($i = 1, ..., r$) and the second characteristic in its j^{th} occurrence ($j = 1, ..., s$).

$$h_{ij}^{e} = \frac{h_{i.}^{o} \cdot h_{.j}^{o}}{n} \qquad i = 1, ..., r \text{ and } j = 1, ..., s$$

<u>Yates's[6] Correction</u>

With $v = 1$ degree of freedom, a continuity correction (Yates's correction) must be carried out:

$$\chi_{corr}^{2} = \sum_{i=1}^{k} \frac{(|h_{i}^{o} - h_{i}^{e}| - 0.5)^{2}}{h_{i}^{e}}$$

or

$$\chi_{corr}^{2} = \sum_{i=1}^{r} \sum_{j=1}^{s} \frac{\left(|h_{ij}^{o} - h_{ij}^{e}| - 0.5\right)^{2}}{h_{ij}^{e}}$$

[6] Frank Yates (1902 - 1994) was an English statistician.

Chapter 4

Probability Calculation

4.1 Terms and Definitions

Definition:

Probability calculation forms the basis of inferential statistics.

Many events of economic decisions are not strictly determined (predictable), but are stochastic, i.e. determined by chance.

Fundamental Terms

(1) *Random Experiment*

A process which is carried out according to specific regulations and can be repeated as often as desired, but whose result is determined by chance, i.e. it cannot be clearly determined in advance.

Example:

Games of chance like lottery or roulette.

(2) *Elementary Event / Event / Sample space*

In every random experiment, a set of possible elementary events exists, so-called elementary events or realisations. An event comprises any subset of the sample space S.

F. W. Peren, *Statistics for Business and Economics*,
https://doi.org/10.1007/978-3-662-64276-4_4

Example:

Throwing a die once.

This random experiment is comprised of six possible *elementary events*: $e_1 = 1,\ e_2 = 2,\ ...,\ e_6 = 6$

The set of all elementary events comprises the so-called sample space S.

4.2 Definitions of Probability

Probability is a measure to quantify the certainty or uncertainty of the occurrence of a certain event in the context of a random experiment.

4.2.1 The Classical Definition of Probabilty

If all elementary events are equally probable, the probability of event A occuring in a given random experiment is determined by the following quotient:

$$P(A) = \frac{\text{number of favourable cases}}{\text{number of all equally probable cases}}$$

with $A =$ {set of all favourable cases}

Example:

What is the probability of event A, that you land on the number 6 when throwing an (untainted) die?

$$P(A) = \frac{1}{6}$$

Remark:

The practical significance of the classical definition of probability is limited since it can only be applied to equally probable events.

4.2.2 The Statistical Definition of Probability

The statistical definition of probability assumes a random experiment consisting of a sequence of independent trials tending towards infinity.

$$P(A) = \lim_{n\to\infty} \frac{h_n(A)}{n} = \lim_{n\to\infty} f_n(A)$$

with: n = number of tries/observations

h_n = absolute frequency of event A

f_n = relative frequency of event A

4.2.3 The Subjective Definition of Probability

Especially in practical decision making, probabilities are determined neither by using the classical nor the statistcal definition of probability. In most cases, the subjective probability of experts are used, usually by using the so-called *Delphi method*. In economics, subjective probabilities are often used in decision models with uncertainty.

4.2.4 Axioms of Probability Calculation

The axiomatic definition of probability does not seek to explain the essence of probability, but rather defines the mathematical properties of probability in terms of three axioms.

Axiom 1:

The probability $P(A)$ of event A of a random experiment is a uniquely determined, real, non-negative number between zero and one:

$$\boxed{0 \leq P(A) \leq 1 \text{ with } P(A) \in \text{real numbers and } A \subset S}$$

Remark:

$P(A)$ is not negative.

Axiom 2:

The probability of the occurrence of all events within a random experiment, i.e. for the entire sample space S, is one:

$$\boxed{P(S) = 1}$$

Axiom 3:

If two events A and B of a random experiment are mutually exclusive, the following applies:

$$\boxed{P(A \cup B) = P(A) + P(B) \text{ with } A \cap B = \{\ \}}$$

4.3 Theorems of Probability Calculation

4.3.1 Theorem of Complementary Events

The sum of probabilities of the random event A and the event complementary to A, $\overline{A}$ ($=$ event not-A), is equal to one:

$$\boxed{P(A \cup \overline{A}) = P(A) + P(\overline{A}) = 1}$$

<u>Example:</u>

What is the probability that in a roll of two dice the sum of both dice is not 4?

$\Rightarrow$ 2 dice of 6 values each, i.e. there are 36 equally probable elementary events (i, j), whereby $i = 1, ..., 6$ and $j = 1, ..., 6$.

Since the elementary events (i, j) are equally probable, the following applies:

$$P(i,\, j) = \frac{1}{36} \quad \text{with} \quad \sum_{i=1}^{6} \sum_{j=1}^{6} P(i,\, j) = 1$$

Event A, that the sum of both dice equals 4, is composed of three elementary events:

$$A = \{(1,\, 3) \cup (2,\, 2) \cup (3,\, 1)\}$$

$$\Rightarrow \quad P(A) = P(1,3) + P(2,2) + P(3,\, 1) = \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{1}{12}$$

$$\Rightarrow \quad P(\overline{A}) = 1 - P(A) = 1 - \frac{1}{12} = \frac{11}{12}$$

4.3.2 The Multiplication Theorem with Independence of Events

If events A and B occur independently from another, the following applies:

$$\boxed{P(A \cap B) = P(A) \cdot P(B)}$$

Example:

A coin with tails on one side is tossed twice. What is the probability of getting tails both times?

Each toss is new and therefore independent of one other.

$$\Rightarrow \quad P(A) = 0.5 \quad \text{with} \quad \begin{aligned} A &= \text{tails and} \\ \overline{A} &= \text{not-tails} \end{aligned}$$

$$\Rightarrow \quad P(A_1 \cap A_2) = P(A_1) \cdot P(A_2) = 0.5 \cdot 0.5 = 0.25$$

The probability of getting tails twice in a row is 0.25; the counter probability is 0.75.

4.3.3 The Addition Theorem

(1) Events A and B are not mutually exclusive.

If A and B are any two events of a random experiment that are not mutually exclusive, the additive (joint) probability of the event $P(A \cup B)$ is calculated as:

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

The probability that either A or B or $A \cap B$, i.e. A and B occur together, is obtained by adding $P(A)$ and $P(B)$ minus the probability of the joint occurrence of A and B.

(2) A and B are mutually exclusive.

In the case where events A and B are mutually exclusive, the following applies:

$$\boxed{P(A \cup B) = P(A) + P(B)}$$

Example:

What is the probability that, if two identical coins are tossed with heads on one side and tails on the other, at least one coin will show tails?

Step 1: Sample space S

$S = \{tt, th, ht, hh\}$ with $t =$ tails and $h =$ heads

$P(tt) = P(th) = P(ht) = P(hh) = \dfrac{1}{4}$

All events are equally probable.

Step 2: Exclusivity / non-exclusivity:

$A =$ event that the 1st coin lands on tails $= \{tt, th\}$

$B =$ event that the 2nd coin lands on tails $= \{tt, ht\}$

$\Rightarrow$ Events A and B are not mutually exclusive if the 1st and 2nd coin shows tails $\{tt\}$.

Step 3: Addition theorem

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Step 4: Calculation

$$P(A \cup B) = \underbrace{\frac{1}{4} + \frac{1}{4}}_{P(A)} + \underbrace{\frac{1}{4} + \frac{1}{4}}_{P(B)} - \underbrace{\frac{1}{4}}_{P(tt)} = \frac{3}{4} = 0.75$$

4.3.4 Conditional Probability

The probability of an event A occuring is often *conditional*, i.e. dependent on the occurrence of another event B. The probability of A given the occurrence of (another) event B is called the *conditional probability* of event A given condition B, $P(A/B)$, with:

$$\boxed{P(A/B) = \frac{P(A \cap B)}{P(B)}}$$

4.3.5 Stochastic Independence

An event A is *stochastically* (taking chance into account) *independent* of event B if the occurrence of A does not depend on the occurrence or non-occurrence of event B.

The following applies:

$$\boxed{P(A/B) = P(A/\overline{B})}$$

4.3.6 The Multiplication Theorem in General Form

If the multiplication theorem in section 4.3.2 is to be generally applied, i.e. event B is conditioned by event A, the following applies:

$$P(A \cap B) = P(A) \cdot P(B/A) = P(B) \cdot P(A/B)$$

Example:

The probability of drawing a king from a pack of cards (52 cards with 4 kings) is: $P(A) = \frac{4}{52} = \frac{1}{13}$. If the drawn card is a king, the probability of drawing a king again is: $P(B) = \frac{3}{51}$. The probability of drawing two kings with two cards is therefore:

$$P(A \cap B) = P(A) \cdot P(B/A) = \frac{1}{13} \cdot \frac{3}{51} \approx 0.0045 \mathrel{\widehat{=}} 0.45\,\% = 4.5‰$$

4.3.7 The Theorem of Total Probability

Let $A_1, A_2, ..., A_n$ be mutually exclusive events that completely fill a sample space S, so that:

$$A_1 \cup A_2 \cup ... \cup A_n = S \text{ and } A_i \cap A_j = \{\ \} \text{ with } i, j = 1, ..., n;\ i \neq j$$

then any event E can be expressed as a union of mutually exclusive events:

$$E = (E \cap A_1) \cup (E \cap A_2) \cup \ldots \cup (E \cap A_n)$$

(1) Application of the addition theorem for mutually exclusive events:

$$\Rightarrow P(E) = P(E \cap A_1) + \ldots + P(E \cap A_n)$$

(2) Application of the general multiplication theorem in the case of dependency of events:

$$\Rightarrow \quad P(E) = P(A_1) \cdot P(E/A_1) + \ldots + P(A_n) \cdot P(E/A_n)$$
$$\Leftrightarrow \quad P(E) = \sum_{i=1}^{n} P(A_i) \cdot P(E/A_i)$$

4.3.8 Bayes' Theorem (Bayes' Rule)

With the help of Bayes' theorem, the conditional probability for any event A_j can be determined from the possible events $A_1, A_2, \ldots, A_n$ of the sample space S under the condition that a certain event E has occurred:

$$P(A_j/E) = \frac{P(A_j \cap E)}{P(E)} = \frac{P(A_j) \cdot P(E/A_j)}{\sum_{i=1}^{n} P(A_i) \cdot P(E/A_i)}$$

with:

$A_1 \cup A_2 \cup ... \cup A_n = S$ and $A_i \cap A_j = \{\}$ with $i, j = 1, ..., n;\ i \neq j$

Since the probability for the occurrence of event A_j refers to an event that has occurred before, $P(A_j/E)$, i.e. can in fact only be calculated after the event E, the probability $P(A_i/E)$ is also called the a posteriori probability. The non-conditional probability $P(A_j)$ is accordingly called the a priori probability.

Example:

A daily production of $1{,}000$ quantity units (QU) of a product is distributed among three machines as follows: M_1 100 QU, M_2 400 QU and M_3 500 QU. The relative error frequencies (error rates) are 0.05 ($5\,\%$) for M_1, 0.04 ($4\,\%$) for M_2 and 0.02 ($2\,\%$) for M_3.

One unit is randomly selected from a daily production. This unit is found to be defective.

What is the probability that this defective unit was produced on machine M_1 or M_2 or M_3?

A_j: QU was produced on machine M_j; $j = 1, 2, 3$

E: QU is defective.

Bayes' Theorem

$$P(A_j/E) = \frac{P(A_j \cap E)}{P(E)} = \frac{P(A_j) \cdot P(E/A_j)}{\sum_{i=1}^{n} P(A_i) \cdot P(E/A_i)}$$

$$P(A_1) = \frac{100}{1{,}000} = 0.1$$

$$P(A_2) = \frac{400}{1{,}000} = 0.4$$

$$P(A_3) = \frac{500}{1{,}000} = 0.5$$

$$\begin{aligned}
\Rightarrow P(A_1/E) &= \frac{0.1 \cdot 0.05}{0.1 \cdot 0.05 + 0.4 \cdot 0.04 + 0.5 \cdot 0.02} = \\
&= \frac{0.1 \cdot 0.05}{0.031} \approx 0.16 \\
\Rightarrow P(A_2/E) &= \frac{0.4 \cdot 0.04}{0.031} \approx 0.52 \\
\Rightarrow P(A_3/E) &= \frac{0.5 \cdot 0.02}{0.031} \approx 0.32
\end{aligned}$$

The probability that the defective unit was drawn from machine M_2 is highest with approximately 0.52 ($= 52\,\%$).

4.3.9 Overview of the Probability Calculation of Mutually Exclusive and Non-Exclusive Events

	$\cup \Rightarrow +$ or	$\cap \Rightarrow \times$ and
Events A and B are mutually exclusive $\Rightarrow$ no common elements $\Rightarrow$ **independence** of events	$P(A \cup B) =$ $= P(A) + P(B)$ see 4.3.3 (2)	$P(A \cap B) =$ $= P(A) \cdot P(B)$ see 4.3.2
Events A and B are <u>not</u> mutually exclusive $\Rightarrow$ there are common elements $\Rightarrow$ **dependence** of events	$P(A \cup B) =$ $= P(A) + P(B)$ $- P(A \cap B)$ see 4.3.3 (1)	$P(A \cap B) =$ $= P(A) \cdot P(B/A)$ $= P(B) \cdot P(A/B)$ see 4.3.6

4.4 Random Variable

4.4.1 The Concept of Random Variables

Definition:

If a variable changes in an unpredictable way, i.e. if the variable takes on its values only depending on chance, it is called a *random variable*. Random variables are usually symbolised with capital letters X, Y, Z, while their values are correspondingly marked with lowercase letters.

Example:

In the random experiment of tossing a coin with tails on one side twice, the frequency of the "tails"-event depends on chance. The values of this random variable X, $X =$ number of tails when a coin is tossed twice, can be $x = 0 \lor x = 1 \lor x = 2$.

4.4.2 The Probability Function of Discrete Random Variables

A probability $P(X = x_i)$ can be assigned to each specific value x_i of a random variable X. The function $f(x_i)$, which indicates the probability of realisation for each value of the random variable, is referred to as the probability function of the random variable X:

$$\boxed{f(x_i) = P(X = x_i)}$$

It has the properties: $f(x_i) \geq 0$ and $\sum\limits_{i=1}^{n} f(x_i) = 1$

with $i = 1, ..., n$

4.4.3 The Distribution Function of Discrete Random Variables

The function value of a distribution function $F(x)$ of a random variable X indicates the probability that the random variable X is at most the value x_i:

$$\boxed{F(x) = P(X \leq x_i)}$$

The distribution function of discrete random variables has the following properties:

(1) for an arbitrary x: $0 \leq F(x) \leq 1$

(2) impossible event: $\lim\limits_{\substack{x\to-\infty \\ \text{or } x\to 0}} F(x) = 0$

(3) certain event: $\lim\limits_{x\to\infty} F(x) = 1$

4.4.4 Probability Density and Distribution Function of Continuous Random Variables

(1) The Probability Density (The Density Function)

If the observed random variable X takes not only on discrete values but (in a certain interval) any value from the range of real numbers, then the probability distribution becomes a so-called probability density or, in other words, a density function.

The density function has the properties:

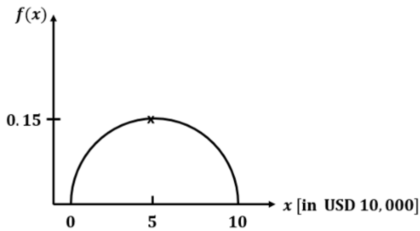
$$f(x) \geq 0 \text{ and } \int_{x_{min}}^{x_{max}} f(x)dx = 1$$

Example:

The daily cash receipts of a retailer fluctuate between $x_{min} = \$0$ and $x_{max} = \$10,000$. By definition, the variable x has 1 million [in cents] values and thus corresponds de facto to a continuous random variable. The probability distribution of the random variable X can be represented by the following density function:

$$f(x) = -0.006x^2 + 0.06x$$
$$f'(x) = -0.012x + 0.06 = 0$$
$$\Rightarrow x_{max} = 5$$

$$f(5) = 0.15$$
$$f(x) = 0$$
$$\Rightarrow x = 0 \ \vee \ x = 10$$



The function $f(x) = -0.006x^2 + 0.06x$ is a density function if the area under the function between x_{min} and x_{max} is equal to one.

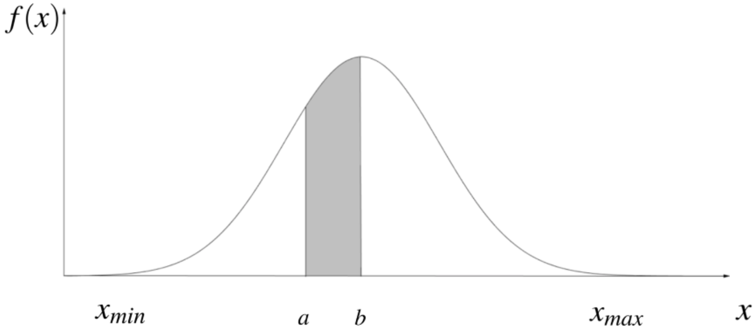
$$\int_{x_{min}}^{x_{max}} f(x)dx = \int_0^{10} (-0.006x^2 + 0.06x)dx =$$

$$= \left[-0.006 \cdot \frac{1}{3}x^3 + 0.06 \cdot \frac{1}{2} \cdot x^2\right]_0^{10} =$$

$$= (-0.002 \cdot 10^3 + 0.03 \cdot 10^2) - 0 = -2 + 3 = 1$$

The probability that the continuous random variable X will take on a value x that lies in the interval $[a, b]$ corresponds to the area, which locates under the considered density function $f(x)$ within the limits a and b:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$



Remark: For continuous random variables, the probability that the random variable X takes on any specific value x is always equal to zero:

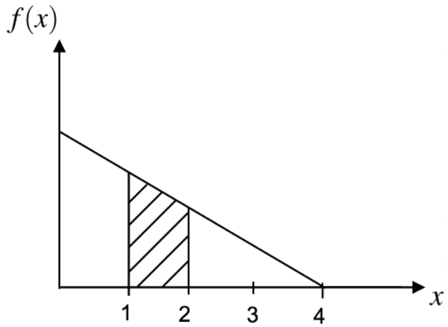
$P(X = x) = 0$

so that: $P(a \leq X \leq b) = P(a < x < b)$

Example:

The continuous random variable X records the delay of a subway at a certain stop and has the following density function in minutes:

$$f(x) = \begin{cases} 0.5 - 0.125x & \text{for } 0 \leq x \leq 4 \ \text{[minutes]} \\ 0 & \text{for } \quad x > 4 \ \text{[minutes]} \end{cases}$$



$f(x)$ is a density function, because:

(1) $f(x) \geq 0$ for all x

(2) $\int\limits_{x_{min}}^{x_{max}} f(x)dx = \int\limits_{0}^{4} (0.5 - 0.125x)dx =$

$$= \left[0.5x - 0.0625 \cdot x^2\right]_0^4 =$$

$$= 2 - 1 = 1$$

What is the probability that the subway will be delayed by one to two minutes?

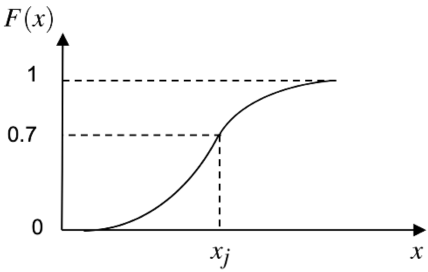
$P(1 \leq x \leq 2) =$ corresponding area under the density function

$$f(x) = \int_1^2 f(x)dx = \int_1^2 (0.5 - 0.125x)dx =$$

$$= \left[0.5x - 0.125 \cdot \frac{1}{2}x^2\right]_1^2 = 0.75 - 0.4375 = 0.3125 = 31.25\,\%$$

(2) The Distribution Function

The distribution function of a continuous random variable X, $F(x)$, is a continuous function:



Interpretation: $x_j : F(x_j) = 0.7$ indicates the probability that the random variable takes on the realisation x_j at most.

Properties of the Distribution Function $F(x)$

(1) $0 \leq F(x) \leq 1$

(2) $F(x)$ is monotonically increasing, i.e. for $x_1 \leq x_2$: $F(x_1) \leq F(x_2)$

(3) $\lim\limits_{x \to \infty} F(x) = 1$

(4) $\lim\limits_{\substack{x \to -\infty \\ \text{or } x \to 0}} F(x) = 0$

(5) $F(x)$ is continuous for all x

(6) The first derivative of the distribution function $F(x)$ yields the density function (= probability function) $f(x)$.

For the density function from the example “delay of a subway” (see section 4.4.4)

$$f(x) = \begin{cases} 0.5 - 0.125x & \text{for } 0 \leq x \leq 4 \ \text{[minutes]} \\ 0 & \text{for } \quad x > 4 \ \text{[minutes]} \end{cases}$$

the following distribution function is obtained

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 0.5x - 0.125 \cdot \frac{1}{2}x^2 & \text{for } 0 \leq x \leq 4 \\ 1 & \text{for } \quad x > 4 \end{cases}$$

Example: Random variable $X =$ delay of a subway

What is the probability that the subway is delayed by one to two minutes?

$$\begin{aligned} P(1 \leq x \leq 2) &= F(2) - F(1) = \\ &= (0.5 \cdot 2 - 0.0625 \cdot 2^2) - (0.5 \cdot 1 - 0.0625 \cdot 1^2) = \\ &= 0.75 - 0.4375 = 0.3125 = 31.25\,\% \end{aligned}$$

4.4.5 Expected Value and Variance of Random Variables

Like the frequency distributions of descriptive statistics, the probability distributions of random variables can be characterised by measures (parameters).

(1) Expected Value $E(X)$

- for <u>discrete</u> random variables:

$$E(X) = \sum_i x_i \cdot P(X = x_i) = \sum_i x_i \cdot f(x_i)$$

- for <u>continuous</u> random variables:

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)dx$$

or for an interval $[x_u \; ; \; x_o]$: $E(X) = \int_{x_u}^{x_o} x \cdot f(x)dx$

(2) Variance $Var(X)$

- for <u>discrete</u> random variables:

$$Var(X) = \sum_i [x_i - E(X)]^2 \; f(x_i) =$$

$$= \sum_i {x_i}^2 \; f(x_i) - [E(X)]^2$$

- for <u>continuous</u> random variables:

$$Var(X) = \int_{-\infty}^{\infty} [x - E(X)]^2 \; f(x)dx =$$

$$= \int_{-\infty}^{\infty} x^2 \; f(x)dx - [E(X)]^2$$

or for an interval $[x_u \; ; \; x_o]$:

$$\int_{x_u}^{x_o} [x - E(X)]^2 \; f(x)dx =$$

$$= \int_{x_u}^{x_o} x^2 \; f(x)dx - [E(X)]^2$$

Example 1:

Random experiment "tossing a coin three times" with tails on one side:

$X =$ number of tails $\Rightarrow$ discrete random variable

$E(X) =$ the number of tails to be expected on average for a larger number of trials

$$E(X) = \sum_i x_i f(x_i) =$$
$$= 0 \cdot 0.125 + 1 \cdot 0.375 + 2 \cdot 0.375 + 3 \cdot 0.125 = 1.5$$

On average, each attempt, i.e. "tossing a coin three times", is expected to result in tails 1.5 times.

$$Var(X) = \sum_i x_i^2 f(x_i) - [E(X)]^2 =$$
$$= 0^2 \cdot 0.125 + 1^2 \cdot 0.375 + 2^2 \cdot 0.375 + 3^2 \cdot 0.125 - 1.5^2 = 0.75$$

Example 2:

Delay of a subway at a certain stop in minutes.

$X =$ delay in minutes $\Rightarrow$ continuous random variable

(1) $f(x) = \begin{cases} 0.5 - 0.125x & \text{for } 0 \leq x \leq 4 \\ 0 & \text{for } \quad x > 4 \end{cases}$

(2) $E(X) = \int_{x_u}^{x_o} x \cdot f(x)dx = \int_0^4 x \cdot (0.5 - 0.125x)dx =$

$$= \int_0^4 (0.5x - 0.125x^2)dx = \left[0.5 \cdot \frac{1}{2}x^2 - 0.125 \cdot \frac{1}{3}x^3\right]_0^4 =$$

$$= F(4) - F(0) = 1.\overline{3} \ \text{[minutes]}$$

On average, a delay of about 1.33 minutes can be expected.

$$\begin{aligned}
Var(x) &= \int_{x_u}^{x_o} x^2 \cdot f(x)dx - [E(X)]^2 = \\
&= \int_0^4 x^2 \cdot (0.5 - 0.125x)dx - (1.3333)^2 = \\
&= \int_0^4 (0.5x^2 - 0.125x^3)dx - (1.3333)^2 = \\
&= \left[0.5 \cdot \frac{1}{3}x^3 - 0.125 \cdot \frac{1}{4}x^4\right]_0^4 - 1.3333^2 = \\
&= F(4) - F(0) - 1.3333^2 = 0.889 \ \ \text{[minutes}^2\text{]}
\end{aligned}$$

Standard deviation $= \sqrt{0.889} = 0.9429$ [minutes]

Appendix A
Statistical Tables

F. W. Peren, *Statistics for Business and Economics*,
https://doi.org/10.1007/978-3-662-64276-4

Binomial Distribution - Probability Mass Function

$$f_B(x \mid n;\ \theta) = \begin{cases} \binom{n}{x}\ \theta^x\,(1-\theta)^{n-x} & \text{for} \quad x = 0,\ 1,\ \ldots,\ n \qquad 0 < \theta < 1 \\ 0 & \text{others} \end{cases}$$

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
1	0	0.9900	0.9500	0.9000	0.8500	0.8000	0.7000	0.6000	0.5000
	1	0.0100	0.0500	0.1000	0.1500	0.2000	0.3000	0.4000	0.5000
2	0	0.9801	0.9025	0.8100	0.7225	0.6400	0.4900	0.3600	0.2500
	1	0.0198	0.0950	0.1800	0.2550	0.3200	0.4200	0.4800	0.5000
	2	0.0001	0.0025	0.0100	0.0225	0.0400	0.0900	0.1600	0.2500
3	0	0.9703	0.8574	0.7290	0.6141	0.5120	0.3430	0.2160	0.1250
	1	0.0294	0.1354	0.2430	0.3251	0.3840	0.4410	0.4320	0.3750
	2	0.0003	0.0071	0.0270	0.0574	0.0960	0.1890	0.2880	0.3750
	3	0.0000	0.0001	0.0010	0.0034	0.0080	0.0270	0.0640	0.1250
4	0	0.9606	0.8145	0.6561	0.5220	0.4096	0.2401	0.1296	0.0625
	1	0.0388	0.1715	0.2916	0.3685	0.4096	0.4116	0.3456	0.2500
	2	0.0006	0.0135	0.0486	0.0975	0.1536	0.2646	0.3456	0.3750
	3	0.0000	0.0005	0.0036	0.0115	0.0256	0.0756	0.1536	0.2500
	4	0.0000	0.0000	0.0001	0.0005	0.0016	0.0081	0.0256	0.0625
5	0	0.9510	0.7738	0.5905	0.4437	0.3277	0.1681	0.0778	0.0313
	1	0.0480	0.2036	0.3281	0.3915	0.4096	0.3602	0.2592	0.1563
	2	0.0010	0.0214	0.0729	0.1382	0.2048	0.3087	0.3456	0.3125
	3	0.0000	0.0011	0.0081	0.0244	0.0512	0.1323	0.2304	0.3125
	4	0.0000	0.0000	0.0005	0.0022	0.0064	0.0284	0.0768	0.1563
	5	0.0000	0.0000	0.0000	0.0001	0.0003	0.0024	0.0102	0.0313
6	0	0.9415	0.7351	0.5314	0.3771	0.2621	0.1176	0.0467	0.0156
	1	0.0571	0.2321	0.3543	0.3993	0.3932	0.3025	0.1866	0.0938
	2	0.0014	0.0305	0.0984	0.1762	0.2458	0.3241	0.3110	0.2344
	3	0.0000	0.0021	0.0146	0.0415	0.0819	0.1852	0.2765	0.3125
	4	0.0000	0.0001	0.0012	0.0055	0.0154	0.0595	0.1382	0.2344
	5	0.0000	0.0000	0.0001	0.0004	0.0015	0.0102	0.0369	0.0938
	6	0.0000	0.0000	0.0000	0.0000	0.0001	0.0007	0.0041	0.0156
7	0	0.9321	0.6983	0.4783	0.3206	0.2097	0.0824	0.0280	0.0078
	1	0.0659	0.2573	0.3720	0.3960	0.3670	0.2471	0.1306	0.0547
	2	0.0020	0.0406	0.1240	0.2097	0.2753	0.3177	0.2613	0.1641
	3	0.0000	0.0036	0.0230	0.0617	0.1147	0.2269	0.2903	0.2734
	4	0.0000	0.0002	0.0026	0.0109	0.0287	0.0972	0.1935	0.2734

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
1	0	0.4000	0.3000	0.2500	0.2000	0.1500	0.1000	0.0500	0.0100
	1	0.6000	0.7000	0.7500	0.8000	0.8500	0.9000	0.9500	0.9900
2	0	0.1600	0.0900	0.0625	0.0400	0.0225	0.0100	0.0025	0.0001
	1	0.4800	0.4200	0.3750	0.3200	0.2550	0.1800	0.0950	0.0198
	2	0.3600	0.4900	0.5625	0.6400	0.7225	0.8100	0.9025	0.9801
3	0	0.0640	0.0270	0.0156	0.0080	0.0034	0.0010	0.0001	0.0000
	1	0.2880	0.1890	0.1406	0.0960	0.0574	0.0270	0.0071	0.0003
	2	0.4320	0.4410	0.4219	0.3840	0.3251	0.2430	0.1354	0.0294
	3	0.2160	0.3430	0.4219	0.5120	0.6141	0.7290	0.8574	0.9703
4	0	0.0256	0.0081	0.0039	0.0016	0.0005	0.0001	0.0000	0.0000
	1	0.1536	0.0756	0.0469	0.0256	0.0115	0.0036	0.0005	0.0000
	2	0.3456	0.2646	0.2109	0.1536	0.0975	0.0486	0.0135	0.0006
	3	0.3456	0.4116	0.4219	0.4096	0.3685	0.2916	0.1715	0.0388
	4	0.1296	0.2401	0.3164	0.4096	0.5220	0.6561	0.8145	0.9606
5	0	0.0102	0.0024	0.0010	0.0003	0.0001	0.0000	0.0000	0.0000
	1	0.0768	0.0284	0.0146	0.0064	0.0022	0.0005	0.0000	0.0000
	2	0.2304	0.1323	0.0879	0.0512	0.0244	0.0081	0.0011	0.0000
	3	0.3456	0.3087	0.2637	0.2048	0.1382	0.0729	0.0214	0.0010
	4	0.2592	0.3602	0.3955	0.4096	0.3915	0.3281	0.2036	0.0480
	5	0.0778	0.1681	0.2373	0.3277	0.4437	0.5905	0.7738	0.9510
6	0	0.0041	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000
	1	0.0369	0.0102	0.0044	0.0015	0.0004	0.0001	0.0000	0.0000
	2	0.1382	0.0595	0.0330	0.0154	0.0055	0.0012	0.0001	0.0000
	3	0.2765	0.1852	0.1318	0.0819	0.0415	0.0146	0.0021	0.0000
	4	0.3110	0.3241	0.2966	0.2458	0.1762	0.0984	0.0305	0.0014
	5	0.1866	0.3025	0.3560	0.3932	0.3993	0.3543	0.2321	0.0571
	6	0.0467	0.1176	0.1780	0.2621	0.3771	0.5314	0.7351	0.9415
7	0	0.0016	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0172	0.0036	0.0013	0.0004	0.0001	0.0000	0.0000	0.0000
	2	0.0774	0.0250	0.0115	0.0043	0.0012	0.0002	0.0000	0.0000
	3	0.1935	0.0972	0.0577	0.0287	0.0109	0.0026	0.0002	0.0000
	4	0.2903	0.2269	0.1730	0.1147	0.0617	0.0230	0.0036	0.0000

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
7	5	0.0000	0.0000	0.0002	0.0012	0.0043	0.0250	0.0774	0.1641
	6	0.0000	0.0000	0.0000	0.0001	0.0004	0.0036	0.0172	0.0547
	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0016	0.0078
8	0	0.9227	0.6634	0.4305	0.2725	0.1678	0.0576	0.0168	0.0039
	1	0.0746	0.2793	0.3826	0.3847	0.3355	0.1977	0.0896	0.0313
	2	0.0026	0.0515	0.1488	0.2376	0.2936	0.2965	0.2090	0.1094
	3	0.0001	0.0054	0.0331	0.0839	0.1468	0.2541	0.2787	0.2188
	4	0.0000	0.0004	0.0046	0.0185	0.0459	0.1361	0.2322	0.2734
	5	0.0000	0.0000	0.0004	0.0026	0.0092	0.0467	0.1239	0.2188
	6	0.0000	0.0000	0.0000	0.0002	0.0011	0.0100	0.0413	0.1094
	7	0.0000	0.0000	0.0000	0.0000	0.0001	0.0012	0.0079	0.0313
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0007	0.0039
9	0	0.9135	0.6302	0.3874	0.2316	0.1342	0.0404	0.0101	0.0020
	1	0.0830	0.2985	0.3874	0.3679	0.3020	0.1556	0.0605	0.0176
	2	0.0034	0.0629	0.1722	0.2597	0.3020	0.2668	0.1612	0.0703
	3	0.0001	0.0077	0.0446	0.1069	0.1762	0.2668	0.2508	0.1641
	4	0.0000	0.0006	0.0074	0.0283	0.0661	0.1715	0.2508	0.2461
	5	0.0000	0.0000	0.0008	0.0050	0.0165	0.0735	0.1672	0.2461
	6	0.0000	0.0000	0.0001	0.0006	0.0028	0.0210	0.0743	0.1641
	7	0.0000	0.0000	0.0000	0.0000	0.0003	0.0039	0.0212	0.0703
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0004	0.0035	0.0176
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003	0.0020
10	0	0.9044	0.5987	0.3487	0.1969	0.1074	0.0282	0.0060	0.0010
	1	0.0914	0.3151	0.3874	0.3474	0.2684	0.1211	0.0403	0.0098
	2	0.0042	0.0746	0.1937	0.2759	0.3020	0.2335	0.1209	0.0439
	3	0.0001	0.0105	0.0574	0.1298	0.2013	0.2668	0.2150	0.1172
	4	0.0000	0.0010	0.0112	0.0401	0.0881	0.2001	0.2508	0.2051
	5	0.0000	0.0001	0.0015	0.0085	0.0264	0.1029	0.2007	0.2461
	6	0.0000	0.0000	0.0001	0.0012	0.0055	0.0368	0.1115	0.2051
	7	0.0000	0.0000	0.0000	0.0001	0.0008	0.0090	0.0425	0.1172
	8	0.0000	0.0000	0.0000	0.0000	0.0001	0.0014	0.0106	0.0439
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0016	0.0098
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0010
11	0	0.8953	0.5688	0.3138	0.1673	0.0859	0.0198	0.0036	0.0005
	1	0.0995	0.3293	0.3835	0.3248	0.2362	0.0932	0.0266	0.0054
	2	0.0050	0.0867	0.2131	0.2866	0.2953	0.1998	0.0887	0.0269
	3	0.0002	0.0137	0.0710	0.1517	0.2215	0.2568	0.1774	0.0806
	4	0.0000	0.0014	0.0158	0.0536	0.1107	0.2201	0.2365	0.1611

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
7	5	0.2613	0.3177	0.3115	0.2753	0.2097	0.1240	0.0406	0.0020
	6	0.1306	0.2471	0.3115	0.3670	0.3960	0.3720	0.2573	0.0659
	7	0.0280	0.0824	0.1335	0.2097	0.3206	0.4783	0.6983	0.9321
8	0	0.0007	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0079	0.0012	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
	2	0.0413	0.0100	0.0038	0.0011	0.0002	0.0000	0.0000	0.0000
	3	0.1239	0.0467	0.0231	0.0092	0.0026	0.0004	0.0000	0.0000
	4	0.2322	0.1361	0.0865	0.0459	0.0185	0.0046	0.0004	0.0000
	5	0.2787	0.2541	0.2076	0.1468	0.0839	0.0331	0.0054	0.0001
	6	0.2090	0.2965	0.3115	0.2936	0.2376	0.1488	0.0515	0.0026
	7	0.0896	0.1977	0.2670	0.3355	0.3847	0.3826	0.2793	0.0746
	8	0.0168	0.0576	0.1001	0.1678	0.2725	0.4305	0.6634	0.9227
9	0	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0035	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0212	0.0039	0.0012	0.0003	0.0000	0.0000	0.0000	0.0000
	3	0.0743	0.0210	0.0087	0.0028	0.0006	0.0001	0.0000	0.0000
	4	0.1672	0.0735	0.0389	0.0165	0.0050	0.0008	0.0000	0.0000
	5	0.2508	0.1715	0.1168	0.0661	0.0283	0.0074	0.0006	0.0000
	6	0.2508	0.2668	0.2336	0.1762	0.1069	0.0446	0.0077	0.0001
	7	0.1612	0.2668	0.3003	0.3020	0.2597	0.1722	0.0629	0.0034
	8	0.0605	0.1556	0.2253	0.3020	0.3679	0.3874	0.2985	0.0830
	9	0.0101	0.0404	0.0751	0.1342	0.2316	0.3874	0.6302	0.9135
10	0	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0016	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0106	0.0014	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
	3	0.0425	0.0090	0.0031	0.0008	0.0001	0.0000	0.0000	0.0000
	4	0.1115	0.0368	0.0162	0.0055	0.0012	0.0001	0.0000	0.0000
	5	0.2007	0.1029	0.0584	0.0264	0.0085	0.0015	0.0001	0.0000
	6	0.2508	0.2001	0.1460	0.0881	0.0401	0.0112	0.0010	0.0000
	7	0.2150	0.2668	0.2503	0.2013	0.1298	0.0574	0.0105	0.0001
	8	0.1209	0.2335	0.2816	0.3020	0.2759	0.1937	0.0746	0.0042
	9	0.0403	0.1211	0.1877	0.2684	0.3474	0.3874	0.3151	0.0914
	10	0.0060	0.0282	0.0563	0.1074	0.1969	0.3487	0.5987	0.9044
11	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0052	0.0005	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0234	0.0037	0.0011	0.0002	0.0000	0.0000	0.0000	0.0000
	4	0.0701	0.0173	0.0064	0.0017	0.0003	0.0000	0.0000	0.0000

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
11	5	0.0000	0.0001	0.0025	0.0132	0.0388	0.1321	0.2207	0.2256
	6	0.0000	0.0000	0.0003	0.0023	0.0097	0.0566	0.1471	0.2256
	7	0.0000	0.0000	0.0000	0.0003	0.0017	0.0173	0.0701	0.1611
	8	0.0000	0.0000	0.0000	0.0000	0.0002	0.0037	0.0234	0.0806
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0052	0.0269
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007	0.0054
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005
12	0	0.8864	0.5404	0.2824	0.1422	0.0687	0.0138	0.0022	0.0002
	1	0.1074	0.3413	0.3766	0.3012	0.2062	0.0712	0.0174	0.0029
	2	0.0060	0.0988	0.2301	0.2924	0.2835	0.1678	0.0639	0.0161
	3	0.0002	0.0173	0.0852	0.1720	0.2362	0.2397	0.1419	0.0537
	4	0.0000	0.0021	0.0213	0.0683	0.1329	0.2311	0.2128	0.1208
	5	0.0000	0.0002	0.0038	0.0193	0.0532	0.1585	0.2270	0.1934
	6	0.0000	0.0000	0.0005	0.0040	0.0155	0.0792	0.1766	0.2256
	7	0.0000	0.0000	0.0000	0.0006	0.0033	0.0291	0.1009	0.1934
	8	0.0000	0.0000	0.0000	0.0001	0.0005	0.0078	0.0420	0.1208
	9	0.0000	0.0000	0.0000	0.0000	0.0001	0.0015	0.0125	0.0537
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0025	0.0161
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003	0.0029
	12	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002
13	0	0.8775	0.5133	0.2542	0.1209	0.0550	0.0097	0.0013	0.0001
	1	0.1152	0.3512	0.3672	0.2774	0.1787	0.0540	0.0113	0.0016
	2	0.0070	0.1109	0.2448	0.2937	0.2680	0.1388	0.0453	0.0095
	3	0.0003	0.0214	0.0997	0.1900	0.2457	0.2181	0.1107	0.0349
	4	0.0000	0.0028	0.0277	0.0838	0.1535	0.2337	0.1845	0.0873
	5	0.0000	0.0003	0.0055	0.0266	0.0691	0.1803	0.2214	0.1571
	6	0.0000	0.0000	0.0008	0.0063	0.0230	0.1030	0.1968	0.2095
	7	0.0000	0.0000	0.0001	0.0011	0.0058	0.0442	0.1312	0.2095
	8	0.0000	0.0000	0.0000	0.0001	0.0011	0.0142	0.0656	0.1571
	9	0.0000	0.0000	0.0000	0.0000	0.0001	0.0034	0.0243	0.0873
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0065	0.0349
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0012	0.0095
	12	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0016
	13	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001
14	0	0.8687	0.4877	0.2288	0.1028	0.0440	0.0068	0.0008	0.0001
	1	0.1229	0.3593	0.3559	0.2539	0.1539	0.0407	0.0073	0.0009
	2	0.0081	0.1229	0.2570	0.2912	0.2501	0.1134	0.0317	0.0056
	3	0.0003	0.0259	0.1142	0.2056	0.2501	0.1943	0.0845	0.0222

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
11	5	0.1471	0.0566	0.0268	0.0097	0.0023	0.0003	0.0000	0.0000
	6	0.2207	0.1321	0.0803	0.0388	0.0132	0.0025	0.0001	0.0000
	7	0.2365	0.2201	0.1721	0.1107	0.0536	0.0158	0.0014	0.0000
	8	0.1774	0.2568	0.2581	0.2215	0.1517	0.0710	0.0137	0.0002
	9	0.0887	0.1998	0.2581	0.2953	0.2866	0.2131	0.0867	0.0050
	10	0.0266	0.0932	0.1549	0.2362	0.3248	0.3835	0.3293	0.0995
	11	0.0036	0.0198	0.0422	0.0859	0.1673	0.3138	0.5688	0.8953
12	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0025	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0125	0.0015	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
	4	0.0420	0.0078	0.0024	0.0005	0.0001	0.0000	0.0000	0.0000
	5	0.1009	0.0291	0.0115	0.0033	0.0006	0.0000	0.0000	0.0000
	6	0.1766	0.0792	0.0401	0.0155	0.0040	0.0005	0.0000	0.0000
	7	0.2270	0.1585	0.1032	0.0532	0.0193	0.0038	0.0002	0.0000
	8	0.2128	0.2311	0.1936	0.1329	0.0683	0.0213	0.0021	0.0000
	9	0.1419	0.2397	0.2581	0.2362	0.1720	0.0852	0.0173	0.0002
	10	0.0639	0.1678	0.2323	0.2835	0.2924	0.2301	0.0988	0.0060
	11	0.0174	0.0712	0.1267	0.2062	0.3012	0.3766	0.3413	0.1074
	12	0.0022	0.0138	0.0317	0.0687	0.1422	0.2824	0.5404	0.8864
13	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0012	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0065	0.0006	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0243	0.0034	0.0009	0.0001	0.0000	0.0000	0.0000	0.0000
	5	0.0656	0.0142	0.0047	0.0011	0.0001	0.0000	0.0000	0.0000
	6	0.1312	0.0442	0.0186	0.0058	0.0011	0.0001	0.0000	0.0000
	7	0.1968	0.1030	0.0559	0.0230	0.0063	0.0008	0.0000	0.0000
	8	0.2214	0.1803	0.1258	0.0691	0.0266	0.0055	0.0003	0.0000
	9	0.1845	0.2337	0.2097	0.1535	0.0838	0.0277	0.0028	0.0000
	10	0.1107	0.2181	0.2517	0.2457	0.1900	0.0997	0.0214	0.0003
	11	0.0453	0.1388	0.2059	0.2680	0.2937	0.2448	0.1109	0.0070
	12	0.0113	0.0540	0.1029	0.1787	0.2774	0.3672	0.3512	0.1152
	13	0.0013	0.0097	0.0238	0.0550	0.1209	0.2542	0.5133	0.8775
14	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0033	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
14	4	0.0000	0.0037	0.0349	0.0998	0.1720	0.2290	0.1549	0.0611
	5	0.0000	0.0004	0.0078	0.0352	0.0860	0.1963	0.2066	0.1222
	6	0.0000	0.0000	0.0013	0.0093	0.0322	0.1262	0.2066	0.1833
	7	0.0000	0.0000	0.0002	0.0019	0.0092	0.0618	0.1574	0.2095
	8	0.0000	0.0000	0.0000	0.0003	0.0020	0.0232	0.0918	0.1833
	9	0.0000	0.0000	0.0000	0.0000	0.0003	0.0066	0.0408	0.1222
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0014	0.0136	0.0611
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0033	0.0222
	12	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0056
	13	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0009
	14	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001
15	0	0.8601	0.4633	0.2059	0.0874	0.0352	0.0047	0.0005	0.0000
	1	0.1303	0.3658	0.3432	0.2312	0.1319	0.0305	0.0047	0.0005
	2	0.0092	0.1348	0.2669	0.2856	0.2309	0.0916	0.0219	0.0032
	3	0.0004	0.0307	0.1285	0.2184	0.2501	0.1700	0.0634	0.0139
	4	0.0000	0.0049	0.0428	0.1156	0.1876	0.2186	0.1268	0.0417
	5	0.0000	0.0006	0.0105	0.0449	0.1032	0.2061	0.1859	0.0916
	6	0.0000	0.0000	0.0019	0.0132	0.0430	0.1472	0.2066	0.1527
	7	0.0000	0.0000	0.0003	0.0030	0.0138	0.0811	0.1771	0.1964
	8	0.0000	0.0000	0.0000	0.0005	0.0035	0.0348	0.1181	0.1964
	9	0.0000	0.0000	0.0000	0.0001	0.0007	0.0116	0.0612	0.1527
	10	0.0000	0.0000	0.0000	0.0000	0.0001	0.0030	0.0245	0.0916
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0074	0.0417
	12	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0016	0.0139
	13	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003	0.0032
	14	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005
	15	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
20	0	0.8179	0.3585	0.1216	0.0388	0.0115	0.0008	0.0000	0.0000
	1	0.1652	0.3774	0.2702	0.1368	0.0576	0.0068	0.0005	0.0000
	2	0.0159	0.1887	0.2852	0.2293	0.1369	0.0278	0.0031	0.0002
	3	0.0010	0.0596	0.1901	0.2428	0.2054	0.0716	0.0123	0.0011
	4	0.0000	0.0133	0.0898	0.1821	0.2182	0.1304	0.0350	0.0046
	5	0.0000	0.0022	0.0319	0.1028	0.1746	0.1789	0.0746	0.0148
	6	0.0000	0.0003	0.0089	0.0454	0.1091	0.1916	0.1244	0.0370
	7	0.0000	0.0000	0.0020	0.0160	0.0545	0.1643	0.1659	0.0739
	8	0.0000	0.0000	0.0004	0.0046	0.0222	0.1144	0.1797	0.1201
	9	0.0000	0.0000	0.0001	0.0011	0.0074	0.0654	0.1597	0.1602
	10	0.0000	0.0000	0.0000	0.0002	0.0020	0.0308	0.1171	0.1762

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.60	0.70	0.75	0.80	0.85	0.90	0.95	0.99
14	4	0.0136	0.0014	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0408	0.0066	0.0018	0.0003	0.0000	0.0000	0.0000	0.0000
	6	0.0918	0.0232	0.0082	0.0020	0.0003	0.0000	0.0000	0.0000
	7	0.1574	0.0618	0.0280	0.0092	0.0019	0.0002	0.0000	0.0000
	8	0.2066	0.1262	0.0734	0.0322	0.0093	0.0013	0.0000	0.0000
	9	0.2066	0.1963	0.1468	0.0860	0.0352	0.0078	0.0004	0.0000
	10	0.1549	0.2290	0.2202	0.1720	0.0998	0.0349	0.0037	0.0000
	11	0.0845	0.1943	0.2402	0.2501	0.2056	0.1142	0.0259	0.0003
	12	0.0317	0.1134	0.1802	0.2501	0.2912	0.2570	0.1229	0.0081
	13	0.0073	0.0407	0.0832	0.1539	0.2539	0.3559	0.3593	0.1229
	14	0.0008	0.0068	0.0178	0.0440	0.1028	0.2288	0.4877	0.8687
15	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0016	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0074	0.0006	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0245	0.0030	0.0007	0.0001	0.0000	0.0000	0.0000	0.0000
	6	0.0612	0.0116	0.0034	0.0007	0.0001	0.0000	0.0000	0.0000
	7	0.1181	0.0348	0.0131	0.0035	0.0005	0.0000	0.0000	0.0000
	8	0.1771	0.0811	0.0393	0.0138	0.0030	0.0003	0.0000	0.0000
	9	0.2066	0.1472	0.0917	0.0430	0.0132	0.0019	0.0000	0.0000
	10	0.1859	0.2061	0.1651	0.1032	0.0449	0.0105	0.0006	0.0000
	11	0.1268	0.2186	0.2252	0.1876	0.1156	0.0428	0.0049	0.0000
	12	0.0634	0.1700	0.2252	0.2501	0.2184	0.1285	0.0307	0.0004
	13	0.0219	0.0916	0.1559	0.2309	0.2856	0.2669	0.1348	0.0092
	14	0.0047	0.0305	0.0668	0.1319	0.2312	0.3432	0.3658	0.1303
	15	0.0005	0.0047	0.0134	0.0352	0.0874	0.2059	0.4633	0.8601
20	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	6	0.0049	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	7	0.0146	0.0010	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0355	0.0039	0.0008	0.0001	0.0000	0.0000	0.0000	0.0000
	9	0.0710	0.0120	0.0030	0.0005	0.0000	0.0000	0.0000	0.0000
	10	0.1171	0.0308	0.0099	0.0020	0.0002	0.0000	0.0000	0.0000

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
20	11	0.0000	0.0000	0.0000	0.0000	0.0005	0.0120	0.0710	0.1602
	12	0.0000	0.0000	0.0000	0.0000	0.0001	0.0039	0.0355	0.1201
	13	0.0000	0.0000	0.0000	0.0000	0.0000	0.0010	0.0146	0.0739
	14	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0049	0.0370
	15	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0013	0.0148
	16	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003	0.0046
	17	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0011
	18	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002
	19	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
30	0	0.7397	0.2146	0.0424	0.0076	0.0012	0.0000	0.0000	0.0000
	1	0.2242	0.3389	0.1413	0.0404	0.0093	0.0003	0.0000	0.0000
	2	0.0328	0.2586	0.2277	0.1034	0.0337	0.0018	0.0000	0.0000
	3	0.0031	0.1270	0.2361	0.1703	0.0785	0.0072	0.0003	0.0000
	4	0.0002	0.0451	0.1771	0.2028	0.1325	0.0208	0.0012	0.0000
	5	0.0000	0.0124	0.1023	0.1861	0.1723	0.0464	0.0041	0.0001
	6	0.0000	0.0027	0.0474	0.1368	0.1795	0.0829	0.0115	0.0006
	7	0.0000	0.0005	0.0180	0.0828	0.1538	0.1219	0.0263	0.0019
	8	0.0000	0.0001	0.0058	0.0420	0.1106	0.1501	0.0505	0.0055
	9	0.0000	0.0000	0.0016	0.0181	0.0676	0.1573	0.0823	0.0133
	10	0.0000	0.0000	0.0004	0.0067	0.0355	0.1416	0.1152	0.0280
	11	0.0000	0.0000	0.0001	0.0022	0.0161	0.1103	0.1396	0.0509
	12	0.0000	0.0000	0.0000	0.0006	0.0064	0.0749	0.1474	0.0806
	13	0.0000	0.0000	0.0000	0.0001	0.0022	0.0444	0.1360	0.1115
	14	0.0000	0.0000	0.0000	0.0000	0.0007	0.0231	0.1101	0.1354
	15	0.0000	0.0000	0.0000	0.0000	0.0002	0.0106	0.0783	0.1445
	16	0.0000	0.0000	0.0000	0.0000	0.0000	0.0042	0.0489	0.1354
	17	0.0000	0.0000	0.0000	0.0000	0.0000	0.0015	0.0269	0.1115
	18	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0129	0.0806
	19	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0054	0.0509
	20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0020	0.0280
	21	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0133
	22	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0055
	23	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0019
	24	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006
	25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001
	26	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Binomial Distribution - Probability Mass Function

n	x	θ 0.60	0.70	0.75	0.80	0.85	0.90	0.95	0.99
20	11	0.1597	0.0654	0.0271	0.0074	0.0011	0.0001	0.0000	0.0000
	12	0.1797	0.1144	0.0609	0.0222	0.0046	0.0004	0.0000	0.0000
	13	0.1659	0.1643	0.1124	0.0545	0.0160	0.0020	0.0000	0.0000
	14	0.1244	0.1916	0.1686	0.1091	0.0454	0.0089	0.0003	0.0000
	15	0.0746	0.1789	0.2023	0.1746	0.1028	0.0319	0.0022	0.0000
	16	0.0350	0.1304	0.1897	0.2182	0.1821	0.0898	0.0133	0.0000
	17	0.0123	0.0716	0.1339	0.2054	0.2428	0.1901	0.0596	0.0010
	18	0.0031	0.0278	0.0669	0.1369	0.2293	0.2852	0.1887	0.0159
	19	0.0005	0.0068	0.0211	0.0576	0.1368	0.2702	0.3774	0.1652
	20	0.0000	0.0008	0.0032	0.0115	0.0388	0.1216	0.3585	0.8179
30	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0020	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	11	0.0054	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	12	0.0129	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	13	0.0269	0.0015	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000
	14	0.0489	0.0042	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000
	15	0.0783	0.0106	0.0019	0.0002	0.0000	0.0000	0.0000	0.0000
	16	0.1101	0.0231	0.0054	0.0007	0.0000	0.0000	0.0000	0.0000
	17	0.1360	0.0444	0.0134	0.0022	0.0001	0.0000	0.0000	0.0000
	18	0.1474	0.0749	0.0291	0.0064	0.0006	0.0000	0.0000	0.0000
	19	0.1396	0.1103	0.0551	0.0161	0.0022	0.0001	0.0000	0.0000
	20	0.1152	0.1416	0.0909	0.0355	0.0067	0.0004	0.0000	0.0000
	21	0.0823	0.1573	0.1298	0.0676	0.0181	0.0016	0.0000	0.0000
	22	0.0505	0.1501	0.1593	0.1106	0.0420	0.0058	0.0001	0.0000
	23	0.0263	0.1219	0.1662	0.1538	0.0828	0.0180	0.0005	0.0000
	24	0.0115	0.0829	0.1455	0.1795	0.1368	0.0474	0.0027	0.0000
	25	0.0041	0.0464	0.1047	0.1723	0.1861	0.1023	0.0124	0.0000
	26	0.0012	0.0208	0.0604	0.1325	0.2028	0.1771	0.0451	0.0002

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
50	0	0.6050	0.0769	0.0052	0.0003	0.0000	0.0000	0.0000	0.0000
	1	0.3056	0.2025	0.0286	0.0026	0.0002	0.0000	0.0000	0.0000
	2	0.0756	0.2611	0.0779	0.0113	0.0011	0.0000	0.0000	0.0000
	3	0.0122	0.2199	0.1386	0.0319	0.0044	0.0000	0.0000	0.0000
	4	0.0015	0.1360	0.1809	0.0661	0.0128	0.0001	0.0000	0.0000
	5	0.0001	0.0658	0.1849	0.1072	0.0295	0.0006	0.0000	0.0000
	6	0.0000	0.0260	0.1541	0.1419	0.0554	0.0018	0.0000	0.0000
	7	0.0000	0.0086	0.1076	0.1575	0.0870	0.0048	0.0000	0.0000
	8	0.0000	0.0024	0.0643	0.1493	0.1169	0.0110	0.0002	0.0000
	9	0.0000	0.0006	0.0333	0.1230	0.1364	0.0220	0.0005	0.0000
	10	0.0000	0.0001	0.0152	0.0890	0.1398	0.0386	0.0014	0.0000
	11	0.0000	0.0000	0.0061	0.0571	0.1271	0.0602	0.0035	0.0000
	12	0.0000	0.0000	0.0022	0.0328	0.1033	0.0838	0.0076	0.0001
	13	0.0000	0.0000	0.0007	0.0169	0.0755	0.1050	0.0147	0.0003
	14	0.0000	0.0000	0.0002	0.0079	0.0499	0.1189	0.0260	0.0008
	15	0.0000	0.0000	0.0001	0.0033	0.0299	0.1223	0.0415	0.0020
	16	0.0000	0.0000	0.0000	0.0013	0.0164	0.1147	0.0606	0.0044
	17	0.0000	0.0000	0.0000	0.0005	0.0082	0.0983	0.0808	0.0087
	18	0.0000	0.0000	0.0000	0.0001	0.0037	0.0772	0.0987	0.0160
	19	0.0000	0.0000	0.0000	0.0000	0.0016	0.0558	0.1109	0.0270
	20	0.0000	0.0000	0.0000	0.0000	0.0006	0.0370	0.1146	0.0419
	21	0.0000	0.0000	0.0000	0.0000	0.0002	0.0227	0.1091	0.0598
	22	0.0000	0.0000	0.0000	0.0000	0.0001	0.0128	0.0959	0.0788
	23	0.0000	0.0000	0.0000	0.0000	0.0000	0.0067	0.0778	0.0960
	24	0.0000	0.0000	0.0000	0.0000	0.0000	0.0032	0.0584	0.1080
	25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0014	0.0405	0.1123
	26	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0259	0.1080
	27	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0154	0.0960
	38	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0084	0.0788
	29	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0043	0.0598
	30	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0020	0.0419
	31	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0009	0.0270
	32	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003	0.0160
	33	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0087
	34	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0044
	35	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0020
	40	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	45	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	50	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Binomial Distribution - Probability Mass Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
50	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	12	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	13	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	14	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	15	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	16	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	17	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	18	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	19	0.0009	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	20	0.0020	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	21	0.0043	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	22	0.0084	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	23	0.0154	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	24	0.0259	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	25	0.0405	0.0014	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	26	0.0584	0.0032	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000
	27	0.0778	0.0067	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000
	38	0.0959	0.0128	0.0016	0.0001	0.0000	0.0000	0.0000	0.0000
	29	0.1091	0.0227	0.0036	0.0002	0.0000	0.0000	0.0000	0.0000
	30	0.1146	0.0370	0.0077	0.0006	0.0000	0.0000	0.0000	0.0000
	31	0.1109	0.0558	0.0148	0.0016	0.0000	0.0000	0.0000	0.0000
	32	0.0987	0.0772	0.0264	0.0037	0.0001	0.0000	0.0000	0.0000
	33	0.0808	0.0983	0.0432	0.0082	0.0005	0.0000	0.0000	0.0000
	34	0.0606	0.1147	0.0648	0.0164	0.0013	0.0000	0.0000	0.0000
	35	0.0415	0.1223	0.0888	0.0299	0.0033	0.0001	0.0000	0.0000
	40	0.0014	0.0386	0.0985	0.1398	0.0890	0.0152	0.0001	0.0000
	45	0.0000	0.0006	0.0049	0.0295	0.1072	0.1849	0.0658	0.0001
	50	0.0000	0.0000	0.0000	0.0000	0.0003	0.0052	0.0769	0.6050

Binomial Distribution - Cumulative Distribution Function

$$F_B(x \mid n;\ \theta) = \sum_{\nu=0}^{x} \binom{n}{\nu}\ \theta^{\nu}\ (1-\theta)^{n-\nu} \qquad 0 < \theta < 1$$

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
1	0	0.9900	0.9500	0.9000	0.8500	0.8000	0.7000	0.6000	0.5000
	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	0	0.9801	0.9025	0.8100	0.7225	0.6400	0.4900	0.3600	0.2500
	1	0.9999	0.9975	0.9900	0.9775	0.9600	0.9100	0.8400	0.7500
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.9703	0.8574	0.7290	0.6141	0.5120	0.3430	0.2160	0.1250
	1	0.9997	0.9928	0.9720	0.9393	0.8960	0.7840	0.6480	0.5000
	2	1.0000	0.9999	0.9990	0.9966	0.9920	0.9730	0.9360	0.8750
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	0	0.9606	0.8145	0.6561	0.5220	0.4096	0.2401	0.1296	0.0625
	1	0.9994	0.9860	0.9477	0.8905	0.8192	0.6517	0.4752	0.3125
	2	1.0000	0.9995	0.9963	0.9880	0.9728	0.9163	0.8208	0.6875
	3	1.0000	1.0000	0.9999	0.9995	0.9984	0.9919	0.9744	0.9375
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0	0.9510	0.7738	0.5905	0.4437	0.3277	0.1681	0.0778	0.0313
	1	0.9990	0.9774	0.9185	0.8352	0.7373	0.5282	0.3370	0.1875
	2	1.0000	0.9988	0.9914	0.9734	0.9421	0.8369	0.6826	0.5000
	3	1.0000	1.0000	0.9995	0.9978	0.9933	0.9692	0.9130	0.8125
	4	1.0000	1.0000	1.0000	0.9999	0.9997	0.9976	0.9898	0.9688
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	0	0.9415	0.7351	0.5314	0.3771	0.2621	0.1176	0.0467	0.0156
	1	0.9985	0.9672	0.8857	0.7765	0.6554	0.4202	0.2333	0.1094
	2	1.0000	0.9978	0.9842	0.9527	0.9011	0.7443	0.5443	0.3438
	3	1.0000	0.9999	0.9987	0.9941	0.9830	0.9295	0.8208	0.6563
	4	1.0000	1.0000	0.9999	0.9996	0.9984	0.9891	0.9590	0.8906
	5	1.0000	1.0000	1.0000	1.0000	0.9999	0.9993	0.9959	0.9844
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	0	0.9321	0.6983	0.4783	0.3206	0.2097	0.0824	0.0280	0.0078
	1	0.9980	0.9556	0.8503	0.7166	0.5767	0.3294	0.1586	0.0625
	2	1.0000	0.9962	0.9743	0.9262	0.8520	0.6471	0.4199	0.2266
	3	1.0000	0.9998	0.9973	0.9879	0.9667	0.8740	0.7102	0.5000
	4	1.0000	1.0000	0.9998	0.9988	0.9953	0.9712	0.9037	0.7734
	5	1.0000	1.0000	1.0000	0.9999	0.9996	0.9962	0.9812	0.9375
	6	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9984	0.9922
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
1	0	0.4000	0.3000	0.2500	0.2000	0.1500	0.1000	0.0500	0.0100
	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	0	0.1600	0.0900	0.0625	0.0400	0.0225	0.0100	0.0025	0.0001
	1	0.6400	0.5100	0.4375	0.3600	0.2775	0.1900	0.0975	0.0199
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.0640	0.0270	0.0156	0.0080	0.0034	0.0010	0.0001	0.0000
	1	0.3520	0.2160	0.1563	0.1040	0.0608	0.0280	0.0073	0.0003
	2	0.7840	0.6570	0.5781	0.4880	0.3859	0.2710	0.1426	0.0297
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	0	0.0256	0.0081	0.0039	0.0016	0.0005	0.0001	0.0000	0.0000
	1	0.1792	0.0837	0.0508	0.0272	0.0120	0.0037	0.0005	0.0000
	2	0.5248	0.3483	0.2617	0.1808	0.1095	0.0523	0.0140	0.0006
	3	0.8704	0.7599	0.6836	0.5904	0.4780	0.3439	0.1855	0.0394
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0	0.0102	0.0024	0.0010	0.0003	0.0001	0.0000	0.0000	0.0000
	1	0.0870	0.0308	0.0156	0.0067	0.0022	0.0005	0.0000	0.0000
	2	0.3174	0.1631	0.1035	0.0579	0.0266	0.0086	0.0012	0.0000
	3	0.6630	0.4718	0.3672	0.2627	0.1648	0.0815	0.0226	0.0010
	4	0.9222	0.8319	0.7627	0.6723	0.5563	0.4095	0.2262	0.0490
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	0	0.0041	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000
	1	0.0410	0.0109	0.0046	0.0016	0.0004	0.0001	0.0000	0.0000
	2	0.1792	0.0705	0.0376	0.0170	0.0059	0.0013	0.0001	0.0000
	3	0.4557	0.2557	0.1694	0.0989	0.0473	0.0159	0.0022	0.0000
	4	0.7667	0.5798	0.4661	0.3446	0.2235	0.1143	0.0328	0.0015
	5	0.9533	0.8824	0.8220	0.7379	0.6229	0.4686	0.2649	0.0585
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	0	0.0016	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0188	0.0038	0.0013	0.0004	0.0001	0.0000	0.0000	0.0000
	2	0.0963	0.0288	0.0129	0.0047	0.0012	0.0002	0.0000	0.0000
	3	0.2898	0.1260	0.0706	0.0333	0.0121	0.0027	0.0002	0.0000
	4	0.5801	0.3529	0.2436	0.1480	0.0738	0.0257	0.0038	0.0000
	5	0.8414	0.6706	0.5551	0.4233	0.2834	0.1497	0.0444	0.0020
	6	0.9720	0.9176	0.8665	0.7903	0.6794	0.5217	0.3017	0.0679
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
8	0	0.9227	0.6634	0.4305	0.2725	0.1678	0.0576	0.0168	0.0039
	1	0.9973	0.9428	0.8131	0.6572	0.5033	0.2553	0.1064	0.0352
	2	0.9999	0.9942	0.9619	0.8948	0.7969	0.5518	0.3154	0.1445
	3	1.0000	0.9996	0.995	0.9786	0.9437	0.8059	0.5941	0.3633
	4	1.0000	1.0000	0.9996	0.9971	0.9896	0.9420	0.8263	0.6367
	5	1.0000	1.0000	1.0000	0.9998	0.9988	0.9887	0.9502	0.8555
	6	1.0000	1.0000	1.0000	1.0000	0.9999	0.9987	0.9915	0.9648
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9993	0.9961
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	0	0.9135	0.6302	0.3874	0.2316	0.1342	0.0404	0.0101	0.0020
	1	0.9966	0.9288	0.7748	0.5995	0.4362	0.1960	0.0705	0.0195
	2	0.9999	0.9916	0.9470	0.8591	0.7382	0.4628	0.2318	0.0898
	3	1.0000	0.9994	0.9917	0.9661	0.9144	0.7297	0.4826	0.2539
	4	1.0000	1.0000	0.9991	0.9944	0.9804	0.9012	0.7334	0.5000
	5	1.0000	1.0000	0.9999	0.9994	0.9969	0.9747	0.9006	0.7461
	6	1.0000	1.0000	1.0000	1.0000	0.9997	0.9957	0.9750	0.9102
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9996	0.9962	0.9805
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9980
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
10	0	0.9044	0.5987	0.3487	0.1969	0.1074	0.0282	0.0060	0.0010
	1	0.9957	0.9139	0.7361	0.5443	0.3758	0.1493	0.0464	0.0107
	2	0.9999	0.9885	0.9298	0.8202	0.6778	0.3828	0.1673	0.0547
	3	1.0000	0.9990	0.9872	0.9500	0.8791	0.6496	0.3823	0.1719
	4	1.0000	0.9999	0.9984	0.9901	0.9672	0.8497	0.6331	0.3770
	5	1.0000	1.0000	0.9999	0.9986	0.9936	0.9527	0.8338	0.6230
	6	1.0000	1.0000	1.0000	0.9999	0.9991	0.9894	0.9452	0.8281
	7	1.0000	1.0000	1.0000	1.0000	0.9999	0.9984	0.9877	0.9453
	8	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9983	0.9893
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9990
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
11	0	0.8953	0.5688	0.3138	0.1673	0.0859	0.0198	0.0036	0.0005
	1	0.9948	0.8981	0.6974	0.4922	0.3221	0.1130	0.0302	0.0059
	2	0.9998	0.9848	0.9104	0.7788	0.6174	0.3127	0.1189	0.0327
	3	1.0000	0.9984	0.9815	0.9306	0.8389	0.5696	0.2963	0.1133
	4	1.0000	0.9999	0.9972	0.9841	0.9496	0.7897	0.5328	0.2744
	5	1.0000	1.0000	0.9997	0.9973	0.9883	0.9218	0.7535	0.5000
	6	1.0000	1.0000	1.0000	0.9997	0.9980	0.9784	0.9006	0.7256
	7	1.0000	1.0000	1.0000	1.0000	0.9998	0.9957	0.9707	0.8867
	8	1.0000	1.0000	1.0000	1.0000	1.0000	0.9994	0.9941	0.9673
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9993	0.9941
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9995
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
8	0	0.0007	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0085	0.0013	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
	2	0.0498	0.0113	0.0042	0.0012	0.0002	0.0000	0.0000	0.0000
	3	0.1737	0.0580	0.0273	0.0104	0.0029	0.0004	0.0000	0.0000
	4	0.4059	0.1941	0.1138	0.0563	0.0214	0.0050	0.0004	0.0000
	5	0.6846	0.4482	0.3215	0.2031	0.1052	0.0381	0.0058	0.0001
	6	0.8936	0.7447	0.6329	0.4967	0.3428	0.1869	0.0572	0.0027
	7	0.9832	0.9424	0.8999	0.8322	0.7275	0.5695	0.3366	0.0773
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	0	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0038	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0250	0.0043	0.0013	0.0003	0.0000	0.0000	0.0000	0.0000
	3	0.0994	0.0253	0.0100	0.0031	0.0006	0.0001	0.0000	0.0000
	4	0.2666	0.0988	0.0489	0.0196	0.0056	0.0009	0.0000	0.0000
	5	0.5174	0.2703	0.1657	0.0856	0.0339	0.0083	0.0006	0.0000
	6	0.7682	0.5372	0.3993	0.2618	0.1409	0.0530	0.0084	0.0001
	7	0.9295	0.8040	0.6997	0.5638	0.4005	0.2252	0.0712	0.0034
	8	0.9899	0.9596	0.9249	0.8658	0.7684	0.6126	0.3698	0.0865
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
10	0	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0017	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0123	0.0016	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
	3	0.0548	0.0106	0.0035	0.0009	0.0001	0.0000	0.0000	0.0000
	4	0.1662	0.0473	0.0197	0.0064	0.0014	0.0001	0.0000	0.0000
	5	0.3669	0.1503	0.0781	0.0328	0.0099	0.0016	0.0001	0.0000
	6	0.6177	0.3504	0.2241	0.1209	0.0500	0.0128	0.0010	0.0000
	7	0.8327	0.6172	0.4744	0.3222	0.1798	0.0702	0.0115	0.0001
	8	0.9536	0.8507	0.7560	0.6242	0.4557	0.2639	0.0861	0.0043
	9	0.9940	0.9718	0.9437	0.8926	0.8031	0.6513	0.4013	0.0956
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
11	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0059	0.0006	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0293	0.0043	0.0012	0.0002	0.0000	0.0000	0.0000	0.0000
	4	0.0994	0.0216	0.0076	0.0020	0.0003	0.0000	0.0000	0.0000
	5	0.2465	0.0782	0.0343	0.0117	0.0027	0.0003	0.0000	0.0000
	6	0.4672	0.2103	0.1146	0.0504	0.0159	0.0028	0.0001	0.0000
	7	0.7037	0.4304	0.2867	0.1611	0.0694	0.0185	0.0016	0.0000
	8	0.8811	0.6873	0.5448	0.3826	0.2212	0.0896	0.0152	0.0002
	9	0.9698	0.8870	0.8029	0.6779	0.5078	0.3026	0.1019	0.0052
	10	0.9964	0.9802	0.9578	0.9141	0.8327	0.6862	0.4312	0.1047
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
12	0	0.8864	0.5404	0.2824	0.1422	0.0687	0.0138	0.0022	0.0002
	1	0.9938	0.8816	0.6590	0.4435	0.2749	0.0850	0.0196	0.0032
	2	0.9998	0.9804	0.8891	0.7358	0.5583	0.2528	0.0834	0.0193
	3	1.0000	0.9978	0.9744	0.9078	0.7946	0.4925	0.2253	0.0730
	4	1.0000	0.9998	0.9957	0.9761	0.9274	0.7237	0.4382	0.1938
	5	1.0000	1.0000	0.9995	0.9954	0.9806	0.8822	0.6652	0.3872
	6	1.0000	1.0000	0.9999	0.9993	0.9961	0.9614	0.8418	0.6128
	7	1.0000	1.0000	1.0000	0.9999	0.9994	0.9905	0.9427	0.8062
	8	1.0000	1.0000	1.0000	1.0000	0.9999	0.9983	0.9847	0.9270
	9	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9972	0.9807
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9968
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
13	0	0.8775	0.5133	0.2542	0.1209	0.0550	0.0097	0.0013	0.0001
	1	0.9928	0.8646	0.6213	0.3983	0.2336	0.0637	0.0126	0.0017
	2	0.9997	0.9755	0.8661	0.6920	0.5017	0.2025	0.0579	0.0112
	3	1.0000	0.9969	0.9658	0.8820	0.7473	0.4206	0.1686	0.0461
	4	1.0000	0.9997	0.9935	0.9658	0.9009	0.6543	0.3530	0.1334
	5	1.0000	1.0000	0.9991	0.9925	0.9700	0.8346	0.5744	0.2905
	6	1.0000	1.0000	0.9999	0.9987	0.9930	0.9376	0.7712	0.5000
	7	1.0000	1.0000	1.0000	0.9998	0.9988	0.9818	0.9023	0.7095
	8	1.0000	1.0000	1.0000	1.0000	0.9998	0.9960	0.9679	0.8666
	9	1.0000	1.0000	1.0000	1.0000	1.0000	0.9993	0.9922	0.9539
	10	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9987	0.9888
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9983
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
14	0	0.8687	0.4877	0.2288	0.1028	0.0440	0.0068	0.0008	0.0001
	1	0.9916	0.8470	0.5846	0.3567	0.1979	0.0475	0.0081	0.0009
	2	0.9997	0.9699	0.8416	0.6479	0.4481	0.1608	0.0398	0.0065
	3	1.0000	0.9958	0.9559	0.8535	0.6982	0.3552	0.1243	0.0287
	4	1.0000	0.9996	0.9908	0.9533	0.8702	0.5842	0.2793	0.0898
	5	1.0000	1.0000	0.9985	0.9885	0.9561	0.7805	0.4859	0.2120
	6	1.0000	1.0000	0.9998	0.9978	0.9884	0.9067	0.6925	0.3953
	7	1.0000	1.0000	1.0000	0.9997	0.9976	0.9685	0.8499	0.6047
	8	1.0000	1.0000	1.0000	1.0000	0.9996	0.9917	0.9417	0.7880
	9	1.0000	1.0000	1.0000	1.0000	1.0000	0.9983	0.9825	0.9102
	10	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9961	0.9713
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9994	0.9935
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9991
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ 0.60	0.70	0.75	0.80	0.85	0.90	0.95	0.99
12	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0028	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0153	0.0017	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
	4	0.0573	0.0095	0.0028	0.0006	0.0001	0.0000	0.0000	0.0000
	5	0.1582	0.0386	0.0143	0.0039	0.0007	0.0001	0.0000	0.0000
	6	0.3348	0.1178	0.0544	0.0194	0.0046	0.0005	0.0000	0.0000
	7	0.5618	0.2763	0.1576	0.0726	0.0239	0.0043	0.0002	0.0000
	8	0.7747	0.5075	0.3512	0.2054	0.0922	0.0256	0.0022	0.0000
	9	0.9166	0.7472	0.6093	0.4417	0.2642	0.1109	0.0196	0.0002
	10	0.9804	0.9150	0.8416	0.7251	0.5565	0.3410	0.1184	0.0062
	11	0.9978	0.9862	0.9683	0.9313	0.8578	0.7176	0.4596	0.1136
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
13	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0013	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0078	0.0007	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0321	0.0040	0.0010	0.0002	0.0000	0.0000	0.0000	0.0000
	5	0.0977	0.0182	0.0056	0.0012	0.0002	0.0000	0.0000	0.0000
	6	0.2288	0.0624	0.0243	0.0070	0.0013	0.0001	0.0000	0.0000
	7	0.4256	0.1654	0.0802	0.0300	0.0075	0.0009	0.0000	0.0000
	8	0.6470	0.3457	0.2060	0.0991	0.0342	0.0065	0.0003	0.0000
	9	0.8314	0.5794	0.4157	0.2527	0.1180	0.0342	0.0031	0.0000
	10	0.9421	0.7975	0.6674	0.4983	0.3080	0.1339	0.0245	0.0003
	11	0.9874	0.9363	0.8733	0.7664	0.6017	0.3787	0.1354	0.0072
	12	0.9987	0.9903	0.9762	0.9450	0.8791	0.7458	0.4867	0.1225
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
14	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0039	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0175	0.0017	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0583	0.0083	0.0022	0.0004	0.0000	0.0000	0.0000	0.0000
	6	0.1501	0.0315	0.0103	0.0024	0.0003	0.0000	0.0000	0.0000
	7	0.3075	0.0933	0.0383	0.0116	0.0022	0.0002	0.0000	0.0000
	8	0.5141	0.2195	0.1117	0.0439	0.0115	0.0015	0.0000	0.0000
	9	0.7207	0.4158	0.2585	0.1298	0.0467	0.0092	0.0004	0.0000
	10	0.8757	0.6448	0.4787	0.3018	0.1465	0.0441	0.0042	0.0000
	11	0.9602	0.8392	0.7189	0.5519	0.3521	0.1584	0.0301	0.0003
	12	0.9919	0.9525	0.8990	0.8021	0.6433	0.4154	0.1530	0.0084
	13	0.9992	0.9932	0.9822	0.9560	0.8972	0.7712	0.5123	0.1313
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
15	0	0.8601	0.4633	0.2059	0.0874	0.0352	0.0047	0.0005	0.0000
	1	0.9904	0.8290	0.5490	0.3186	0.1671	0.0353	0.0052	0.0005
	2	0.9996	0.9638	0.8159	0.6042	0.3980	0.1268	0.0271	0.0037
	3	1.0000	0.9945	0.9444	0.8227	0.6482	0.2969	0.0905	0.0176
	4	1.0000	0.9994	0.9873	0.9383	0.8358	0.5155	0.2173	0.0592
	5	1.0000	0.9999	0.9978	0.9832	0.9389	0.7216	0.4032	0.1509
	6	1.0000	1.0000	0.9997	0.9964	0.9819	0.8689	0.6098	0.3036
	7	1.0000	1.0000	1.0000	0.9994	0.9958	0.9500	0.7869	0.5000
	8	1.0000	1.0000	1.0000	0.9999	0.9992	0.9848	0.9050	0.6964
	9	1.0000	1.0000	1.0000	1.0000	0.9999	0.9963	0.9662	0.8491
	10	1.0000	1.0000	1.0000	1.0000	1.0000	0.9993	0.9907	0.9408
	11	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9981	0.9824
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9963
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9995
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
20	0	0.8179	0.3585	0.1216	0.0388	0.0115	0.0008	0.0000	0.0000
	1	0.9831	0.7358	0.3917	0.1756	0.0692	0.0076	0.0005	0.0000
	2	0.9990	0.9245	0.6769	0.4049	0.2061	0.0355	0.0036	0.0002
	3	1.0000	0.9841	0.8670	0.6477	0.4114	0.1071	0.0160	0.0013
	4	1.0000	0.9974	0.9568	0.8298	0.6296	0.2375	0.0510	0.0059
	5	1.0000	0.9997	0.9887	0.9327	0.8042	0.4164	0.1256	0.0207
	6	1.0000	1.0000	0.9976	0.9781	0.9133	0.6080	0.2500	0.0577
	7	1.0000	1.0000	0.9996	0.9941	0.9679	0.7723	0.4159	0.1316
	8	1.0000	1.0000	0.9999	0.9987	0.9900	0.8867	0.5956	0.2517
	9	1.0000	1.0000	1.0000	0.9998	0.9974	0.9520	0.7553	0.4119
	10	1.0000	1.0000	1.0000	1.0000	0.9994	0.9829	0.8725	0.5881
	11	1.0000	1.0000	1.0000	1.0000	0.9999	0.9949	0.9435	0.7483
	12	1.0000	1.0000	1.0000	1.0000	1.0000	0.9987	0.9790	0.8684
	13	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9935	0.9423
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9984	0.9793
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9941
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9987
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ 0.60	0.70	0.75	0.80	0.85	0.90	0.95	0.99
15	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0019	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0093	0.0007	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0338	0.0037	0.0008	0.0001	0.0000	0.0000	0.0000	0.0000
	6	0.0950	0.0152	0.0042	0.0008	0.0001	0.0000	0.0000	0.0000
	7	0.2131	0.0500	0.0173	0.0042	0.0006	0.0000	0.0000	0.0000
	8	0.3902	0.1311	0.0566	0.0181	0.0036	0.0003	0.0000	0.0000
	9	0.5968	0.2784	0.1484	0.0611	0.0168	0.0022	0.0001	0.0000
	10	0.7827	0.4845	0.3135	0.1642	0.0617	0.0127	0.0006	0.0000
	11	0.9095	0.7031	0.5387	0.3518	0.1773	0.0556	0.0055	0.0000
	12	0.9729	0.8732	0.7639	0.6020	0.3958	0.1841	0.0362	0.0004
	13	0.9948	0.9647	0.9198	0.8329	0.6814	0.4510	0.1710	0.0096
	14	0.9995	0.9953	0.9866	0.9648	0.9126	0.7941	0.5367	0.1399
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
20	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0016	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	6	0.0065	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	7	0.0210	0.0013	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0565	0.0051	0.0009	0.0001	0.0000	0.0000	0.0000	0.0000
	9	0.1275	0.0171	0.0039	0.0006	0.0000	0.0000	0.0000	0.0000
	10	0.2447	0.0480	0.0139	0.0026	0.0002	0.0000	0.0000	0.0000
	11	0.4044	0.1133	0.0409	0.0100	0.0013	0.0001	0.0000	0.0000
	12	0.5841	0.2277	0.1018	0.0321	0.0059	0.0004	0.0000	0.0000
	13	0.7500	0.3920	0.2142	0.0867	0.0219	0.0024	0.0000	0.0000
	14	0.8744	0.5836	0.3828	0.1958	0.0673	0.0113	0.0003	0.0000
	15	0.9490	0.7625	0.5852	0.3704	0.1702	0.0432	0.0026	0.0000
	16	0.9840	0.8929	0.7748	0.5886	0.3523	0.1330	0.0159	0.0000
	17	0.9964	0.9645	0.9087	0.7939	0.5951	0.3231	0.0755	0.0010
	18	0.9995	0.9924	0.9757	0.9308	0.8244	0.6083	0.2642	0.0169
	19	1.0000	0.9992	0.9968	0.9885	0.9612	0.8784	0.6415	0.1821
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
30	0	0.7397	0.2146	0.0424	0.0076	0.0012	0.0000	0.0000	0.0000
	1	0.9639	0.5535	0.1837	0.0480	0.0105	0.0003	0.0000	0.0000
	2	0.9967	0.8122	0.4114	0.1514	0.0442	0.0021	0.0000	0.0000
	3	0.9998	0.9392	0.6474	0.3217	0.1227	0.0093	0.0003	0.0000
	4	1.0000	0.9844	0.8245	0.5245	0.2552	0.0302	0.0015	0.0000
	5	1.0000	0.9967	0.9268	0.7106	0.4275	0.0766	0.0057	0.0002
	6	1.0000	0.9994	0.9742	0.8474	0.6070	0.1595	0.0172	0.0007
	7	1.0000	0.9999	0.9922	0.9302	0.7608	0.2814	0.0435	0.0026
	8	1.0000	1.0000	0.9980	0.9722	0.8713	0.4315	0.0940	0.0081
	9	1.0000	1.0000	0.9995	0.9903	0.9389	0.5888	0.1763	0.0214
	10	1.0000	1.0000	0.9999	0.9971	0.9744	0.7304	0.2915	0.0494
	11	1.0000	1.0000	1.0000	0.9992	0.9905	0.8407	0.4311	0.1002
	12	1.0000	1.0000	1.0000	0.9998	0.9969	0.9155	0.5785	0.1808
	13	1.0000	1.0000	1.0000	1.0000	0.9991	0.9599	0.7145	0.2923
	14	1.0000	1.0000	1.0000	1.0000	0.9998	0.9831	0.8246	0.4278
	15	1.0000	1.0000	1.0000	1.0000	0.9999	0.9936	0.9029	0.5722
	16	1.0000	1.0000	1.0000	1.0000	1.0000	0.9979	0.9519	0.7077
	17	1.0000	1.0000	1.0000	1.0000	1.0000	0.9994	0.9788	0.8192
	18	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9917	0.8998
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9971	0.9506
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9991	0.9786
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9919
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9974
	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9993
	24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998
	25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	26	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
50	0	0.6050	0.0769	0.0052	0.0003	0.0000	0.0000	0.0000	0.0000
	1	0.9106	0.2794	0.0338	0.0029	0.0002	0.0000	0.0000	0.0000
	2	0.9862	0.5405	0.1117	0.0142	0.0013	0.0000	0.0000	0.0000
	3	0.9984	0.7604	0.2503	0.0460	0.0057	0.0000	0.0000	0.0000
	4	0.9999	0.8964	0.4312	0.1121	0.0185	0.0002	0.0000	0.0000
	5	1.0000	0.9622	0.6161	0.2194	0.0480	0.0007	0.0000	0.0000
	6	1.0000	0.9882	0.7702	0.3613	0.1034	0.0025	0.0000	0.0000
	7	1.0000	0.9968	0.8779	0.5188	0.1904	0.0073	0.0001	0.0000
	8	1.0000	0.9992	0.9421	0.6681	0.3073	0.0183	0.0002	0.0000
	9	1.0000	0.9998	0.9755	0.7911	0.4437	0.0402	0.0008	0.0000
	10	1.0000	1.0000	0.9906	0.8801	0.5836	0.0789	0.0022	0.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
30	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0009	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0029	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	11	0.0083	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	12	0.0212	0.0006	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	13	0.0481	0.0021	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000
	14	0.0971	0.0064	0.0008	0.0001	0.0000	0.0000	0.0000	0.0000
	15	0.1754	0.0169	0.0027	0.0002	0.0000	0.0000	0.0000	0.0000
	16	0.2855	0.0401	0.0082	0.0009	0.0000	0.0000	0.0000	0.0000
	17	0.4215	0.0845	0.0216	0.0031	0.0002	0.0000	0.0000	0.0000
	18	0.5689	0.1593	0.0507	0.0095	0.0008	0.0000	0.0000	0.0000
	19	0.7085	0.2696	0.1057	0.0256	0.0029	0.0001	0.0000	0.0000
	20	0.8237	0.4112	0.1966	0.0611	0.0097	0.0005	0.0000	0.0000
	21	0.9060	0.5685	0.3264	0.1287	0.0278	0.0020	0.0000	0.0000
	22	0.9565	0.7186	0.4857	0.2392	0.0698	0.0078	0.0001	0.0000
	23	0.9828	0.8405	0.6519	0.3930	0.1526	0.0258	0.0006	0.0000
	24	0.9943	0.9234	0.7974	0.5725	0.2894	0.0732	0.0033	0.0000
	25	0.9985	0.9698	0.9021	0.7448	0.4755	0.1755	0.0156	0.0000
	26	0.9997	0.9907	0.9626	0.8773	0.6783	0.3526	0.0608	0.0002
50	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.01	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
50	11	1.0000	1.0000	0.9968	0.9372	0.7107	0.1390	0.0057	0.0000
	12	1.0000	1.0000	0.9990	0.9699	0.8139	0.2229	0.0133	0.0002
	13	1.0000	1.0000	0.9997	0.9868	0.8894	0.3279	0.0280	0.0005
	14	1.0000	1.0000	0.9999	0.9947	0.9393	0.4468	0.0540	0.0013
	15	1.0000	1.0000	1.0000	0.9981	0.9692	0.5692	0.0955	0.0033
	16	1.0000	1.0000	1.0000	0.9993	0.9856	0.6839	0.1561	0.0077
	17	1.0000	1.0000	1.0000	0.9998	0.9937	0.7822	0.2369	0.0164
	18	1.0000	1.0000	1.0000	0.9999	0.9975	0.8594	0.3356	0.0325
	19	1.0000	1.0000	1.0000	1.0000	0.9991	0.9152	0.4465	0.0595
	20	1.0000	1.0000	1.0000	1.0000	0.9997	0.9522	0.5610	0.1013
	21	1.0000	1.0000	1.0000	1.0000	0.9999	0.9749	0.6701	0.1611
	22	1.0000	1.0000	1.0000	1.0000	1.0000	0.9877	0.7660	0.2399
	23	1.0000	1.0000	1.0000	1.0000	1.0000	0.9944	0.8438	0.3359
	24	1.0000	1.0000	1.0000	1.0000	1.0000	0.9976	0.9022	0.4439
	25	1.0000	1.0000	1.0000	1.0000	1.0000	0.9991	0.9427	0.5561
	26	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9686	0.6641
	27	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9840	0.7601
	38	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9924	0.8389
	29	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9966	0.8987
	30	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9986	0.9405
	31	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9995	0.9675
	32	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9836
	33	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9923
	34	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9967
	35	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9987
	40	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	45	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Binomial Distribution - Cumulative Distribution Function

n	x	θ							
		0.60	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**0.99**
50	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	12	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	13	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	14	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	15	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	16	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	17	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	18	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	19	0.0014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	20	0.0034	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	21	0.0076	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	22	0.0160	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	23	0.0314	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	24	0.0573	0.0009	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	25	0.0978	0.0024	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	26	0.1562	0.0056	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000
	27	0.2340	0.0123	0.0010	0.0000	0.0000	0.0000	0.0000	0.0000
	38	0.3299	0.0251	0.0026	0.0001	0.0000	0.0000	0.0000	0.0000
	29	0.4390	0.0478	0.0063	0.0003	0.0000	0.0000	0.0000	0.0000
	30	0.5535	0.0848	0.0139	0.0009	0.0000	0.0000	0.0000	0.0000
	31	0.6644	0.1406	0.0287	0.0025	0.0001	0.0000	0.0000	0.0000
	32	0.7631	0.2178	0.0551	0.0063	0.0002	0.0000	0.0000	0.0000
	33	0.8439	0.3161	0.0983	0.0144	0.0007	0.0000	0.0000	0.0000
	34	0.9045	0.4308	0.1631	0.0308	0.0019	0.0000	0.0000	0.0000
	35	0.9460	0.5532	0.2519	0.0607	0.0053	0.0001	0.0000	0.0000
	40	0.9992	0.9598	0.8363	0.5563	0.2089	0.0245	0.0002	0.0000
	45	1.0000	0.9998	0.9979	0.9815	0.8879	0.5688	0.1036	0.0001
	50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Hypergeometric Distribution - Probability Mass Function

$$f_H(x \mid N;\, n;\, M) = \begin{cases} \dfrac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} & \text{for} \quad x = 0,\, 1,\, \ldots,\, n \\ 0 & \text{others} \end{cases}$$

For $M > n$ the following applies: $f_H\,(x \mid N;\, n;\, M) = f_H\,(x \mid N;\, M;\, n)$

N	n	M	x	f_H
2	1	1	0	0.5000
2	1	1	1	0.5000
3	1	1	0	0.6667
3	1	1	1	0.3333
3	2	1	0	0.3333
3	2	1	1	0.6667
3	2	2	1	0.6667
3	2	2	2	0.3333
4	1	1	0	0.7500
4	1	1	1	0.2500
4	2	1	0	0.5000
4	2	1	1	0.5000
4	2	2	0	0.1667
4	2	2	1	0.6667
4	2	2	2	0.1667
4	3	1	0	0.2500
4	3	1	1	0.7500
4	3	2	1	0.5000
4	3	2	2	0.5000
4	3	3	2	0.7500
4	3	3	3	0.2500
5	1	1	0	0.8000
5	1	1	1	0.2000
5	2	1	0	0.6000
5	2	1	1	0.4000
5	2	2	0	0.3000
5	2	2	1	0.6000
5	2	2	2	0.1000
5	3	1	0	0.4000
5	3	1	1	0.6000
5	3	2	0	0.1000
5	3	2	1	0.6000
5	3	2	2	0.3000
5	3	3	1	0.3000
5	3	3	2	0.6000
5	3	3	3	0.1000
5	4	1	0	0.2000
5	4	1	1	0.8000
5	4	2	1	0.4000
5	4	2	2	0.6000
5	4	3	2	0.6000
5	4	3	3	0.4000
5	4	4	3	0.8000
5	4	4	4	0.2000
6	1	1	0	0.8333
6	1	1	1	0.1667
6	2	1	0	0.6667
6	2	1	1	0.3333
6	2	2	0	0.4000
6	2	2	1	0.5333
6	2	2	2	0.0667
6	3	1	0	0.5000
6	3	1	1	0.5000
6	3	2	0	0.2000
6	3	2	1	0.6000
6	3	2	2	0.2000
6	3	3	0	0.0500
6	3	3	1	0.4500
6	3	3	2	0.4500
6	3	3	3	0.0500
6	4	1	0	0.3333
6	4	1	1	0.6667
6	4	2	0	0.0667
6	4	2	1	0.5333
6	4	2	2	0.4000
6	4	3	1	0.2000
6	4	3	2	0.6000
6	4	3	3	0.2000
6	4	4	2	0.4000
6	4	4	3	0.5333
6	4	4	4	0.0667
6	5	1	0	0.1667
6	5	1	1	0.8333
6	5	2	1	0.3333
6	5	2	2	0.6667
6	5	3	2	0.5000
6	5	3	3	0.5000
6	5	4	3	0.6667
6	5	4	4	0.3333
6	5	5	4	0.8333
6	5	5	5	0.1667
7	1	1	0	0.8571
7	1	1	1	0.1429
7	2	1	0	0.7143
7	2	1	1	0.2857
7	2	2	0	0.4762
7	2	2	1	0.4762
7	2	2	2	0.0476
7	3	1	0	0.5714
7	3	1	1	0.4286

Hypergeometric Distribution - Probability Mass Function

N	n	M	x	f_H
7	3	2	0	0.2857
7	3	2	1	0.5714
7	3	2	2	0.1429
7	3	3	0	0.1143
7	3	3	1	0.5143
7	3	3	2	0.3429
7	3	3	3	0.0286
7	4	1	0	0.4286
7	4	1	1	0.5714
7	4	2	0	0.1429
7	4	2	1	0.5714
7	4	2	2	0.2857
7	4	3	0	0.0286
7	4	3	1	0.3429
7	4	3	2	0.5143
7	4	3	3	0.1143
7	4	4	1	0.1143
7	4	4	2	0.5143
7	4	4	3	0.3429
7	4	4	4	0.0286
7	5	1	0	0.2857
7	5	1	1	0.7143
7	5	2	0	0.0476
7	5	2	1	0.4762
7	5	2	2	0.4762
7	5	3	1	0.1429
7	5	3	2	0.5714
7	5	3	3	0.2857
7	5	4	2	0.2857
7	5	4	3	0.5714
7	5	4	4	0.1429
7	5	5	3	0.4762
7	5	5	4	0.4762
7	5	5	5	0.0476
7	6	1	0	0.1429
7	6	1	1	0.8571
7	6	2	1	0.2857
7	6	2	2	0.7143
7	6	3	2	0.4286
7	6	3	3	0.5714
7	6	4	3	0.5714
7	6	4	4	0.4286
7	6	5	4	0.7143
7	6	5	5	0.2857
7	6	6	5	0.8571

N	n	M	x	f_H
7	6	6	6	0.1429
8	1	1	0	0.8750
8	1	1	1	0.1250
8	2	1	0	0.7500
8	2	1	1	0.2500
8	2	2	0	0.5357
8	2	2	1	0.4286
8	2	2	2	0.0357
8	3	1	0	0.6250
8	3	1	1	0.3750
8	3	2	0	0.3571
8	3	2	1	0.5357
8	3	2	2	0.1071
8	3	3	0	0.1786
8	3	3	1	0.5357
8	3	3	2	0.2679
8	3	3	3	0.0179
8	4	1	0	0.5000
8	4	1	1	0.5000
8	4	2	0	0.2143
8	4	2	1	0.5714
8	4	2	2	0.2143
8	4	3	0	0.0714
8	4	3	1	0.4286
8	4	3	2	0.4286
8	4	3	3	0.0714
8	4	4	0	0.0143
8	4	4	1	0.2286
8	4	4	2	0.5143
8	4	4	3	0.2286
8	4	4	4	0.0143
8	5	1	0	0.3750
8	5	1	1	0.6250
8	5	2	0	0.1071
8	5	2	1	0.5357
8	5	2	2	0.3571
8	5	3	0	0.0179
8	5	3	1	0.2679
8	5	3	2	0.5357
8	5	3	3	0.1786
8	5	4	1	0.0714
8	5	4	2	0.4286
8	5	4	3	0.4286
8	5	4	4	0.0714
8	5	5	2	0.1786

N	n	M	x	f_H
8	5	5	3	0.5357
8	5	5	4	0.2679
8	5	5	5	0.0179
8	6	1	0	0.2500
8	6	1	1	0.7500
8	6	2	0	0.0357
8	6	2	1	0.4286
8	6	2	2	0.5357
8	6	3	1	0.1071
8	6	3	2	0.5357
8	6	3	3	0.3571
8	6	4	2	0.2143
8	6	4	3	0.5714
8	6	4	4	0.2143
8	6	5	3	0.3571
8	6	5	4	0.5357
8	6	5	5	0.1071
8	6	6	4	0.5357
8	6	6	5	0.4286
8	6	6	6	0.0357
8	7	1	0	0.1250
8	7	1	1	0.8750
8	7	2	1	0.2500
8	7	2	2	0.7500
8	7	3	2	0.3750
8	7	3	3	0.6250
8	7	4	3	0.5000
8	7	4	4	0.5000
8	7	5	4	0.6250
8	7	5	5	0.3750
8	7	6	5	0.7500
8	7	6	6	0.2500
8	7	7	6	0.8750
8	7	7	7	0.1250
9	1	1	0	0.8889
9	1	1	1	0.1111
9	2	1	0	0.7778
9	2	1	1	0.2222
9	2	2	0	0.5833
9	2	2	1	0.3889
9	2	2	2	0.0278
9	3	1	0	0.6667
9	3	1	1	0.3333
9	3	2	0	0.4167
9	3	2	1	0.5000

Hypergeometric Distribution - Probability Mass Function

N	n	M	x	f_H
9	3	2	2	0.0833
9	3	3	0	0.2381
9	3	3	1	0.5357
9	3	3	2	0.2143
9	3	3	3	0.0119
9	4	1	0	0.5556
9	4	1	1	0.4444
9	4	2	0	0.2778
9	4	2	1	0.5556
9	4	2	2	0.1667
9	4	3	0	0.1190
9	4	3	1	0.4762
9	4	3	2	0.3571
9	4	3	3	0.0476
9	4	4	0	0.0397
9	4	4	1	0.3175
9	4	4	2	0.4762
9	4	4	3	0.1587
9	4	4	4	0.0079
9	5	1	0	0.4444
9	5	1	1	0.5556
9	5	2	0	0.1667
9	5	2	1	0.5556
9	5	2	2	0.2778
9	5	3	0	0.0476
9	5	3	1	0.3571
9	5	3	2	0.4762
9	5	3	3	0.1190
9	5	4	0	0.0079
9	5	4	1	0.1587
9	5	4	2	0.4762
9	5	4	3	0.3175
9	5	4	4	0.0397
9	5	5	1	0.0397
9	5	5	2	0.3175
9	5	5	3	0.4762
9	5	5	4	0.1587
9	5	5	5	0.0079
9	6	1	0	0.3333
9	6	1	1	0.6667
9	6	2	0	0.0833
9	6	2	1	0.5000
9	6	2	2	0.4167
9	6	3	0	0.0119
9	6	3	1	0.2143

N	n	M	x	f_H
9	6	3	2	0.5357
9	6	3	3	0.2381
9	6	4	1	0.0476
9	6	4	2	0.3571
9	6	4	3	0.4762
9	6	4	4	0.1190
9	6	5	2	0.1190
9	6	5	3	0.4762
9	6	5	4	0.3571
9	6	5	5	0.0476
9	6	6	3	0.2381
9	6	6	4	0.5357
9	6	6	5	0.2143
9	6	6	6	0.0119
9	7	1	0	0.2222
9	7	1	1	0.7778
9	7	2	0	0.0278
9	7	2	1	0.3889
9	7	2	2	0.5833
9	7	3	1	0.0833
9	7	3	2	0.5000
9	7	3	3	0.4167
9	7	4	2	0.1667
9	7	4	3	0.5556
9	7	4	4	0.2778
9	7	5	3	0.2778
9	7	5	4	0.5556
9	7	5	5	0.1667
9	7	6	4	0.4167
9	7	6	5	0.5000
9	7	6	6	0.0833
9	7	7	5	0.5833
9	7	7	6	0.3889
9	7	7	7	0.0278
9	8	1	0	0.1111
9	8	1	1	0.8889
9	8	2	1	0.2222
9	8	2	2	0.7778
9	8	3	2	0.3333
9	8	3	3	0.6667
9	8	4	3	0.4444
9	8	4	4	0.5556
9	8	5	4	0.5556
9	8	5	5	0.4444
9	8	6	5	0.6667

N	n	M	x	f_H
9	8	6	6	0.3333
9	8	7	6	0.7778
9	8	7	7	0.2222
9	8	8	7	0.8889
9	8	8	8	0.1111
10	1	1	0	0.9000
10	1	1	1	0.1000
10	2	1	0	0.8000
10	2	1	1	0.2000
10	2	2	0	0.6222
10	2	2	1	0.3556
10	2	2	2	0.0222
10	3	1	0	0.7000
10	3	1	1	0.3000
10	3	2	0	0.4667
10	3	2	1	0.4667
10	3	2	2	0.0667
10	3	3	0	0.2917
10	3	3	1	0.5250
10	3	3	2	0.1750
10	3	3	3	0.0083
10	4	1	0	0.6000
10	4	1	1	0.4000
10	4	2	0	0.3333
10	4	2	1	0.5333
10	4	2	2	0.1333
10	4	3	0	0.1667
10	4	3	1	0.5000
10	4	3	2	0.3000
10	4	3	3	0.0333
10	4	4	0	0.0714
10	4	4	1	0.3810
10	4	4	2	0.4286
10	4	4	3	0.1143
10	4	4	4	0.0048
10	5	1	0	0.5000
10	5	1	1	0.5000
10	5	2	0	0.2222
10	5	2	1	0.5556
10	5	2	2	0.2222
10	5	3	0	0.0833
10	5	3	1	0.4167
10	5	3	2	0.4167
10	5	3	3	0.0833
10	5	4	0	0.0238

Hypergeometric Distribution - Probability Mass Function

N	n	M	x	f_H	N	n	M	x	f_H	N	n	M	x	f_H
10	5	4	1	0.2381	10	7	4	3	0.5000	10	9	5	4	0.5000
10	5	4	2	0.4762	10	7	4	4	0.1667	10	9	5	5	0.5000
10	5	4	3	0.2381	10	7	5	2	0.0833	10	9	6	5	0.6000
10	5	4	4	0.0238	10	7	5	3	0.4167	10	9	6	6	0.4000
10	5	5	0	0.0040	10	7	5	4	0.4167	10	9	7	6	0.7000
10	5	5	1	0.0992	10	7	5	5	0.0833	10	9	7	7	0.3000
10	5	5	2	0.3968	10	7	6	3	0.1667	10	9	8	7	0.8000
10	5	5	3	0.3968	10	7	6	4	0.5000	10	9	8	8	0.2000
10	5	5	4	0.0992	10	7	6	5	0.3000	10	9	9	8	0.9000
10	5	5	5	0.0040	10	7	6	6	0.0333	10	9	9	9	0.1000
10	6	1	0	0.4000	10	7	7	4	0.2917	11	1	1	0	0.9091
10	6	1	1	0.6000	10	7	7	5	0.5250	11	1	1	1	0.0909
10	6	2	0	0.1333	10	7	7	6	0.1750	11	2	1	0	0.8182
10	6	2	1	0.5333	10	7	7	7	0.0083	11	2	1	1	0.1818
10	6	2	2	0.3333	10	8	1	0	0.2000	11	2	2	0	0.6545
10	6	3	0	0.0333	10	8	1	1	0.8000	11	2	2	1	0.3273
10	6	3	1	0.3000	10	8	2	0	0.0222	11	2	2	2	0.0182
10	6	3	2	0.5000	10	8	2	1	0.3556	11	3	1	0	0.7273
10	6	3	3	0.1667	10	8	2	2	0.6222	11	3	1	1	0.2727
10	6	4	0	0.0048	10	8	3	1	0.0667	11	3	2	0	0.5091
10	6	4	1	0.1143	10	8	3	2	0.4667	11	3	2	1	0.4364
10	6	4	2	0.4286	10	8	3	3	0.4667	11	3	2	2	0.0545
10	6	4	3	0.3810	10	8	4	2	0.1333	11	3	3	0	0.3394
10	6	4	4	0.0714	10	8	4	3	0.5333	11	3	3	1	0.5091
10	6	5	1	0.0238	10	8	4	4	0.3333	11	3	3	2	0.1455
10	6	5	2	0.2381	10	8	5	3	0.2222	11	3	3	3	0.0061
10	6	5	3	0.4762	10	8	5	4	0.5556	11	4	1	0	0.6364
10	6	5	4	0.2381	10	8	5	5	0.2222	11	4	1	1	0.3636
10	6	5	5	0.0238	10	8	6	4	0.3333	11	4	2	0	0.3818
10	6	6	2	0.0714	10	8	6	5	0.5333	11	4	2	1	0.5091
10	6	6	3	0.3810	10	8	6	6	0.1333	11	4	2	2	0.1091
10	6	6	4	0.4286	10	8	7	5	0.4667	11	4	3	0	0.2121
10	6	6	5	0.1143	10	8	7	6	0.4667	11	4	3	1	0.5091
10	6	6	6	0.0048	10	8	7	7	0.0667	11	4	3	2	0.2545
10	7	1	0	0.3000	10	8	8	6	0.6222	11	4	3	3	0.0242
10	7	1	1	0.7000	10	8	8	7	0.3556	11	4	4	0	0.1061
10	7	2	0	0.0667	10	8	8	8	0.0222	11	4	4	1	0.4242
10	7	2	1	0.4667	10	9	1	0	0.1000	11	4	4	2	0.3818
10	7	2	2	0.4667	10	9	1	1	0.9000	11	4	4	3	0.0848
10	7	3	0	0.0083	10	9	2	1	0.2000	11	4	4	4	0.0030
10	7	3	1	0.1750	10	9	2	2	0.8000	11	5	1	0	0.5455
10	7	3	2	0.5250	10	9	3	2	0.3000	11	5	1	1	0.4545
10	7	3	3	0.2917	10	9	3	3	0.7000	11	5	2	0	0.2727
10	7	4	1	0.0333	10	9	4	3	0.4000	11	5	2	1	0.5455
10	7	4	2	0.3000	10	9	4	4	0.6000	11	5	2	2	0.1818

Hypergeometric Distribution - Probability Mass Function

N	n	M	x	f_H
11	5	3	0	0.1212
11	5	3	1	0.4545
11	5	3	2	0.3636
11	5	3	3	0.0606
11	5	4	0	0.0455
11	5	4	1	0.3030
11	5	4	2	0.4545
11	5	4	3	0.1818
11	5	4	4	0.0152
11	5	5	0	0.0130
11	5	5	1	0.1623
11	5	5	2	0.4329
11	5	5	3	0.3247
11	5	5	4	0.0649
11	5	5	5	0.0022
11	6	1	0	0.4545
11	6	1	1	0.5455
11	6	2	0	0.1818
11	6	2	1	0.5455
11	6	2	2	0.2727
11	6	3	0	0.0606
11	6	3	1	0.3636
11	6	3	2	0.4545
11	6	3	3	0.1212
11	6	4	0	0.0152
11	6	4	1	0.1818
11	6	4	2	0.4545
11	6	4	3	0.3030
11	6	4	4	0.0455
11	6	5	0	0.0022
11	6	5	1	0.0649
11	6	5	2	0.3247
11	6	5	3	0.4329
11	6	5	4	0.1623
11	6	5	5	0.0130
11	6	6	1	0.0130
11	6	6	2	0.1623
11	6	6	3	0.4329
11	6	6	4	0.3247
11	6	6	5	0.0649
11	6	6	6	0.0022
11	7	1	0	0.3636
11	7	1	1	0.6364
11	7	2	0	0.1091
11	7	2	1	0.5091
11	7	2	2	0.3818
11	7	3	0	0.0242
11	7	3	1	0.2545
11	7	3	2	0.5091
11	7	3	3	0.2121
11	7	4	0	0.0030
11	7	4	1	0.0848
11	7	4	2	0.3818
11	7	4	3	0.4242
11	7	4	4	0.1061
11	7	5	1	0.0152
11	7	5	2	0.1818
11	7	5	3	0.4545
11	7	5	4	0.3030
11	7	5	5	0.0455
11	7	6	2	0.0455
11	7	6	3	0.3030
11	7	6	4	0.4545
11	7	6	5	0.1818
11	7	6	6	0.0152
11	7	7	3	0.1061
11	7	7	4	0.4242
11	7	7	5	0.3818
11	7	7	6	0.0848
11	7	7	7	0.0030
11	8	1	0	0.2727
11	8	1	1	0.7273
11	8	2	0	0.0545
11	8	2	1	0.4364
11	8	2	2	0.5091
11	8	3	0	0.0061
11	8	3	1	0.1455
11	8	3	2	0.5091
11	8	3	3	0.3394
11	8	4	1	0.0242
11	8	4	2	0.2545
11	8	4	3	0.5091
11	8	4	4	0.2121
11	8	5	2	0.0606
11	8	5	3	0.3636
11	8	5	4	0.4545
11	8	5	5	0.1212
11	8	6	3	0.1212
11	8	6	4	0.4545
11	8	6	5	0.3636
11	8	6	6	0.0606
11	8	7	4	0.2121
11	8	7	5	0.5091
11	8	7	6	0.2545
11	8	7	7	0.0242
11	8	8	5	0.3394
11	8	8	6	0.5091
11	8	8	7	0.1455
11	8	8	8	0.0061
11	9	1	0	0.1818
11	9	1	1	0.8182
11	9	2	0	0.0182
11	9	2	1	0.3273
11	9	2	2	0.6545
11	9	3	1	0.0545
11	9	3	2	0.4364
11	9	3	3	0.5091
11	9	4	2	0.1091
11	9	4	3	0.5091
11	9	4	4	0.3818
11	9	5	3	0.1818
11	9	5	4	0.5455
11	9	5	5	0.2727
11	9	6	4	0.2727
11	9	6	5	0.5455
11	9	6	6	0.1818
11	9	7	5	0.3818
11	9	7	6	0.5091
11	9	7	7	0.1091
11	9	8	6	0.5091
11	9	8	7	0.4364
11	9	8	8	0.0545
11	9	9	7	0.6545
11	9	9	8	0.3273
11	9	9	9	0.0182
11	10	1	0	0.0909
11	10	1	1	0.9091
11	10	2	1	0.1818
11	10	2	2	0.8182
11	10	3	2	0.2727
11	10	3	3	0.7273
11	10	4	3	0.3636
11	10	4	4	0.6364
11	10	5	4	0.4545
11	10	5	5	0.5455

Hypergeometric Distribution - Probability Mass Function

N	n	M	x	f_H
11	10	6	5	0.5455
11	10	6	6	0.4545
11	10	7	6	0.6364
11	10	7	7	0.3636
11	10	8	7	0.7273
11	10	8	8	0.2727
11	10	9	8	0.8182
11	10	9	9	0.1818
11	10	10	9	0.9091
11	10	10	10	0.0909
12	1	1	0	0.9167
12	1	1	1	0.0833
12	2	1	0	0.8333
12	2	1	1	0.1667
12	2	2	0	0.6818
12	2	2	1	0.3030
12	2	2	2	0.0152
12	3	1	0	0.7500
12	3	1	1	0.2500
12	3	2	0	0.5455
12	3	2	1	0.4091
12	3	2	2	0.0455
12	3	3	0	0.3818
12	3	3	1	0.4909
12	3	3	2	0.1227
12	3	3	3	0.0045
12	4	1	0	0.6667
12	4	1	1	0.3333
12	4	2	0	0.4242
12	4	2	1	0.4848
12	4	2	2	0.0909
12	4	3	0	0.2545
12	4	3	1	0.5091
12	4	3	2	0.2182
12	4	3	3	0.0182
12	4	4	0	0.1414
12	4	4	1	0.4525
12	4	4	2	0.3394
12	4	4	3	0.0646
12	4	4	4	0.0020
12	5	1	0	0.5833
12	5	1	1	0.4167
12	5	2	0	0.3182
12	5	2	1	0.5303
12	5	2	2	0.1515

N	n	M	x	f_H
12	5	3	0	0.1591
12	5	3	1	0.4773
12	5	3	2	0.3182
12	5	3	3	0.0455
12	5	4	0	0.0707
12	5	4	1	0.3535
12	5	4	2	0.4242
12	5	4	3	0.1414
12	5	4	4	0.0101
12	5	5	0	0.0265
12	5	5	1	0.2210
12	5	5	2	0.4419
12	5	5	3	0.2652
12	5	5	4	0.0442
12	5	5	5	0.0013
12	6	1	0	0.5000
12	6	1	1	0.5000
12	6	2	0	0.2273
12	6	2	1	0.5455
12	6	2	2	0.2273
12	6	3	0	0.0909
12	6	3	1	0.4091
12	6	3	2	0.4091
12	6	3	3	0.0909
12	6	4	0	0.0303
12	6	4	1	0.2424
12	6	4	2	0.4545
12	6	4	3	0.2424
12	6	4	4	0.0303
12	6	5	0	0.0076
12	6	5	1	0.1136
12	6	5	2	0.3788
12	6	5	3	0.3788
12	6	5	4	0.1136
12	6	5	5	0.0076
12	6	6	0	0.0011
12	6	6	1	0.0390
12	6	6	2	0.2435
12	6	6	3	0.4329
12	6	6	4	0.2435
12	6	6	5	0.0390
12	6	6	6	0.0011
12	7	1	0	0.4167
12	7	1	1	0.5833
12	7	2	0	0.1515

N	n	M	x	f_H
12	7	2	1	0.5303
12	7	2	2	0.3182
12	7	3	0	0.0455
12	7	3	1	0.3182
12	7	3	2	0.4773
12	7	3	3	0.1591
12	7	4	0	0.0101
12	7	4	1	0.1414
12	7	4	2	0.4242
12	7	4	3	0.3535
12	7	4	4	0.0707
12	7	5	0	0.0013
12	7	5	1	0.0442
12	7	5	2	0.2652
12	7	5	3	0.4419
12	7	5	4	0.2210
12	7	5	5	0.0265
12	7	6	1	0.0076
12	7	6	2	0.1136
12	7	6	3	0.3788
12	7	6	4	0.3788
12	7	6	5	0.1136
12	7	6	6	0.0076
12	7	7	2	0.0265
12	7	7	3	0.2210
12	7	7	4	0.4419
12	7	7	5	0.2652
12	7	7	6	0.0442
12	7	7	7	0.0013
12	8	1	0	0.3333
12	8	1	1	0.6667
12	8	2	0	0.0909
12	8	2	1	0.4848
12	8	2	2	0.4242
12	8	3	0	0.0182
12	8	3	1	0.2182
12	8	3	2	0.5091
12	8	3	3	0.2545
12	8	4	0	0.0020
12	8	4	1	0.0646
12	8	4	2	0.3394
12	8	4	3	0.4525
12	8	4	4	0.1414
12	8	5	1	0.0101
12	8	5	2	0.1414

Hypergeometric Distribution - Probability Mass Function

N	n	M	x	f_H
12	8	5	3	0.4242
12	8	5	4	0.3535
12	8	5	5	0.0707
12	8	6	2	0.0303
12	8	6	3	0.2424
12	8	6	4	0.4545
12	8	6	5	0.2424
12	8	6	6	0.0303
12	8	7	3	0.0707
12	8	7	4	0.3535
12	8	7	5	0.4242
12	8	7	6	0.1414
12	8	7	7	0.0101
12	8	8	4	0.1414
12	8	8	5	0.4525
12	8	8	6	0.3394
12	8	8	7	0.0646
12	8	8	8	0.0020
12	9	1	0	0.2500
12	9	1	1	0.7500
12	9	2	0	0.0455
12	9	2	1	0.4091
12	9	2	2	0.5455
12	9	3	0	0.0045
12	9	3	1	0.1227
12	9	3	2	0.4909
12	9	3	3	0.3818
12	9	4	1	0.0182
12	9	4	2	0.2182
12	9	4	3	0.5091
12	9	4	4	0.2545
12	9	5	2	0.0455
12	9	5	3	0.3182
12	9	5	4	0.4773
12	9	5	5	0.1591
12	9	6	3	0.0909
12	9	6	4	0.4091
12	9	6	5	0.4091
12	9	6	6	0.0909
12	9	7	4	0.1591
12	9	7	5	0.4773
12	9	7	6	0.3182
12	9	7	7	0.0455
12	9	8	5	0.2545
12	9	8	6	0.5091
12	9	8	7	0.2182
12	9	8	8	0.0182
12	9	9	6	0.3818
12	9	9	7	0.4909
12	9	9	8	0.1227
12	9	9	9	0.0045
12	10	1	0	0.1667
12	10	1	1	0.8333
12	10	2	0	0.0152
12	10	2	1	0.3030
12	10	2	2	0.6818
12	10	3	1	0.0455
12	10	3	2	0.4091
12	10	3	3	0.5455
12	10	4	2	0.0909
12	10	4	3	0.4848
12	10	4	4	0.4242
12	10	5	3	0.1515
12	10	5	4	0.5303
12	10	5	5	0.3182
12	10	6	4	0.2273
12	10	6	5	0.5455
12	10	6	6	0.2273
12	10	7	5	0.3182
12	10	7	6	0.5303
12	10	7	7	0.1515
12	10	8	6	0.4242
12	10	8	7	0.4848
12	10	8	8	0.0909
12	10	9	7	0.5455
12	10	9	8	0.4091
12	10	9	9	0.0455
12	10	10	8	0.6818
12	10	10	9	0.3030
12	10	10	10	0.0152
12	11	1	0	0.0833
12	11	1	1	0.9167
12	11	2	1	0.1667
12	11	2	2	0.8333
12	11	3	2	0.2500
12	11	3	3	0.7500
12	11	4	3	0.3333
12	11	4	4	0.6667
12	11	5	4	0.4167
12	11	5	5	0.5833
12	11	6	5	0.5000
12	11	6	6	0.5000
12	11	7	6	0.5833
12	11	7	7	0.4167
12	11	8	7	0.6667
12	11	8	8	0.3333
12	11	9	8	0.7500
12	11	9	9	0.2500
12	11	10	9	0.8333
12	11	10	10	0.1667
12	11	11	10	0.9167
12	11	11	11	0.0833

Hypergeometric Distribution - Cumulative Distribution Function

$$F_H(x \mid N;\, n;\, M) = \sum_{v=0}^{x} \frac{\binom{M}{v}\binom{N-M}{n-v}}{\binom{N}{n}} \qquad \text{for} \quad x = 0, 1, ..., n$$

For $M > n$ the following applies: $F_H\,(x \mid N;\, n;\, M) = F_H\,(x \mid N;\, M;\, n)$

N	n	M	x	F_H
2	1	1	0	0.5000
2	1	1	1	1.0000
3	1	1	0	0.6667
3	1	1	1	1.0000
3	2	1	0	0.3333
3	2	1	1	1.0000
3	2	2	1	0.6667
3	2	2	2	1.0000
4	1	1	0	0.7500
4	1	1	1	1.0000
4	2	1	0	0.5000
4	2	1	1	1.0000
4	2	2	0	0.1667
4	2	2	1	0.8333
4	2	2	2	1.0000
4	3	1	0	0.2500
4	3	1	1	1.0000
4	3	2	1	0.5000
4	3	2	2	1.0000
4	3	3	2	0.7500
4	3	3	3	1.0000
5	1	1	0	0.8000
5	1	1	1	1.0000
5	2	1	0	0.6000
5	2	1	1	1.0000
5	2	2	0	0.3000
5	2	2	1	0.9000
5	2	2	2	1.0000
5	3	1	0	0.4000
5	3	1	1	1.0000

N	n	M	x	F_H
5	3	2	0	0.1000
5	3	2	1	0.7000
5	3	2	2	1.0000
5	3	3	1	0.3000
5	3	3	2	0.9000
5	3	3	3	1.0000
5	4	1	0	0.2000
5	4	1	1	1.0000
5	4	2	1	0.4000
5	4	2	2	1.0000
5	4	3	2	0.6000
5	4	3	3	1.0000
5	4	4	3	0.8000
5	4	4	4	1.0000
6	1	1	0	0.8333
6	1	1	1	1.0000
6	2	1	0	0.6667
6	2	1	1	1.0000
6	2	2	0	0.4000
6	2	2	1	0.9333
6	2	2	2	1.0000
6	3	1	0	0.5000
6	3	1	1	1.0000
6	3	2	0	0.2000
6	3	2	1	0.8000
6	3	2	2	1.0000
6	3	3	0	0.0500
6	3	3	1	0.5000
6	3	3	2	0.9500
6	3	3	3	1.0000

N	n	M	x	F_H
6	4	1	0	0.3333
6	4	1	1	1.0000
6	4	2	0	0.0667
6	4	2	1	0.6000
6	4	2	2	1.0000
6	4	3	1	0.2000
6	4	3	2	0.8000
6	4	3	3	1.0000
6	4	4	2	0.4000
6	4	4	3	0.9333
6	4	4	4	1.0000
6	5	1	0	0.1667
6	5	1	1	1.0000
6	5	2	1	0.3333
6	5	2	2	1.0000
6	5	3	2	0.5000
6	5	3	3	1.0000
6	5	4	3	0.6667
6	5	4	4	1.0000
6	5	5	4	0.8333
6	5	5	5	1.0000
7	1	1	0	0.8571
7	1	1	1	1.0000
7	2	1	0	0.7143
7	2	1	1	1.0000
7	2	2	0	0.4762
7	2	2	1	0.9524
7	2	2	2	1.0000
7	3	1	0	0.5714
7	3	1	1	1.0000

Hypergeometric Distribution - Cumulative Distribution Function

N	n	M	x	F_H
7	3	2	0	0.2857
7	3	2	1	0.8571
7	3	2	2	1.0000
7	3	3	0	0.1143
7	3	3	1	0.6286
7	3	3	2	0.9714
7	3	3	3	1.0000
7	4	1	0	0.4286
7	4	1	1	1.0000
7	4	2	0	0.1429
7	4	2	1	0.7143
7	4	2	2	1.0000
7	4	3	0	0.0286
7	4	3	1	0.3714
7	4	3	2	0.8857
7	4	3	3	1.0000
7	4	4	1	0.1143
7	4	4	2	0.6286
7	4	4	3	0.9714
7	4	4	4	1.0000
7	5	1	0	0.2857
7	5	1	1	1.0000
7	5	2	0	0.0476
7	5	2	1	0.5238
7	5	2	2	1.0000
7	5	3	1	0.1429
7	5	3	2	0.7143
7	5	3	3	1.0000
7	5	4	2	0.2857
7	5	4	3	0.8571
7	5	4	4	1.0000
7	5	5	3	0.4762
7	5	5	4	0.9524
7	5	5	5	1.0000
7	6	1	0	0.1429
7	6	1	1	1.0000
7	6	2	1	0.2857
7	6	2	2	1.0000
7	6	3	2	0.4286
7	6	3	3	1.0000
7	6	4	3	0.5714
7	6	4	4	1.0000
7	6	5	4	0.7143
7	6	5	5	1.0000
7	6	6	5	0.8571
7	6	6	6	1.0000
8	1	1	0	0.8750
8	1	1	1	1.0000
8	2	1	0	0.7500
8	2	1	1	1.0000
8	2	2	0	0.5357
8	2	2	1	0.9643
8	2	2	2	1.0000
8	3	1	0	0.6250
8	3	1	1	1.0000
8	3	2	0	0.3571
8	3	2	1	0.8929
8	3	2	2	1.0000
8	3	3	0	0.1786
8	3	3	1	0.7143
8	3	3	2	0.9821
8	3	3	3	1.0000
8	4	1	0	0.5000
8	4	1	1	1.0000
8	4	2	0	0.2143
8	4	2	1	0.7857
8	4	2	2	1.0000
8	4	3	0	0.0714
8	4	3	1	0.5000
8	4	3	2	0.9286
8	4	3	3	1.0000
8	4	4	0	0.0143
8	4	4	1	0.2429
8	4	4	2	0.7571
8	4	4	3	0.9857
8	4	4	4	1.0000
8	5	1	0	0.3750
8	5	1	1	1.0000
8	5	2	0	0.1071
8	5	2	1	0.6429
8	5	2	2	1.0000
8	5	3	0	0.0179
8	5	3	1	0.2857
8	5	3	2	0.8214
8	5	3	3	1.0000
8	5	4	1	0.0714
8	5	4	2	0.5000
8	5	4	3	0.9286
8	5	4	4	1.0000
8	5	5	2	0.1786
8	5	5	3	0.7143
8	5	5	4	0.9821
8	5	5	5	1.0000
8	6	1	0	0.2500
8	6	1	1	1.0000
8	6	2	0	0.0357
8	6	2	1	0.4643
8	6	2	2	1.0000
8	6	3	1	0.1071
8	6	3	2	0.6429
8	6	3	3	1.0000
8	6	4	2	0.2143
8	6	4	3	0.7857
8	6	4	4	1.0000
8	6	5	3	0.3571
8	6	5	4	0.8929
8	6	5	5	1.0000
8	6	6	4	0.5357
8	6	6	5	0.9643
8	6	6	6	1.0000
8	7	1	0	0.1250
8	7	1	1	1.0000
8	7	2	1	0.2500
8	7	2	2	1.0000
8	7	3	2	0.3750
8	7	3	3	1.0000
8	7	4	3	0.5000
8	7	4	4	1.0000
8	7	5	4	0.6250
8	7	5	5	1.0000
8	7	6	5	0.7500
8	7	6	6	1.0000
8	7	7	6	0.8750
8	7	7	7	1.0000
9	1	1	0	0.8889
9	1	1	1	1.0000
9	2	1	0	0.7778
9	2	1	1	1.0000
9	2	2	0	0.5833
9	2	2	1	0.9722
9	2	2	2	1.0000
9	3	1	0	0.6667
9	3	1	1	1.0000
9	3	2	0	0.4167
9	3	2	1	0.9167

Hypergeometric Distribution - Cumulative Distribution Function

N	n	M	x	F_H
9	3	2	2	1.0000
9	3	3	0	0.2381
9	3	3	1	0.7738
9	3	3	2	0.9881
9	3	3	3	1.0000
9	4	1	0	0.5556
9	4	1	1	1.0000
9	4	2	0	0.2778
9	4	2	1	0.8333
9	4	2	2	1.0000
9	4	3	0	0.1190
9	4	3	1	0.5952
9	4	3	2	0.9524
9	4	3	3	1.0000
9	4	4	0	0.0397
9	4	4	1	0.3571
9	4	4	2	0.8333
9	4	4	3	0.9921
9	4	4	4	1.0000
9	5	1	0	0.4444
9	5	1	1	1.0000
9	5	2	0	0.1667
9	5	2	1	0.7222
9	5	2	2	1.0000
9	5	3	0	0.0476
9	5	3	1	0.4048
9	5	3	2	0.8810
9	5	3	3	1.0000
9	5	4	0	0.0079
9	5	4	1	0.1667
9	5	4	2	0.6429
9	5	4	3	0.9603
9	5	4	4	1.0000
9	5	5	1	0.0397
9	5	5	2	0.3571
9	5	5	3	0.8333
9	5	5	4	0.9921
9	5	5	5	1.0000
9	6	1	0	0.3333
9	6	1	1	1.0000
9	6	2	0	0.0833
9	6	2	1	0.5833
9	6	2	2	1.0000
9	6	3	0	0.0119
9	6	3	1	0.2262
9	6	3	2	0.7619
9	6	3	3	1.0000
9	6	4	1	0.0476
9	6	4	2	0.4048
9	6	4	3	0.8810
9	6	4	4	1.0000
9	6	5	2	0.1190
9	6	5	3	0.5952
9	6	5	4	0.9524
9	6	5	5	1.0000
9	6	6	3	0.2381
9	6	6	4	0.7738
9	6	6	5	0.9881
9	6	6	6	1.0000
9	7	1	0	0.2222
9	7	1	1	1.0000
9	7	2	0	0.0278
9	7	2	1	0.4167
9	7	2	2	1.0000
9	7	3	1	0.0833
9	7	3	2	0.5833
9	7	3	3	1.0000
9	7	4	2	0.1667
9	7	4	3	0.7222
9	7	4	4	1.0000
9	7	5	3	0.2778
9	7	5	4	0.8333
9	7	5	5	1.0000
9	7	6	4	0.4167
9	7	6	5	0.9167
9	7	6	6	1.0000
9	7	7	5	0.5833
9	7	7	6	0.9722
9	7	7	7	1.0000
9	8	1	0	0.1111
9	8	1	1	1.0000
9	8	2	1	0.2222
9	8	2	2	1.0000
9	8	3	2	0.3333
9	8	3	3	1.0000
9	8	4	3	0.4444
9	8	4	4	1.0000
9	8	5	4	0.5556
9	8	5	5	1.0000
9	8	6	5	0.6667
9	8	6	6	1.0000
9	8	7	6	0.7778
9	8	7	7	1.0000
9	8	8	7	0.8889
9	8	8	8	1.0000
10	1	1	0	0.9000
10	1	1	1	1.0000
10	2	1	0	0.8000
10	2	1	1	1.0000
10	2	2	0	0.6222
10	2	2	1	0.9778
10	2	2	2	1.0000
10	3	1	0	0.7000
10	3	1	1	1.0000
10	3	2	0	0.4667
10	3	2	1	0.9333
10	3	2	2	1.0000
10	3	3	0	0.2917
10	3	3	1	0.8167
10	3	3	2	0.9917
10	3	3	3	1.0000
10	4	1	0	0.6000
10	4	1	1	1.0000
10	4	2	0	0.3333
10	4	2	1	0.8667
10	4	2	2	1.0000
10	4	3	0	0.1667
10	4	3	1	0.6667
10	4	3	2	0.9667
10	4	3	3	1.0000
10	4	4	0	0.0714
10	4	4	1	0.4524
10	4	4	2	0.8810
10	4	4	3	0.9952
10	4	4	4	1.0000
10	5	1	0	0.5000
10	5	1	1	1.0000
10	5	2	0	0.2222
10	5	2	1	0.7778
10	5	2	2	1.0000
10	5	3	0	0.0833
10	5	3	1	0.5000
10	5	3	2	0.9167
10	5	3	3	1.0000
10	5	4	0	0.0238

Hypergeometric Distribution - Cumulative Distribution Function

N	n	M	x	F_H
10	5	4	1	0.2619
10	5	4	2	0.7381
10	5	4	3	0.9762
10	5	4	4	1.0000
10	5	5	0	0.0040
10	5	5	1	0.1032
10	5	5	2	0.5000
10	5	5	3	0.8968
10	5	5	4	0.9960
10	5	5	5	1.0000
10	6	1	0	0.4000
10	6	1	1	1.0000
10	6	2	0	0.1333
10	6	2	1	0.6667
10	6	2	2	1.0000
10	6	3	0	0.0333
10	6	3	1	0.3333
10	6	3	2	0.8333
10	6	3	3	1.0000
10	6	4	0	0.0048
10	6	4	1	0.1190
10	6	4	2	0.5476
10	6	4	3	0.9286
10	6	4	4	1.0000
10	6	5	1	0.0238
10	6	5	2	0.2619
10	6	5	3	0.7381
10	6	5	4	0.9762
10	6	5	5	1.0000
10	6	6	2	0.0714
10	6	6	3	0.4524
10	6	6	4	0.8810
10	6	6	5	0.9952
10	6	6	6	1.0000
10	7	1	0	0.3000
10	7	1	1	1.0000
10	7	2	0	0.0667
10	7	2	1	0.5333
10	7	2	2	1.0000
10	7	3	0	0.0083
10	7	3	1	0.1833
10	7	3	2	0.7083
10	7	3	3	1.0000
10	7	4	1	0.0333
10	7	4	2	0.3333

N	n	M	x	F_H
10	7	4	3	0.8333
10	7	4	4	1.0000
10	7	5	2	0.0833
10	7	5	3	0.5000
10	7	5	4	0.9167
10	7	5	5	1.0000
10	7	6	3	0.1667
10	7	6	4	0.6667
10	7	6	5	0.9667
10	7	6	6	1.0000
10	7	7	4	0.2917
10	7	7	5	0.8167
10	7	7	6	0.9917
10	7	7	7	1.0000
10	8	1	0	0.2000
10	8	1	1	1.0000
10	8	2	0	0.0222
10	8	2	1	0.3778
10	8	2	2	1.0000
10	8	3	1	0.0667
10	8	3	2	0.5333
10	8	3	3	1.0000
10	8	4	2	0.1333
10	8	4	3	0.6667
10	8	4	4	1.0000
10	8	5	3	0.2222
10	8	5	4	0.7778
10	8	5	5	1.0000
10	8	6	4	0.3333
10	8	6	5	0.8667
10	8	6	6	1.0000
10	8	7	5	0.4667
10	8	7	6	0.9333
10	8	7	7	1.0000
10	8	8	6	0.6222
10	8	8	7	0.9778
10	8	8	8	1.0000
10	9	1	0	0.1000
10	9	1	1	1.0000
10	9	2	1	0.2000
10	9	2	2	1.0000
10	9	3	2	0.3000
10	9	3	3	1.0000
10	9	4	3	0.4000
10	9	4	4	1.0000

N	n	M	x	F_H
10	9	5	4	0.5000
10	9	5	5	1.0000
10	9	6	5	0.6000
10	9	6	6	1.0000
10	9	7	6	0.7000
10	9	7	7	1.0000
10	9	8	7	0.8000
10	9	8	8	1.0000
10	9	9	8	0.9000
10	9	9	9	1.0000
11	1	1	0	0.9091
11	1	1	1	1.0000
11	2	1	0	0.8182
11	2	1	1	1.0000
11	2	2	0	0.6545
11	2	2	1	0.9818
11	2	2	2	1.0000
11	3	1	0	0.7273
11	3	1	1	1.0000
11	3	2	0	0.5091
11	3	2	1	0.9455
11	3	2	2	1.0000
11	3	3	0	0.3394
11	3	3	1	0.8485
11	3	3	2	0.9939
11	3	3	3	1.0000
11	4	1	0	0.6364
11	4	1	1	1.0000
11	4	2	0	0.3818
11	4	2	1	0.8909
11	4	2	2	1.0000
11	4	3	0	0.2121
11	4	3	1	0.7212
11	4	3	2	0.9758
11	4	3	3	1.0000
11	4	4	0	0.1061
11	4	4	1	0.5303
11	4	4	2	0.9121
11	4	4	3	0.9970
11	4	4	4	1.0000
11	5	1	0	0.5455
11	5	1	1	1.0000
11	5	2	0	0.2727
11	5	2	1	0.8182
11	5	2	2	1.0000

Hypergeometric Distribution - Cumulative Distribution Function

N	n	M	x	F_H
11	5	3	0	0.1212
11	5	3	1	0.5758
11	5	3	2	0.9394
11	5	3	3	1.0000
11	5	4	0	0.0455
11	5	4	1	0.3485
11	5	4	2	0.8030
11	5	4	3	0.9848
11	5	4	4	1.0000
11	5	5	0	0.0130
11	5	5	1	0.1753
11	5	5	2	0.6082
11	5	5	3	0.9329
11	5	5	4	0.9978
11	5	5	5	1.0000
11	6	1	0	0.4545
11	6	1	1	1.0000
11	6	2	0	0.1818
11	6	2	1	0.7273
11	6	2	2	1.0000
11	6	3	0	0.0606
11	6	3	1	0.4242
11	6	3	2	0.8788
11	6	3	3	1.0000
11	6	4	0	0.0152
11	6	4	1	0.1970
11	6	4	2	0.6515
11	6	4	3	0.9545
11	6	4	4	1.0000
11	6	5	0	0.0022
11	6	5	1	0.0671
11	6	5	2	0.3918
11	6	5	3	0.8247
11	6	5	4	0.9870
11	6	5	5	1.0000
11	6	6	1	0.0130
11	6	6	2	0.1753
11	6	6	3	0.6082
11	6	6	4	0.9329
11	6	6	5	0.9978
11	6	6	6	1.0000
11	7	1	0	0.3636
11	7	1	1	1.0000
11	7	2	0	0.1091
11	7	2	1	0.6182
11	7	2	2	1.0000
11	7	3	0	0.0242
11	7	3	1	0.2788
11	7	3	2	0.7879
11	7	3	3	1.0000
11	7	4	0	0.0030
11	7	4	1	0.0879
11	7	4	2	0.4697
11	7	4	3	0.8939
11	7	4	4	1.0000
11	7	5	1	0.0152
11	7	5	2	0.1970
11	7	5	3	0.6515
11	7	5	4	0.9545
11	7	5	5	1.0000
11	7	6	2	0.0455
11	7	6	3	0.3485
11	7	6	4	0.8030
11	7	6	5	0.9848
11	7	6	6	1.0000
11	7	7	3	0.1061
11	7	7	4	0.5303
11	7	7	5	0.9121
11	7	7	6	0.9970
11	7	7	7	1.0000
11	8	1	0	0.2727
11	8	1	1	1.0000
11	8	2	0	0.0545
11	8	2	1	0.4909
11	8	2	2	1.0000
11	8	3	0	0.0061
11	8	3	1	0.1515
11	8	3	2	0.6606
11	8	3	3	1.0000
11	8	4	1	0.0242
11	8	4	2	0.2788
11	8	4	3	0.7879
11	8	4	4	1.0000
11	8	5	2	0.0606
11	8	5	3	0.4242
11	8	5	4	0.8788
11	8	5	5	1.0000
11	8	6	3	0.1212
11	8	6	4	0.5758
11	8	6	5	0.9394
11	8	6	6	1.0000
11	8	7	4	0.2121
11	8	7	5	0.7212
11	8	7	6	0.9758
11	8	7	7	1.0000
11	8	8	5	0.3394
11	8	8	6	0.8485
11	8	8	7	0.9939
11	8	8	8	1.0000
11	9	1	0	0.1818
11	9	1	1	1.0000
11	9	2	0	0.0182
11	9	2	1	0.3455
11	9	2	2	1.0000
11	9	3	1	0.0545
11	9	3	2	0.4909
11	9	3	3	1.0000
11	9	4	2	0.1091
11	9	4	3	0.6182
11	9	4	4	1.0000
11	9	5	3	0.1818
11	9	5	4	0.7273
11	9	5	5	1.0000
11	9	6	4	0.2727
11	9	6	5	0.8182
11	9	6	6	1.0000
11	9	7	5	0.3818
11	9	7	6	0.8909
11	9	7	7	1.0000
11	9	8	6	0.5091
11	9	8	7	0.9455
11	9	8	8	1.0000
11	9	9	7	0.6545
11	9	9	8	0.9818
11	9	9	9	1.0000
11	10	1	0	0.0909
11	10	1	1	1.0000
11	10	2	1	0.1818
11	10	2	2	1.0000
11	10	3	2	0.2727
11	10	3	3	1.0000
11	10	4	3	0.3636
11	10	4	4	1.0000
11	10	5	4	0.4545
11	10	5	5	1.0000

Hypergeometric Distribution - Cumulative Distribution Function

N	n	M	x	F_H
11	10	6	5	0.5455
11	10	6	6	1.0000
11	10	7	6	0.6364
11	10	7	7	1.0000
11	10	8	7	0.7273
11	10	8	8	1.0000
11	10	9	8	0.8182
11	10	9	9	1.0000
11	10	10	9	0.9091
11	10	10	10	1.0000
12	1	1	0	0.9167
12	1	1	1	1.0000
12	2	1	0	0.8333
12	2	1	1	1.0000
12	2	2	0	0.6818
12	2	2	1	0.9848
12	2	2	2	1.0000
12	3	1	0	0.7500
12	3	1	1	1.0000
12	3	2	0	0.5455
12	3	2	1	0.9545
12	3	2	2	1.0000
12	3	3	0	0.3818
12	3	3	1	0.8727
12	3	3	2	0.9955
12	3	3	3	1.0000
12	4	1	0	0.6667
12	4	1	1	1.0000
12	4	2	0	0.4242
12	4	2	1	0.9091
12	4	2	2	1.0000
12	4	3	0	0.2545
12	4	3	1	0.7636
12	4	3	2	0.9818
12	4	3	3	1.0000
12	4	4	0	0.1414
12	4	4	1	0.5939
12	4	4	2	0.9333
12	4	4	3	0.9980
12	4	4	4	1.0000
12	5	1	0	0.5833
12	5	1	1	1.0000
12	5	2	0	0.3182
12	5	2	1	0.8485
12	5	2	2	1.0000
12	5	3	0	0.1591
12	5	3	1	0.6364
12	5	3	2	0.9545
12	5	3	3	1.0000
12	5	4	0	0.0707
12	5	4	1	0.4242
12	5	4	2	0.8485
12	5	4	3	0.9899
12	5	4	4	1.0000
12	5	5	0	0.0265
12	5	5	1	0.2475
12	5	5	2	0.6894
12	5	5	3	0.9545
12	5	5	4	0.9987
12	5	5	5	1.0000
12	6	1	0	0.5000
12	6	1	1	1.0000
12	6	2	0	0.2273
12	6	2	1	0.7727
12	6	2	2	1.0000
12	6	3	0	0.0909
12	6	3	1	0.5000
12	6	3	2	0.9091
12	6	3	3	1.0000
12	6	4	0	0.0303
12	6	4	1	0.2727
12	6	4	2	0.7273
12	6	4	3	0.9697
12	6	4	4	1.0000
12	6	5	0	0.0076
12	6	5	1	0.1212
12	6	5	2	0.5000
12	6	5	3	0.8788
12	6	5	4	0.9924
12	6	5	5	1.0000
12	6	6	0	0.0011
12	6	6	1	0.0400
12	6	6	2	0.2835
12	6	6	3	0.7165
12	6	6	4	0.9600
12	6	6	5	0.9989
12	6	6	6	1.0000
12	7	1	0	0.4167
12	7	1	1	1.0000
12	7	2	0	0.1515
12	7	2	1	0.6818
12	7	2	2	1.0000
12	7	3	0	0.0455
12	7	3	1	0.3636
12	7	3	2	0.8409
12	7	3	3	1.0000
12	7	4	0	0.0101
12	7	4	1	0.1515
12	7	4	2	0.5758
12	7	4	3	0.9293
12	7	4	4	1.0000
12	7	5	0	0.0013
12	7	5	1	0.0455
12	7	5	2	0.3106
12	7	5	3	0.7525
12	7	5	4	0.9735
12	7	5	5	1.0000
12	7	6	1	0.0076
12	7	6	2	0.1212
12	7	6	3	0.5000
12	7	6	4	0.8788
12	7	6	5	0.9924
12	7	6	6	1.0000
12	7	7	2	0.0265
12	7	7	3	0.2475
12	7	7	4	0.6894
12	7	7	5	0.9545
12	7	7	6	0.9987
12	7	7	7	1.0000
12	8	1	0	0.3333
12	8	1	1	1.0000
12	8	2	0	0.0909
12	8	2	1	0.5758
12	8	2	2	1.0000
12	8	3	0	0.0182
12	8	3	1	0.2364
12	8	3	2	0.7455
12	8	3	3	1.0000
12	8	4	0	0.0020
12	8	4	1	0.0667
12	8	4	2	0.4061
12	8	4	3	0.8586
12	8	4	4	1.0000
12	8	5	1	0.0101
12	8	5	2	0.1515

Hypergeometric Distribution - Cumulative Distribution Function

N	n	M	x	F_H
12	8	5	3	0.5758
12	8	5	4	0.9293
12	8	5	5	1.0000
12	8	6	2	0.0303
12	8	6	3	0.2727
12	8	6	4	0.7273
12	8	6	5	0.9697
12	8	6	6	1.0000
12	8	7	3	0.0707
12	8	7	4	0.4242
12	8	7	5	0.8485
12	8	7	6	0.9899
12	8	7	7	1.0000
12	8	8	4	0.1414
12	8	8	5	0.5939
12	8	8	6	0.9333
12	8	8	7	0.9980
12	8	8	8	1.0000
12	9	1	0	0.2500
12	9	1	1	1.0000
12	9	2	0	0.0455
12	9	2	1	0.4545
12	9	2	2	1.0000
12	9	3	0	0.0045
12	9	3	1	0.1273
12	9	3	2	0.6182
12	9	3	3	1.0000
12	9	4	1	0.0182
12	9	4	2	0.2364
12	9	4	3	0.7455
12	9	4	4	1.0000
12	9	5	2	0.0455
12	9	5	3	0.3636
12	9	5	4	0.8409
12	9	5	5	1.0000
12	9	6	3	0.0909
12	9	6	4	0.5000
12	9	6	5	0.9091
12	9	6	6	1.0000
12	9	7	4	0.1591
12	9	7	5	0.6364
12	9	7	6	0.9545
12	9	7	7	1.0000
12	9	8	5	0.2545
12	9	8	6	0.7636
12	9	8	7	0.9818
12	9	8	8	1.0000
12	9	9	6	0.3818
12	9	9	7	0.8727
12	9	9	8	0.9955
12	9	9	9	1.0000
12	10	1	0	0.1667
12	10	1	1	1.0000
12	10	2	0	0.0152
12	10	2	1	0.3182
12	10	2	2	1.0000
12	10	3	1	0.0455
12	10	3	2	0.4545
12	10	3	3	1.0000
12	10	4	2	0.0909
12	10	4	3	0.5758
12	10	4	4	1.0000
12	10	5	3	0.1515
12	10	5	4	0.6818
12	10	5	5	1.0000
12	10	6	4	0.2273
12	10	6	5	0.7727
12	10	6	6	1.0000
12	10	7	5	0.3182
12	10	7	6	0.8485
12	10	7	7	1.0000
12	10	8	6	0.4242
12	10	8	7	0.9091
12	10	8	8	1.0000
12	10	9	7	0.5455
12	10	9	8	0.9545
12	10	9	9	1.0000
12	10	10	8	0.6818
12	10	10	9	0.9848
12	10	10	10	1.0000
12	11	1	0	0.0833
12	11	1	1	1.0000
12	11	2	1	0.1667
12	11	2	2	1.0000
12	11	3	2	0.2500
12	11	3	3	1.0000
12	11	4	3	0.3333
12	11	4	4	1.0000
12	11	5	4	0.4167
12	11	5	5	1.0000
12	11	6	5	0.5000
12	11	6	6	1.0000
12	11	7	6	0.5833
12	11	7	7	1.0000
12	11	8	7	0.6667
12	11	8	8	1.0000
12	11	9	8	0.7500
12	11	9	9	1.0000
12	11	10	9	0.8333
12	11	10	10	1.0000
12	11	11	10	0.9167
12	11	11	11	1.0000

Poisson Distribution - Probability Mass Function

$$f_P(x\,|\,\mu) = \begin{cases} \dfrac{\mu^x e^{-\mu}}{x!} & \text{for } x = 0,\ 1,\ ... \quad \text{with } \mu > 0 \qquad e = 2.71828... \\ 0 & \text{others} \end{cases}$$

x	μ								
	0.001	**0.005**	**0.010**	**0.015**	**0.020**	**0.025**	**0.030**	**0.035**	**0.040**
0	0.9990	0.9950	0.9900	0.9851	0.9802	0.9753	0.9704	0.9656	0.9608
1	0.0010	0.0050	0.0099	0.0148	0.0196	0.0244	0.0291	0.0338	0.0384
2	0.0000	0.0000	0.0000	0.0001	0.0002	0.0003	0.0004	0.0006	0.0008

x	μ								
	0.045	**0.050**	**0.055**	**0.060**	**0.065**	**0.070**	**0.075**	**0.080**	**0.085**
0	0.9560	0.9512	0.9465	0.9418	0.9371	0.9324	0.9277	0.9231	0.9185
1	0.0430	0.0476	0.0521	0.0565	0.0609	0.0653	0.0696	0.0738	0.0781
2	0.0010	0.0012	0.0014	0.0017	0.0020	0.0023	0.0026	0.0030	0.0033
3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001	0.0001

x	μ								
	0.090	**0.100**	**0.150**	**0.200**	**0.300**	**0.400**	**0.500**	**0.600**	**0.700**
0	0.9139	0.9048	0.8607	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966
1	0.0823	0.0905	0.1291	0.1637	0.2222	0.2681	0.3033	0.3293	0.3476
2	0.0037	0.0045	0.0097	0.0164	0.0333	0.0536	0.0758	0.0988	0.1217
3	0.0001	0.0002	0.0005	0.0011	0.0033	0.0072	0.0126	0.0198	0.0284
4	0.0000	0.0000	0.0000	0.0001	0.0003	0.0007	0.0016	0.0030	0.0050
5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0004	0.0007
6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001

x	μ								
	0.800	**0.900**	**1.000**	**1.100**	**1.200**	**1.300**	**1.400**	**1.500**	**1.600**
0	0.4493	0.4066	0.3679	0.3329	0.3012	0.2725	0.2466	0.2231	0.2019
1	0.3595	0.3659	0.3679	0.3662	0.3614	0.3543	0.3452	0.3347	0.3230
2	0.1438	0.1647	0.1839	0.2014	0.2169	0.2303	0.2417	0.2510	0.2584
3	0.0383	0.0494	0.0613	0.0738	0.0867	0.0998	0.1128	0.1255	0.1378
4	0.0077	0.0111	0.0153	0.0203	0.0260	0.0324	0.0395	0.0471	0.0551
5	0.0012	0.0020	0.0031	0.0045	0.0062	0.0084	0.0111	0.0141	0.0176
6	0.0002	0.0003	0.0005	0.0008	0.0012	0.0018	0.0026	0.0035	0.0047
7	0.0000	0.0000	0.0001	0.0001	0.0002	0.0003	0.0005	0.0008	0.0011
8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001	0.0002

Poisson Distribution - Probability Mass Function

x	μ								
	1.700	**1.800**	**1.900**	**2.000**	**2.100**	**2.200**	**2.300**	**2.400**	**2.500**
0	0.1827	0.1653	0.1496	0.1353	0.1225	0.1108	0.1003	0.0907	0.0821
1	0.3106	0.2975	0.2842	0.2707	0.2572	0.2438	0.2306	0.2177	0.2052
2	0.2640	0.2678	0.2700	0.2707	0.2700	0.2681	0.2652	0.2613	0.2565
3	0.1496	0.1607	0.1710	0.1804	0.1890	0.1966	0.2033	0.2090	0.2138
4	0.0636	0.0723	0.0812	0.0902	0.0992	0.1082	0.1169	0.1254	0.1336
5	0.0216	0.0260	0.0309	0.0361	0.0417	0.0476	0.0538	0.0602	0.0668
6	0.0061	0.0078	0.0098	0.0120	0.0146	0.0174	0.0206	0.0241	0.0278
7	0.0015	0.0020	0.0027	0.0034	0.0044	0.0055	0.0068	0.0083	0.0099
8	0.0003	0.0005	0.0006	0.0009	0.0011	0.0015	0.0019	0.0025	0.0031
9	0.0001	0.0001	0.0001	0.0002	0.0003	0.0004	0.0005	0.0007	0.0009
10	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001	0.0002	0.0002

x	μ								
	2.600	**2.700**	**2.800**	**2.900**	**3.000**	**3.100**	**3.200**	**3.300**	**3.400**
0	0.0743	0.0672	0.0608	0.0550	0.0498	0.0450	0.0408	0.0369	0.0334
1	0.1931	0.1815	0.1703	0.1596	0.1494	0.1397	0.1304	0.1217	0.1135
2	0.2510	0.2450	0.2384	0.2314	0.2240	0.2165	0.2087	0.2008	0.1929
3	0.2176	0.2205	0.2225	0.2237	0.2240	0.2237	0.2226	0.2209	0.2186
4	0.1414	0.1488	0.1557	0.1622	0.1680	0.1733	0.1781	0.1823	0.1858
5	0.0735	0.0804	0.0872	0.0940	0.1008	0.1075	0.1140	0.1203	0.1264
6	0.0319	0.0362	0.0407	0.0455	0.0504	0.0555	0.0608	0.0662	0.0716
7	0.0118	0.0139	0.0163	0.0188	0.0216	0.0246	0.0278	0.0312	0.0348
8	0.0038	0.0047	0.0057	0.0068	0.0081	0.0095	0.0111	0.0129	0.0148
9	0.0011	0.0014	0.0018	0.0022	0.0027	0.0033	0.0040	0.0047	0.0056
10	0.0003	0.0004	0.0005	0.0006	0.0008	0.0010	0.0013	0.0016	0.0019
11	0.0001	0.0001	0.0001	0.0002	0.0002	0.0003	0.0004	0.0005	0.0006
12	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001	0.0001	0.0002

x	μ								
	3.500	**3.600**	**3.700**	**3.800**	**3.900**	**4.000**	**4.500**	**5.000**	**5.500**
0	0.0302	0.0273	0.0247	0.0224	0.0202	0.0183	0.0111	0.0067	0.0041
1	0.1057	0.0984	0.0915	0.0850	0.0789	0.0733	0.0500	0.0337	0.0225
2	0.1850	0.1771	0.1692	0.1615	0.1539	0.1465	0.1125	0.0842	0.0618
3	0.2158	0.2125	0.2087	0.2046	0.2001	0.1954	0.1687	0.1404	0.1133
4	0.1888	0.1912	0.1931	0.1944	0.1951	0.1954	0.1898	0.1755	0.1558
5	0.1322	0.1377	0.1429	0.1477	0.1522	0.1563	0.1708	0.1755	0.1714
6	0.0771	0.0826	0.0881	0.0936	0.0989	0.1042	0.1281	0.1462	0.1571
7	0.0385	0.0425	0.0466	0.0508	0.0551	0.0595	0.0824	0.1044	0.1234

Poisson Distribution - Probability Mass Function

x	μ								
	3.500	**3.600**	**3.700**	**3.800**	**3.900**	**4.000**	**4.500**	**5.000**	**5.500**
8	0.0169	0.0191	0.0215	0.0241	0.0269	0.0298	0.0463	0.0653	0.0849
9	0.0066	0.0076	0.0089	0.0102	0.0116	0.0132	0.0232	0.0363	0.0519
10	0.0023	0.0028	0.0033	0.0039	0.0045	0.0053	0.0104	0.0181	0.0285
11	0.0007	0.0009	0.0011	0.0013	0.0016	0.0019	0.0043	0.0082	0.0143
12	0.0002	0.0003	0.0003	0.0004	0.0005	0.0006	0.0016	0.0034	0.0065
13	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002	0.0006	0.0013	0.0028
14	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0005	0.0011
15	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0004
16	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001

x	μ								
	6.000	**6.500**	**7.000**	**7.500**	**8.000**	**8.500**	**9.000**	**9.500**	**10.000**
0	0.0025	0.0015	0.0009	0.0006	0.0003	0.0002	0.0001	0.0001	0.0000
1	0.0149	0.0098	0.0064	0.0041	0.0027	0.0017	0.0011	0.0007	0.0005
2	0.0446	0.0318	0.0223	0.0156	0.0107	0.0074	0.0050	0.0034	0.0023
3	0.0892	0.0688	0.0521	0.0389	0.0286	0.0208	0.0150	0.0107	0.0076
4	0.1339	0.1118	0.0912	0.0729	0.0573	0.0443	0.0337	0.0254	0.0189
5	0.1606	0.1454	0.1277	0.1094	0.0916	0.0752	0.0607	0.0483	0.0378
6	0.1606	0.1575	0.1490	0.1367	0.1221	0.1066	0.0911	0.0764	0.0631
7	0.1377	0.1462	0.1490	0.1465	0.1396	0.1294	0.1171	0.1037	0.0901
8	0.1033	0.1188	0.1304	0.1373	0.1396	0.1375	0.1318	0.1232	0.1126
9	0.0688	0.0858	0.1014	0.1144	0.1241	0.1299	0.1318	0.1300	0.1251
10	0.0413	0.0558	0.0710	0.0858	0.0993	0.1104	0.1186	0.1235	0.1251
11	0.0225	0.0330	0.0452	0.0585	0.0722	0.0853	0.0970	0.1067	0.1137
12	0.0113	0.0179	0.0263	0.0366	0.0481	0.0604	0.0728	0.0844	0.0948
13	0.0052	0.0089	0.0142	0.0211	0.0296	0.0395	0.0504	0.0617	0.0729
14	0.0022	0.0041	0.0071	0.0113	0.0169	0.0240	0.0324	0.0419	0.0521
15	0.0009	0.0018	0.0033	0.0057	0.0090	0.0136	0.0194	0.0265	0.0347
16	0.0003	0.0007	0.0014	0.0026	0.0045	0.0072	0.0109	0.0157	0.0217
17	0.0001	0.0003	0.0006	0.0012	0.0021	0.0036	0.0058	0.0088	0.0128
18	0.0000	0.0001	0.0002	0.0005	0.0009	0.0017	0.0029	0.0046	0.0071
19	0.0000	0.0000	0.0001	0.0002	0.0004	0.0008	0.0014	0.0023	0.0037
20	0.0000	0.0000	0.0000	0.0001	0.0002	0.0003	0.0006	0.0011	0.0019
21	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001	0.0003	0.0005	0.0009
22	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001	0.0002	0.0004
23	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002
24	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001

Poisson Distribution - Cumulative Distribution Function

$$F_P(x \mid \mu) = \sum_{\nu=0}^{x} \frac{\mu^\nu e^{-\mu}}{\nu!} \qquad \text{with } e = 2.71828...$$

x	μ								
	0.001	**0.005**	**0.010**	**0.015**	**0.020**	**0.025**	**0.030**	**0.035**	**0.040**
0	0.9990	0.9950	0.9900	0.9851	0.9802	0.9753	0.9704	0.9656	0.9608
1	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9996	0.9994	0.9992
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

x	μ								
	0.045	**0.050**	**0.055**	**0.060**	**0.065**	**0.070**	**0.075**	**0.080**	**0.085**
0	0.9560	0.9512	0.9465	0.9418	0.9371	0.9324	0.9277	0.9231	0.9185
1	0.9990	0.9988	0.9985	0.9983	0.9980	0.9977	0.9973	0.9970	0.9966
2	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999
3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

x	μ								
	0.090	**0.100**	**0.150**	**0.200**	**0.300**	**0.400**	**0.500**	**0.600**	**0.700**
0	0.9139	0.9048	0.8607	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966
1	0.9962	0.9953	0.9898	0.9825	0.9631	0.9384	0.9098	0.8781	0.8442
2	0.9999	0.9998	0.9995	0.9989	0.9964	0.9921	0.9856	0.9769	0.9659
3	1.0000	1.0000	1.0000	0.9999	0.9997	0.9992	0.9982	0.9966	0.9942
4	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9992
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

x	μ								
	0.800	**0.900**	**1.000**	**1.100**	**1.200**	**1.300**	**1.400**	**1.500**	**1.600**
0	0.4493	0.4066	0.3679	0.3329	0.3012	0.2725	0.2466	0.2231	0.2019
1	0.8088	0.7725	0.7358	0.6990	0.6626	0.6268	0.5918	0.5578	0.5249
2	0.9526	0.9371	0.9197	0.9004	0.8795	0.8571	0.8335	0.8088	0.7834
3	0.9909	0.9865	0.9810	0.9743	0.9662	0.9569	0.9463	0.9344	0.9212
4	0.9986	0.9977	0.9963	0.9946	0.9923	0.9893	0.9857	0.9814	0.9763
5	0.9998	0.9997	0.9994	0.9990	0.9985	0.9978	0.9968	0.9955	0.9940
6	1.0000	1.0000	0.9999	0.9999	0.9997	0.9996	0.9994	0.9991	0.9987
7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Poisson Distribution - Cumulative Distribution Function

x	μ								
	1.700	**1.800**	**1.900**	**2.000**	**2.100**	**2.200**	**2.300**	**2.400**	**2.500**
0	0.1827	0.1653	0.1496	0.1353	0.1225	0.1108	0.1003	0.0907	0.0821
1	0.4932	0.4628	0.4337	0.4060	0.3796	0.3546	0.3309	0.3084	0.2873
2	0.7572	0.7306	0.7037	0.6767	0.6496	0.6227	0.5960	0.5697	0.5438
3	0.9068	0.8913	0.8747	0.8571	0.8386	0.8194	0.7993	0.7787	0.7576
4	0.9704	0.9636	0.9559	0.9473	0.9379	0.9275	0.9162	0.9041	0.8912
5	0.9920	0.9896	0.9868	0.9834	0.9796	0.9751	0.9700	0.9643	0.9580
6	0.9981	0.9974	0.9966	0.9955	0.9941	0.9925	0.9906	0.9884	0.9858
7	0.9996	0.9994	0.9992	0.9989	0.9985	0.9980	0.9974	0.9967	0.9958
8	0.9999	0.9999	0.9998	0.9998	0.9997	0.9995	0.9994	0.9991	0.9989
9	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9998	0.9997
10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

x	μ								
	2.600	**2.700**	**2.800**	**2.900**	**3.000**	**3.100**	**3.200**	**3.300**	**3.400**
0	0.0743	0.0672	0.0608	0.0550	0.0498	0.0450	0.0408	0.0369	0.0334
1	0.2674	0.2487	0.2311	0.2146	0.1991	0.1847	0.1712	0.1586	0.1468
2	0.5184	0.4936	0.4695	0.4460	0.4232	0.4012	0.3799	0.3594	0.3397
3	0.7360	0.7141	0.6919	0.6696	0.6472	0.6248	0.6025	0.5803	0.5584
4	0.8774	0.8629	0.8477	0.8318	0.8153	0.7982	0.7806	0.7626	0.7442
5	0.9510	0.9433	0.9349	0.9258	0.9161	0.9057	0.8946	0.8829	0.8705
6	0.9828	0.9794	0.9756	0.9713	0.9665	0.9612	0.9554	0.9490	0.9421
7	0.9947	0.9934	0.9919	0.9901	0.9881	0.9858	0.9832	0.9802	0.9769
8	0.9985	0.9981	0.9976	0.9969	0.9962	0.9953	0.9943	0.9931	0.9917
9	0.9996	0.9995	0.9993	0.9991	0.9989	0.9986	0.9982	0.9978	0.9973
10	0.9999	0.9999	0.9998	0.9998	0.9997	0.9996	0.9995	0.9994	0.9992
11	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999	0.9998	0.9998
12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

x	μ								
	3.500	**3.600**	**3.700**	**3.800**	**3.900**	**4.000**	**4.500**	**5.000**	**5.500**
0	0.0302	0.0273	0.0247	0.0224	0.0202	0.0183	0.0111	0.0067	0.0041
1	0.1359	0.1257	0.1162	0.1074	0.0992	0.0916	0.0611	0.0404	0.0266
2	0.3208	0.3027	0.2854	0.2689	0.2531	0.2381	0.1736	0.1247	0.0884
3	0.5366	0.5152	0.4942	0.4735	0.4532	0.4335	0.3423	0.2650	0.2017
4	0.7254	0.7064	0.6872	0.6678	0.6484	0.6288	0.5321	0.4405	0.3575
5	0.8576	0.8441	0.8301	0.8156	0.8006	0.7851	0.7029	0.6160	0.5289

Poisson Distribution - Cumulative Distribution Function

x	μ 3.500	3.600	3.700	3.800	3.900	4.000	4.500	5.000	5.500
6	0.9347	0.9267	0.9182	0.9091	0.8995	0.8893	0.8311	0.7622	0.6860
7	0.9733	0.9692	0.9648	0.9599	0.9546	0.9489	0.9134	0.8666	0.8095
8	0.9901	0.9883	0.9863	0.9840	0.9815	0.9786	0.9597	0.9319	0.8944
9	0.9967	0.9960	0.9952	0.9942	0.9931	0.9919	0.9829	0.9682	0.9462
10	0.9990	0.9987	0.9984	0.9981	0.9977	0.9972	0.9933	0.9863	0.9747
11	0.9997	0.9996	0.9995	0.9994	0.9993	0.9991	0.9976	0.9945	0.9890
12	0.9999	0.9999	0.9999	0.9998	0.9998	0.9997	0.9992	0.9980	0.9955
13	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9993	0.9983
14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994
15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998
16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

x	μ 6.000	6.500	7.000	7.500	8.000	8.500	9.000	9.500	10.000
0	0.0025	0.0015	0.0009	0.0006	0.0003	0.0002	0.0001	0.0001	0.0000
1	0.0174	0.0113	0.0073	0.0047	0.0030	0.0019	0.0012	0.0008	0.0005
2	0.0620	0.0430	0.0296	0.0203	0.0138	0.0093	0.0062	0.0042	0.0028
3	0.1512	0.1118	0.0818	0.0591	0.0424	0.0301	0.0212	0.0149	0.0103
4	0.2851	0.2237	0.1730	0.1321	0.0996	0.0744	0.0550	0.0403	0.0293
5	0.4457	0.3690	0.3007	0.2414	0.1912	0.1496	0.1157	0.0885	0.0671
6	0.6063	0.5265	0.4497	0.3782	0.3134	0.2562	0.2068	0.1649	0.1301
7	0.7440	0.6728	0.5987	0.5246	0.4530	0.3856	0.3239	0.2687	0.2202
8	0.8472	0.7916	0.7291	0.6620	0.5925	0.5231	0.4557	0.3918	0.3328
9	0.9161	0.8774	0.8305	0.7764	0.7166	0.6530	0.5874	0.5218	0.4579
10	0.9574	0.9332	0.9015	0.8622	0.8159	0.7634	0.7060	0.6453	0.5830
11	0.9799	0.9661	0.9467	0.9208	0.8881	0.8487	0.8030	0.7520	0.6968
12	0.9912	0.9840	0.9730	0.9573	0.9362	0.9091	0.8758	0.8364	0.7916
13	0.9964	0.9929	0.9872	0.9784	0.9658	0.9486	0.9261	0.8981	0.8645
14	0.9986	0.9970	0.9943	0.9897	0.9827	0.9726	0.9585	0.9400	0.9165
15	0.9995	0.9988	0.9976	0.9954	0.9918	0.9862	0.9780	0.9665	0.9513
16	0.9998	0.9996	0.9990	0.9980	0.9963	0.9934	0.9889	0.9823	0.9730
17	0.9999	0.9998	0.9996	0.9992	0.9984	0.9970	0.9947	0.9911	0.9857
18	1.0000	0.9999	0.9999	0.9997	0.9993	0.9987	0.9976	0.9957	0.9928
19	1.0000	1.0000	1.0000	0.9999	0.9997	0.9995	0.9989	0.9980	0.9965
20	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9991	0.9984
21	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9993
22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997
23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999
24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Standard Normal Distribution - Probability Density Function

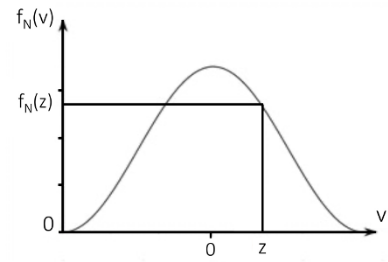
$$f_N(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}}$$

with $\pi = 3.14159...$ $\quad e = 2.71828...$

The following applies:

$$f_N(-z) = f_N(z)$$

$$-\infty < z < +\infty$$



z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.00	0.39894	0.39894	0.39894	0.39894	0.39894	0.39894	0.39894	0.39893	0.39893	0.39893
0.01	0.39892	0.39892	0.39891	0.39891	0.39890	0.39890	0.39889	0.39888	0.39888	0.39887
0.02	0.39886	0.39885	0.39885	0.39884	0.39883	0.39882	0.39881	0.39880	0.39879	0.39877
0.03	0.39876	0.39875	0.39874	0.39873	0.39871	0.39870	0.39868	0.39867	0.39865	0.39864
0.04	0.39862	0.39861	0.39859	0.39857	0.39856	0.39854	0.39852	0.39850	0.39848	0.39846
0.05	0.39844	0.39842	0.39840	0.39838	0.39836	0.39834	0.39832	0.39829	0.39827	0.39825
0.06	0.39822	0.39820	0.39818	0.39815	0.39813	0.39810	0.39807	0.39805	0.39802	0.39799
0.07	0.39797	0.39794	0.39791	0.39788	0.39785	0.39782	0.39779	0.39776	0.39773	0.39770
0.08	0.39767	0.39764	0.39760	0.39757	0.39754	0.39750	0.39747	0.39744	0.39740	0.39737
0.09	0.39733	0.39729	0.39726	0.39722	0.39718	0.39715	0.39711	0.39707	0.39703	0.39699
0.10	0.39695	0.39691	0.39687	0.39683	0.39679	0.39675	0.39671	0.39667	0.39662	0.39658
0.11	0.39654	0.39649	0.39645	0.39640	0.39636	0.39631	0.39627	0.39622	0.39617	0.39613
0.12	0.39608	0.39603	0.39598	0.39594	0.39589	0.39584	0.39579	0.39574	0.39569	0.39564
0.13	0.39559	0.39553	0.39548	0.39543	0.39538	0.39532	0.39527	0.39522	0.39516	0.39511
0.14	0.39505	0.39500	0.39494	0.39488	0.39483	0.39477	0.39471	0.39466	0.39460	0.39454
0.15	0.39448	0.39442	0.39436	0.39430	0.39424	0.39418	0.39412	0.39406	0.39399	0.39393
0.16	0.39387	0.39381	0.39374	0.39368	0.39361	0.39355	0.39348	0.39342	0.39335	0.39329
0.17	0.39322	0.39315	0.39308	0.39302	0.39295	0.39288	0.39281	0.39274	0.39267	0.39260
0.18	0.39253	0.39246	0.39239	0.39232	0.39225	0.39217	0.39210	0.39203	0.39195	0.39188
0.19	0.39181	0.39173	0.39166	0.39158	0.39151	0.39143	0.39135	0.39128	0.39120	0.39112
0.20	0.39104	0.39096	0.39089	0.39081	0.39073	0.39065	0.39057	0.39049	0.39041	0.39032
0.21	0.39024	0.39016	0.39008	0.38999	0.38991	0.38983	0.38974	0.38966	0.38957	0.38949
0.22	0.38940	0.38932	0.38923	0.38915	0.38906	0.38897	0.38888	0.38880	0.38871	0.38862
0.23	0.38853	0.38844	0.38835	0.38826	0.38817	0.38808	0.38799	0.38789	0.38780	0.38771
0.24	0.38762	0.38752	0.38743	0.38734	0.38724	0.38715	0.38705	0.38696	0.38686	0.38676
0.25	0.38667	0.38657	0.38647	0.38638	0.38628	0.38618	0.38608	0.38598	0.38588	0.38578
0.26	0.38568	0.38558	0.38548	0.38538	0.38528	0.38518	0.38508	0.38497	0.38487	0.38477
0.27	0.38466	0.38456	0.38445	0.38435	0.38424	0.38414	0.38403	0.38393	0.38382	0.38371
0.28	0.38361	0.38350	0.38339	0.38328	0.38317	0.38306	0.38296	0.38285	0.38274	0.38263
0.29	0.38251	0.38240	0.38229	0.38218	0.38207	0.38196	0.38184	0.38173	0.38162	0.38150

Standard Normal Distribution - Probability Density Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.30	0.38139	0.38127	0.38116	0.38104	0.38093	0.38081	0.38070	0.38058	0.38046	0.38034
0.31	0.38023	0.38011	0.37999	0.37987	0.37975	0.37963	0.37951	0.37939	0.37927	0.37915
0.32	0.37903	0.37891	0.37879	0.37867	0.37854	0.37842	0.37830	0.37817	0.37805	0.37793
0.33	0.37780	0.37768	0.37755	0.37743	0.37730	0.37717	0.37705	0.37692	0.37679	0.37667
0.34	0.37654	0.37641	0.37628	0.37615	0.37602	0.37589	0.37576	0.37563	0.37550	0.37537
0.35	0.37524	0.37511	0.37498	0.37484	0.37471	0.37458	0.37445	0.37431	0.37418	0.37405
0.36	0.37391	0.37378	0.37364	0.37351	0.37337	0.37323	0.37310	0.37296	0.37282	0.37269
0.37	0.37255	0.37241	0.37227	0.37213	0.37199	0.37186	0.37172	0.37158	0.37144	0.37129
0.38	0.37115	0.37101	0.37087	0.37073	0.37059	0.37044	0.37030	0.37016	0.37002	0.36987
0.39	0.36973	0.36958	0.36944	0.36929	0.36915	0.36900	0.36886	0.36871	0.36856	0.36842
0.40	0.36827	0.36812	0.36797	0.36783	0.36768	0.36753	0.36738	0.36723	0.36708	0.36693
0.41	0.36678	0.36663	0.36648	0.36633	0.36618	0.36603	0.36587	0.36572	0.36557	0.36542
0.42	0.36526	0.36511	0.36496	0.36480	0.36465	0.36449	0.36434	0.36418	0.36403	0.36387
0.43	0.36371	0.36356	0.36340	0.36324	0.36309	0.36293	0.36277	0.36261	0.36245	0.36229
0.44	0.36213	0.36198	0.36182	0.36166	0.36150	0.36133	0.36117	0.36101	0.36085	0.36069
0.45	0.36053	0.36036	0.36020	0.36004	0.35988	0.35971	0.35955	0.35938	0.35922	0.35906
0.46	0.35889	0.35873	0.35856	0.35839	0.35823	0.35806	0.35789	0.35773	0.35756	0.35739
0.47	0.35723	0.35706	0.35689	0.35672	0.35655	0.35638	0.35621	0.35604	0.35587	0.35570
0.48	0.35553	0.35536	0.35519	0.35502	0.35485	0.35468	0.35450	0.35433	0.35416	0.35399
0.49	0.35381	0.35364	0.35347	0.35329	0.35312	0.35294	0.35277	0.35259	0.35242	0.35224
0.50	0.35207	0.35189	0.35171	0.35154	0.35136	0.35118	0.35100	0.35083	0.35065	0.35047
0.51	0.35029	0.35011	0.34993	0.34975	0.34958	0.34940	0.34922	0.34904	0.34885	0.34867
0.52	0.34849	0.34831	0.34813	0.34795	0.34777	0.34758	0.34740	0.34722	0.34703	0.34685
0.53	0.34667	0.34648	0.34630	0.34612	0.34593	0.34575	0.34556	0.34538	0.34519	0.34500
0.54	0.34482	0.34463	0.34445	0.34426	0.34407	0.34388	0.34370	0.34351	0.34332	0.34313
0.55	0.34294	0.34276	0.34257	0.34238	0.34219	0.34200	0.34181	0.34162	0.34143	0.34124
0.56	0.34105	0.34085	0.34066	0.34047	0.34028	0.34009	0.33990	0.33970	0.33951	0.33932
0.57	0.33912	0.33893	0.33874	0.33854	0.33835	0.33815	0.33796	0.33777	0.33757	0.33738
0.58	0.33718	0.33698	0.33679	0.33659	0.33640	0.33620	0.33600	0.33581	0.33561	0.33541
0.59	0.33521	0.33502	0.33482	0.33462	0.33442	0.33422	0.33402	0.33382	0.33362	0.33342
0.60	0.33322	0.33302	0.33282	0.33262	0.33242	0.33222	0.33202	0.33182	0.33162	0.33142
0.61	0.33121	0.33101	0.33081	0.33061	0.33040	0.33020	0.33000	0.32980	0.32959	0.32939
0.62	0.32918	0.32898	0.32878	0.32857	0.32837	0.32816	0.32796	0.32775	0.32754	0.32734
0.63	0.32713	0.32693	0.32672	0.32651	0.32631	0.32610	0.32589	0.32569	0.32548	0.32527
0.64	0.32506	0.32485	0.32465	0.32444	0.32423	0.32402	0.32381	0.32360	0.32339	0.32318
0.65	0.32297	0.32276	0.32255	0.32234	0.32213	0.32192	0.32171	0.32150	0.32129	0.32108
0.66	0.32086	0.32065	0.32044	0.32023	0.32002	0.31980	0.31959	0.31938	0.31916	0.31895
0.67	0.31874	0.31852	0.31831	0.31810	0.31788	0.31767	0.31745	0.31724	0.31702	0.31681
0.68	0.31659	0.31638	0.31616	0.31595	0.31573	0.31551	0.31530	0.31508	0.31487	0.31465
0.69	0.31443	0.31421	0.31400	0.31378	0.31356	0.31334	0.31313	0.31291	0.31269	0.31247

Standard Normal Distribution - Probability Density Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.70	0.31225	0.31204	0.31182	0.31160	0.31138	0.31116	0.31094	0.31072	0.31050	0.31028
0.71	0.31006	0.30984	0.30962	0.30940	0.30918	0.30896	0.30874	0.30852	0.30829	0.30807
0.72	0.30785	0.30763	0.30741	0.30719	0.30696	0.30674	0.30652	0.30630	0.30607	0.30585
0.73	0.30563	0.30540	0.30518	0.30496	0.30473	0.30451	0.30429	0.30406	0.30384	0.30361
0.74	0.30339	0.30316	0.30294	0.30272	0.30249	0.30227	0.30204	0.30181	0.30159	0.30136
0.75	0.30114	0.30091	0.30069	0.30046	0.30023	0.30001	0.29978	0.29955	0.29933	0.29910
0.76	0.29887	0.29865	0.29842	0.29819	0.29796	0.29774	0.29751	0.29728	0.29705	0.29682
0.77	0.29659	0.29637	0.29614	0.29591	0.29568	0.29545	0.29522	0.29499	0.29476	0.29453
0.78	0.29431	0.29408	0.29385	0.29362	0.29339	0.29316	0.29293	0.29270	0.29246	0.29223
0.79	0.29200	0.29177	0.29154	0.29131	0.29108	0.29085	0.29062	0.29039	0.29015	0.28992
0.80	0.28969	0.28946	0.28923	0.28900	0.28876	0.28853	0.28830	0.28807	0.28783	0.28760
0.81	0.28737	0.28714	0.28690	0.28667	0.28644	0.28620	0.28597	0.28574	0.28550	0.28527
0.82	0.28504	0.28480	0.28457	0.28433	0.28410	0.28387	0.28363	0.28340	0.28316	0.28293
0.83	0.28269	0.28246	0.28223	0.28199	0.28176	0.28152	0.28129	0.28105	0.28081	0.28058
0.84	0.28034	0.28011	0.27987	0.27964	0.27940	0.27917	0.27893	0.27869	0.27846	0.27822
0.85	0.27798	0.27775	0.27751	0.27728	0.27704	0.27680	0.27657	0.27633	0.27609	0.27586
0.86	0.27562	0.27538	0.27514	0.27491	0.27467	0.27443	0.27419	0.27396	0.27372	0.27348
0.87	0.27324	0.27301	0.27277	0.27253	0.27229	0.27205	0.27182	0.27158	0.27134	0.27110
0.88	0.27086	0.27063	0.27039	0.27015	0.26991	0.26967	0.26943	0.26919	0.26896	0.26872
0.89	0.26848	0.26824	0.26800	0.26776	0.26752	0.26728	0.26704	0.26680	0.26656	0.26632
0.90	0.26609	0.26585	0.26561	0.26537	0.26513	0.26489	0.26465	0.26441	0.26417	0.26393
0.91	0.26369	0.26345	0.26321	0.26297	0.26273	0.26249	0.26225	0.26201	0.26177	0.26153
0.92	0.26129	0.26105	0.26081	0.26056	0.26032	0.26008	0.25984	0.25960	0.25936	0.25912
0.93	0.25888	0.25864	0.25840	0.25816	0.25792	0.25768	0.25744	0.25719	0.25695	0.25671
0.94	0.25647	0.25623	0.25599	0.25575	0.25551	0.25527	0.25502	0.25478	0.25454	0.25430
0.95	0.25406	0.25382	0.25358	0.25333	0.25309	0.25285	0.25261	0.25237	0.25213	0.25189
0.96	0.25164	0.25140	0.25116	0.25092	0.25068	0.25044	0.25019	0.24995	0.24971	0.24947
0.97	0.24923	0.24899	0.24874	0.24850	0.24826	0.24802	0.24778	0.24754	0.24729	0.24705
0.98	0.24681	0.24657	0.24633	0.24608	0.24584	0.24560	0.24536	0.24512	0.24487	0.24463
0.99	0.24439	0.24415	0.24391	0.24366	0.24342	0.24318	0.24294	0.24270	0.24245	0.24221
1.00	0.24197	0.24173	0.24149	0.24124	0.24100	0.24076	0.24052	0.24028	0.24003	0.23979
1.01	0.23955	0.23931	0.23907	0.23883	0.23858	0.23834	0.23810	0.23786	0.23762	0.23737
1.02	0.23713	0.23689	0.23665	0.23641	0.23616	0.23592	0.23568	0.23544	0.23520	0.23496
1.03	0.23471	0.23447	0.23423	0.23399	0.23375	0.23351	0.23326	0.23302	0.23278	0.23254
1.04	0.23230	0.23206	0.23181	0.23157	0.23133	0.23109	0.23085	0.23061	0.23036	0.23012
1.05	0.22988	0.22964	0.22940	0.22916	0.22892	0.22868	0.22843	0.22819	0.22795	0.22771
1.06	0.22747	0.22723	0.22699	0.22675	0.22651	0.22626	0.22602	0.22578	0.22554	0.22530
1.07	0.22506	0.22482	0.22458	0.22434	0.22410	0.22386	0.22362	0.22338	0.22313	0.22289
1.08	0.22265	0.22241	0.22217	0.22193	0.22169	0.22145	0.22121	0.22097	0.22073	0.22049
1.09	0.22025	0.22001	0.21977	0.21953	0.21929	0.21905	0.21881	0.21857	0.21833	0.21809

Standard Normal Distribution - Probability Density Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
1.10	0.21785	0.21761	0.21737	0.21713	0.21689	0.21665	0.21642	0.21618	0.21594	0.21570
1.11	0.21546	0.21522	0.21498	0.21474	0.21450	0.21426	0.21402	0.21379	0.21355	0.21331
1.12	0.21307	0.21283	0.21259	0.21235	0.21212	0.21188	0.21164	0.21140	0.21116	0.21092
1.13	0.21069	0.21045	0.21021	0.20997	0.20973	0.20950	0.20926	0.20902	0.20878	0.20855
1.14	0.20831	0.20807	0.20783	0.20760	0.20736	0.20712	0.20688	0.20665	0.20641	0.20617
1.15	0.20594	0.20570	0.20546	0.20523	0.20499	0.20475	0.20452	0.20428	0.20404	0.20381
1.16	0.20357	0.20334	0.20310	0.20286	0.20263	0.20239	0.20216	0.20192	0.20168	0.20145
1.17	0.20121	0.20098	0.20074	0.20051	0.20027	0.20004	0.19980	0.19957	0.19933	0.19910
1.18	0.19886	0.19863	0.19839	0.19816	0.19793	0.19769	0.19746	0.19722	0.19699	0.19675
1.19	0.19652	0.19629	0.19605	0.19582	0.19559	0.19535	0.19512	0.19489	0.19465	0.19442
1.20	0.19419	0.19395	0.19372	0.19349	0.19325	0.19302	0.19279	0.19256	0.19232	0.19209
1.21	0.19186	0.19163	0.19140	0.19116	0.19093	0.19070	0.19047	0.19024	0.19001	0.18977
1.22	0.18954	0.18931	0.18908	0.18885	0.18862	0.18839	0.18816	0.18793	0.18770	0.18747
1.23	0.18724	0.18701	0.18678	0.18654	0.18631	0.18609	0.18586	0.18563	0.18540	0.18517
1.24	0.18494	0.18471	0.18448	0.18425	0.18402	0.18379	0.18356	0.18333	0.18311	0.18288
1.25	0.18265	0.18242	0.18219	0.18196	0.18174	0.18151	0.18128	0.18105	0.18083	0.18060
1.26	0.18037	0.18014	0.17992	0.17969	0.17946	0.17924	0.17901	0.17878	0.17856	0.17833
1.27	0.17810	0.17788	0.17765	0.17743	0.17720	0.17697	0.17675	0.17652	0.17630	0.17607
1.28	0.17585	0.17562	0.17540	0.17517	0.17495	0.17472	0.17450	0.17427	0.17405	0.17383
1.29	0.17360	0.17338	0.17315	0.17293	0.17271	0.17248	0.17226	0.17204	0.17181	0.17159
1.30	0.17137	0.17115	0.17092	0.17070	0.17048	0.17026	0.17003	0.16981	0.16959	0.16937
1.31	0.16915	0.16893	0.16870	0.16848	0.16826	0.16804	0.16782	0.16760	0.16738	0.16716
1.32	0.16694	0.16672	0.16650	0.16628	0.16606	0.16584	0.16562	0.16540	0.16518	0.16496
1.33	0.16474	0.16452	0.16430	0.16408	0.16386	0.16365	0.16343	0.16321	0.16299	0.16277
1.34	0.16256	0.16234	0.16212	0.16190	0.16168	0.16147	0.16125	0.16103	0.16082	0.16060
1.35	0.16038	0.16017	0.15995	0.15973	0.15952	0.15930	0.15909	0.15887	0.15866	0.15844
1.36	0.15822	0.15801	0.15779	0.15758	0.15737	0.15715	0.15694	0.15672	0.15651	0.15629
1.37	0.15608	0.15587	0.15565	0.15544	0.15523	0.15501	0.15480	0.15459	0.15437	0.15416
1.38	0.15395	0.15374	0.15352	0.15331	0.15310	0.15289	0.15268	0.15246	0.15225	0.15204
1.39	0.15183	0.15162	0.15141	0.15120	0.15099	0.15078	0.15057	0.15036	0.15015	0.14994
1.40	0.14973	0.14952	0.14931	0.14910	0.14889	0.14868	0.14847	0.14826	0.14806	0.14785
1.41	0.14764	0.14743	0.14722	0.14701	0.14681	0.14660	0.14639	0.14618	0.14598	0.14577
1.42	0.14556	0.14536	0.14515	0.14494	0.14474	0.14453	0.14433	0.14412	0.14392	0.14371
1.43	0.14350	0.14330	0.14309	0.14289	0.14268	0.14248	0.14228	0.14207	0.14187	0.14166
1.44	0.14146	0.14126	0.14105	0.14085	0.14065	0.14044	0.14024	0.14004	0.13984	0.13963
1.45	0.13943	0.13923	0.13903	0.13882	0.13862	0.13842	0.13822	0.13802	0.13782	0.13762
1.46	0.13742	0.13722	0.13702	0.13682	0.13662	0.13642	0.13622	0.13602	0.13582	0.13562
1.47	0.13542	0.13522	0.13502	0.13482	0.13462	0.13442	0.13423	0.13403	0.13383	0.13363
1.48	0.13344	0.13324	0.13304	0.13284	0.13265	0.13245	0.13225	0.13206	0.13186	0.13166
1.49	0.13147	0.13127	0.13108	0.13088	0.13069	0.13049	0.13030	0.13010	0.12991	0.12971

Standard Normal Distribution - Probability Density Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
1.50	0.12952	0.12932	0.12913	0.12894	0.12874	0.12855	0.12835	0.12816	0.12797	0.12778
1.51	0.12758	0.12739	0.12720	0.12701	0.12681	0.12662	0.12643	0.12624	0.12605	0.12586
1.52	0.12566	0.12547	0.12528	0.12509	0.12490	0.12471	0.12452	0.12433	0.12414	0.12395
1.53	0.12376	0.12357	0.12338	0.12320	0.12301	0.12282	0.12263	0.12244	0.12225	0.12207
1.54	0.12188	0.12169	0.12150	0.12132	0.12113	0.12094	0.12075	0.12057	0.12038	0.12020
1.55	0.12001	0.11982	0.11964	0.11945	0.11927	0.11908	0.11890	0.11871	0.11853	0.11834
1.56	0.11816	0.11797	0.11779	0.11761	0.11742	0.11724	0.11705	0.11687	0.11669	0.11651
1.57	0.11632	0.11614	0.11596	0.11578	0.11559	0.11541	0.11523	0.11505	0.11487	0.11469
1.58	0.11450	0.11432	0.11414	0.11396	0.11378	0.11360	0.11342	0.11324	0.11306	0.11288
1.59	0.11270	0.11253	0.11235	0.11217	0.11199	0.11181	0.11163	0.11145	0.11128	0.11110
1.60	0.11092	0.11074	0.11057	0.11039	0.11021	0.11004	0.10986	0.10968	0.10951	0.10933
1.61	0.10915	0.10898	0.10880	0.10863	0.10845	0.10828	0.10810	0.10793	0.10775	0.10758
1.62	0.10741	0.10723	0.10706	0.10688	0.10671	0.10654	0.10637	0.10619	0.10602	0.10585
1.63	0.10567	0.10550	0.10533	0.10516	0.10499	0.10482	0.10464	0.10447	0.10430	0.10413
1.64	0.10396	0.10379	0.10362	0.10345	0.10328	0.10311	0.10294	0.10277	0.10260	0.10243
1.65	0.10226	0.10210	0.10193	0.10176	0.10159	0.10142	0.10126	0.10109	0.10092	0.10075
1.66	0.10059	0.10042	0.10025	0.10009	0.09992	0.09975	0.09959	0.09942	0.09926	0.09909
1.67	0.09893	0.09876	0.09860	0.09843	0.09827	0.09810	0.09794	0.09777	0.09761	0.09745
1.68	0.09728	0.09712	0.09696	0.09679	0.09663	0.09647	0.09630	0.09614	0.09598	0.09582
1.69	0.09566	0.09550	0.09533	0.09517	0.09501	0.09485	0.09469	0.09453	0.09437	0.09421
1.70	0.09405	0.09389	0.09373	0.09357	0.09341	0.09325	0.09309	0.09293	0.09278	0.09262
1.71	0.09246	0.09230	0.09214	0.09199	0.09183	0.09167	0.09151	0.09136	0.09120	0.09104
1.72	0.09089	0.09073	0.09057	0.09042	0.09026	0.09011	0.08995	0.08980	0.08964	0.08949
1.73	0.08933	0.08918	0.08902	0.08887	0.08872	0.08856	0.08841	0.08826	0.08810	0.08795
1.74	0.08780	0.08764	0.08749	0.08734	0.08719	0.08703	0.08688	0.08673	0.08658	0.08643
1.75	0.08628	0.08613	0.08598	0.08583	0.08567	0.08552	0.08537	0.08522	0.08508	0.08493
1.76	0.08478	0.08463	0.08448	0.08433	0.08418	0.08403	0.08388	0.08374	0.08359	0.08344
1.77	0.08329	0.08315	0.08300	0.08285	0.08270	0.08256	0.08241	0.08227	0.08212	0.08197
1.78	0.08183	0.08168	0.08154	0.08139	0.08125	0.08110	0.08096	0.08081	0.08067	0.08052
1.79	0.08038	0.08024	0.08009	0.07995	0.07981	0.07966	0.07952	0.07938	0.07923	0.07909
1.80	0.07895	0.07881	0.07867	0.07852	0.07838	0.07824	0.07810	0.07796	0.07782	0.07768
1.81	0.07754	0.07740	0.07726	0.07712	0.07698	0.07684	0.07670	0.07656	0.07642	0.07628
1.82	0.07614	0.07600	0.07587	0.07573	0.07559	0.07545	0.07531	0.07518	0.07504	0.07490
1.83	0.07477	0.07463	0.07449	0.07436	0.07422	0.07408	0.07395	0.07381	0.07368	0.07354
1.84	0.07341	0.07327	0.07314	0.07300	0.07287	0.07273	0.07260	0.07247	0.07233	0.07220
1.85	0.07206	0.07193	0.07180	0.07167	0.07153	0.07140	0.07127	0.07114	0.07100	0.07087
1.86	0.07074	0.07061	0.07048	0.07035	0.07022	0.07008	0.06995	0.06982	0.06969	0.06956
1.87	0.06943	0.06930	0.06917	0.06904	0.06892	0.06879	0.06866	0.06853	0.06840	0.06827
1.88	0.06814	0.06802	0.06789	0.06776	0.06763	0.06751	0.06738	0.06725	0.06712	0.06700
1.89	0.06687	0.06674	0.06662	0.06649	0.06637	0.06624	0.06612	0.06599	0.06587	0.06574

Standard Normal Distribution - Probability Density Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
1.90	0.06562	0.06549	0.06537	0.06524	0.06512	0.06499	0.06487	0.06475	0.06462	0.06450
1.91	0.06438	0.06425	0.06413	0.06401	0.06389	0.06376	0.06364	0.06352	0.06340	0.06328
1.92	0.06316	0.06304	0.06291	0.06279	0.06267	0.06255	0.06243	0.06231	0.06219	0.06207
1.93	0.06195	0.06183	0.06171	0.06159	0.06148	0.06136	0.06124	0.06112	0.06100	0.06088
1.94	0.06077	0.06065	0.06053	0.06041	0.06029	0.06018	0.06006	0.05994	0.05983	0.05971
1.95	0.05959	0.05948	0.05936	0.05925	0.05913	0.05902	0.05890	0.05879	0.05867	0.05856
1.96	0.05844	0.05833	0.05821	0.05810	0.05798	0.05787	0.05776	0.05764	0.05753	0.05742
1.97	0.05730	0.05719	0.05708	0.05697	0.05685	0.05674	0.05663	0.05652	0.05641	0.05629
1.98	0.05618	0.05607	0.05596	0.05585	0.05574	0.05563	0.05552	0.05541	0.05530	0.05519
1.99	0.05508	0.05497	0.05486	0.05475	0.05464	0.05453	0.05442	0.05432	0.05421	0.05410
2.00	0.05399	0.05388	0.05378	0.05367	0.05356	0.05345	0.05335	0.05324	0.05313	0.05303
2.01	0.05292	0.05281	0.05271	0.05260	0.05250	0.05239	0.05228	0.05218	0.05207	0.05197
2.02	0.05186	0.05176	0.05165	0.05155	0.05145	0.05134	0.05124	0.05113	0.05103	0.05093
2.03	0.05082	0.05072	0.05062	0.05052	0.05041	0.05031	0.05021	0.05011	0.05000	0.04990
2.04	0.04980	0.04970	0.04960	0.04950	0.04939	0.04929	0.04919	0.04909	0.04899	0.04889
2.05	0.04879	0.04869	0.04859	0.04849	0.04839	0.04829	0.04819	0.04810	0.04800	0.04790
2.06	0.04780	0.04770	0.04760	0.04750	0.04741	0.04731	0.04721	0.04711	0.04702	0.04692
2.07	0.04682	0.04673	0.04663	0.04653	0.04644	0.04634	0.04624	0.04615	0.04605	0.04596
2.08	0.04586	0.04577	0.04567	0.04558	0.04548	0.04539	0.04529	0.04520	0.04510	0.04501
2.09	0.04491	0.04482	0.04473	0.04463	0.04454	0.04445	0.04435	0.04426	0.04417	0.04408
2.10	0.04398	0.04389	0.04380	0.04371	0.04362	0.04352	0.04343	0.04334	0.04325	0.04316
2.11	0.04307	0.04298	0.04289	0.04280	0.04271	0.04261	0.04252	0.04243	0.04235	0.04226
2.12	0.04217	0.04208	0.04199	0.04190	0.04181	0.04172	0.04163	0.04154	0.04146	0.04137
2.13	0.04128	0.04119	0.04110	0.04102	0.04093	0.04084	0.04075	0.04067	0.04058	0.04049
2.14	0.04041	0.04032	0.04023	0.04015	0.04006	0.03998	0.03989	0.03981	0.03972	0.03964
2.15	0.03955	0.03947	0.03938	0.03930	0.03921	0.03913	0.03904	0.03896	0.03887	0.03879
2.16	0.03871	0.03862	0.03854	0.03846	0.03837	0.03829	0.03821	0.03813	0.03804	0.03796
2.17	0.03788	0.03780	0.03771	0.03763	0.03755	0.03747	0.03739	0.03731	0.03722	0.03714
2.18	0.03706	0.03698	0.03690	0.03682	0.03674	0.03666	0.03658	0.03650	0.03642	0.03634
2.19	0.03626	0.03618	0.03610	0.03602	0.03595	0.03587	0.03579	0.03571	0.03563	0.03555
2.20	0.03547	0.03540	0.03532	0.03524	0.03516	0.03509	0.03501	0.03493	0.03485	0.03478
2.21	0.03470	0.03462	0.03455	0.03447	0.03440	0.03432	0.03424	0.03417	0.03409	0.03402
2.22	0.03394	0.03387	0.03379	0.03372	0.03364	0.03357	0.03349	0.03342	0.03334	0.03327
2.23	0.03319	0.03312	0.03305	0.03297	0.03290	0.03283	0.03275	0.03268	0.03261	0.03253
2.24	0.03246	0.03239	0.03232	0.03224	0.03217	0.03210	0.03203	0.03195	0.03188	0.03181
2.25	0.03174	0.03167	0.03160	0.03153	0.03146	0.03138	0.03131	0.03124	0.03117	0.03110
2.26	0.03103	0.03096	0.03089	0.03082	0.03075	0.03068	0.03061	0.03054	0.03047	0.03041
2.27	0.03034	0.03027	0.03020	0.03013	0.03006	0.02999	0.02993	0.02986	0.02979	0.02972
2.28	0.02965	0.02959	0.02952	0.02945	0.02939	0.02932	0.02925	0.02918	0.02912	0.02905
2.29	0.02898	0.02892	0.02885	0.02879	0.02872	0.02865	0.02859	0.02852	0.02846	0.02839

Standard Normal Distribution - Probability Density Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
2.30	0.02833	0.02826	0.02820	0.02813	0.02807	0.02800	0.02794	0.02787	0.02781	0.02775
2.31	0.02768	0.02762	0.02755	0.02749	0.02743	0.02736	0.02730	0.02724	0.02717	0.02711
2.32	0.02705	0.02699	0.02692	0.02686	0.02680	0.02674	0.02667	0.02661	0.02655	0.02649
2.33	0.02643	0.02636	0.02630	0.02624	0.02618	0.02612	0.02606	0.02600	0.02594	0.02588
2.34	0.02582	0.02576	0.02570	0.02564	0.02558	0.02552	0.02546	0.02540	0.02534	0.02528
2.35	0.02522	0.02516	0.02510	0.02504	0.02498	0.02492	0.02486	0.02481	0.02475	0.02469
2.36	0.02463	0.02457	0.02452	0.02446	0.02440	0.02434	0.02428	0.02423	0.02417	0.02411
2.37	0.02406	0.02400	0.02394	0.02389	0.02383	0.02377	0.02372	0.02366	0.02360	0.02355
2.38	0.02349	0.02344	0.02338	0.02332	0.02327	0.02321	0.02316	0.02310	0.02305	0.02299
2.39	0.02294	0.02288	0.02283	0.02277	0.02272	0.02266	0.02261	0.02256	0.02250	0.02245
2.40	0.02239	0.02234	0.02229	0.02223	0.02218	0.02213	0.02207	0.02202	0.02197	0.02192
2.41	0.02186	0.02181	0.02176	0.02170	0.02165	0.02160	0.02155	0.02150	0.02144	0.02139
2.42	0.02134	0.02129	0.02124	0.02119	0.02113	0.02108	0.02103	0.02098	0.02093	0.02088
2.43	0.02083	0.02078	0.02073	0.02068	0.02063	0.02058	0.02053	0.02048	0.02043	0.02038
2.44	0.02033	0.02028	0.02023	0.02018	0.02013	0.02008	0.02003	0.01998	0.01993	0.01989
2.45	0.01984	0.01979	0.01974	0.01969	0.01964	0.01960	0.01955	0.01950	0.01945	0.01940
2.46	0.01936	0.01931	0.01926	0.01921	0.01917	0.01912	0.01907	0.01903	0.01898	0.01893
2.47	0.01888	0.01884	0.01879	0.01875	0.01870	0.01865	0.01861	0.01856	0.01851	0.01847
2.48	0.01842	0.01838	0.01833	0.01829	0.01824	0.01820	0.01815	0.01811	0.01806	0.01802
2.49	0.01797	0.01793	0.01788	0.01784	0.01779	0.01775	0.01770	0.01766	0.01762	0.01757
2.50	0.01753	0.01748	0.01744	0.01740	0.01735	0.01731	0.01727	0.01722	0.01718	0.01714
2.60	0.01358	0.01355	0.01351	0.01348	0.01344	0.01341	0.01337	0.01334	0.01330	0.01327
2.70	0.01042	0.01039	0.01036	0.01034	0.01031	0.01028	0.01025	0.01023	0.01020	0.01017
2.80	0.00792	0.00789	0.00787	0.00785	0.00783	0.00781	0.00778	0.00776	0.00774	0.00772
2.90	0.00595	0.00594	0.00592	0.00590	0.00588	0.00587	0.00585	0.00583	0.00582	0.00580
3.00	0.00443	0.00442	0.00441	0.00439	0.00438	0.00437	0.00435	0.00434	0.00433	0.00431
3.10	0.00327	0.00326	0.00325	0.00324	0.00323	0.00322	0.00321	0.00320	0.00319	0.00318
3.20	0.00238	0.00238	0.00237	0.00236	0.00235	0.00235	0.00234	0.00233	0.00232	0.00232
3.30	0.00172	0.00172	0.00171	0.00171	0.00170	0.00169	0.00169	0.00168	0.00168	0.00167
3.40	0.00123	0.00123	0.00122	0.00122	0.00122	0.00121	0.00121	0.00120	0.00120	0.00120
3.50	0.00087	0.00087	0.00087	0.00086	0.00086	0.00086	0.00085	0.00085	0.00085	0.00085
3.60	0.00061	0.00061	0.00061	0.00061	0.00060	0.00060	0.00060	0.00060	0.00059	0.00059
3.70	0.00042	0.00042	0.00042	0.00042	0.00042	0.00042	0.00042	0.00041	0.00041	0.00041
3.80	0.00029	0.00029	0.00029	0.00029	0.00029	0.00029	0.00029	0.00028	0.00028	0.00028

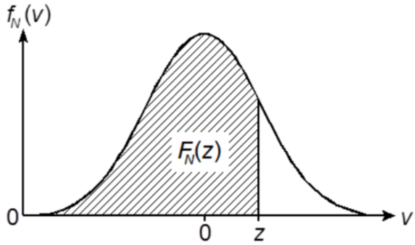
Standard Normal Distribution - Cumulative Distribution Function

$$F_N(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{\frac{-v^2}{2}} \, dv$$

with $\pi = 3.14159...$
$e = 2.71828...$

The following applies:

$F_N(-z) = 1 - F_N(z)$
$-\infty < z < +\infty$



z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.00	0.5000	0.5004	0.5008	0.5012	0.5016	0.5020	0.5024	0.5028	0.5032	0.5036
0.01	0.5040	0.5044	0.5048	0.5052	0.5056	0.5060	0.5064	0.5068	0.5072	0.5076
0.02	0.5080	0.5084	0.5088	0.5092	0.5096	0.5100	0.5104	0.5108	0.5112	0.5116
0.03	0.5120	0.5124	0.5128	0.5132	0.5136	0.5140	0.5144	0.5148	0.5152	0.5156
0.04	0.5160	0.5164	0.5168	0.5171	0.5175	0.5179	0.5183	0.5187	0.5191	0.5195
0.05	0.5199	0.5203	0.5207	0.5211	0.5215	0.5219	0.5223	0.5227	0.5231	0.5235
0.06	0.5239	0.5243	0.5247	0.5251	0.5255	0.5259	0.5263	0.5267	0.5271	0.5275
0.07	0.5279	0.5283	0.5287	0.5291	0.5295	0.5299	0.5303	0.5307	0.5311	0.5315
0.08	0.5319	0.5323	0.5327	0.5331	0.5335	0.5339	0.5343	0.5347	0.5351	0.5355
0.09	0.5359	0.5363	0.5367	0.5370	0.5374	0.5378	0.5382	0.5386	0.5390	0.5394
0.10	0.5398	0.5402	0.5406	0.5410	0.5414	0.5418	0.5422	0.5426	0.5430	0.5434
0.11	0.5438	0.5442	0.5446	0.5450	0.5454	0.5458	0.5462	0.5466	0.5470	0.5474
0.12	0.5478	0.5482	0.5486	0.5489	0.5493	0.5497	0.5501	0.5505	0.5509	0.5513
0.13	0.5517	0.5521	0.5525	0.5529	0.5533	0.5537	0.5541	0.5545	0.5549	0.5553
0.14	0.5557	0.5561	0.5565	0.5569	0.5572	0.5576	0.5580	0.5584	0.5588	0.5592
0.15	0.5596	0.5600	0.5604	0.5608	0.5612	0.5616	0.5620	0.5624	0.5628	0.5632
0.16	0.5636	0.5640	0.5643	0.5647	0.5651	0.5655	0.5659	0.5663	0.5667	0.5671
0.17	0.5675	0.5679	0.5683	0.5687	0.5691	0.5695	0.5699	0.5702	0.5706	0.5710
0.18	0.5714	0.5718	0.5722	0.5726	0.5730	0.5734	0.5738	0.5742	0.5746	0.5750
0.19	0.5753	0.5757	0.5761	0.5765	0.5769	0.5773	0.5777	0.5781	0.5785	0.5789
0.20	0.5793	0.5797	0.5800	0.5804	0.5808	0.5812	0.5816	0.5820	0.5824	0.5828
0.21	0.5832	0.5836	0.5839	0.5843	0.5847	0.5851	0.5855	0.5859	0.5863	0.5867
0.22	0.5871	0.5875	0.5878	0.5882	0.5886	0.5890	0.5894	0.5898	0.5902	0.5906
0.23	0.5910	0.5913	0.5917	0.5921	0.5925	0.5929	0.5933	0.5937	0.5941	0.5944
0.24	0.5948	0.5952	0.5956	0.5960	0.5964	0.5968	0.5972	0.5975	0.5979	0.5983
0.25	0.5987	0.5991	0.5995	0.5999	0.6003	0.6006	0.6010	0.6014	0.6018	0.6022
0.26	0.6026	0.6030	0.6033	0.6037	0.6041	0.6045	0.6049	0.6053	0.6057	0.6060
0.27	0.6064	0.6068	0.6072	0.6076	0.6080	0.6083	0.6087	0.6091	0.6095	0.6099
0.28	0.6103	0.6106	0.6110	0.6114	0.6118	0.6122	0.6126	0.6129	0.6133	0.6137
0.29	0.6141	0.6145	0.6149	0.6152	0.6156	0.6160	0.6164	0.6168	0.6171	0.6175

Standard Normal Distribution - Cumulative Distribution Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.30	0.6179	0.6183	0.6187	0.6191	0.6194	0.6198	0.6202	0.6206	0.6210	0.6213
0.31	0.6217	0.6221	0.6225	0.6229	0.6232	0.6236	0.6240	0.6244	0.6248	0.6251
0.32	0.6255	0.6259	0.6263	0.6267	0.6270	0.6274	0.6278	0.6282	0.6285	0.6289
0.33	0.6293	0.6297	0.6301	0.6304	0.6308	0.6312	0.6316	0.6319	0.6323	0.6327
0.34	0.6331	0.6334	0.6338	0.6342	0.6346	0.6350	0.6353	0.6357	0.6361	0.6365
0.35	0.6368	0.6372	0.6376	0.6380	0.6383	0.6387	0.6391	0.6395	0.6398	0.6402
0.36	0.6406	0.6410	0.6413	0.6417	0.6421	0.6424	0.6428	0.6432	0.6436	0.6439
0.37	0.6443	0.6447	0.6451	0.6454	0.6458	0.6462	0.6465	0.6469	0.6473	0.6477
0.38	0.6480	0.6484	0.6488	0.6491	0.6495	0.6499	0.6503	0.6506	0.6510	0.6514
0.39	0.6517	0.6521	0.6525	0.6528	0.6532	0.6536	0.6539	0.6543	0.6547	0.6551
0.40	0.6554	0.6558	0.6562	0.6565	0.6569	0.6573	0.6576	0.6580	0.6584	0.6587
0.41	0.6591	0.6595	0.6598	0.6602	0.6606	0.6609	0.6613	0.6617	0.6620	0.6624
0.42	0.6628	0.6631	0.6635	0.6639	0.6642	0.6646	0.6649	0.6653	0.6657	0.6660
0.43	0.6664	0.6668	0.6671	0.6675	0.6679	0.6682	0.6686	0.6689	0.6693	0.6697
0.44	0.6700	0.6704	0.6708	0.6711	0.6715	0.6718	0.6722	0.6726	0.6729	0.6733
0.45	0.6736	0.6740	0.6744	0.6747	0.6751	0.6754	0.6758	0.6762	0.6765	0.6769
0.46	0.6772	0.6776	0.6780	0.6783	0.6787	0.6790	0.6794	0.6798	0.6801	0.6805
0.47	0.6808	0.6812	0.6815	0.6819	0.6823	0.6826	0.6830	0.6833	0.6837	0.6840
0.48	0.6844	0.6847	0.6851	0.6855	0.6858	0.6862	0.6865	0.6869	0.6872	0.6876
0.49	0.6879	0.6883	0.6886	0.6890	0.6893	0.6897	0.6901	0.6904	0.6908	0.6911
0.50	0.6915	0.6918	0.6922	0.6925	0.6929	0.6932	0.6936	0.6939	0.6943	0.6946
0.51	0.6950	0.6953	0.6957	0.6960	0.6964	0.6967	0.6971	0.6974	0.6978	0.6981
0.52	0.6985	0.6988	0.6992	0.6995	0.6999	0.7002	0.7006	0.7009	0.7013	0.7016
0.53	0.7019	0.7023	0.7026	0.7030	0.7033	0.7037	0.7040	0.7044	0.7047	0.7051
0.54	0.7054	0.7057	0.7061	0.7064	0.7068	0.7071	0.7075	0.7078	0.7082	0.7085
0.55	0.7088	0.7092	0.7095	0.7099	0.7102	0.7106	0.7109	0.7112	0.7116	0.7119
0.56	0.7123	0.7126	0.7129	0.7133	0.7136	0.7140	0.7143	0.7146	0.7150	0.7153
0.57	0.7157	0.7160	0.7163	0.7167	0.7170	0.7174	0.7177	0.7180	0.7184	0.7187
0.58	0.7190	0.7194	0.7197	0.7201	0.7204	0.7207	0.7211	0.7214	0.7217	0.7221
0.59	0.7224	0.7227	0.7231	0.7234	0.7237	0.7241	0.7244	0.7247	0.7251	0.7254
0.60	0.7257	0.7261	0.7264	0.7267	0.7271	0.7274	0.7277	0.7281	0.7284	0.7287
0.61	0.7291	0.7294	0.7297	0.7301	0.7304	0.7307	0.7311	0.7314	0.7317	0.7320
0.62	0.7324	0.7327	0.7330	0.7334	0.7337	0.7340	0.7343	0.7347	0.7350	0.7353
0.63	0.7357	0.7360	0.7363	0.7366	0.7370	0.7373	0.7376	0.7379	0.7383	0.7386
0.64	0.7389	0.7392	0.7396	0.7399	0.7402	0.7405	0.7409	0.7412	0.7415	0.7418
0.65	0.7422	0.7425	0.7428	0.7431	0.7434	0.7438	0.7441	0.7444	0.7447	0.7451
0.66	0.7454	0.7457	0.7460	0.7463	0.7467	0.7470	0.7473	0.7476	0.7479	0.7483
0.67	0.7486	0.7489	0.7492	0.7495	0.7498	0.7502	0.7505	0.7508	0.7511	0.7514
0.68	0.7517	0.7521	0.7524	0.7527	0.7530	0.7533	0.7536	0.7540	0.7543	0.7546
0.69	0.7549	0.7552	0.7555	0.7558	0.7562	0.7565	0.7568	0.7571	0.7574	0.7577

Standard Normal Distribution - Cumulative Distribution Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.70	0.7580	0.7583	0.7587	0.7590	0.7593	0.7596	0.7599	0.7602	0.7605	0.7608
0.71	0.7611	0.7615	0.7618	0.7621	0.7624	0.7627	0.7630	0.7633	0.7636	0.7639
0.72	0.7642	0.7645	0.7649	0.7652	0.7655	0.7658	0.7661	0.7664	0.7667	0.7670
0.73	0.7673	0.7676	0.7679	0.7682	0.7685	0.7688	0.7691	0.7694	0.7697	0.7700
0.74	0.7704	0.7707	0.7710	0.7713	0.7716	0.7719	0.7722	0.7725	0.7728	0.7731
0.75	0.7734	0.7737	0.7740	0.7743	0.7746	0.7749	0.7752	0.7755	0.7758	0.7761
0.76	0.7764	0.7767	0.7770	0.7773	0.7776	0.7779	0.7782	0.7785	0.7788	0.7791
0.77	0.7794	0.7796	0.7799	0.7802	0.7805	0.7808	0.7811	0.7814	0.7817	0.7820
0.78	0.7823	0.7826	0.7829	0.7832	0.7835	0.7838	0.7841	0.7844	0.7847	0.7849
0.79	0.7852	0.7855	0.7858	0.7861	0.7864	0.7867	0.7870	0.7873	0.7876	0.7879
0.80	0.7881	0.7884	0.7887	0.7890	0.7893	0.7896	0.7899	0.7902	0.7905	0.7907
0.81	0.7910	0.7913	0.7916	0.7919	0.7922	0.7925	0.7927	0.7930	0.7933	0.7936
0.82	0.7939	0.7942	0.7945	0.7947	0.7950	0.7953	0.7956	0.7959	0.7962	0.7964
0.83	0.7967	0.7970	0.7973	0.7976	0.7979	0.7981	0.7984	0.7987	0.7990	0.7993
0.84	0.7995	0.7998	0.8001	0.8004	0.8007	0.8009	0.8012	0.8015	0.8018	0.8021
0.85	0.8023	0.8026	0.8029	0.8032	0.8034	0.8037	0.8040	0.8043	0.8046	0.8048
0.86	0.8051	0.8054	0.8057	0.8059	0.8062	0.8065	0.8068	0.8070	0.8073	0.8076
0.87	0.8078	0.8081	0.8084	0.8087	0.8089	0.8092	0.8095	0.8098	0.8100	0.8103
0.88	0.8106	0.8108	0.8111	0.8114	0.8117	0.8119	0.8122	0.8125	0.8127	0.8130
0.89	0.8133	0.8135	0.8138	0.8141	0.8143	0.8146	0.8149	0.8151	0.8154	0.8157
0.90	0.8159	0.8162	0.8165	0.8167	0.8170	0.8173	0.8175	0.8178	0.8181	0.8183
0.91	0.8186	0.8189	0.8191	0.8194	0.8196	0.8199	0.8202	0.8204	0.8207	0.8210
0.92	0.8212	0.8215	0.8217	0.8220	0.8223	0.8225	0.8228	0.8230	0.8233	0.8236
0.93	0.8238	0.8241	0.8243	0.8246	0.8248	0.8251	0.8254	0.8256	0.8259	0.8261
0.94	0.8264	0.8266	0.8269	0.8272	0.8274	0.8277	0.8279	0.8282	0.8284	0.8287
0.95	0.8289	0.8292	0.8295	0.8297	0.8300	0.8302	0.8305	0.8307	0.8310	0.8312
0.96	0.8315	0.8317	0.8320	0.8322	0.8325	0.8327	0.8330	0.8332	0.8335	0.8337
0.97	0.8340	0.8342	0.8345	0.8347	0.8350	0.8352	0.8355	0.8357	0.8360	0.8362
0.98	0.8365	0.8367	0.8370	0.8372	0.8374	0.8377	0.8379	0.8382	0.8384	0.8387
0.99	0.8389	0.8392	0.8394	0.8396	0.8399	0.8401	0.8404	0.8406	0.8409	0.8411
1.00	0.8413	0.8416	0.8418	0.8421	0.8423	0.8426	0.8428	0.8430	0.8433	0.8435
1.01	0.8438	0.8440	0.8442	0.8445	0.8447	0.8449	0.8452	0.8454	0.8457	0.8459
1.02	0.8461	0.8464	0.8466	0.8468	0.8471	0.8473	0.8476	0.8478	0.8480	0.8483
1.03	0.8485	0.8487	0.8490	0.8492	0.8494	0.8497	0.8499	0.8501	0.8504	0.8506
1.04	0.8508	0.8511	0.8513	0.8515	0.8518	0.8520	0.8522	0.8525	0.8527	0.8529
1.05	0.8531	0.8534	0.8536	0.8538	0.8541	0.8543	0.8545	0.8547	0.8550	0.8552
1.06	0.8554	0.8557	0.8559	0.8561	0.8563	0.8566	0.8568	0.8570	0.8572	0.8575
1.07	0.8577	0.8579	0.8581	0.8584	0.8586	0.8588	0.8590	0.8593	0.8595	0.8597
1.08	0.8599	0.8602	0.8604	0.8606	0.8608	0.8610	0.8613	0.8615	0.8617	0.8619
1.09	0.8621	0.8624	0.8626	0.8628	0.8630	0.8632	0.8635	0.8637	0.8639	0.8641

Standard Normal Distribution - Cumulative Distribution Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
1.10	0.8643	0.8646	0.8648	0.8650	0.8652	0.8654	0.8656	0.8659	0.8661	0.8663
1.11	0.8665	0.8667	0.8669	0.8671	0.8674	0.8676	0.8678	0.8680	0.8682	0.8684
1.12	0.8686	0.8689	0.8691	0.8693	0.8695	0.8697	0.8699	0.8701	0.8703	0.8706
1.13	0.8708	0.8710	0.8712	0.8714	0.8716	0.8718	0.8720	0.8722	0.8724	0.8726
1.14	0.8729	0.8731	0.8733	0.8735	0.8737	0.8739	0.8741	0.8743	0.8745	0.8747
1.15	0.8749	0.8751	0.8753	0.8755	0.8757	0.8760	0.8762	0.8764	0.8766	0.8768
1.16	0.8770	0.8772	0.8774	0.8776	0.8778	0.8780	0.8782	0.8784	0.8786	0.8788
1.17	0.8790	0.8792	0.8794	0.8796	0.8798	0.8800	0.8802	0.8804	0.8806	0.8808
1.18	0.8810	0.8812	0.8814	0.8816	0.8818	0.8820	0.8822	0.8824	0.8826	0.8828
1.19	0.8830	0.8832	0.8834	0.8836	0.8838	0.8840	0.8842	0.8843	0.8845	0.8847
1.20	0.8849	0.8851	0.8853	0.8855	0.8857	0.8859	0.8861	0.8863	0.8865	0.8867
1.21	0.8869	0.8871	0.8872	0.8874	0.8876	0.8878	0.8880	0.8882	0.8884	0.8886
1.22	0.8888	0.8890	0.8891	0.8893	0.8895	0.8897	0.8899	0.8901	0.8903	0.8905
1.23	0.8907	0.8908	0.8910	0.8912	0.8914	0.8916	0.8918	0.8920	0.8921	0.8923
1.24	0.8925	0.8927	0.8929	0.8931	0.8933	0.8934	0.8936	0.8938	0.8940	0.8942
1.25	0.8944	0.8945	0.8947	0.8949	0.8951	0.8953	0.8954	0.8956	0.8958	0.8960
1.26	0.8962	0.8963	0.8965	0.8967	0.8969	0.8971	0.8972	0.8974	0.8976	0.8978
1.27	0.8980	0.8981	0.8983	0.8985	0.8987	0.8988	0.8990	0.8992	0.8994	0.8996
1.28	0.8997	0.8999	0.9001	0.9003	0.9004	0.9006	0.9008	0.9010	0.9011	0.9013
1.29	0.9015	0.9016	0.9018	0.9020	0.9022	0.9023	0.9025	0.9027	0.9029	0.9030
1.30	0.9032	0.9034	0.9035	0.9037	0.9039	0.9041	0.9042	0.9044	0.9046	0.9047
1.31	0.9049	0.9051	0.9052	0.9054	0.9056	0.9057	0.9059	0.9061	0.9062	0.9064
1.32	0.9066	0.9067	0.9069	0.9071	0.9072	0.9074	0.9076	0.9077	0.9079	0.9081
1.33	0.9082	0.9084	0.9086	0.9087	0.9089	0.9091	0.9092	0.9094	0.9096	0.9097
1.34	0.9099	0.9100	0.9102	0.9104	0.9105	0.9107	0.9108	0.9110	0.9112	0.9113
1.35	0.9115	0.9117	0.9118	0.9120	0.9121	0.9123	0.9125	0.9126	0.9128	0.9129
1.36	0.9131	0.9132	0.9134	0.9136	0.9137	0.9139	0.9140	0.9142	0.9143	0.9145
1.37	0.9147	0.9148	0.9150	0.9151	0.9153	0.9154	0.9156	0.9157	0.9159	0.9161
1.38	0.9162	0.9164	0.9165	0.9167	0.9168	0.9170	0.9171	0.9173	0.9174	0.9176
1.39	0.9177	0.9179	0.9180	0.9182	0.9183	0.9185	0.9186	0.9188	0.9189	0.9191
1.40	0.9192	0.9194	0.9195	0.9197	0.9198	0.9200	0.9201	0.9203	0.9204	0.9206
1.41	0.9207	0.9209	0.9210	0.9212	0.9213	0.9215	0.9216	0.9218	0.9219	0.9221
1.42	0.9222	0.9223	0.9225	0.9226	0.9228	0.9229	0.9231	0.9232	0.9234	0.9235
1.43	0.9236	0.9238	0.9239	0.9241	0.9242	0.9244	0.9245	0.9246	0.9248	0.9249
1.44	0.9251	0.9252	0.9253	0.9255	0.9256	0.9258	0.9259	0.9261	0.9262	0.9263
1.45	0.9265	0.9266	0.9267	0.9269	0.9270	0.9272	0.9273	0.9274	0.9276	0.9277
1.46	0.9279	0.9280	0.9281	0.9283	0.9284	0.9285	0.9287	0.9288	0.9289	0.9291
1.47	0.9292	0.9294	0.9295	0.9296	0.9298	0.9299	0.9300	0.9302	0.9303	0.9304
1.48	0.9306	0.9307	0.9308	0.9310	0.9311	0.9312	0.9314	0.9315	0.9316	0.9318
1.49	0.9319	0.9320	0.9322	0.9323	0.9324	0.9325	0.9327	0.9328	0.9329	0.9331

Standard Normal Distribution - Cumulative Distribution Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
1.50	0.9332	0.9333	0.9335	0.9336	0.9337	0.9338	0.9340	0.9341	0.9342	0.9344
1.51	0.9345	0.9346	0.9347	0.9349	0.9350	0.9351	0.9352	0.9354	0.9355	0.9356
1.52	0.9357	0.9359	0.9360	0.9361	0.9362	0.9364	0.9365	0.9366	0.9367	0.9369
1.53	0.9370	0.9371	0.9372	0.9374	0.9375	0.9376	0.9377	0.9379	0.9380	0.9381
1.54	0.9382	0.9383	0.9385	0.9386	0.9387	0.9388	0.9389	0.9391	0.9392	0.9393
1.55	0.9394	0.9395	0.9397	0.9398	0.9399	0.9400	0.9401	0.9403	0.9404	0.9405
1.56	0.9406	0.9407	0.9409	0.9410	0.9411	0.9412	0.9413	0.9414	0.9416	0.9417
1.57	0.9418	0.9419	0.9420	0.9421	0.9423	0.9424	0.9425	0.9426	0.9427	0.9428
1.58	0.9429	0.9431	0.9432	0.9433	0.9434	0.9435	0.9436	0.9437	0.9439	0.9440
1.59	0.9441	0.9442	0.9443	0.9444	0.9445	0.9446	0.9448	0.9449	0.9450	0.9451
1.60	0.9452	0.9453	0.9454	0.9455	0.9456	0.9458	0.9459	0.9460	0.9461	0.9462
1.61	0.9463	0.9464	0.9465	0.9466	0.9467	0.9468	0.9470	0.9471	0.9472	0.9473
1.62	0.9474	0.9475	0.9476	0.9477	0.9478	0.9479	0.9480	0.9481	0.9482	0.9483
1.63	0.9484	0.9486	0.9487	0.9488	0.9489	0.9490	0.9491	0.9492	0.9493	0.9494
1.64	0.9495	0.9496	0.9497	0.9498	0.9499	0.9500	0.9501	0.9502	0.9503	0.9504
1.65	0.9505	0.9506	0.9507	0.9508	0.9509	0.9510	0.9511	0.9512	0.9513	0.9514
1.66	0.9515	0.9516	0.9517	0.9518	0.9519	0.9520	0.9521	0.9522	0.9523	0.9524
1.67	0.9525	0.9526	0.9527	0.9528	0.9529	0.9530	0.9531	0.9532	0.9533	0.9534
1.68	0.9535	0.9536	0.9537	0.9538	0.9539	0.9540	0.9541	0.9542	0.9543	0.9544
1.69	0.9545	0.9546	0.9547	0.9548	0.9549	0.9550	0.9551	0.9552	0.9552	0.9553
1.70	0.9554	0.9555	0.9556	0.9557	0.9558	0.9559	0.9560	0.9561	0.9562	0.9563
1.71	0.9564	0.9565	0.9566	0.9566	0.9567	0.9568	0.9569	0.9570	0.9571	0.9572
1.72	0.9573	0.9574	0.9575	0.9576	0.9576	0.9577	0.9578	0.9579	0.9580	0.9581
1.73	0.9582	0.9583	0.9584	0.9585	0.9585	0.9586	0.9587	0.9588	0.9589	0.9590
1.74	0.9591	0.9592	0.9592	0.9593	0.9594	0.9595	0.9596	0.9597	0.9598	0.9599
1.75	0.9599	0.9600	0.9601	0.9602	0.9603	0.9604	0.9605	0.9605	0.9606	0.9607
1.76	0.9608	0.9609	0.9610	0.9610	0.9611	0.9612	0.9613	0.9614	0.9615	0.9616
1.77	0.9616	0.9617	0.9618	0.9619	0.9620	0.9621	0.9621	0.9622	0.9623	0.9624
1.78	0.9625	0.9625	0.9626	0.9627	0.9628	0.9629	0.9630	0.9630	0.9631	0.9632
1.79	0.9633	0.9634	0.9634	0.9635	0.9636	0.9637	0.9638	0.9638	0.9639	0.9640
1.80	0.9641	0.9641	0.9642	0.9643	0.9644	0.9645	0.9645	0.9646	0.9647	0.9648
1.81	0.9649	0.9649	0.9650	0.9651	0.9652	0.9652	0.9653	0.9654	0.9655	0.9655
1.82	0.9656	0.9657	0.9658	0.9658	0.9659	0.9660	0.9661	0.9662	0.9662	0.9663
1.83	0.9664	0.9664	0.9665	0.9666	0.9667	0.9667	0.9668	0.9669	0.9670	0.9670
1.84	0.9671	0.9672	0.9673	0.9673	0.9674	0.9675	0.9676	0.9676	0.9677	0.9678
1.85	0.9678	0.9679	0.9680	0.9681	0.9681	0.9682	0.9683	0.9683	0.9684	0.9685
1.86	0.9686	0.9686	0.9687	0.9688	0.9688	0.9689	0.9690	0.9690	0.9691	0.9692
1.87	0.9693	0.9693	0.9694	0.9695	0.9695	0.9696	0.9697	0.9697	0.9698	0.9699
1.88	0.9699	0.9700	0.9701	0.9701	0.9702	0.9703	0.9704	0.9704	0.9705	0.9706
1.89	0.9706	0.9707	0.9708	0.9708	0.9709	0.9710	0.9710	0.9711	0.9712	0.9712

Standard Normal Distribution - Cumulative Distribution Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
1.90	0.9713	0.9713	0.9714	0.9715	0.9715	0.9716	0.9717	0.9717	0.9718	0.9719
1.91	0.9719	0.9720	0.9721	0.9721	0.9722	0.9723	0.9723	0.9724	0.9724	0.9725
1.92	0.9726	0.9726	0.9727	0.9728	0.9728	0.9729	0.9729	0.9730	0.9731	0.9731
1.93	0.9732	0.9733	0.9733	0.9734	0.9734	0.9735	0.9736	0.9736	0.9737	0.9737
1.94	0.9738	0.9739	0.9739	0.9740	0.9741	0.9741	0.9742	0.9742	0.9743	0.9744
1.95	0.9744	0.9745	0.9745	0.9746	0.9746	0.9747	0.9748	0.9748	0.9749	0.9749
1.96	0.9750	0.9751	0.9751	0.9752	0.9752	0.9753	0.9754	0.9754	0.9755	0.9755
1.97	0.9756	0.9756	0.9757	0.9758	0.9758	0.9759	0.9759	0.9760	0.9760	0.9761
1.98	0.9761	0.9762	0.9763	0.9763	0.9764	0.9764	0.9765	0.9765	0.9766	0.9766
1.99	0.9767	0.9768	0.9768	0.9769	0.9769	0.9770	0.9770	0.9771	0.9771	0.9772
2.00	0.9772	0.9773	0.9774	0.9774	0.9775	0.9775	0.9776	0.9776	0.9777	0.9777
2.01	0.9778	0.9778	0.9779	0.9779	0.9780	0.9780	0.9781	0.9782	0.9782	0.9783
2.02	0.9783	0.9784	0.9784	0.9785	0.9785	0.9786	0.9786	0.9787	0.9787	0.9788
2.03	0.9788	0.9789	0.9789	0.9790	0.9790	0.9791	0.9791	0.9792	0.9792	0.9793
2.04	0.9793	0.9794	0.9794	0.9795	0.9795	0.9796	0.9796	0.9797	0.9797	0.9798
2.05	0.9798	0.9799	0.9799	0.9800	0.9800	0.9801	0.9801	0.9802	0.9802	0.9803
2.06	0.9803	0.9803	0.9804	0.9804	0.9805	0.9805	0.9806	0.9806	0.9807	0.9807
2.07	0.9808	0.9808	0.9809	0.9809	0.9810	0.9810	0.9811	0.9811	0.9811	0.9812
2.08	0.9812	0.9813	0.9813	0.9814	0.9814	0.9815	0.9815	0.9816	0.9816	0.9816
2.09	0.9817	0.9817	0.9818	0.9818	0.9819	0.9819	0.9820	0.9820	0.9820	0.9821
2.10	0.9821	0.9822	0.9822	0.9823	0.9823	0.9824	0.9824	0.9824	0.9825	0.9825
2.11	0.9826	0.9826	0.9827	0.9827	0.9827	0.9828	0.9828	0.9829	0.9829	0.9830
2.12	0.9830	0.9830	0.9831	0.9831	0.9832	0.9832	0.9832	0.9833	0.9833	0.9834
2.13	0.9834	0.9835	0.9835	0.9835	0.9836	0.9836	0.9837	0.9837	0.9837	0.9838
2.14	0.9838	0.9839	0.9839	0.9839	0.9840	0.9840	0.9841	0.9841	0.9841	0.9842
2.15	0.9842	0.9843	0.9843	0.9843	0.9844	0.9844	0.9845	0.9845	0.9845	0.9846
2.16	0.9846	0.9847	0.9847	0.9847	0.9848	0.9848	0.9848	0.9849	0.9849	0.9850
2.17	0.9850	0.9850	0.9851	0.9851	0.9851	0.9852	0.9852	0.9853	0.9853	0.9853
2.18	0.9854	0.9854	0.9854	0.9855	0.9855	0.9856	0.9856	0.9856	0.9857	0.9857
2.19	0.9857	0.9858	0.9858	0.9858	0.9859	0.9859	0.9860	0.9860	0.9860	0.9861
2.20	0.9861	0.9861	0.9862	0.9862	0.9862	0.9863	0.9863	0.9863	0.9864	0.9864
2.21	0.9864	0.9865	0.9865	0.9866	0.9866	0.9866	0.9867	0.9867	0.9867	0.9868
2.22	0.9868	0.9868	0.9869	0.9869	0.9869	0.9870	0.9870	0.9870	0.9871	0.9871
2.23	0.9871	0.9872	0.9872	0.9872	0.9873	0.9873	0.9873	0.9874	0.9874	0.9874
2.24	0.9875	0.9875	0.9875	0.9876	0.9876	0.9876	0.9876	0.9877	0.9877	0.9877
2.25	0.9878	0.9878	0.9878	0.9879	0.9879	0.9879	0.9880	0.9880	0.9880	0.9881
2.26	0.9881	0.9881	0.9882	0.9882	0.9882	0.9882	0.9883	0.9883	0.9883	0.9884
2.27	0.9884	0.9884	0.9885	0.9885	0.9885	0.9885	0.9886	0.9886	0.9886	0.9887
2.28	0.9887	0.9887	0.9888	0.9888	0.9888	0.9888	0.9889	0.9889	0.9889	0.9890
2.29	0.9890	0.9890	0.9890	0.9891	0.9891	0.9891	0.9892	0.9892	0.9892	0.9892

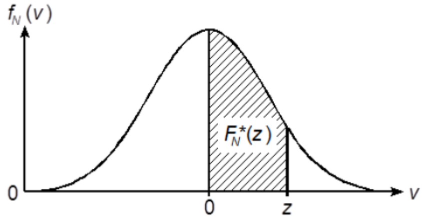
Standard Normal Distribution - Cumulative Distribution Function

z	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
2.30	0.9893	0.9893	0.9893	0.9894	0.9894	0.9894	0.9894	0.9895	0.9895	0.9895
2.31	0.9896	0.9896	0.9896	0.9896	0.9897	0.9897	0.9897	0.9897	0.9898	0.9898
2.32	0.9898	0.9899	0.9899	0.9899	0.9899	0.9900	0.9900	0.9900	0.9900	0.9901
2.33	0.9901	0.9901	0.9901	0.9902	0.9902	0.9902	0.9903	0.9903	0.9903	0.9903
2.34	0.9904	0.9904	0.9904	0.9904	0.9905	0.9905	0.9905	0.9905	0.9906	0.9906
2.35	0.9906	0.9906	0.9907	0.9907	0.9907	0.9907	0.9908	0.9908	0.9908	0.9908
2.36	0.9909	0.9909	0.9909	0.9909	0.9910	0.9910	0.9910	0.9910	0.9911	0.9911
2.37	0.9911	0.9911	0.9912	0.9912	0.9912	0.9912	0.9912	0.9913	0.9913	0.9913
2.38	0.9913	0.9914	0.9914	0.9914	0.9914	0.9915	0.9915	0.9915	0.9915	0.9916
2.39	0.9916	0.9916	0.9916	0.9916	0.9917	0.9917	0.9917	0.9917	0.9918	0.9918
2.40	0.9918	0.9918	0.9918	0.9919	0.9919	0.9919	0.9919	0.9920	0.9920	0.9920
2.41	0.9920	0.9920	0.9921	0.9921	0.9921	0.9921	0.9922	0.9922	0.9922	0.9922
2.42	0.9922	0.9923	0.9923	0.9923	0.9923	0.9923	0.9924	0.9924	0.9924	0.9924
2.43	0.9925	0.9925	0.9925	0.9925	0.9925	0.9926	0.9926	0.9926	0.9926	0.9926
2.44	0.9927	0.9927	0.9927	0.9927	0.9927	0.9928	0.9928	0.9928	0.9928	0.9928
2.45	0.9929	0.9929	0.9929	0.9929	0.9929	0.9930	0.9930	0.9930	0.9930	0.9930
2.46	0.9931	0.9931	0.9931	0.9931	0.9931	0.9931	0.9932	0.9932	0.9932	0.9932
2.47	0.9932	0.9933	0.9933	0.9933	0.9933	0.9933	0.9934	0.9934	0.9934	0.9934
2.48	0.9934	0.9934	0.9935	0.9935	0.9935	0.9935	0.9935	0.9936	0.9936	0.9936
2.49	0.9936	0.9936	0.9936	0.9937	0.9937	0.9937	0.9937	0.9937	0.9938	0.9938
2.50	0.9938	0.9938	0.9938	0.9938	0.9939	0.9939	0.9939	0.9939	0.9939	0.9939
2.60	0.9953	0.9954	0.9954	0.9954	0.9954	0.9954	0.9954	0.9954	0.9954	0.9955
2.70	0.9965	0.9965	0.9966	0.9966	0.9966	0.9966	0.9966	0.9966	0.9966	0.9966
2.80	0.9974	0.9975	0.9975	0.9975	0.9975	0.9975	0.9975	0.9975	0.9975	0.9975
2.90	0.9981	0.9981	0.9981	0.9982	0.9982	0.9982	0.9982	0.9982	0.9982	0.9982
3.00	0.9987	0.9987	0.9987	0.9987	0.9987	0.9987	0.9987	0.9987	0.9987	0.9987
3.10	0.9990	0.9990	0.9990	0.9990	0.9990	0.9990	0.9991	0.9991	0.9991	0.9991
3.20	0.9993	0.9993	0.9993	0.9993	0.9993	0.9993	0.9993	0.9993	0.9993	0.9993
3.30	0.9995	0.9995	0.9995	0.9995	0.9995	0.9995	0.9995	0.9995	0.9995	0.9995
3.40	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997
3.50	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.60	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.70	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.80	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999

Standard Normal Distribution - One-sided Confidence Intervals

The following applies:

$$
\begin{aligned}
F_N^*(z) &= F_N(z) - 0.5 \\
&= F_N^*(-z) \\
F_N^*(0) &= 0 \\
0 \leq z &< +\infty
\end{aligned}
$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986

Standard Normal Distribution - One-sided Confidence Intervals

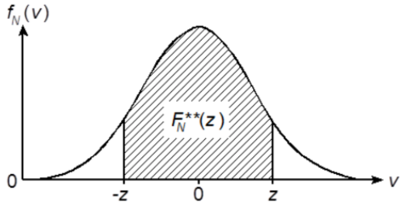
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

Standard Normal Distribution - Two-sided, Symmetric Confidence Intervals

The following applies:

$$\begin{aligned} F_N^{**}(z) &= F_N(z) - F_N(-z) \\ &= 2F_N(z) - 1 \\ &= 2F_N^{*}(z) \end{aligned}$$

$-\infty < z < +\infty$



$F_N^{*}(z)$ see also standard normal distribution, one-sided confidence intervals.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0080	0.0160	0.0239	0.0319	0.0399	0.0478	0.0558	0.0638	0.0717
0.1	0.0797	0.0876	0.0955	0.1034	0.1113	0.1192	0.1271	0.1350	0.1428	0.1507
0.2	0.1585	0.1663	0.1741	0.1819	0.1897	0.1974	0.2051	0.2128	0.2205	0.2282
0.3	0.2358	0.2434	0.2510	0.2586	0.2661	0.2737	0.2812	0.2886	0.2961	0.3035
0.4	0.3108	0.3182	0.3255	0.3328	0.3401	0.3473	0.3545	0.3616	0.3688	0.3759
0.5	0.3829	0.3899	0.3969	0.4039	0.4108	0.4177	0.4245	0.4313	0.4381	0.4448
0.6	0.4515	0.4581	0.4647	0.4713	0.4778	0.4843	0.4907	0.4971	0.5035	0.5098
0.7	0.5161	0.5223	0.5285	0.5346	0.5407	0.5467	0.5527	0.5587	0.5646	0.5705
0.8	0.5763	0.5821	0.5878	0.5935	0.5991	0.6047	0.6102	0.6157	0.6211	0.6265
0.9	0.6319	0.6372	0.6424	0.6476	0.6528	0.6579	0.6629	0.6680	0.6729	0.6778
1.0	0.6827	0.6875	0.6923	0.6970	0.7017	0.7063	0.7109	0.7154	0.7199	0.7243
1.1	0.7287	0.7330	0.7373	0.7415	0.7457	0.7499	0.7540	0.7580	0.7620	0.7660
1.2	0.7699	0.7737	0.7775	0.7813	0.7850	0.7887	0.7923	0.7959	0.7995	0.8029
1.3	0.8064	0.8098	0.8132	0.8165	0.8198	0.8230	0.8262	0.8293	0.8324	0.8355
1.4	0.8385	0.8415	0.8444	0.8473	0.8501	0.8529	0.8557	0.8584	0.8611	0.8638
1.5	0.8664	0.8690	0.8715	0.8740	0.8764	0.8789	0.8812	0.8836	0.8859	0.8882
1.6	0.8904	0.8926	0.8948	0.8969	0.8990	0.9011	0.9031	0.9051	0.9070	0.9090
1.7	0.9109	0.9127	0.9146	0.9164	0.9181	0.9199	0.9216	0.9233	0.9249	0.9265
1.8	0.9281	0.9297	0.9312	0.9328	0.9342	0.9357	0.9371	0.9385	0.9399	0.9412
1.9	0.9426	0.9439	0.9451	0.9464	0.9476	0.9488	0.9500	0.9512	0.9523	0.9534
2.0	0.9545	0.9556	0.9566	0.9576	0.9586	0.9596	0.9606	0.9615	0.9625	0.9634
2.1	0.9643	0.9651	0.9660	0.9668	0.9676	0.9684	0.9692	0.9700	0.9707	0.9715
2.2	0.9722	0.9729	0.9736	0.9743	0.9749	0.9756	0.9762	0.9768	0.9774	0.9780
2.3	0.9786	0.9791	0.9797	0.9802	0.9807	0.9812	0.9817	0.9822	0.9827	0.9832
2.4	0.9836	0.9840	0.9845	0.9849	0.9853	0.9857	0.9861	0.9865	0.9869	0.9872
2.5	0.9876	0.9879	0.9883	0.9886	0.9889	0.9892	0.9895	0.9898	0.9901	0.9904
2.6	0.9907	0.9909	0.9912	0.9915	0.9917	0.9920	0.9922	0.9924	0.9926	0.9929
2.7	0.9931	0.9933	0.9935	0.9937	0.9939	0.9940	0.9942	0.9944	0.9946	0.9947
2.8	0.9949	0.9950	0.9952	0.9953	0.9955	0.9956	0.9958	0.9959	0.9960	0.9961
2.9	0.9963	0.9964	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972

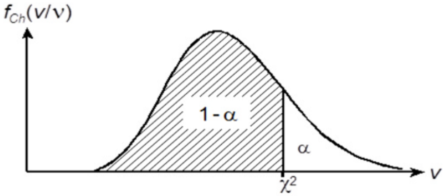
Standard Normal Distribution - Two-sided, Symmetric Confidence Intervals

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
3.0	0.9973	0.9974	0.9975	0.9976	0.9976	0.9977	0.9978	0.9979	0.9979	0.9980
3.1	0.9981	0.9981	0.9982	0.9983	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986
3.2	0.9986	0.9987	0.9987	0.9988	0.9988	0.9988	0.9989	0.9989	0.9990	0.9990
3.3	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.4	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995	0.9995
3.5	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997	0.9997
3.6	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	0.9998	0.9998
3.7	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999

Chi-square Distribution - Distribution Function

The corresponding values of χ^2 for the parameters v and $(1-a)$ in form of a distribution function are given.

The following is applicable for χ^2:



$$W(0 < X^2 \leq \chi^2) = F_{CH}\left(\frac{\chi^2}{\nu}\right)$$
$$= 1-\alpha$$

with $X^2 =$ random variable

v	$1-\alpha$								
	0.001	**0.005**	**0.010**	**0.025**	**0.050**	**0.100**	**0.250**	**0.400**	**0.500**
1	0.000	0.000	0.000	0.001	0.004	0.016	0.102	0.275	0.455
2	0.002	0.010	0.020	0.051	0.103	0.211	0.575	1.022	1.386
3	0.024	0.072	0.115	0.216	0.352	0.5840	1.213	1.869	2.366
4	0.091	0.207	0.297	0.484	0.711	1.064	1.923	2.753	3.357
5	0.210	0.412	0.554	0.831	1.145	1.610	2.675	3.655	4.351
6	0.381	0.676	0.872	1.237	1.635	2.204	3.455	4.570	5.348
7	0.598	0.989	1.239	1.690	2.167	2.833	4.255	5.493	6.346
8	0.857	1.344	1.646	2.180	2.733	3.490	5.071	6.423	7.344
9	1.152	1.735	2.088	2.700	3.325	4.168	5.899	7.357	8.343
10	1.479	2.156	2.558	3.247	3.940	4.865	6.737	8.295	9.342
11	1.834	2.603	3.053	3.816	4.575	5.578	7.584	9.237	10.341
12	2.214	3.074	3.571	4.404	5.226	6.304	8.438	10.182	11.340
13	2.617	3.565	4.107	5.009	5.892	7.042	9.299	11.129	12.340
14	3.041	4.075	4.660	5.629	6.571	7.790	10.165	12.078	13.339
15	3.483	4.601	5.229	6.262	7.261	8.547	11.037	13.030	14.339
16	3.942	5.142	5.812	6.908	7.962	9.312	11.912	13.983	15.338
17	4.416	5.697	6.408	7.564	8.672	10.085	12.792	14.937	16.338
18	4.905	6.265	7.015	8.231	9.390	10.865	13.675	15.893	17.338
19	5.407	6.844	7.633	8.907	10.117	11.651	14.562	16.850	18.338
20	5.921	7.434	8.260	9.591	10.851	12.443	15.452	17.809	19.337
21	6.447	8.034	8.897	10.283	11.591	13.240	16.344	18.768	20.337
22	6.983	8.643	9.542	10.982	12.338	14.041	17.240	19.729	21.337
23	7.529	9.260	10.196	11.689	13.091	14.848	18.137	20.690	22.337
24	8.085	9.886	10.856	12.401	13.848	15.659	19.037	21.652	23.337
25	8.649	10.520	11.524	13.120	14.611	16.473	19.939	22.616	24.337

Chi-square Distribution - Distribution Function

ν	$1-\alpha$								
	0.001	**0.005**	**0.010**	**0.025**	**0.050**	**0.100**	**0.250**	**0.400**	**0.500**
30	11.588	13.787	14.953	16.791	18.493	20.599	24.478	27.442	29.336
35	14.688	17.192	18.509	20.569	22.465	24.797	29.054	32.282	34.336
40	17.916	20.707	22.164	24.433	26.509	29.051	33.660	37.134	39.335
45	21.251	24.311	25.901	28.366	30.612	33.350	38.291	41.995	44.335
50	24.674	27.991	29.707	32.357	34.764	37.689	42.942	46.864	49.335
60	31.738	35.534	37.485	40.482	43.188	46.459	52.294	56.620	59.335
70	39.036	43.275	45.442	48.758	51.739	55.329	61.698	66.396	69.334
80	46.520	51.172	53.540	57.153	60.391	64.278	71.145	76.188	79.334
90	54.155	59.196	61.754	65.647	69.126	73.291	80.625	85.993	89.334
100	61.918	67.328	70.065	74.222	77.929	82.358	90.133	95.808	99.334

Chi-square Distribution - Distribution Function

ν	$1-\alpha$								
	0.600	**0.750**	**0.800**	**0.850**	**0.900**	**0.950**	**0.975**	**0.980**	**0.990**
1	0.708	1.323	1.642	2.072	2.706	3.841	5.024	5.412	6.635
2	1.833	2.773	3.219	3.794	4.605	5.991	7.378	7.824	9.210
3	2.946	4.108	4.642	5.317	6.251	7.815	9.348	9.837	11.345
4	4.045	5.385	5.989	6.745	7.779	9.488	11.143	11.668	13.277
5	5.132	6.626	7.289	8.115	9.236	11.070	12.833	13.388	15.086
6	6.211	7.841	8.558	9.446	10.645	12.592	14.449	15.033	16.812
7	7.283	9.037	9.803	10.748	12.017	14.067	16.013	16.622	18.475
8	8.351	10.219	11.030	12.027	13.362	15.507	17.535	18.168	20.090
9	9.414	11.389	12.242	13.288	14.684	16.919	19.023	19.679	21.666
10	10.473	12.549	13.442	14.534	15.987	18.307	20.483	21.161	23.209
11	11.530	13.701	14.631	15.767	17.275	19.675	21.920	22.618	24.725
12	12.584	14.845	15.812	16.989	18.549	21.026	23.337	24.054	26.217
13	13.636	15.984	16.985	18.202	19.812	22.362	24.736	25.472	27.688
14	14.685	17.117	18.151	19.406	21.064	23.685	26.119	26.873	29.141
15	15.733	18.245	19.311	20.603	22.307	24.996	27.488	28.259	30.578
16	16.780	19.369	20.465	21.793	23.542	26.296	28.845	29.633	32.000
17	17.824	20.489	21.615	22.977	24.769	27.587	30.191	30.995	33.409
18	18.868	21.605	22.760	24.155	25.989	28.869	31.526	32.346	34.805
19	19.910	22.718	23.900	25.329	27.204	30.144	32.852	33.687	36.191
20	20.951	23.828	25.038	26.498	28.412	31.410	34.170	35.020	37.566
21	21.991	24.935	26.171	27.662	29.615	32.671	35.479	36.343	38.932
22	23.031	26.039	27.301	28.822	30.813	33.924	36.781	37.659	40.289
23	24.069	27.141	28.429	29.979	32.007	35.172	38.076	38.968	41.638
24	25.106	28.241	29.553	31.132	33.196	36.415	39.364	40.270	42.980
25	26.143	29.339	30.675	32.282	34.382	37.652	40.646	41.566	44.314
30	31.316	34.800	36.250	37.990	40.256	43.773	46.979	47.962	50.892
35	36.475	40.223	41.778	43.640	46.059	49.802	53.203	54.244	57.342
40	41.622	45.616	47.269	49.244	51.805	55.758	59.342	60.436	63.691
45	46.761	50.985	52.729	54.810	57.505	61.656	65.410	66.555	69.957
50	51.892	56.334	58.164	60.346	63.167	67.505	71.420	72.613	76.154
60	62.135	66.981	68.972	71.341	74.397	79.082	83.298	84.580	88.380
70	72.358	77.577	79.715	82.255	85.527	90.530	95.020	96.390	100.430
80	82.566	88.130	90.410	93.110	96.580	101.880	106.630	108.070	112.330
90	92.760	98.650	101.050	103.900	107.570	113.150	118.140	119.650	124.120
100	102.950	109.140	111.670	114.660	118.500	124.340	129.560	131.140	135.810

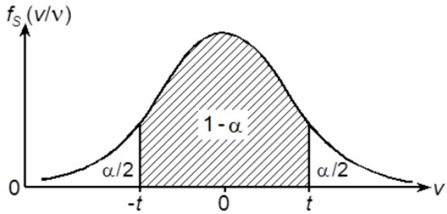
Student's t-Distribution - Two-sided, Symmetric Confidence Intervals

The corresponding values of t for the parameters v and $(1-a)$ in form of two-sided, symmetric confidence intervals are given.

The following is applicable for t:

$W(-t < T \leq t) = 1 - \alpha$

with T = random variable



v	$1-\alpha$								
	0.100	**0.150**	**0.200**	**0.250**	**0.300**	**0.350**	**0.400**	**0.450**	**0.500**
1	0.158	0.240	0.325	0.414	0.510	0.613	0.727	0.854	1.000
2	0.142	0.215	0.289	0.365	0.445	0.528	0.617	0.713	0.816
3	0.137	0.206	0.277	0.349	0.424	0.502	0.584	0.671	0.765
4	0.134	0.202	0.271	0.341	0.414	0.490	0.569	0.652	0.741
5	0.132	0.199	0.267	0.337	0.408	0.482	0.559	0.641	0.727
6	0.131	0.197	0.265	0.334	0.404	0.477	0.553	0.633	0.718
7	0.130	0.196	0.263	0.331	0.402	0.474	0.549	0.628	0.711
8	0.130	0.195	0.262	0.330	0.399	0.471	0.546	0.624	0.706
9	0.129	0.195	0.261	0.329	0.398	0.469	0.543	0.621	0.703
10	0.129	0.194	0.260	0.328	0.397	0.468	0.542	0.619	0.700
11	0.129	0.194	0.260	0.327	0.396	0.466	0.540	0.617	0.697
12	0.128	0.193	0.259	0.326	0.395	0.465	0.539	0.615	0.695
13	0.128	0.193	0.259	0.325	0.394	0.464	0.538	0.614	0.694
14	0.128	0.193	0.258	0.325	0.393	0.464	0.537	0.613	0.692
15	0.128	0.192	0.258	0.325	0.393	0.463	0.536	0.612	0.691
16	0.128	0.192	0.258	0.324	0.392	0.462	0.535	0.611	0.690
17	0.128	0.192	0.257	0.324	0.392	0.462	0.534	0.610	0.689
18	0.127	0.192	0.257	0.324	0.392	0.461	0.534	0.609	0.688
19	0.127	0.192	0.257	0.323	0.391	0.461	0.533	0.609	0.688
20	0.127	0.192	0.257	0.323	0.391	0.461	0.533	0.608	0.687
21	0.127	0.191	0.257	0.323	0.391	0.460	0.532	0.608	0.686
22	0.127	0.191	0.256	0.323	0.390	0.460	0.532	0.607	0.686
23	0.127	0.191	0.256	0.322	0.390	0.460	0.532	0.607	0.685
24	0.127	0.191	0.256	0.322	0.390	0.460	0.531	0.606	0.685
25	0.127	0.191	0.256	0.322	0.390	0.459	0.531	0.606	0.684

Student's t-Distribution - Two-sided, Symmetric Confidence Intervals

ν	$1-\alpha$								
	0.100	**0.150**	**0.200**	**0.250**	**0.300**	**0.350**	**0.400**	**0.450**	**0.500**
26	0.127	0.191	0.256	0.322	0.390	0.459	0.531	0.606	0.684
27	0.127	0.191	0.256	0.322	0.389	0.459	0.531	0.605	0.684
28	0.127	0.191	0.256	0.322	0.389	0.459	0.530	0.605	0.683
29	0.127	0.191	0.256	0.322	0.389	0.459	0.530	0.605	0.683
30	0.127	0.191	0.256	0.322	0.389	0.458	0.530	0.605	0.683
40	0.126	0.190	0.255	0.321	0.388	0.457	0.529	0.603	0.681
50	0.126	0.190	0.255	0.320	0.388	0.457	0.528	0.602	0.679
100	0.126	0.190	0.254	0.320	0.386	0.455	0.526	0.600	0.677
150	0.126	0.189	0.254	0.319	0.386	0.455	0.526	0.599	0.676
∞	0.126	0.189	0.253	0.319	0.385	0.454	0.524	0.598	0.675

Student's t-Distribution - Two-sided, Symmetric Confidence Intervals

ν	$1-\alpha$								
	0.600	**0.700**	**0.800**	**0.850**	**0.900**	**0.950**	**0.975**	**0.990**	**0.995**
1	1.376	1.963	3.078	4.165	6.314	12.706	25.452	63.657	127.321
2	1.061	1.386	1.886	2.282	2.920	4.303	6.205	9.925	14.089
3	0.978	1.250	1.638	1.924	2.353	3.182	4.177	5.841	7.453
4	0.941	1.190	1.533	1.778	2.132	2.776	3.495	4.604	5.598
5	0.920	1.156	1.476	1.699	2.015	2.571	3.163	4.032	4.773
6	0.906	1.134	1.440	1.650	1.943	2.447	2.969	3.707	4.317
7	0.896	1.119	1.415	1.617	1.895	2.365	2.841	3.499	4.029
8	0.889	1.108	1.397	1.592	1.860	2.306	2.752	3.355	3.833
9	0.883	1.100	1.383	1.574	1.833	2.262	2.685	3.250	3.690
10	0.879	1.093	1.372	1.559	1.812	2.228	2.634	3.169	3.581
11	0.876	1.088	1.363	1.548	1.796	2.201	2.593	3.106	3.497
12	0.873	1.083	1.356	1.538	1.782	2.179	2.560	3.055	3.428
13	0.870	1.079	1.350	1.530	1.771	2.160	2.533	3.012	3.372
14	0.868	1.076	1.345	1.523	1.761	2.145	2.510	2.977	3.326
15	0.866	1.074	1.341	1.517	1.753	2.131	2.490	2.947	3.286
16	0.865	1.071	1.337	1.512	1.746	2.120	2.473	2.921	3.252
17	0.863	1.069	1.333	1.508	1.740	2.110	2.458	2.898	3.222
18	0.862	1.067	1.330	1.504	1.734	2.101	2.445	2.878	3.197
19	0.861	1.066	1.328	1.500	1.729	2.093	2.433	2.861	3.174
20	0.860	1.064	1.325	1.497	1.725	2.086	2.423	2.845	3.153
21	0.859	1.063	1.323	1.494	1.721	2.080	2.414	2.831	3.135
22	0.858	1.061	1.321	1.492	1.717	2.074	2.405	2.819	3.119
23	0.858	1.060	1.319	1.489	1.714	2.069	2.398	2.807	3.104
24	0.857	1.059	1.318	1.487	1.711	2.064	2.391	2.797	3.091
25	0.856	1.058	1.316	1.485	1.708	2.060	2.385	2.787	3.078
26	0.856	1.058	1.315	1.483	1.706	2.056	2.379	2.779	3.067
27	0.855	1.057	1.314	1.482	1.703	2.052	2.373	2.771	3.057
28	0.855	1.056	1.313	1.480	1.701	2.048	2.368	2.763	3.047
29	0.854	1.055	1.311	1.479	1.699	2.045	2.364	2.756	3.038
30	0.854	1.055	1.310	1.477	1.697	2.042	2.360	2.750	3.030
40	0.851	1.050	1.303	1.468	1.684	2.021	2.329	2.704	2.971
50	0.849	1.047	1.299	1.462	1.676	2.009	2.311	2.678	2.937
100	0.845	1.042	1.290	1.451	1.660	1.984	2.276	2.626	2.871
150	0.844	1.040	1.287	1.447	1.655	1.976	2.264	2.609	2.849
∞	0.842	1.036	1.282	1.440	1.645	1.960	2.242	2.576	2.808

Student's t-Distribution - Distribution Function

The corresponding values of t for the parameters v and $(1-a)$ in form of a distribution function are given.

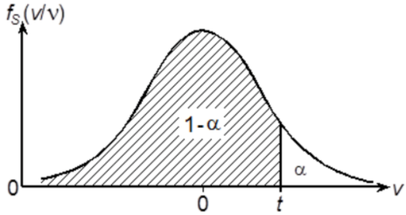
The following is applicable for t:

$$W(-\infty < T \le t) = F_S\left(\frac{t}{v}\right) = 1-\alpha$$

with $T =$ random variable

The following applies:

$$F_S\left(\frac{-t}{v}\right) = 1 - F_S\left(\frac{t}{v}\right)$$



v	$1-\alpha$								
	0.600	**0.700**	**0.800**	**0.900**	**0.950**	**0.975**	**0.990**	**0.995**	**0.999**
1	0.325	0.727	1.376	3.078	6.314	12.706	31.821	63.657	318.309
2	0.289	0.617	1.061	1.886	2.920	4.303	6.965	9.925	22.327
3	0.277	0.584	0.978	1.638	2.353	3.182	4.541	5.841	10.215
4	0.271	0.569	0.941	1.533	2.132	2.776	3.747	4.604	7.173
5	0.267	0.559	0.920	1.476	2.015	2.571	3.365	4.032	5.893
6	0.265	0.553	0.906	1.440	1.943	2.447	3.143	3.707	5.208
7	0.263	0.549	0.896	1.415	1.895	2.365	2.998	3.499	4.785
8	0.262	0.546	0.889	1.397	1.860	2.306	2.896	3.355	4.501
9	0.261	0.543	0.883	1.383	1.833	2.262	2.821	3.250	4.297
10	0.260	0.542	0.879	1.372	1.812	2.228	2.764	3.169	4.144
11	0.260	0.540	0.876	1.363	1.796	2.201	2.718	3.106	4.025
12	0.259	0.539	0.873	1.356	1.782	2.179	2.681	3.055	3.930
13	0.259	0.538	0.870	1.350	1.771	2.160	2.650	3.012	3.852
14	0.258	0.537	0.868	1.345	1.761	2.145	2.624	2.977	3.787
15	0.258	0.536	0.866	1.341	1.753	2.131	2.602	2.947	3.733
16	0.258	0.535	0.865	1.337	1.746	2.120	2.583	2.921	3.686
17	0.257	0.534	0.863	1.333	1.740	2.110	2.567	2.898	3,646
18	0.257	0.534	0.862	1.330	1.734	2.101	2.552	2.878	3.610
19	0.257	0.533	0.861	1.328	1.729	2.093	2.539	2.861	3.579
20	0.257	0.533	0.860	1.325	1.725	2.086	2.528	2.845	3.552
21	0.257	0.532	0.859	1.323	1.721	2.080	2.518	2.831	3.527
22	0.256	0.532	0.858	1.321	1.717	2.074	2.508	2.819	3.505
23	0.256	0.532	0.858	1.319	1.714	2.069	2.500	2.807	3.485
24	0.256	0.531	0.857	1.318	1.711	2.064	2.492	2.797	3.467
25	0.256	0.531	0.856	1.316	1.708	2.060	2.485	2.787	3.450
30	0.256	0.530	0.854	1.310	1.697	2.042	2.457	2.750	3.385
40	0.255	0.529	0.851	1.303	1.684	2.021	2.423	2.704	3.307
50	0.255	0.528	0.849	1.299	1.676	2.009	2.403	2.678	3.261
100	0.254	0.526	0.845	1.290	1.660	1.984	2.364	2.626	3.174
150	0.254	0.526	0.844	1.287	1.655	1.976	2.351	2.609	3.145
∞	0.253	0.524	0.842	1.282	1.645	1.960	2.326	2.576	3.090

F-Distribution - Distribution Function with $\alpha = 0.05$

The corresponding values of F_c for the parameters ν_1 and ν_2 in form of a F-distribution function with $(1-a) = 0.95$ are given.

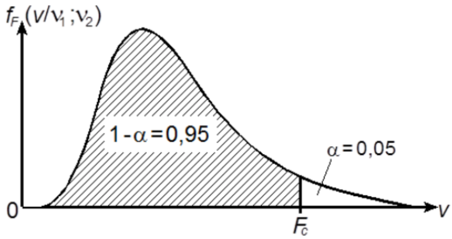
The following is applicable for F_c:

$$W(0 < F \leq F_c) = F\left(\frac{F_c}{\nu_1;\, \nu_2}\right)$$
$$= 1 - \alpha = 0.95$$

with $F =$ random variable

and $F_{\alpha;\, \nu_1;\, \nu_2} = \dfrac{1}{F_{1-\alpha;\, \nu_1;\, \nu_2}}$

with $\nu_1 = n_1 - 1$
and $\nu_2 = n_2 - 1$



ν_2	ν_1										
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	242.98
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.76
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.94
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.70
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.60
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.31
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.10
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.94
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.72
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.51
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.46
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.41
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.34
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.31
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.28
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.26
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.24
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.22
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.20

F-Distribution - Distribution Function with $\alpha = 0.05$

v_2	v_1										
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.17
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.15
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.14
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.13
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.04
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03	1.99
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.95
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07	2.02	1.97	1.93
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95	1.91
90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94	1.90
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.89
150	3.90	3.06	2.66	2.43	2.27	2.16	2.07	2.00	1.94	1.89	1.85
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93	1.88	1.84
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.79

F-Distribution - Distribution Function with $\alpha = 0.05$

ν_2	ν_1										
	12	**13**	**14**	**15**	**20**	**30**	**40**	**50**	**100**	**200**	∞
1	243.91	244.69	245.36	245.95	248.01	250.1	251.14	251.77	253.04	253.68	254.31
2	19.41	19.42	19.42	19.43	19.45	19.46	19.47	19.48	19.49	19.49	19.50
3	8.74	8.73	8.71	8.70	8.66	8.62	8.59	8.58	8.55	8.54	8.53
4	5.91	5.89	5.87	5.86	5.80	5.75	5.72	5.70	5.66	5.65	5.63
5	4.68	4.66	4.64	4.62	4.56	4.50	4.46	4.44	4.41	4.39	4.37
6	4.00	3.98	3.96	3.94	3.87	3.81	3.77	3.75	3.71	3.69	3.67
7	3.57	3.55	3.53	3.51	3.44	3.38	3.34	3.32	3.27	3.25	3.23
8	3.28	3.26	3.24	3.22	3.15	3.08	3.04	3.02	2.97	2.95	2.93
9	3.07	3.05	3.03	3.01	2.94	2.86	2.83	2.80	2.76	2.73	2.71
10	2.91	2.89	2.86	2.85	2.77	2.70	2.66	2.64	2.59	2.56	2.54
11	2.79	2.76	2.74	2.72	2.65	2.57	2.53	2.51	2.46	2.43	2.40
12	2.69	2.66	2.64	2.62	2.54	2.47	2.43	2.40	2.35	2.32	2.30
13	2.60	2.58	2.55	2.53	2.46	2.38	2.34	2.31	2.26	2.23	2.21
14	2.53	2.51	2.48	2.46	2.39	2.31	2.27	2.24	2.19	2.16	2.13
15	2.48	2.45	2.42	2.40	2.33	2.25	2.20	2.18	2.12	2.10	2.07
16	2.42	2.40	2.37	2.35	2.28	2.19	2.15	2.12	2.07	2.04	2.01
17	2.38	2.35	2.33	2.31	2.23	2.15	2.10	2.08	2.02	1.99	1.96
18	2.34	2.31	2.29	2.27	2.19	2.11	2.06	2.04	1.98	1.95	1.92
19	2.31	2.28	2.26	2.23	2.16	2.07	2.03	2.00	1.94	1.91	1.88
20	2.28	2.25	2.22	2.20	2.12	2.04	1.99	1.97	1.91	1.88	1.84
21	2.25	2.22	2.20	2.18	2.10	2.01	1.96	1.94	1.88	1.84	1.81
22	2.23	2.20	2.17	2.15	2.07	1.98	1.94	1.91	1.85	1.82	1.78
23	2.20	2.18	2.15	2.13	2.05	1.96	1.91	1.88	1.82	1.79	1.76
24	2.18	2.15	2.13	2.11	2.03	1.94	1.89	1.86	1.80	1.77	1.73
25	2.16	2.14	2.11	2.09	2.01	1.92	1.87	1.84	1.78	1.75	1.71
26	2.15	2.12	2.09	2.07	1.99	1.90	1.85	1.82	1.76	1.73	1.69
27	2.13	2.10	2.08	2.06	1.97	1.88	1.84	1.81	1.74	1.71	1.67
28	2.12	2.09	2.06	2.04	1.96	1.87	1.82	1.79	1.73	1.69	1.65
29	2.10	2.08	2.05	2.03	1.94	1.85	1.81	1.77	1.71	1.67	1.64
30	2.09	2.06	2.04	2.01	1.93	1.84	1.79	1.76	1.70	1.66	1.62
40	2.00	1.97	1.95	1.92	1.84	1.74	1.69	1.66	1.59	1.55	1.51
50	1.95	1.92	1.89	1.87	1.78	1.69	1.63	1.60	1.52	1.48	1.44
60	1.92	1.89	1.86	1.84	1.75	1.65	1.59	1.56	1.48	1.44	1.39
70	1.89	1.86	1.84	1.81	1.72	1.62	1.57	1.53	1.45	1.40	1.35
80	1.88	1.84	1.82	1.79	1.70	1.60	1.54	1.51	1.43	1.38	1.32
90	1.86	1.83	1.80	1.78	1.69	1.59	1.53	1.49	1.41	1.36	1.30
100	1.85	1.82	1.79	1.77	1.68	1.57	1.52	1.48	1.39	1.34	1.28
150	1.82	1.79	1.76	1.73	1.64	1.54	1.48	1.44	1.34	1.29	1.22
200	1.80	1.77	1.74	1.72	1.62	1.52	1.46	1.41	1.32	1.26	1.19
∞	1.75	1.72	1.69	1.67	1.57	1.46	1.39	1.35	1.24	1.17	1.01

F-Distribution - Distribution Function with $\alpha = 0.01$

The corresponding values of F_c for the parameters ν_1 and ν_2 in form of a F-distribution function with $(1-a) = 0.99$ are given.

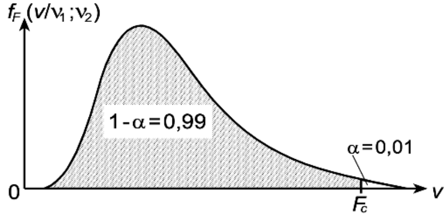
The following is applicable for F_c:

$$W(0 < F \leq F_c) = F\left(\frac{F_c}{\nu_1;\ \nu_2}\right) = 1 - \alpha = 0.99$$

with F = random variable

and $F_{\alpha;\,\nu_1;\,\nu_2} = \dfrac{1}{F_{1-\alpha;\,\nu_1;\,\nu_2}}$

with $\nu_1 = n_1 - 1$
and $\nu_2 = n_2 - 1$



ν_2	ν_1										
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
1	4052	4999	5403	5625	5764	5859	5928	5981	6022	6056	6083
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.41
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.13
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.45
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.96
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.79
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.54
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.73
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.18
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.77
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.46
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.22
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	4.02
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.86
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.73
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.62
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.52
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.43
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.36
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.29
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.24
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.18
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.14
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.09
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	3.06

F-Distribution - Distribution Function with $\alpha = 0.01$

ν_2	ν_1										
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	3.02
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.99
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.96
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.93
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.91
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.73
50	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89	2.78	2.70	2.63
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.56
70	7.01	4.92	4.07	3.60	3.29	3.07	2.91	2.78	2.67	2.59	2.51
80	6.96	4.88	4.04	3.56	3.26	3.04	2.87	2.74	2.64	2.55	2.48
90	6.93	4.85	4.01	3.53	3.23	3.01	2.84	2.72	2.61	2.52	2.45
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50	2.43
150	6.81	4.75	3.91	3.45	3.14	2.92	2.76	2.63	2.53	2.44	2.37
200	6.76	4.71	3.88	3.41	3.11	2.89	2.73	2.60	2.50	2.41	2.34
∞	6.64	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.25

F-Distribution - Distribution Function with $\alpha = 0.01$

ν_2	ν_1										
	12	**13**	**14**	**15**	**20**	**30**	**40**	**50**	**100**	**200**	∞
1	6106	6126	6143	6157	6209	6261	6287	6303	6334	6350	6366
2	99.42	99.42	99.43	99.43	99.45	99.47	99.47	99.48	99.49	99.49	99.50
3	27.05	26.98	26.92	26.87	26.69	26.50	26.41	26.35	26.24	26.18	26.13
4	14.37	14.31	14.25	14.20	14.02	13.84	13.75	13.69	13.58	13.52	13.46
5	9.89	9.82	9.77	9.72	9.55	9.38	9.29	9.24	9.13	9.08	9.02
6	7.72	7.66	7.60	7.56	7.40	7.23	7.14	7.09	6.99	6.93	6.88
7	6.47	6.41	6.36	6.31	6.16	5.99	5.91	5.86	5.75	5.70	5.65
8	5.67	5.61	5.56	5.52	5.36	5.20	5.12	5.07	4.96	4.91	4.86
9	5.11	5.05	5.01	4.96	4.81	4.65	4.57	4.52	4.41	4.36	4.31
10	4.71	4.65	4.60	4.56	4.41	4.25	4.17	4.12	4.01	3.96	3.91
11	4.40	4.34	4.29	4.25	4.10	3.94	3.86	3.81	3.71	3.66	3.60
12	4.16	4.10	4.05	4.01	3.86	3.70	3.62	3.57	3.47	3.41	3.36
13	3.96	3.91	3.86	3.82	3.66	3.51	3.43	3.38	3.27	3.22	3.17
14	3.80	3.75	3.70	3.66	3.51	3.35	3.27	3.22	3.11	3.06	3.00
15	3.67	3.61	3.56	3.52	3.37	3.21	3.13	3.08	2.98	2.92	2.87
16	3.55	3.50	3.45	3.41	3.26	3.10	3.02	2.97	2.86	2.81	2.75
17	3.46	3.40	3.35	3.31	3.16	3.00	2.92	2.87	2.76	2.71	2.65
18	3.37	3.32	3.27	3.23	3.08	2.92	2.84	2.78	2.68	2.62	2.57
19	3.30	3.24	3.19	3.15	3.00	2.84	2.76	2.71	2.60	2.55	2.49
20	3.23	3.18	3.13	3.09	2.94	2.78	2.69	2.64	2.54	2.48	2.42
21	3.17	3.12	3.07	3.03	2.88	2.72	2.64	2.58	2.48	2.42	2.36
22	3.12	3.07	3.02	2.98	2.83	2.67	2.58	2.53	2.42	2.36	2.31
23	3.07	3.02	2.97	2.93	2.78	2.62	2.54	2.48	2.37	2.32	2.26
24	3.03	2.98	2.93	2.89	2.74	2.58	2.49	2.44	2.33	2.27	2.21
25	2.99	2.94	2.89	2.85	2.70	2.54	2.45	2.40	2.29	2.23	2.17
26	2.96	2.90	2.86	2.81	2.66	2.50	2.42	2.36	2.25	2.19	2.13
27	2.93	2.87	2.82	2.78	2.63	2.47	2.38	2.33	2.22	2.16	2.10
28	2.90	2.84	2.79	2.75	2.60	2.44	2.35	2.30	2.19	2.13	2.06
29	2.87	2.81	2.77	2.73	2.57	2.41	2.33	2.27	2.16	2.10	2.03
30	2.84	2.79	2.74	2.70	2.55	2.39	2.30	2.25	2.13	2.07	2.01
40	2.66	2.61	2.56	2.52	2.37	2.20	2.11	2.06	1.94	1.87	1.80
50	2.56	2.51	2.46	2.42	2.27	2.10	2.01	1.95	1.82	1.76	1.68
60	2.50	2.44	2.39	2.35	2.20	2.03	1.94	1.88	1.75	1.68	1.60
70	2.45	2.40	2.35	2.31	2.15	1.98	1.89	1.83	1.70	1.62	1.54
80	2.42	2.36	2.31	2.27	2.12	1.94	1.85	1.79	1.65	1.58	1.49
90	2.39	2.33	2.29	2.24	2.09	1.92	1.82	1.76	1.62	1.55	1.46
100	2.37	2.31	2.27	2.22	2.07	1.89	1.80	1.74	1.60	1.52	1.43
150	2.31	2.25	2.20	2.16	2.00	1.83	1.73	1.66	1.52	1.43	1.33
200	2.27	2.22	2.17	2.13	1.97	1.79	1.69	1.63	1.48	1.39	1.28
∞	2.18	2.13	2.08	2.04	1.88	1.70	1.59	1.52	1.36	1.25	1.01

Appendix B
Bibliography

F. W. Peren, *Statistics for Business and Economics*,
https://doi.org/10.1007/978-3-662-64276-4

[1] Arnold, V.I. (2001): Gewöhnliche Differentialgleichungen, 2nd ed., Berlin, Heidelberg & New York.

[2] Asser, G. (1988): Grundbegriffe der Mathematik - Mengen, Abbildungen, natürliche Zahlen, 5th ed., Leipzig.

[3] Bamberg, G. & Baur, F. (2007): Statistik, Munich & Vienna.

[4] Banerjee, K.S. (1977): On the Factorial Approach Providing the True Cost of Living Index, Göttingen.

[5] Bär, G. (2001): Geometrie, 2nd ed., Stuttgart, Leipzig & Wiesbaden.

[6] Bartsch, H.-J. (2004): Taschenbuch mathematischer Formeln, 20th ed., Munich & Vienna.

[7] Beyer, W.H. (1976): Handbook of Tables for Probability and Statistics, Cleveland (Ohio).

[8] Biermann, B. (2002): Die Mathematik von Zinsinstrumenten, Munich.

[9] Biess, G. (1979): Graphentheorie, Frankfurt am Main.

[10] Bleymüller, J. & Gehlert, G. (2011): Statistische Formeln, Tabellen und Statistik-Software, 12th ed., Munich.

[11] Bleymüller, J. (2012): Statistik für Wirtschaftswissenschaftler, 16th ed., Munich.

[12] Bohley, P. (1998): Formeln, Rechenregeln, EDV und Tabellen zur Statistik, 6th ed., Munich & Vienna.

[13] Bomsdorf, E.; Gröhn, E.; Mosler, K. & Schmid, F. (2006): Definitionen, Formeln und Tabellen zur Statistik, 5th ed., Cologne.

[14] Bosch, K. (1998): Statistik-Taschenbuch, Munich & Vienna.

[15] Bosch, K. (2007): Finanzmathematik, Munich & Vienna.

[16] Brehmer, S. & Haar, H. (1972): Differentialformen und Vektor-analysis, Berlin.

[17] Bröcker, T. (1992): Analysis III, 1st ed., Zurich.

[18] Bronstein, I.N.; Semendjajew, K.A.; Musiol, G. & Mühlig, H. (2005): Taschenbuch der Mathematik, 6th ed., Frankfurt am Main.

[19] Bröse, K. & Schmetzke, R. (1985): Tabellen- und Formelsamm lung zur Finanz- und Lebensversicherungsmathematik, Karlsruhe.

[20] Bücker, R. (2002): Mathematik für Wirtschaftswissenschaftler, 6th ed., Munich.

[21] Bücker, R. (2003): Statistik für Wirtschaftswissenschaftler, 5th ed., Munich.

[22] Castillo, E.; Gutièrrez, J.M. & Hadi, A.S. (1997): Expert Systems and Probabilistic Network Models, New York.

[23] Clement, R. & Peren, F.W. (2017): Peren-Clement-Index: Bewertung von Direktinvestitionen durch eine simultane Erfassung von Makroebene und Unternehmensebene, Wiesbaden.

[24] Clement, R. & Peren, F.W. (2019): Peren-Clement Index – PCI 2.0. Evaluation of Foreign Direct Investments through Simultaneous Assessment at the Macro and Corporate Levels, Passau.

[25] Collatz, L. (1966): Numerical Treatment of Differential Equations, Berlin, Heidelberg & New York.

[26] Courant, R. (1972): Vorlesungen über Differential- und Integral-rechnung, Vol. 1 & 2, 4th ed., Berlin, Heidelberg & New York.

[27] Cox, D.R. & Wermuth, N. (1995): Multivariate Dependencies, London et al.

[28] Drabek, P. & Kufner, A. (1996): Integralgleichungen, Stuttgart & Leipzig.

[29] Eichholz, W. & Vilkner, E. (2013): Taschenbuch für Wirtschaftsmathematik, 6th updated ed., Munich.

[30] Engeln-Müllges, G. & Reutter, F. (1987): Formelsammlung zur Numerischen Mathematik mit C-Programmen, Mannheim, Vienna & Zurich.

[31] Fetzer, A. & Fränkel, H. (1995): Mathematik Lehrbuch für Fachhochschulen, Vol. 1 & 2, Berlin, Heidelberg & New York.

[32] Fischer, W. & Lieb, I. (2003): Funktionentheorie, 8th ed., Wiesbaden.

[33] Fisz, M. (1988): Wahrscheinlichkeitsrechnung und mathematische Statistik, Berlin.

[34] Görke, L. (1974): Mengen – Relationen – Funktionen, Frankfurt am Main.

[35] Gottwald, S. (1995): Mehrwertige Logik - Eine Einführung in Theorie und Anwendung, Munich.

[36] Grundmann, W. & Luderer, B. (2003): Formelsammlung – Finanzmathematik, Versicherungsmathematik, Wertpapieranalyse, 2nd ed., Stuttgart, Leipzig & Wiesbaden.

[37] Grundmann, W. & Luderer, B. (2009): Finanz- und Versicherungsmathematik, 3rd updated ed., Stuttgart, Leipzig & Wiesbaden.

[38] Hartung, J.; Elpelt, B. & Klösener, K.-H. (2005): Statistik – Lehr- und Handbuch der angewandten Statistik, 14th ed., Munich & Vienna.

[39] Hippmann, H.-D. (1995): Formelsammlung Statistik, Statistische Grundbegriffe, Formeln, Schaubilder und Tabellen, Stuttgart.

[40] Joos, G. & Richter, E.W. (1994): Höhere Mathematik für den Praktiker, Frankfurt am Main.

[41] Klingbeil, E. (1988): Variationsrechnung, Mannheim, Vienna & Zurich.

[42] Klingenberg, W. (1992): Lineare Algebra und Geometrie, Berlin, Heidelberg & New York.

[43] Knopp, K. (1996): Theorie und Anwendung der unendlichen Reihen, 6th ed., Berlin, Heidelberg & New York.

[44] Koller, S. (2012): Neue graphische Tafeln zur Beurteilung statistischer Zahlen, reprint of the 4th ed. from 1969, Darmstadt.

[45] Kosmol, P. (1992): Methoden zur numerischen Behandlung nichtlinearer Gleichungen und Optimierungsaufgaben, Leipzig.

[46] Lippe, P. von der (2001): Chain Indices, A Study in Price Index Theory, Stuttgart.

[47] Lippe, P. von der (2007): Index Theory and Price Statistics, Frankfurt am Main.

[48] Luderer, B.; Nollau, V. & Vetters, K. (2005): Mathematische Formeln für Wirtschaftswissenschaftler, 5th ed., Stuttgart & Leipzig.

[49] Pakusch, C.; Peren, F.W. & Shakoor, M.A. (2016): The PCI - A Global Risk Index for the Simultaneous Assessment of Macro and Company Individual Investment Risks. In: Journal of Business Strategies, 33(2), p. 154-173.

[50] Pakusch, C.; Peren, F.W. & Shakoor, M.A. (2018): Peren-Clement-Index – eine exemplarische Fallstudie. In: Gadatsch, A. et al. (Hrsg.): Nachhaltiges Wirtschaften im digitalen Zeitalter: Innovation - Steuerung - Compliance, Wiesbaden, p. 105-117.

[51] Papoulis, A. (1962): The Fourier Integral and its Applications, McGraw-Hill.

[52] Pearl, J. (1992): Probabilistic Reasoning in Intelligent Systems, 2nd ed., San Mateo.

[53] Peren, F.W. (1986): Einkommen, Konsum und Ersparnis der privaten Haushalte in der Bundesrepublik Deutschland seit 1970: Analyse unter Verwendung makroökonomischer Konsumfunktionen, Frankfurt am Main.

[54] Peren, F.W. & Clement, R. (2019): Peren-Clement Index – PCI 2.0: Evaluation of Foreign Direct Investments through Simulta neous Assessment at the Macro and Corporate Levels, Passau.

[55] Peren, F.W. (2020a): Formelsammlung Wirtschaftsmathematik, 3rd ed., Berlin & Heidelberg.

[56] Peren, F.W. (2020b): Formelsammlung Wirtschaftsstatistik, 3rd ed., Berlin & Heidelberg.

[57] Peren, F.W. (2021): Math for Business and Economics. Compendium of Essential Formulas, Berlin & Heidelberg.

[58] Rinne, H. (1982): Statistische Formelsammlung, Frankfurt am Main.

[59] Rinne, H. (2003): Taschenbuch der Statistik für Wirtschafts- und Sozialwissenschaften, 3rd ed., Frankfurt am Main.

[60] Sauerbier, T. & Voß, W. (2006): Kleine Formelsammlung Statistik, 3rd ed., Leipzig.

[61] Scheid, H. & Frommer, A. (2006): Zahlentheorie, 4th ed., Mannheim, Vienna & Zurich.

[62] Schneider, W.; Kornrumpf, J. & Mohr, W. (1995): Statistische Methodenlehre – Definitions- und Formelsammlung zur deskriptiven und induktiven Statistik mit Erläuterungen, Vienna & Munich.

[63] Schütte, K. (1966): Index mathematischer Tafelwerke und Tabellen, Munich.

[64] Schwarze, J. (2005): Grundlagen der Statistik – Band 1 und 2 – Neue Wirtschaftsbriefe, 10th ed., Herne & Berlin.

[65] Schwetlick, H. & Kretschmar, H. (1991): Numerische Verfahren für Naturwissenschaftler und Ingenieure, Leipzig.

[66] Sieber, H. (1995): Mathematische Begriffe und Formeln, Stuttgart.

[67] Spirtes, P.; Glymour, C. & Scheines, R. (1993): Causation, Prediction and Search – Lecture Notes in Statistics 81, New York.

[68] Stiefel, E. & Fässler, A. (1992): Gruppentheoretische Methoden und ihre Anwendungen, Basel.

[69] Toda, M. (1989): Nonlinear Waves and Solitons, Tokyo & Dordrecht.

[70] Vianelli, S. (1959): Prontuari per Calcoli Statistici, Bologna.

[71] Vogel, F. (2005): Beschreibende und schließende Statistik – Formeln, Definitionen, Erläuterungen, Stichwörter und Tabellen, Munich & Vienna.

[72] Voß, W. (2000): Taschenbuch der Statistik, Leipzig.

[73] Wechler, W. (1992): Universal Algebra for Computer Scientists, Berlin, Heidelberg & New York.

[74] Wobst, R. (1997): Methoden, Risiken und Nutzen der Datenverschlüsselung, Boston.

[75] Wolfram, S. (1999): Mathematica, 4th ed., Cambridge.

[76] Yosida, K. (2008): Functional Analysis, reprint of the 6th ed. from 1980, Berlin, Heidelberg & New York.

Index

F. W. Peren, *Statistics for Business and Economics*,
https://doi.org/10.1007/978-3-662-64276-4